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Development of new post hoc methods for structured data

Développement de nouvelles méthodes post hoc pour données structurées

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Titre: Développement de nouvelles méthodes post hoc pour données structurées
Mots clés: Tests multiples, méthodes post hoc, p-valeur, e-valeur, statistiques.

Résumé: Ce manuscrit présente mes contributions à la théorie des tests multiples. Il introduit de nouvelles méthodologies pour contrôler les fausses découvertes grâce à des bornes de confiance et des procédures de contrôle du False Discovery Rate (FDR). La plupart de ces procédures assurent un recouvrement "post hoc", ce qui signifie qu'elles préservent les garanties statistiques même après l'intervention du praticien — comme le p-hacking ou le HARKing — et offrent une plus grande flexibilité en lui permettant de sélectionner l'ensemble de rejets de manière adaptative. De plus, ces méthodes exploitent également la structure des données ; elles vont au-delà de l'hypothèse i.i.d. traditionnelle en prenant en compte l'hétérogénéité, l'échangeabilité ou la structure en arbre des données afin d'améliorer la puissance statistique.

Title: Development of new post hoc methods for structured data
Keywords: Multiple testing, post hoc methods, p-value, e-value, statistics.

Abstract: This manuscript presents my contributions to multiple testing theory. It introduces new methodologies for controlling false discoveries through confidence bounds and False Discovery Rate (FDR) controlling procedures. Most of these procedures achieve "post hoc" coverage, meaning that they preserve statistical guarantees even after practitioner intervention — such as p-hacking or HARKing — and offer greater flexibility by allowing practitioners to select the rejection set adaptively. Furthermore, these methods also exploit the data structure; they go beyond the traditional i.i.d. assumption by accounting for heterogeneity, exchangeability or tree-structured data to improve statistical power.

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Contents

1 Introduction (Français) 1

1.1 Motivations . . . 2
1.2 Tests d'hypothèses simples . . . 2
1.2.1 Modèle statistique . . . 2
1.2.2 p-valeurs et e-valeurs . . . 3
1.2.3 Tests d'hypothèses discrets et test exact de Fisher . . . 5
1.3 Théorie des tests multiples . . . 7
1.3.1 Modèle formel . . . 7
1.3.2 Family-wise error rate (FWER) et procédures de closed testing . . . 10
1.3.3 False discovery rate (FDR) et procédure de BH . . . 14
1.3.4 Enveloppes de confiance sur le nombre de fausses découvertes . . . 17
1.3.5 Prédiction conforme et tests multiples . . . 19
1.4 Contributions . . . 21
1.4.1 Enveloppes de confiance pour les fausses découvertes avec des données hétérogènes . . . 22
1.4.2 Contrôle conforme du FDR et bornes conformes sur le FDP pour la prédiction d'arêtes dans un graphe . . . 23
1.4.3 p-valeurs composées hétérogènes et closure principle . . . 23
1.4.4 Sur les enveloppes de confiance fortement cohérentes pour les fausses découvertes . . . 23
1.4.5 Contributions additionnelles . . . 24

2 Introduction (English) 25

2.1 Motivations . . . 26
2.2 Simple hypothesis testing . . . 26
2.2.1 Formal framework . . . 26
2.2.2 p-values and e-values . . . 27
2.2.3 Discrete hypothesis testing and Fisher's exact test . . . 29
2.3 Multiple testing framework . . . 31
2.3.1 Formal framework . . . 31
2.3.2 Family-wise error rate (FWER) and closed testing procedures . . . 33
2.3.3 False discovery rate (FDR) and BH procedure . . . 37
2.3.4 Confidence envelopes on the number of false discoveries . . . 39
2.3.5 Conformal Prediction and multiple testing . . . 42
2.4 Contributions . . . 44
2.4.1 Confidence envelopes for the false discoveries with heterogeneous data . . . 44
2.4.2 FDR control and FDP bounds for conformal link prediction . . . 45
2.4.3 Heterogeneous compound p-values and closure principle . . . 45

2.4.4 On strongly coherent confidence envelopes for the false discoveries 45
2.4.5 Additional contributions . . . 46

3 Confidence envelopes for the false discoveries with heterogeneous data 47
3.1 Introduction . . . 48
3.1.1 Overview of goals and motivations . . . 48
3.1.2 Confidence envelopes in the context of selective inference . . . 50
3.1.3 Contributions and chapter organization . . . 51
3.2 Multiple testing and confidence bound formalism . . . 52
3.2.1 Mathematical notation . . . 52
3.2.2 Classical multiple testing formalism . . . 52
3.2.3 Confidence bounds . . . 54
3.2.4 Heterogeneity . . . 55
3.3 Review and extension of existing theory . . . 56
3.3.1 Homogeneous concentration inequalities . . . 57
3.3.2 Confidence envelopes by the inversion procedure . . . 59
3.3.3 Reference families, JER and interpolation . . . 60
3.4 General shortcuts for the inversion procedure . . . 65
3.4.1 Shortcuts for the inversion procedure . . . 66
3.4.2 Applications to homogeneous inequalities . . . 67
3.5 Heterogeneous probabilistic inequalities and associated local test confidence envelopes . . . 70
3.5.1 Bretagnolle inequality and applications . . . 71
3.5.2 Heterogeneous Simes-like inequality . . . 73
3.6 Heterogeneous confidence envelopes using reference families . . . 77
3.6.1 Top-k setting . . . 77
3.6.2 Simultaneous k-FWER . . . 79
3.6.3 Deterministic setting . . . 81
3.7 Numerical experiments . . . 81
3.7.1 Simulation setting . . . 81
3.7.2 Analyses for top-k setting . . . 82
3.7.3 Analyses for k-FWER setting . . . 83
3.7.4 Comparison with other homogeneous bounds . . . 84
3.8 Conclusion and perspectives . . . 85

4 FDR control and FDP bounds for conformal link prediction 89
4.1 Introduction . . . 89
4.2 Conformal link prediction . . . 90
4.2.1 Model and notation . . . 90
4.3 Conformal link prediction procedure . . . 91
4.4 FDR and FDP control guarantees . . . 93
4.5 Discussion . . . 97

5 Heterogeneous compound p-values and closure principle 99
5.1 Introduction . . . 100
5.1.1 Background . . . 100
5.1.2 Contributions . . . 101
5.2 Preliminaries . . . 101
5.2.1 Model and definitions . . . 101
5.2.2 DBY procedure (Döhler, 2018) and Heyse procedure (Heyse, 2011) 102

5.2.3 Compound p-values, compound e-values and calibrators . . . 102
5.2.4 Compound p-values and multiple testing framework . . . 103
5.2.5 The e-Closure Principle (Xu et al., 2025) for false discovery rate control 104
5.3 Heterogeneous compound p-values and Heyse and DBY procedures . . 104
5.3.1 Heterogeneous locally compound p-values . . . 104
5.3.2 Applications to the heterogeneous multiple testing framework . . 105
5.4 Heterogeneity and the e-Closure Principle . . . 106
5.4.1 Local heterogeneous e-values for the e-Closure Principle . . . 106
5.4.2 Closed DBY procedure . . . 107
5.5 Conclusion . . . 108

6 On strongly coherent confidence envelopes for the false discoveries 111
6.1 Introduction . . . 112
6.1.1 Motivations . . . 112
6.1.2 Model and notations . . . 112
6.2 Literature review . . . 113
6.2.1 Interpolation devices from the JER literature . . . 113
6.2.2 Interpolation device from Goeman et al. (2021) . . . 114
6.2.3 Discussion on Xu et al. (2024a) procedure . . . 115
6.3 Interpolation operators and strong coherence . . . 116
6.3.1 Interpolation operators from the JER literature . . . 116
6.3.2 Strong coherence . . . 118
6.4 Local confidence bounds: a uniform improvement of Durand et al. (2020) methods . . . 119
6.4.1 Considering local confidence bounds in Durand et al. (2020) methods 119
6.4.2 Simulations . . . 121
6.5 Conclusion . . . 123

7 Conclusion and perspectives 125

A Supplementary material for Chapter 3 133
A.1 Van Zuijlen inequality and confidence envelope . . . 134
A.2 Bridging some gaps in the literature . . . 135
A.3 Complements on the top-k setting . . . 135
A.3.1 Top-k with ties . . . 135
A.3.2 Links between the top-k and k-FWER settings . . . 136
A.4 Uniformization of super-uniform p-values . . . 139
A.5 Complements on Katsevich and Ramdas (2020) and Meah et al. (2024) confidence envelopes . . . 141
A.6 Proofs of Chapter 3 . . . 143
A.6.1 Proof of Lemma 3.3.1 . . . 143
A.6.2 Proofs of Proposition 3.3.2 and Proposition 3.3.3 . . . 143
A.6.3 Proof of Proposition 3.3.6 . . . 143
A.6.4 Proof of Theorem 3.3.14 . . . 143
A.6.5 Proof of Proposition 3.3.18 . . . 144
A.6.6 Proof of Proposition 3.3.23 . . . 144
A.6.7 Proof of Lemma 3.4.1 . . . 145
A.6.8 Proof of Proposition 3.4.2 . . . 145
A.6.9 Proof of Proposition 3.4.3 . . . 146
A.6.10 Proof of Proposition 3.4.4 . . . 146

A.6.11 Proof of Lemma 3.4.5 . . . 147
A.6.12 Proof of Corollary 3.4.8 . . . 147
A.6.13 Proof of Proposition 3.4.9 . . . 147
A.6.14 Proof of Proposition 3.4.11 . . . 149
A.6.15 Proofs of Lemma 3.4.12 and Proposition 3.4.14 . . . 150
A.6.16 Proof of Proposition 3.5.1 . . . 151
A.6.17 Proof of Corollary 3.5.2 . . . 151
A.6.18 Proof of Lemma 3.5.4 . . . 151
A.6.19 Proof of Proposition 3.5.6 . . . 152
A.6.20 Proof of Lemma 3.5.12 . . . 152
A.6.21 Proof of Corollary 3.5.15 . . . 152
A.6.22 Proof of Lemma 3.5.17 . . . 153
A.6.23 Proof of Example 3.6.3 . . . 154
A.6.24 Proof of Proposition 3.6.5 . . . 155
A.6.25 Proof of Proposition 3.6.9 and Remark 3.6.10 . . . 155

B Supplementary material for Chapter 6 **157**
B.1 Deferred proofs from Chapter 6 . . . 157
B.1.1 Proof of Proposition 6.3.3 . . . 157
B.1.2 Proof of Proposition 6.2.5 . . . 158
B.1.3 Proof of Proposition 6.3.4 . . . 159
B.2 An example of a coherent confidence envelope that cannot be written as a closed testing procedure . . . 159

C A note on $\mathrm{BH}+$ procedure from Chen (2020) **161**
C.1 The $\mathrm{BH}+$ procedure . . . 161
C.2 Generalization of the $\mathrm{BH}+$ procedure . . . 162

D How to fission a Bernoulli variable? **165**

Chapter 1

Introduction (Français)

Contents

1.1 Motivations . . . 2
1.2 Tests d'hypothèses simples . . . 2
1.2.1 Modèle statistique . . . 2
1.2.2 p-valeurs et e-valeurs . . . 3
1.2.3 Tests d'hypothèses discrets et test exact de Fisher . . . 5
1.3 Théorie des tests multiples . . . 7
1.3.1 Modèle formel . . . 7
1.3.2 Family-wise error rate (FWER) et procédures de closed testing . . . 10
1.3.3 False discovery rate (FDR) et procédure de BH . . . 14
1.3.4 Enveloppes de confiance sur le nombre de fausses découvertes . . . 17
1.3.5 Prédiction conforme et tests multiples . . . 19
1.4 Contributions . . . 21
1.4.1 Enveloppes de confiance pour les fausses découvertes avec des données hétérogènes . . . 22
1.4.2 Contrôle conforme du FDR et bornes conformes sur le FDP pour la prédiction d'arêtes dans un graphe . . . 23
1.4.3 p-valeurs composées hétérogènes et closure principle . . . 23
1.4.4 Sur les enveloppes de confiance fortement cohérentes pour les fausses découvertes . . . 23
1.4.5 Contributions additionnelles . . . 24

Ce chapitre vise à fournir une introduction autonome à la théorie des tests multiples et décrit la plupart des concepts et outils de base qui seront utilisés dans la suite de cette thèse. Il est organisé comme suit : la Section 1.1 donne quelques motivations pour les tests multiples, puis la Section 1.2 décrit le cadre d'un test d'hypothèse simple ainsi que les notions de p-valeurs et de e-valeurs. La Section 1.3 introduit le cadre des tests multiples et les notions fondamentales de taux d'erreur issues de la littérature sur les tests multiples. Elle fournit également un bref aperçu des procédures classiques de tests multiples. Enfin, cette section présente deux thèmes plus récents : la construction d'enveloppes de confiance sur la proportion de fausses découvertes, et les liens entre les tests multiples et la prédiction conforme. Ce chapitre se conclut par la Section 1.4 qui donne un aperçu des contributions présentées les chapitres suivants de cette thèse.

1.1 Motivations

Les tests multiples interviennent dans de nombreux domaines scientifiques, tels que la génomique, les essais cliniques, les neurosciences, l'astrophysique ou les sciences sociales. Ces domaines impliquent souvent le test simultané d'un grand nombre d'hypothèses, ce qui peut entraîner un risque accru de fausses découvertes. Par exemple, si l'on souhaite déterminer quelles substances sont responsables d'une maladie particulière, on peut tester l'effet d'un grand nombre de substances et, *statistiquement*, certaines d'entre elles apparaîtront comme significatives par hasard, même si elles n'ont aucun effet réel. Nous décrivons brièvement ci-dessous certaines des applications mentionnées ci-dessus.

Génomique Un exemple typique de tests multiples en génomique est constitué par les *genome-wide association studies* (GWAS), qui forment un domaine très actif de la littérature en bio-informatique depuis les années 2000, à la suite du développement des technologies de génotypage. Elles visent à identifier des variants génétiques (SNP, pour *single nucleotide polymorphisms*) qui sont associés à un trait ou une maladie particulière (*par exemple* le diabète, les maladies auto-immunes ou la schizophrénie). Dans une étude GWAS, aucune hypothèse a priori n'étant spécifiée, les chercheurs testent des centaines de milliers, voire des millions de SNP pour étudier leur association avec le trait d'intérêt (Dudbridge and Gusnanto, 2008; Visscher et al., 2017).

Essais cliniques Les essais cliniques sont des études de recherche qui visent à évaluer la sécurité et l'efficacité de nouveaux traitements ou interventions médicales sur des sujets humains. Par exemple, Cui et al. (2021), Chapitre 11, décrivent un essai clinique qui vise à évaluer l'effet d'un nouveau médicament en fonction de sa dose. L'essai implique de tester plusieurs hypothèses correspondant à différentes doses du médicament, et l'objectif est d'identifier les doses qui sont efficaces tout en contrôlant le risque de fausses découvertes.

Crise de la réplicabilité Les tests multiples ont été sous-utilisés ou mal employés dans divers domaines avec des pratiques telles que le *p-hacking* ou le *HARKing* (Hypothesizing After the Results are Known), contribuant en partie à la crise de la réplicabilité (Ioannidis, 2005a; Camerer et al., 2018), c'est-à-dire l'incapacité à reproduire les résultats d'études publiées. Liu and Shiraito (2023) ont souligné que la plupart des études en sciences politiques ne corrigeaient pas les tests multiples alors qu'elles auraient dû le faire, et ont suggéré des méthodologies pour corriger la multiplicité des tests dans ce domaine.

1.2 Tests d'hypothèses simples

Cette section donne une introduction à la théorie des tests d'hypothèses simples et à deux statistiques fondamentales dans ce cadre : la p-valeur et la e-valeur.

1.2.1 Modèle statistique

Nous considérons un modèle statistique $(\mathfrak{F}, \mathcal{X})$ où $\mathfrak{F}$ est une famille de mesures de probabilité sur un espace mesurable $(\mathcal{X}, \Xi)$. Nous observons une variable aléatoire X,

définie sur un espace de probabilité abstrait $(\Omega, \Sigma, \mathbb{P})$, suivant une loi P_0 qui appartient à la famille de lois $\mathfrak{F}$. Un test d'hypothèse vise à décider si une hypothèse a priori sur la loi P_0 est raisonnable. Plus formellement, un test est une statistique binaire $\varphi(X) \in \{0, 1\}$ basée sur l'observation X. Il renvoie 1 si l'hypothèse a priori sur la loi des données, c'est-à-dire $P_0 \in H_0 \subseteq \mathfrak{F}$, est rejetée, et 0 sinon. L'ensemble H_0 est appelé l'hypothèse nulle et nous définissons un autre sous-ensemble de lois, H_1, appelé l'hypothèse alternative. Notons que l'hypothèse alternative n'est pas nécessairement le complémentaire de l'hypothèse nulle. Classiquement, deux types d'erreurs sont considérés dans ce cadre : l'erreur de type I, qui est le supremum des probabilités de rejeter l'hypothèse nulle alors qu'elle est vraie, et l'erreur de type II, qui est le supremum des probabilités d'accepter l'hypothèse nulle alors qu'elle est fausse. Formellement, l'erreur de type I d'un test φ est définie comme $\sup_{P \in H_0} \mathbb{E}_{X \sim P}\big[\varphi(X)\big] = \sup_{P \in H_0} \mathbb{P}_{X \sim P}\big(\varphi(X) = 1\big)$ et l'erreur de type II est définie comme $\sup_{P \in H_1} \mathbb{E}_{X \sim P}\big[1 - \varphi(X)\big] = \sup_{P \in H_1} \mathbb{P}_{X \sim P}\big(\varphi(X) = 0\big)$. Le test est dit être un test de niveau α si son erreur de type I est d'au plus α. Remarquons que si H_1 et H_0 ne sont pas disjoints, alors l'erreur de type II est d'au moins $1 - \alpha$, il est donc moins intéressant statistiquement de considérer de tels problèmes. Dans le cas gaussien unilatéral, $\mathfrak{F}$ est l'ensemble de toutes les lois gaussiennes de variance 1, c'est-à-dire $\mathfrak{F} = \big\{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\big\}$. Un problème de test courant dans ce cas consiste à tester $H_0 : \text{“ } \mu \leq 0 \text{ ”}$ contre $H_1 : \text{“ } \mu > 0 \text{ ”}$, c'est-à-dire à tester si la moyenne de la loi P_0 est positive ou non. Si nous définissons $\mathbf{\Phi}$ comme la fonction de répartition de la loi normale centrée réduite $\mathcal{N}(0, 1)$, alors le test $\varphi = \mathbb{1}\big\{X > \mathbf{\Phi}^{-1}(1 - \alpha)\big\}$, où $\mathbb{1}\{\cdot\}$ est la fonction indicatrice, est un test de niveau α pour ce problème.

1.2.2 p-valeurs et e-valeurs

Définir un test pour un niveau $\alpha \in [0, 1]$ peut être trop restrictif en pratique, car cela nécessite de spécifier le niveau de signification a priori. À la place, il est courant de définir une p-valeur $p(X)$, qui est une variable aléatoire à valeurs dans $[0, 1]$ *sur-uniforme* sous l'hypothèse nulle.

Définition 1.2.1 (Sur-uniformité)**.** *Soit T une variable aléatoire. On dit que T est sur-uniforme si pour tout $\alpha \in [0, 1]$, $\mathbb{P}(T \leq \alpha) \leq \alpha$.*

Nous définissons formellement une p-valeur comme suit :

Définition 1.2.2 (p-valeur)**.** *Soit $p : \mathcal{X} \to [0, 1]$ une statistique. On dit que p est une p-valeur pour l'hypothèse nulle H_0 si pour tout $P \in H_0$, $X \sim P$ implique que $p(X)$ est sur-uniforme.*

Ainsi, une p-valeur définit une famille de tests de niveau α par $\varphi_\alpha = \mathbb{1}\big\{p(X) \leq \alpha\big\}$ pour $\alpha \in [0, 1]$. Généralement, si H_0 est une hypothèse simple — c'est-à-dire $H_0 = \{P\}$ pour $P \in \mathfrak{F}$ — et que X prend ses valeurs dans $\mathbb{R}$, nous pouvons alors définir une p-valeur comme $p(X) = F(X)$, où F est la fonction de répartition de P. La fonction de répartition F de X sous P est définie par $F(x) = \mathbb{P}_{X \sim P}(X \leq x)$ pour tout $x \in \mathbb{R}$. Cette p-valeur présente un grand intérêt car elle est, en un sens, la “meilleure” p-valeur que l'on puisse définir pour ce problème de test. En effet, soit G une fonction croissante de $\mathbb{R}$ dans $[0, 1]$ telle que $G(X)$ soit sur-uniforme sous P. Alors, pour tout $x \in \mathbb{R}$, nous avons $\mathbb{P}_{X \sim P}(X \leq x) \leq \mathbb{P}_{X \sim P}\big(G(X) \leq G(x)\big)$ car G est croissante. Comme $G(X)$ est sur-uniforme sous P, nous avons $\mathbb{P}_{X \sim P}\big(G(X) \leq G(x)\big) \leq G(x)$. Ainsi, en combinant ces deux inégalités, nous avons $F(x) \leq G(x)$ pour tout $x \in \mathbb{R}$. Par conséquent, toute autre p-valeur $G(X)$ construite à partir d'une fonction croissante G est plus grande que $F(X)$.

De même, si G est décroissante et que $G(X)$ est sur-uniforme sous P, alors $G(x) \geq 1 - F(x)$ pour tout $x \in \mathbb{R}$. En effet, pour tout $x \in \mathbb{R}$, nous avons

$$1 - F(x) \leq \mathbb{P}_{X\sim P}(X \geq x) \leq \mathbb{P}_{X\sim P}\big(G(X) \leq G(x)\big) \leq G(x).$$

Pour en revenir au cas gaussien unilatéral, une p-valeur pour le problème de test H_0 : " $\mu \leq 0$ " contre H_1 : " $\mu > 0$ " est donnée par $p(X) = 1 - \mathbf{\Phi}(X)$. Notons que $p(X)$ est une p-valeur car pour tout $\alpha \in [0, 1]$,

$$\begin{aligned}
\sup_{P\in H_0} \mathbb{P}_{X\sim P}\big(p(X) \leq \alpha\big) &= \sup_{\mu\leq 0} \mathbb{P}_{X\sim\mathcal{N}(\mu,1)}\big(1 - \mathbf{\Phi}(X) \leq \alpha\big) \\
&= \sup_{\mu\leq 0} \mathbb{P}_{X\sim\mathcal{N}(\mu,1)}\big(X \geq \mathbf{\Phi}^{-1}(1-\alpha)\big) \\
&= \sup_{\mu\leq 0} \mathbb{P}_{X\sim\mathcal{N}(\mu,1)}\big(X - \mu \geq \mathbf{\Phi}^{-1}(1-\alpha) - \mu\big) \\
&= \sup_{\mu\leq 0} 1 - \mathbf{\Phi}\big(\mathbf{\Phi}^{-1}(1-\alpha) - \mu\big) \\
&= 1 - \mathbf{\Phi}\big(\mathbf{\Phi}^{-1}(1-\alpha)\big) = \alpha,
\end{aligned}$$

d'où pour tout $P \in H_0$, $\mathbb{P}_{X\sim P}\big(p(X) \leq \alpha\big) \leq \alpha$.

Un autre formalisme récent pour définir une famille de tests consiste à construire une e-valeur $e(X)$, qui est une variable aléatoire positive telle que

$$\sup_{P\in H_0} \mathbb{E}_{X\sim P}\big[e(X)\big] \leq 1.$$

Ainsi, une e-valeur définit une famille de tests via $\varphi_\alpha = \mathbb{1}\big\{e(X) \geq 1/\alpha\big\}$ pour $\alpha \in [0, 1]$ en utilisant l'inégalité de Markov. Remarquons que pour une e-valeur $e(X)$, la variable aléatoire $1/e(X) \wedge 1$ est une p-valeur. Cependant, il n'est pas vrai en général que pour une p-valeur $p(X)$, la variable aléatoire $1/p(X)$ soit une e-valeur — par exemple lorsque $p(X) \sim \mathcal{U}\big([0, 1]\big)$, car dans ce cas $1/p(X)$ a une espérance infinie. Une e-valeur classique, lors du test d'une hypothèse nulle simple $H_0 = \{P\}$ contre une hypothèse alternative simple $H_1 = \{Q\}$ telle que $Q \ll P$, est le rapport de vraisemblance $e(X) = \frac{\mathrm{d}Q}{\mathrm{d}P}(X)$, qui est la dérivée de Radon-Nikodym de Q par rapport à P. En effet, nous avons $\mathbb{E}_{X\sim P}\big[e(X)\big] = \mathbb{E}_{X\sim P}\big[\frac{\mathrm{d}Q}{\mathrm{d}P}(X)\big] = \int \frac{\mathrm{d}Q}{\mathrm{d}P}(x)\mathrm{d}P(x) = Q(\mathcal{X}) = 1$. À partir de cet exemple simple, nous pouvons construire une e-valeur pour le cas gaussien unilatéral, donnée par $e(X) = \exp(\mu X - \mu^2/2)$, qui est le rapport de vraisemblance entre la loi normale centrée réduite et la loi normale de moyenne $\mu > 0$ et de variance 1. En effet, pour tout $\nu \leq 0$, nous avons

$$\begin{aligned}
\mathbb{E}_{X\sim\mathcal{N}(\nu,1)}\big[e(X)\big] &= \mathbb{E}_{X\sim\mathcal{N}(\nu,1)}\Big[\exp\big(\mu X - \mu^2/2\big)\Big] \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\big(\mu x - \mu^2/2\big) \exp\big(-(x-\nu)^2/2\big) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\big(\mu x - \mu^2/2 - x^2/2 + \nu x - \nu^2/2\big) dx \\
&= \frac{\exp(\mu\nu)}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\Big(-\big(x - (\mu+\nu)\big)^2/2\Big) dx \\
&= \exp(\mu\nu) \leq 1.
\end{aligned}$$

Remarquons que la e-valeur $e(X)$ est une variable aléatoire positive, et que la p-valeur $1/e(X) \wedge 1$ est strictement sur-uniforme lorsque $\nu \leq 0$. C'est un phénomène général

: comme expliqué précédemment, si $p(X) \sim \mathcal{U}\big([0,1]\big)$, alors $1/p(X)$ n'est pas une e-valeur, et ainsi, pour une e-valeur $e(X)$, la p-valeur $1/e(X) \wedge 1$ est strictement sur-uniforme, faute de quoi $e(X)$ ne serait pas une e-valeur. Par conséquent, la famille de tests définie par la e-valeur $e(X)$, $\varphi_\alpha = \mathbb{1}\big\{e(X) \geq 1/\alpha\big\}$, est telle que pour tout $\alpha \in [0,1]$, φ_α est un test de niveau α, mais son erreur de type I est strictement inférieure à α. Il en résulte que cette famille de tests sera toujours plus conservative qu'une famille de tests définie par une p-valeur particulière, car ces dernières ont la possibilité d'être exactement uniformes sous l'hypothèse nulle. À ce stade, la justification des e-valeurs pour tester une unique hypothèse nulle peut sembler contestable, mais elles possèdent certaines propriétés pratiques qui les rendent intéressantes, en particulier dans le cadre des tests multiples. L'une de ces propriétés est que l'ensemble des e-valeurs est équivalent à l'ensemble des p-valeurs post-hoc (Koning, 2025) via la bijection $e \mapsto 1/e \wedge 1$. Ce sous-ensemble de p-valeurs possède la propriété que pour toute variable aléatoire $\hat{\alpha}$ telle que $\hat{\alpha} \in (0,1]$, nous avons

$$\mathbb{E}_{X\sim P}\left[\frac{\mathbb{1}\big\{p(X) \leq \hat{\alpha}\big\}}{\hat{\alpha}}\right] \leq 1, \quad \forall P \in H_0, \tag{1.1}$$

ou de manière équivalente,

$$\mathbb{E}_{X\sim P}\left[\sup_{\alpha\in(0,1]} \frac{\mathbb{1}\big\{p(X) \leq \alpha\big\}}{\alpha}\right] \leq 1, \quad \forall P \in H_0. \tag{1.2}$$

Nous verrons dans la Section 1.3.3 que cette propriété s'étend au cadre des tests multiples, en particulier pour les méthodes de contrôle du FDR basées sur les e-valeurs.

1.2.3 Tests d'hypothèses discrets et test exact de Fisher

Les méthodes statistiques pour données discrètes sont utilisées lorsque la variable réponse est constituée de comptages ou de résultats catégoriels plutôt que de mesures continues. Les exemples typiques incluent les résultats binaires (succès/échec), les tables de contingence ou les données de comptage. Dans ces contextes, les tests classiques fondés sur des hypothèses de normalité sont souvent inadaptés, et les méthodes spécifiquement conçues pour les lois discrètes sont préférées. Par conséquent, les p-valeurs dans ce cadre sont souvent discrètes, ce qui peut conduire à des tests conservateurs. Nous illustrerons ce fait avec l'une des procédures les plus largement utilisées, le *test exact de Fisher* (Fisher, 1934), qui évalue l'association entre deux variables catégorielles dans une table de contingence 2×2.

Nous introduisons d'abord le cadre formel du test exact de Fisher, puis nous discutons du problème de la discrétisation dans ce contexte. Nous suivons ici le formalisme de Agresti (2013) dans un *modèle multinomial*. Soient X et Y deux variables catégorielles, où X possède $I \in \mathbb{N}$ catégories et Y possède $J \in \mathbb{N}$ catégories. Ainsi X est une variable aléatoire telle que $X \in \{1,\dots,I\}$ et Y est une variable aléatoire telle que $Y \in \{1,\dots,J\}$. Les observations sont un ensemble de n couples $\big(X^\ell, Y^\ell\big)_{\ell\in\{1,\dots,n\}}$, ou de manière équivalente un ensemble de n matrices binaires de même loi $Z^1,\dots,Z^n \sim P$; pour tout $\ell \in \{1,\dots,n\}$ $Z^\ell = \big(Z^\ell_{ij}\big)_{\substack{i\in\{1,\dots,I\}\\ j\in\{1,\dots,J\}}}$ et nous avons pour tout $i \in \{1,\dots,I\}$ et $j \in \{1,\dots,J\}$,

$$Z^\ell_{ij} = \begin{cases} 1 \text{ si } (X_\ell, Y_\ell) = (i,j) \\ 0 \text{ sinon} \end{cases}.$$

Généralement, la quantité observée est la matrice de contingence $T = \sum_{\ell=1}^{n} Z^{\ell}$. Nous définissons les quantités suivantes pour tout $i \in \{1, \dots, I\}$ et $j \in \{1, \dots, J\}$:

$$\pi_{ij} = \mathbb{P}(Y = j, X = i),\ \pi_{i+} = \mathbb{P}(X = i) = \sum_{j=1}^{J} \pi_{ij},\ \pi_{+j} = \mathbb{P}(Y = j) = \sum_{i=1}^{I} \pi_{ij},$$

$$n_{ij} = \sum_{\ell=1}^{n} Z_{ij}^{\ell},\ n_{i+} = \sum_{j=1}^{J} n_{ij},\ n_{+j} = \sum_{i=1}^{I} n_{ij},\ n = \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij}.$$

Les variables n_{ij} représentent les effectifs des observations dans chaque cellule de la table de contingence, et les variables n_{i+} et n_{+j} représentent respectivement les totaux des lignes et des colonnes — également appelés effectifs marginaux. Ceci est représenté dans la table de contingence suivante :

	$Y = 1$	$Y = 2$	Total
$X = 1$	n_{11}	n_{12}	n_{1+}
$X = 2$	n_{21}	n_{22}	n_{2+}
Total	n_{+1}	n_{+2}	n

Par souci de simplicité, nous supposons ici que $I = 2 = J$ puisqu'il s'agit du cadre du test exact de Fisher, mais les résultats peuvent être étendus au cas général.

Nous supposons que les données suivent un *un modèle multinomial par lignes*, c'est-à-dire que pour tout $i \in \{1, 2\}$, n_{i+} est fixé, et le couple (n_{i1}, n_{i2}) suit une loi multinomiale de paramètres n_{i+} et $(\pi_{i1}/\pi_{i+}, \pi_{i2}/\pi_{i+})$, ce qui est équivalent à dire que $n_{i1} \sim \mathcal{B}(n_{i+}, \pi_{i1}/\pi_{i+})$ et $n_{i2} = n_{i+} - n_{i1}$, où $\mathcal{B}(n, p)$ est la loi binomiale de paramètres n et p. De plus, cette hypothèse implique que n_{11} et n_{21} sont indépendants. Ensuite, nous souhaitons tester l'hypothèse nulle d'indépendance entre X et Y. Dans ce cas, H_0 correspond à l'ensemble des distributions du modèle multinomial par lignes tels que, pour tout $i \in \{1, 2\}$ et $j \in \{1, 2\}$, $\pi_{ij} = \pi_{i+}\pi_{+j}$, et l'hypothèse alternative est l'ensemble des distributions du modèle ne vérifiant pas ces conditions, c'est-à-dire où X et Y ne sont pas indépendants.

Soit $P \in H_0$, nous raisonnons conditionnellement à l'effectif n_{+1}. Remarquons que $n_{+1} + n_{+2} = n$, et que la loi de n_{11} est suffisante pour déterminer la loi de Z sous l'hypothèse nulle, car elle fixe les valeurs de n_{12}, n_{21} et n_{22} étant donné que n_{1+}, n_{2+} et n_{+1} sont fixés. Soient $k \in \{0, \dots, n_{1+}\}$ et $a \in \{0, \dots, n\}$, la probabilité d'observer $n_{11} = k$ sous l'hypothèse nulle est :

$$\begin{aligned}
\mathbb{P}(n_{11} = k \mid n_{+1} = a) =& \frac{\mathbb{P}(n_{11} = k, n_{+1} = a)}{\mathbb{P}(n_{+1} = a)} \\
=& \frac{\mathbb{P}(n_{11} = k, n_{21} = a - k)}{\mathbb{P}(n_{+1} = a)} = \frac{\mathbb{P}(n_{11} = k)\mathbb{P}(n_{21} = a - k)}{\mathbb{P}(n_{+1} = a)} \\
=& \frac{\binom{n_{1+}}{k}\left(\frac{\pi_{11}}{\pi_{1+}}\right)^{k}\left(1 - \frac{\pi_{11}}{\pi_{1+}}\right)^{n_{1+}-k}\binom{n_{2+}}{a-k}\left(\frac{\pi_{21}}{\pi_{2+}}\right)^{a-k}\left(1 - \frac{\pi_{21}}{\pi_{2+}}\right)^{n_{2+}-(a-k)}}{\binom{n}{a}(\pi_{+1})^{a}(1 - \pi_{+1})^{n-a}} \qquad (1.3) \\
=& \frac{\binom{n_{1+}}{k}\binom{n_{2+}}{a-k}}{\binom{n}{a}} = \frac{\binom{n_{1+}}{k}\binom{n-n_{1+}}{a-k}}{\binom{n}{a}},
\end{aligned}$$

où (1.3) vient du fait que $n_{+1} = n_{11} + n_{12}$ suit une loi binomiale de paramètres n et π_{+1}.

Ainsi, conditionnellement à $(n_{+1} = a)$, la probabilité d'observer la table

	$Y = 1$	$Y = 2$	Total
$X = 1$	k	$n_{1+} - k$	n_{1+}
$X = 2$	$a - k$	$n_{2+} - (a - k)$	n_{2+}
Total	a	$n - a$	n

sous H_0 est donnée par la loi hypergéométrique de paramètres n, n_{1+} et a. Notons que sous d'autres hypothèses sur la distribution des données de contingence, telles que l'*échantillonnage multinomial* ou l'*échantillonnage de Poisson*, la loi de n_{11} conditionnellement aux marginales reste hypergéométrique (voir la Section 3.5.1 de Agresti, 2013). La p-valeur unilatérale est alors obtenue en sommant les probabilités de toutes les tables qui sont au moins aussi extrêmes que celle observée — disons $k \in \{0, 1, \dots, n\}$ — c'est-à-dire $p = \mathbb{P}(n_{11} \leq k \mid n_{1+}, n_{2+}, n)$. Pour obtenir la p-valeur bilatérale, nous sommons les probabilités de toutes les tables qui sont au moins aussi extrêmes que celle observée dans les deux directions, c'est-à-dire

$$p(k) = \sum_{\substack{i \in \{0,\dots,n\} \text{tels que} \\ \mathbb{P}(n_{11}=i \mid n_{+1}=a) \leq \mathbb{P}(n_{11}=k \mid n_{+1}=a)}} \mathbb{P}(n_{11} = i \mid n_{+1} = a).$$

Enfin, remarquons que les p-valeurs obtenues à partir du test exact de Fisher sont discrètes, et que leur support peut changer en fonction des valeurs des marginales. Cette observation est importante dans le cadre des tests multiples, où nous pouvons vouloir appliquer une procédure de tests multiples à un ensemble de p-valeurs obtenues à partir du test exact de Fisher avec différents marginales. Ce cadre particulier, où les lois marginales des p-valeurs varient sous l'hypothèse nulle, est appelé *tests multiples hétérogènes*. Le test exact de Fisher sera utilisé par exemple dans les simulations du Chapitre 3 et du Chapitre 6.

L'aspect discret des données peut conduire à des p-valeurs conservatrices, ce qui peut réduire la puissance du test, en particulier lorsque le niveau α du test n'appartient pas au support de la p-valeur. Pour remédier à ce problème, diverses méthodes ont été proposées pour améliorer la puissance des tests discrets, telles que les techniques de randomisation. Nous donnons un exemple de technique de randomisation dans l'Appendice A.4. Cependant, ces méthodes peuvent être inadaptées dans certains contextes, en particulier lorsque le praticien souhaite obtenir des résultats reproductibles. Ce fait motive la prise en compte du cadre des tests multiples hétérogènes car il permet des procédures de test plus puissantes sans recourir à une randomisation externe.

1.3 Théorie des tests multiples

Cette section donne une introduction à la théorie classique des tests multiples.

1.3.1 Modèle formel

Dans le cadre des tests multiples, nous considérons le même modèle statistique $(\mathfrak{F}, \mathcal{X})$ que dans Section 1.2.1, mais nous nous intéressons au test de $m \in \mathbb{N}$ hypothèses nulles $H_1, \dots, H_m \subseteq \mathfrak{F}$. Pour tout $P \in \mathfrak{F}$, nous notons $\mathcal{H}_0(P) = \{i \in [\![1, m]\!] : P \in H_i\}$ l'ensemble des hypothèses nulles vérifiées par P et $\mathcal{H}_1(P) = \mathcal{H}_0^c$ l'ensemble des hypothèses nulles qui ne sont pas vérifiées par P, où S^c désigne le complémentaire de

S dans $[\![1, m]\!]$ pour tout $S \subseteq [\![1, m]\!]$. Nous définissons également la notion d'*hypothèse d'intersection* H_S comme

$$H_S = \bigcap_{i \in S} H_i. \tag{1.4}$$

Un cas particulier d'hypothèse d'intersection est l'*hypothèse nulle globale*

$$H_{[\![1,m]\!]} = \bigcap_{i \in [\![1,m]\!]} H_i,$$

et nous disons que nous sommes *sous la nulle globale* si nous considérons que $P \in H_{[\![1,m]\!]}$. Notons que $H_{[\![1,m]\!]}$ peut être vide, mais dans la plupart des situations pratiques, ce n'est pas le cas. Par exemple, dans le cadre gaussien multivarié, $X = (X_1, \ldots, X_m)$ est un vecteur gaussien de dimension m et nous souhaitons tester les hypothèses nulles H_i : " $X_i \sim \mathcal{N}(\mu_i, 1), \mu_i \leq 0$ " pour tout $i \in [\![1, m]\!]$. Dans ce cas, l'hypothèse nulle globale vérifie

$$\left\{ \bigotimes_{1 \leq i \leq m} \mathcal{N}(\mu_i, 1) : \mu_1, \ldots, \mu_m \leq 0 \right\} \subseteq H_{[\![1,m]\!]}.$$

Une procédure de tests multiples R est une fonction qui prend en entrée l'observation X et renvoie un sous-ensemble de $[\![1, m]\!]$, que l'on appelle l'ensemble des hypothèses nulles rejetées. Nous cherchons à contrôler le nombre de fausses découvertes, c'est-à-dire le nombre de vraies hypothèses nulles qui sont rejetées à tort. Pour une loi P fixée, nous définissons le nombre de fausses découvertes d'une procédure R par

$$V\big(R(X)\big) = |\mathcal{H}_0(P) \cap R(X)|, \tag{1.5}$$

ainsi la fonction V dépend de P à travers l'ensemble $\mathcal{H}_0(P)$. L'objectif principal des tests multiples est de contrôler le nombre de fausses découvertes dues à la multiplicité des tests ; comme nous l'illustrons avec l'exemple suivant : si nous testons de manière indépendante m vraies hypothèses nulles au niveau α, la probabilité de rejeter au moins une vraie hypothèse nulle est de $1 - (1 - \alpha)^m$, ce qui est bien plus grand que α lorsque m est grand, puisque $1 - (1 - \alpha)^m \to 1$ quand $m \to \infty$. Il est donc nécessaire d'utiliser des métriques d'erreur alternatives pour contrôler le nombre de fausses découvertes.

Dans la suite, selon le contexte, nous considérerons soit des p-valeurs, soit des e-valeurs pour chaque hypothèse nulle. C'est-à-dire que nous considérerons m p-valeurs $p_1(X), \ldots, p_m(X)$ telles que pour tout $\alpha \in [0, 1]$,

$$\sup_{P \in H_i} \mathbb{P}_{X \sim P}\big(p_i(X) \leq \alpha\big) \leq \alpha, \quad i \in [\![1, m]\!], \tag{1.6}$$

ou m e-valeurs $e_1(X), \ldots, e_m(X)$ telles que

$$\sup_{P \in H_i} \mathbb{E}_{X \sim P}\big[e_i(X)\big] \leq 1, \quad i \in [\![1, m]\!]. \tag{1.7}$$

Nous considérerons, dans la suite, certaines structures de dépendance entre la famille de p-valeurs ou de e-valeurs. Nous commençons par deux hypothèses de dépendance classiques sur les p-valeurs, qui sont l'hypothèse d'indépendance et l'hypothèse wPRDS (*weakly Positive Regression Dependence on a Subset*). L'hypothèse d'indépendance énonce que les p-valeurs correspondant aux vraies hypothèses nulles — c'est-à-dire les p-valeurs $p_i(X)$ pour $i \in \mathcal{H}_0(P)$ — sont des variables aléatoires indépendantes, et que ces p-valeurs sont indépendantes des p-valeurs qui correspondent aux fausses hypothèses nulles :

Hypothèse 1.1 (Indépendance). *Pour tout $P \in \mathfrak{F}$, $(p_i)_{i\in\mathcal{H}_0(P)}$ est un vecteur de variables aléatoires indépendantes et est indépendant de $(p_i)_{i\in\mathcal{H}_1(P)}$.*

Cette hypothèse est très forte et peut ne pas être vérifiée en pratique. Une autre hypothèse classique est l'hypothèse wPRDS (Finner et al., 2009; Roquain, 2011), qui énonce que les p-valeurs sont positivement dépendantes en un certain sens.

Hypothèse 1.2 (wPRDS). *Soit $D \subseteq [0,1]^m$. On dit que D est un ensemble croissant si pour tout $u \in D$, pour tout $v \in [0,1]^m$ tel que $v \geq u$ composante par composante, on a $v \in D$. On dit que les p-valeurs $(p_i)_{i\in[\![1,m]\!]}$ sont wPRDS si elles satisfont la condition suivante : pour tout $P \in \mathfrak{F}$, pour tout $i \in \mathcal{H}_0(P)$, pour tout ensemble mesurable croissant $D \subseteq [0,1]^m$, la fonction*

$$u \mapsto \mathbb{P}_{X\sim P}\big((p_j)_{j\in[\![1,m]\!]} \in D \mid p_i \leq u\big)$$

est croissante sur l'ensemble $\big\{u : 0 \leq u \leq 1, \mathbb{P}_{X\sim P}(p_i \leq u) > 0\big\}$.

Cette dernière hypothèse est plus faible que l'hypothèse d'indépendance, ce qui signifie que si les p-valeurs satisfont l'Hypothèse 1.1, alors elles satisfont également l'Hypothèse 1.2, mais la réciproque n'est pas vraie en general. Par exemple, dans le cas gaussien multivarié, si $X = (X_1, \ldots, X_m) \sim \mathcal{N}(\mu, \Sigma)$ avec $\Sigma_{i,j} \geq 0$ pour tous $i, j \in [\![1, m]\!]$, alors les p-valeurs $p_i(X) = 1 - \mathbf{\Phi}\big(X_i/\sqrt{\Sigma_{i,i}}\big)$ satisfont l'Hypothèse 1.2 mais pas nécessairement Hypothèse 1.1 (voir Giraud, 2015, Exercice 10.6.3). Notons qu'historiquement, une autre hypothèse, appelée dépendance par régression positive sur un sous-ensemble (PRDS), a été introduite parBenjamini and Yekutieli (2001), où le conditionnement "$p_i \leq u$" est remplacé par "$p_i = u$" dans l'Hypothèse 1.2. Cependant, bien que l'hypothèse PRDS soit plus forte que l'hypothèse wPRDS (voir la preuve de la Proposition 3.6 dans Blanchard and Roquain, 2008 par exemple), l'hypothèse wPRDS est une condition suffisante pour le contrôle du FDR de la procédure de BH, comme nous le verrons dans la Section 2.3.3. Supposer que l'une de ces deux hypothèses est vérifiée revient à restreindre la famille de lois $\mathfrak{F}$ à un sous-modèle $\mathfrak{F}' \subseteq \mathfrak{F}$ tel que pour tout $P \in \mathfrak{F}'$,les p-valeurs satisfont l'Hypothèse 1.1 ou l'Hypothèse 1.2. Dans le cas du modèle gaussien unilatéral multivarié, supposer l'Hypothèse 1.1 implique que

$$H_{[\![1,m]\!]} = \left\{ \bigotimes_{1\leq i\leq m} \mathcal{N}(\mu_i, 1) : \mu_1, \ldots, \mu_m \leq 0 \right\},$$

par exemple. Contrairement au cadre des tests d'hypothèses simples, où l'erreur de type I est la métrique d'erreur naturelle à contrôler, dans le cadre des tests multiples, il n'y a pas de métrique d'erreur naturelle à contrôler. Devrions-nous contrôler la probabilité de rejeter au moins une vraie hypothèse nulle, l'espérance de la proportion de vraies hypothèses nulles rejetées, ou autre chose ?

Il sera parfois pratique de trier les p-valeurs par ordre croissant, afin de déterminer quelles sont les hypothèses nulles à rejeter. À cette fin, nous notons les statistiques d'ordre des p-valeurs $p_1, \ldots, p_m$ par $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$.

Dans la suite, nous introduisons les métriques d'erreur les plus classiquement utilisées dans la littérature des tests multiples, et nous donnons quelques exemples de procédures contrôlant ces métriques d'erreur. Ensuite, nous introduirons la notion d'enveloppe de confiance, qui est une généralisation des procédures de contrôle du FWER, et nous donnerons une zoologie des enveloppes de confiance les plus utilisées. Enfin, nous introduirons le cadre de la prédiction conforme et fournirons quelques liens de cette théorie avec la théorie des tests multiples.

1.3.2 Family-wise error rate (FWER) et procédures de closed testing

Le Family-Wise Error Rate (FWER) est la métrique d'erreur la plus classique à contrôler dans le cadre des tests multiples. Sa définition formelle peut être remontée à Tukey (1953) (Benjamini and Braun, 2002), mais le problème de la réalisation de tests multiples simultanés avait déjà été repéré par Bonferroni (1936), par exemple. Nous définissons le FWER d'une procédure R comme la probabilité de rejeter au moins une vraie hypothèse nulle, c'est-à-dire que, pour tout $P \in \mathfrak{F}$,

$$\mathrm{FWER}_P(R) = \mathbb{P}_{X\sim P}\big(\exists i \in \mathcal{H}_0 : i \in R(X)\big) = \mathbb{P}_{X\sim P}\Big(V\big(R(X)\big) > 0\Big). \tag{1.8}$$

Nous disons qu'une procédure R *contrôle fortement* le FWER au niveau $\alpha \in [0,1]$ si pour tout $P \in \mathfrak{F}$, $\mathrm{FWER}_P(R) \leq \alpha$, et que R *contrôle faiblement* le FWER au niveau α si $\mathrm{FWER}_P(R) \leq \alpha$ pour tout $P \in H_{[\![1,m]\!]}$, ou, de manière équivalente,

$$\mathbb{P}_{X\sim P}\Big(R(X) \neq \varnothing\Big) \leq \alpha \text{ pour tout } P \in H_{[\![1,m]\!]}.$$

Il existe plusieurs procédures contrôlant le FWER fortement; la plus classique est la procédure de Bonferroni, qui rejette toutes les hypothèses nulles H_i telles que la p-valeur correspondante $p_i(X)$ est inférieure à α/m pour un niveau de signification $\alpha \in [0,1]$ donné. La procédure de Bonferroni contrôle le FWER au niveau α, sans aucune hypothèse sur la structure de dépendance des p-valeurs. Néanmoins, il est connu que la procédure de Bonferroni est conservatrice. Elle peut être améliorée en utilisant les procédures de closed testing introduites par Marcus et al. (1976), et largement étudiées dans la littérature sur les tests multiples (Sonnemann and Finner, 1988; Sonnemann, 1982; Sonnemann, 2008; Romano et al., 2011). Pour définir une procédure de closed testing, nous devons introduire la notion de *tests locaux*, qui est une famille de tests $\varphi = (\varphi_S)_{S\subseteq[\![1,m]\!]}$ associée à la famille des hypothèses d'intersection H_S. Ensuite, la méthode du closed testing définit une nouvelle famille de tests $\psi = (\psi_i)_{i\in[\![1,m]\!]}$, appelés *tests locaux effectifs*, comme suit :

$$\forall S \subseteq [\![1,m]\!], \psi_S(X) = \min_{S\subseteq T\subseteq[\![1,m]\!]} \varphi_T(X). \tag{1.9}$$

Marcus et al. (1976) a montré que, si la famille de tests φ est telle que pour tout $P \in \mathfrak{F}$, $\varphi_{\mathcal{H}_0(P)}$ est un test de niveau α, alors la famille de tests ψ satisfait la propriété suivante :

$$\forall P \in \mathfrak{F}, \mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big) \leq \alpha. \tag{1.10}$$

Notons que (1.10) est vérifiée lorsque φ est une famille de tests locaux de niveau α — c'est-à-dire que, pour tout $S \subseteq [\![1,m]\!]$, φ_S est un test de niveau α pour l'hypothèse d'intersection H_S — puisque cela implique que pour tout $P \in \mathfrak{F}$, $\mathbb{P}_{X\sim P}\big(\varphi_{\mathcal{H}_0(P)}(X) = 1\big) \leq \alpha$. La preuve de (1.10) est directe : s'il existe $S \subseteq \mathcal{H}_0(P)$ tel que $\psi_S(X) = 1$, alors $\varphi_{\mathcal{H}_0(P)}(X) = 1$, sinon cela contredirait le fait que $\psi_S(X) = 1$. Ainsi, nous avons, pour tout $P \in \mathfrak{F}$, $\mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big) \leq \mathbb{P}_{X\sim P}\big(\varphi_{\mathcal{H}_0(P)}(X) = 1\big) \leq \alpha$. Notons qu'ici, nous avons déjà considéré le formalisme adapté à la construction d'enveloppes de confiance sur le nombre de fausses découvertes décrite par Goeman et al. (2021) et introduite pour la première fois par Goeman and Solari (2011), mais la procédure du closed testing pour le contrôle fort du FWER est un cas particulier de cette construction, où nous ne considérons que les tests locaux ψ_S pour $S \subseteq [\![1,m]\!]$ tels que $|S| = 1$. En effet, si nous notons la procédure de tests multiples

$$R_\varphi(X) = \big\{i \in [\![1,m]\!] : \psi_{\{i\}}(X) = 1\big\}, \tag{1.11}$$

nous avons

$$\begin{aligned}\sup_{P\in\mathfrak{F}} \mathrm{FWER}_P\left(R_\varphi(X)\right) &= \sup_{P\in\mathfrak{F}} \mathbb{P}_{X\sim P}\big(\exists i \in \mathcal{H}_0(P) : \psi_{\{i\}}(X) = 1\big)\\ &\leq \sup_{P\in\mathfrak{F}} \mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big)\\ &\leq \alpha,\end{aligned}$$

où la dernière inégalité découle de (1.10).

Soit $\alpha \in (0,1)$, le closed testing permet d'améliorer la procédure de Bonferroni au niveau α en utilisant les tests locaux φ_α définis par :

$$\forall S \subseteq [\![1,m]\!], \varphi_{\alpha,S} = \mathbb{1}\left\{\min_{i\in S} p_i \leq \frac{\alpha}{|S|}\right\}. \tag{1.12}$$

Ainsi, φ_α est une famille de tests locaux de niveau α puisque pour $S \subseteq [\![1,m]\!]$, pour tout $P \in H_S$,

$$\begin{aligned}\mathbb{P}_{X\sim P}(\varphi_{\alpha,S}(X) = 1) &= \mathbb{P}_{X\sim P}\left(\inf_{i\in S} p_i(X) \leq \frac{\alpha}{|S|}\right)\\ &= \mathbb{P}_{X\sim P}\left(\exists i \in S : p_i(X) \leq \frac{\alpha}{|S|}\right)\\ &\leq \sum_{i\in S} \mathbb{P}_{X\sim P}\left(p_i(X) \leq \frac{\alpha}{|S|}\right) \qquad (1.13)\\ &\leq \sum_{i\in S} \frac{\alpha}{|S|} = \alpha, \qquad (1.14)\end{aligned}$$

où (1.13) provient de l'inégalité de Boole et (1.14) est vérifiée car $S \subseteq \mathcal{H}_0(P)$ et les p-valeurs sous l'hypothèse nulle sont sur-uniformes. Ensuite, la procédure de closed testing définit des tests locaux effectifs $\psi_\alpha = (\psi_{\alpha,S})_{S\subseteq[\![1,m]\!]}$ par (1.9) avec $\varphi = (\varphi_{\alpha,S})_{S\subseteq[\![1,m]\!]}$. Il reste à expliciter la procédure de tests multiples $R_{\varphi_\alpha}(X) = \big\{i \in [\![1,m]\!] : \psi_{\alpha,\{i\}}(X) = 1\big\}$ pour ce choix de tests locaux. La proposition suivante énonce que la procédure R_{φ_α} est la procédure de Holm (Holm, 1979) :

Proposition 1.3.1. *La procédure de tests multiples R_{φ_α} est la procédure de Holm au niveau α, c'est-à-dire*

$$R_{\varphi_\alpha}(X) = R_\alpha^{\mathrm{Holm}} := \left\{i \in [\![1,m]\!] : p_i(X) \leq \frac{\alpha}{m-k^*+1}\right\}, \tag{1.15}$$

où $k^* = \max\big\{k \in [\![1,m]\!] : \forall i \leq k, p_{(i)}(X) \leq \frac{\alpha}{m-i+1}\big\} \cup \{0\}$.

Nous présentons ci-dessous la preuve de cette proposition.

Proof. Soit $\tilde{k} = |R_{\varphi_\alpha}(X)|$. Nous montrons tout d'abord que $\tilde{R}_\alpha^{\mathrm{Holm}}(X) = R_{\varphi_\alpha}(X)$, où $\tilde{R}_\alpha^{\mathrm{Holm}}(X)$ est définie avec " $\tilde{k}$ " au lieu de " k^* " dans la définition de $R_\alpha^{\mathrm{Holm}}(X)$.
Soit $i \in R^c_{\varphi_\alpha}$, alors $\psi_{\alpha,\{i\}}(X) = 0$, donc il existe $T \subseteq [\![1,m]\!]$ tel que $i \in T$ et $\varphi_{\alpha,T}(X) = 0$. De plus, si $j \in R_{\varphi_\alpha}(X)$, alors pour tout $T' \subseteq [\![1,m]\!]$ tel que $j \in T'$, $\varphi_{\alpha,T'}(X) = 1$. Ainsi, $T \subseteq R^c_{\varphi_\alpha}$, et $\varphi_{\alpha,T}(X) = 0$, d'où $\inf_{j\in T} p_j(X) > \frac{\alpha}{|T|}$, en particulier pour $j = i$, nous avons $p_i(X) > \frac{\alpha}{|T|} \geq \frac{\alpha}{|R^c_{\varphi_\alpha}|} = \frac{\alpha}{m-\tilde{k}} > \frac{\alpha}{m-\tilde{k}+1}$, ce qui implique

$$R^c_{\varphi_\alpha} \subseteq \left\{i \in [\![1,m]\!] : p_i(X) \geq \frac{\alpha}{m-\tilde{k}}\right\}.$$

Par conséquent, nous avons

$$\tilde{R}_\alpha^{\mathrm{Holm}}(X) \subseteq R_{\varphi_\alpha}(X).$$

Réciproquement, si $i \in R_{\varphi_\alpha}(X)$, alors $\psi_{\alpha,\{i\}}(X) = 1$, donc $\varphi_{\alpha,\{i\}\cup R^c_{\varphi_\alpha}}(X) = 1$, par conséquent

$$\inf_{j\in\{i\}\cup R^c_{\varphi_\alpha}} p_j(X) \leq \frac{\alpha}{m-\tilde{k}+1},$$

d'où $p_i(X) \leq \frac{\alpha}{m-\tilde{k}+1}$, car pour tout $j \in R^c_{\varphi_\alpha}$, nous avons $p_j(X) > \frac{\alpha}{m-\tilde{k}+1}$ d'après le paragraphe précédent. Ainsi, $R_{\varphi_\alpha}(X) \subseteq \tilde{R}_\alpha^{\mathrm{Holm}}(X)$, et nous concluons que

$$R_{\varphi_\alpha}(X) = \tilde{R}_\alpha^{\mathrm{Holm}}(X).$$

Il nous reste à montrer que $\tilde{k} = k^*$. Remarquons que $p_{(\tilde{k}+1)}(X) > \frac{\alpha}{m-\tilde{k}}$ d'après la preuve ci-dessus car R_{φ_α} est l'ensemble des indices d'hypothèses correspondant aux $\tilde{k}$ plus petites p-valeurs, par conséquent l'hypothèse correspondant à la p-valeur $p_{(\tilde{k}+1)}(X)$ appartient à $R^c_{\varphi_\alpha}$, sans quoi cela contredirait la définition de $\tilde{k}$. Par conséquent, $k^* \leq \tilde{k}$, car k^* est tel que $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$ pour tout $i \leq k^*$.
Réciproquement, nous montrons que si $i \leq \tilde{k}$, alors $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$.
Supposons par l'absurde que $p_{(i)}(X) > \frac{\alpha}{m-i+1}$, et notons $j \in [\![1, m]\!]$ l'indice de l'hypothèse correspondant à la p-valeur $p_{(i)}(X)$. Alors l'ensemble T défini par

$$T = \left\{\ell \in [\![1, m]\!] : p_\ell(X) > \frac{\alpha}{m-i+1}\right\}$$

satisfait $|T| \geq m - i + 1$ car $p_{(i)}(X) > \frac{\alpha}{m-i+1}$. Ainsi $\varphi_{\alpha,T}(X) = 0$ car

$$\min_{\ell\in T} p_\ell(X) > \alpha/(m-i+1) \geq \alpha/|T|,$$

ce qui implique que $\psi_{\alpha,\{j\}}(X) = 0$, ce qui contredit le fait que

$$j \in R_{\varphi_\alpha}(X) = \big\{\ell \in [\![1, m]\!] : \psi_{\alpha,\{\ell\}}(X) = 1\big\}.$$

D'où $k^* \geq \tilde{k}$ par définition de k^* comme le plus grand entier vérifiant que pour tout $i \leq k^*$, $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$.

Par conséquent, $\tilde{k} = k^*$. □

Tout comme la procédure de Bonferroni, la procédure de Holm contrôle fortement le FWER au niveau α sans aucune hypothèse sur la structure de dépendance des p-valeurs, et elle est plus puissante que la procédure de Bonferroni, car la procédure de Bonferroni est un sous-ensemble de la procédure de Holm. La procédure de Holm est un cas particulier d'une classe plus générale de procédures appelées procédures *step-down*, qui sont définies comme suit :

Définition 1.3.2 (Procédure Step-down)**.** *Une procédure step-down est une procédure de tests multiples R telle qu'il existe une suite de seuils $\tau_1 \leq \tau_2 \leq \ldots \leq \tau_m$ telle que*

$$R(X) = \mathrm{SD}(\tau) = \big\{i \in [\![1, m]\!] : p_i(X) \leq \tau_{k^*}\big\}, \tag{1.16}$$

où $k^* = \max\big\{k \in [\![1, m]\!] : \forall i \leq k, p_{(i)}(X) \leq \tau_i\big\} \cup \{0\}$.

La suite de seuils $\tau = (\tau_1, \dots, \tau_m)$ est appelée les valeurs critiques de la procédure step-down. La procédure de Holm est une procédure step-down ayant pour valeurs critiques $\tau_i = \alpha/(m - i + 1)$ pour $i \in [\![1, m]\!]$. Notons que la preuve historique par Holm (1979) du contrôle fort du FWER de la procédure de Holm n'utilisait pas de closed testing. Les procédures de closed testing jouent un rôle central parmi les procédures contrôlant le FWER, car toute procédure contrôlant le FWER qui est *cohérente* peut s'exprimer sous la forme d'une procédure de closed testing (Sonnemann, 1982). Gabriel (1969) a historiquement défini la notion de procédures cohérentes comme des procédures telles que si une hypothèse H_i n'est pas rejetée avec $i \in [\![1, m]\!]$, alors toute hypothèse H_j, $j \in [\![1, m]\!]$ telle que $H_i \subset H_j$ ne doit pas non plus être rejetée, où $\subset$ désigne la relation d'inclusion ensembliste stricte. Notons que cette notion de cohérence ne restreint pas l'espace des procédures de tests multiples s'il n'y a pas d'hypothèses nulles emboîtées.

Enfin, nous présentons une dernière procédure classique contrôlant le FWER, qui est la procédure de Simes (Simes, 1986). Cette procédure assure un contrôle faible du FWER, sous l'hypothèse PRDS sur les p-valeurs — bien qu'historiquement, Simes (1986) ait prouvé sa validité sous l'Hypothèse 1.1. Elle est définie comme suit pour un niveau de signification α fixé au préalable :

$$R_\alpha^{\mathrm{Sim}}(X) = \left\{ i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha k^*}{m} \right\}, \tag{1.17}$$

où $k^* = \max\left\{ k \in [\![1, m]\!] : p_{(k)}(X) \leq \frac{\alpha k}{m} \right\} \cup \{0\}$. La procédure de Simes est un cas particulier d'une procédure step-up avec les valeurs critiques $\tau = \left(\frac{k\alpha}{m}\right)_{1 \leq k \leq m}$, qui est définie comme suit :

Définition 1.3.3 (procédure step-up)**.** *Une procédure step-up est une procédure de tests multiples R telle qu'il existe une famille de valeurs critiques $\tau = (\tau_1, \dots, \tau_m)$ telle que*

$$R(X) = \mathrm{SU}(\tau) = \left\{ i \in [\![1, m]\!] : p_i(X) \leq \tau_{k^*} \right\}, \tag{1.18}$$

où $k^* = \max\left\{ k \in [\![1, m]\!] : p_{(k)}(X) \leq \tau_k \right\} \cup \{0\}$.

Ainsi, la procédure step-up diffère de la procédure step-down dans la définition de k^*, et nous avons toujours $\mathrm{SD}(\tau) \subseteq \mathrm{SU}(\tau)$. La validité de la procédure de Simes sous l'hypothèse nulle globale est une conséquence de l'inégalité de Simes (Seeger, 1968; Simes, 1986), qui énonce que, pour tout $P \in H_{[\![1,m]\!]}$ tel que les p-valeurs satisfont l'Hypothèse 1.2 — bien qu'elle ait été historiquement prouvée sous l'Hypothèse 1.1,

$$\mathbb{P}_{X \sim P}\left(\exists k \in [\![1, m]\!] : p_{(k)}(X) \leq \frac{\alpha k}{m} \right) \leq \alpha. \tag{1.19}$$

Cette inégalité a été une pierre angulaire pour construire des enveloppes de confiance sur le nombre de fausses découvertes, comme nous le verrons dans la Section 1.3.4. Notons qu'il existe des généralisations de cette inégalité, voir par exemple Sarkar (2008). Bien que cette procédure ne présente pas d'intérêt pratique en soi — car elle n'est valide que sous $H_{[\![1,m]\!]}$ et que, en pratique, $\mathcal{H}_0(P_0)$ est inconnu — elle constitue un élément de base pour la construction de la procédure de Hommel (Hommel, 1988), qui est la procédure de closed testing associée aux tests locaux de Simes. La procédure de Hommel consiste à rejeter les hypothèses suivantes, pour un niveau de signification α fixé au préalable :

$$R_\alpha^{\mathrm{Hom}}(X) = \left\{ i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha}{\hat{m}_0} \right\}, \tag{1.20}$$

où $\hat{m}_0 = \max\{k \in [\![1,m]\!] : \forall i \leq k, p_{(n-k+i)} > \frac{i\alpha}{k}\} \cup \{0\}$ et $\frac{\alpha}{0} = +\infty$ par convention. La procédure de Hommel contrôle le FWER au niveau α sous une version plus faible de l'hypothèse wPRDS sur les p-valeurs ; elle nécessite seulement que, pour tout $P \in \mathfrak{F}$, pour tout ensemble croissant $D \subseteq [0,1]^{\mathcal{H}_0(P)}$, $u \mapsto \mathbb{P}_{X\sim P}\big((p_j)_{j\in\mathcal{H}_0(P)} \in D \mid p_i \leq u\big)$ soit croissante pour tout $i \in \mathcal{H}_0(P)$ sur l'ensemble $\big\{u : 0 \leq u \leq 1, \mathbb{P}_{X\sim P}(p_i \leq u) > 0\big\}$. La procédure de Hommel peut être considérée comme une version adaptative de la procédure de Bonferroni dans le sens suivant : si nous connaissions le nombre de vraies hypothèses nulles m_0, nous pourrions utiliser la procédure de Bonferroni avec des valeurs critiques α/m_0 au lieu de α/m, ce qui serait une procédure plus puissante. Ici, au lieu de m_0 qui est inconnu, nous utilisons l'entier aléatoire $\hat{m}_0$ qui est un *sur-estimateur* de m_0 avec une probabilité d'au moins $1-\alpha$. Cette observation est particulièrement importante dans la construction d'enveloppes de confiance sur le nombre de fausses découvertes, voir la Proposition 3.3.12 pour une généralisation de ce sur-estimateur dans ce contexte, où la définition de $\hat{m}_0$ est exactement la même que celle que nous obtenons ci-dessus avec $\ell_{i:n} = \frac{\alpha i}{n}$.

1.3.3 False discovery rate (FDR) et procédure de BH

Le *False Discovery Rate* (FDR) a été popularisé par l'article fondateur de Benjamini and Hochberg (1995) comme une métrique d'erreur alternative au FWER, moins conservatrice et plus puissante en pratique. Il est défini comme l'espérance de la proportion de vraies hypothèses nulles rejetées, c'est-à-dire que, pour toute procédure de tests multiples R, le FDR de R est défini, pour tout $P \in \mathfrak{F}$, par

$$\mathrm{FDR}_P(R) = \mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\right]. \tag{1.21}$$

Notons que l'idée de considérer la proportion de fausses découvertes sur le nombre total de rejets était déjà présente dans les travaux de Eklund and Seeger (1965) et Seeger (1966) — même si ce n'était pas en espérance. Nous disons qu'une procédure R assure le contrôle du FDR au niveau $\alpha \in [0,1]$ si pour tout $P \in \mathfrak{F}$, $\mathrm{FDR}_P(R) \leq \alpha$. Cette métrique d'erreur peut être vue comme une généralisation du FWER sous l'hypothèse nulle globale, c'est-à-dire :

$$\forall P \in H_{[\![1,m]\!]}, \mathrm{FDR}_P(R) = \mathrm{FWER}_P(R). \tag{1.22}$$

Nous prouvons cette propriété ci-dessous :

Proof. Soit $P \in H_{[\![1,m]\!]}$, nous avons $V\big(R(X)\big) = |R(X)|$, d'où

$$\begin{aligned}
&\mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\right] \\
&= \mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\mathbb{1}\{|R(X)| > 0\}\right] + \mathbb{E}_{X\sim P}\left[\frac{|R(X)|}{|R(X)| \vee 1}\mathbb{1}\{|R(X)| = 0\}\right] \\
&= \mathbb{P}_{X\sim P}\big(V\big(R(X)\big) > 0\big) + 0 = \mathbb{P}_{X\sim P}\big(V\big(R(X)\big) > 0\big).
\end{aligned}$$

□

Nous introduisons d'abord des procédures contrôlant le FDR en utilisant des p-valeurs. La procédure la plus classique contrôlant le FDR est la procédure de Benjamini-Hochberg (BH) (Benjamini and Hochberg, 1995), qui est une procédure step-up définie par les valeurs critiques $\tau = \left(\frac{k\alpha}{m}\right)_{1\leq k\leq m}$ pour un niveau α fixé au préalable. Nous notons cette procédure R_α^{BH}, et nous avons le résultat suivant :

Proposition 1.3.4. *Supposons que l'Hypothèse 1.2 soit vérifiée. Soient $\alpha \in (0,1]$ et $P \in \mathfrak{F}$. Alors la procédure de BH R_α^{BH} contrôle le FDR_P au niveau $\frac{m_0(P)}{m}\alpha$.*

Remarquons que la procédure de BH est exactement la même que la procédure de Simes, mais l'article fondateur de Benjamini and Hochberg (1995) a prouvé qu'elle contrôle le FDR au niveau $\frac{m_0}{m}\alpha$ sous l'Hypothèse 1.1, puis Benjamini and Yekutieli (2001) a étendu ce résultat sous l'hypothèse PRDS. Enfin, à ma connaissance, on peut faire remonter la preuve du contrôle du FDR de la procédure de BH sous l'Hypothèse 1.2 au Lemme 3.2 de Blanchard and Roquain (2008), bien que les auteurs n'aient pas énoncé explicitement l'Hypothèse 1.2. Notons qu'il existe certaines généralisations de la procédure de BH, par exemple la procédure de Heyse (Heyse, 2011), qui est une procédure step-up adaptée à des p-valeurs *hétérogènes*, c'est-à-dire des p-valeurs qui ne partagent pas la même loi marginale sous l'hypothèse nulle. Cependant, la procédure de Heyse ne contrôle pas le FDR en général, comme l'ont montré Heller and Gur (2014) ou Döhler et al. (2018). Nous donnons une définition formelle de la procédure de Heyse dans la Section 5.2.2 et abordons ce problème dans la Section 5.3.2.

La deuxième procédure contrôlant le FDR que nous introduisons est la procédure de Benjamini-Yekutieli (BY) (Benjamini and Yekutieli, 2001), qui est une procédure step-up définie par les valeurs critiques $\tau = \left(\frac{k\alpha}{mh_m}\right)_{1\leq k\leq m}$ pour un niveau α fixé au préalable, où $h_m = \sum_{i=1}^m 1/i$. Nous notons cette procédure R_α^{BY}, et nous avons le résultat suivant :

Proposition 1.3.5. *Soient $\alpha \in (0,1]$ et $P \in \mathfrak{F}$. Alors la procédure BY, R_α^{BY}, contrôle le FDR_P au niveau $\frac{m_0(P)}{m}\alpha$.*

Remarquons que la procédure BY contrôle le FDR au niveau $\frac{m_0(P)}{m}\alpha$ sans aucune hypothèse sur la structure de dépendance des p-valeurs, mais au prix d'une perte de puissance par rapport à la procédure de BH. En effet, $R_\alpha^{\mathrm{BY}} = R_{\alpha/h_m}^{\mathrm{BH}}$, et $h_m \sim \log(m)$, d'où les valeurs critiques de la procédure de BY correspondent à celles de la procédure de BH divisées par un facteur logarithmique, ce qui est significatif lorsque m est grand. Notons que la procédure de BY est un cas particulier d'une classe plus générale de procédures issues de Blanchard and Roquain (2008) conçues pour contrôler le FDR sous une structure de dépendance arbitraire des p-valeurs. La procédure de BY, tout comme la procédure de BH avec la procédure de Heyse, a été généralisée au cas des p-valeurs hétérogènes par la procédure DBY (Döhler, 2018). Nous donnons une définition formelle de la procédure DBY dans la Section 5.2.2 et étendons la validité de la procédure DBY dans la Section 5.3.2.

La dernière procédure (non adaptative) contrôlant le FDR que nous introduisons est la procédure e-BH (Wang and Ramdas, 2022), qui est conçue pour contrôler le FDR en utilisant des e-valeurs au lieu de p-valeurs. Il s'agit, comme pour la procédure de BH, d'une procédure définie par ce qui suit :

$$R_\alpha^{e\text{-BH}}(X) = \left\{ i \in [\![1,m]\!] : e_i(X) \geq \frac{m}{\alpha k^*} \right\}, \tag{1.23}$$

où $k^* = \max\{k \in [\![1,m]\!] : e_{[k]}(X) \geq \frac{m}{\alpha k}\} \cup \{0\}$, $e_{[k]}(X)$ est la k-ième plus grande e-valeur, et nous posons $\frac{m}{\alpha k^*} = \infty$ si $k^* = 0$. Notons que la procédure e-BH est équivalente à appliquer la procédure de BH aux p-valeurs $p_i(X) = (1/e_i(X)) \wedge 1$, pour tout $i \in [\![1,m]\!]$. Ainsi, la différence entre la procédure de BH et la procédure e-BH provient de la loi marginale des p-valeurs sous l'hypothèse nulle ; dans le cas de e-BH, nous supposons que les p-valeurs utilisées ne sont pas seulement distribuées de façon sur-uniforme, mais qu'elles dérivent également de e-valeurs, ce qui est une hypothèse plus forte (voir la Section 1.2.2). Nous avons le résultat suivant :

Proposition 1.3.6. *Soient $\alpha \in (0,1]$ et $P \in \mathfrak{F}$. Alors la procédure e-BH $R_\alpha^{e\text{-BH}}$ contrôle le FDR_P au niveau $\frac{m_0(P)}{m}\alpha$. De plus, $R^{e\text{-BH}}$ contrôle le FDR de manière post-hoc, c'est-à-dire que*

$$\forall P \in \mathfrak{F}, \mathbb{E}_{X\sim P}\left[\sup_{\alpha\in(0,1]} \frac{V\left(R_\alpha^{e\text{-BH}}\right)}{\alpha|R_\alpha^{e\text{-BH}}| \vee 1}\right] \leq \frac{m_0(P)}{m}. \tag{1.24}$$

Ainsi, la procédure e-BH satisfait une propriété analogue à la propriété (1.2), mais propre au cadre des tests multiples. Notons que la procédure e-BH peut être utilisée avec des e-valeurs *composées* (*compound* en anglais) (voir Definition 5.2.5 pour une définition formelle), au lieu de e-valeurs classiques, et la borne du contrôle sur le FDR_P, ainsi que le contrôle post-hoc, devient alors 1 au lieu de $\frac{m_0(P)}{m}$. De nouvelles procédures contrôlant le FDR combinent des e-valeurs avec un principe de closed testing similaire à celui présenté dans la subsection 1.3.2, comme dans Xu et al. (2025) ou Goeman (2026), afin d'obtenir une amélioration pratique par rapport aux procédures BY et BH. Nous détaillons le closed testing pour le contrôle du FDR dans la Section 5.2.5.

Enfin, nous considérons la classe des procédures adaptatives introduite par Storey (2002) (avec des résultats prouvés formellement plus tard par Storey et al., 2004). Storey (2002) a proposé d'estimer le nombre de vraies hypothèses nulles m_0 par

$$\hat{m}_0 = \frac{|\{i \in [\![1,m]\!] : p_i(X) > \lambda\}| + 1}{1-\lambda}$$

pour un $\lambda \in (0,1)$ fixé au préalable, puis de rejeter selon la procédure step-up ayant pour valeurs critiques $\frac{k\alpha}{\hat{m}_0} \wedge \lambda$ pour $k \in [\![1,m]\!]$, que nous noterons R_α^{St}. Storey et al. (2004) a énoncé le résultat suivant :

Proposition 1.3.7. *Soit $P \in \mathfrak{F}$. Supposons que les $(p_i(X))_{i\in\mathcal{H}_0(P)}$ sont indépendantes et que pour tout $i \in \mathcal{H}_0(P)$, $p_i(X) \sim U(0,1)$. Soit $\alpha \in (0,1]$. Alors la procédure de Storey R_α^{St} contrôle le FDR_P au niveau $\left(1-\lambda^{m_0(P)}\right)\alpha \leq \alpha$.*

Remarquons que l'hypothèse d'indépendance sur les p-valeurs sous l'hypothèse nulle est légèrement différente de l'Hypothèse 1.1, car la preuve repose sur des propriétés de martingale de $\sum_{i\in\mathcal{H}_0(P)} \mathbb{1}\{p_i(X) \leq t\}$. À la suite de ce travail, de nombreux articles ont été consacrés au traitement des structures de dépendance, en particulier Blanchard and Roquain (2009), qui a étendu la théorie du FDR adaptatif à diverses structures de dépendance, et sous l'hypothèse de sur-uniformité des p-valeurs sous l'hypothèse nulle au lieu de l'uniformité. Cet article a notamment montré qu'une légère modification de la procédure de Storey, la procédure step-up ayant pour valeurs critiques $\frac{k\alpha}{\hat{m}_0}$ pour $k \in [\![1,m]\!]$ (sans minimisation avec λ donc), contrôle le FDR au niveau α sous l'hypothèse que toutes les p-valeurs sont indépendantes, ce qui est une hypothèse indispensable pour l'argument de type *leave-one-out* utilisé dans la preuve. Récemment, plusieurs articles se ont considéré le problème du contrôle adaptatif du

FDR, et ceci par exemple en considérant des données discrètes, comme Döhler and Meah (2026), afin de gagner en puissance par rapport aux procédures initiales, ou en considérant des e-valeurs composées, comme Ignatiadis et al. (2026), afin d'obtenir une amélioration uniforme par rapport à diverses procédures adaptatives.

1.3.4 Enveloppes de confiance sur le nombre de fausses découvertes

Les enveloppes de confiance sur le nombre de fausses découvertes ont été popularisées par Goeman and Solari (2011), mais la construction de telles enveloppes peut être remontée à Genovese and Wasserman (2004). Par comparaison avec les procédures de tests multiples classiques, que nous avons illustrées à travers le contrôle du FWER et du FDR, les enveloppes de confiance sur le nombre de fausses découvertes offrent un cadre plus flexible pour contrôler le nombre de fausses découvertes, permettant de sélectionner n'importe quel sous-ensemble d'hypothèses après avoir observé les données, tout en garantissant une borne supérieure valide sur le nombre de fausses découvertes. Cela permet notamment de sélectionner les hypothèses à rejeter de manière post-hoc, c'est-à-dire après avoir observé les données. Cette propriété permet en particulier aux chercheurs d'explorer les données et de sélectionner dynamiquement n'importe quel sous-ensemble d'hypothèses, tout en apportant une réponse à certaines pratiques inappropriées de la communauté scientifique, telles que le *p-hacking*, c'est-à-dire la pratique consistant à sélectionner des hypothèses à rejeter après observation des données, puis à ne rapporter comme significatives que les hypothèses sélectionnées.

Une enveloppe de confiance sur le nombre de fausses découvertes est une fonction aléatoire $\hat{V} : \mathcal{P}(\llbracket 1, m \rrbracket) \to \llbracket 1, m \rrbracket$ — où $\mathcal{P}(\llbracket 1, m \rrbracket)$ désigne l'ensemble des parties de $\llbracket 1, m \rrbracket$ — telle que pour un niveau $\alpha \in (0, 1]$ fixé au préalable,

$$\forall P \in \mathfrak{F}, \mathbb{P}_{X\sim P}\big(\forall S \subseteq \llbracket 1, m \rrbracket : V(S) \leq \hat{V}(S)\big) \geq 1 - \alpha. \tag{1.25}$$

Ici, nous voyons déjà pourquoi la notion d'enveloppe de confiance sur le nombre de fausses découvertes est plus flexible que les procédures de tests multiples classiques : ces méthodes ne prescrivent pas un sous-ensemble spécifique d'hypothèses à rejeter, mais fournissent plutôt une borne supérieure valide sur le nombre de fausses découvertes pour tout sous-ensemble d'hypothèses, permettant à l'utilisateur d'explorer les données et de choisir n'importe quel sous-ensemble d'hypothèses à rejeter.

Notons que ce type de procédures peut être considéré comme une généralisation des méthodes de contrôle du FWER ; si nous considérons une procédure de tests multiples R qui contrôle fortement le FWER au niveau α, nous pouvons définir une enveloppe de confiance $\hat{V}$ sur le nombre de fausses découvertes comme suit :

$$\forall S \subseteq \llbracket 1, m \rrbracket, \hat{V}(S) = |R(X)^c \cap S|, \tag{1.26}$$

ce qui constitue une enveloppe de confiance sur le nombre de fausses découvertes au niveau α. En effet, pour tout $P \in \mathfrak{F}$, nous avons

$$\begin{aligned}
&\mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : V(S) > \hat{V}(S)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : |R(X) \cap \mathcal{H}_0(P) \cap S| > |R(X)^c \cap \mathcal{H}_0(P)^c \cap S|\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : |R(X) \cap \mathcal{H}_0(P) \cap S| > 0\big) \\
&= \mathbb{P}_{X\sim P}\big(|R(X) \cap \mathcal{H}_0(P)| > 0\big) = \mathrm{FWER}_P\left(R(X)\right) \leq \alpha
\end{aligned}$$

Ainsi, toute procédure de tests multiples contrôlant fortement le FWER au niveau α peut être utilisée pour construire une enveloppe de confiance sur le nombre de fausses découvertes au niveau α. Réciproquement, une enveloppe de confiance sur le nombre de fausses découvertes $\hat{V}$ au niveau α fournit une procédure de tests multiples qui assure un contrôle fort du FWER au niveau α comme suit :

$$R(X) = \big\{i \in [\![1, m]\!] : \hat{V}(\{i\}) = 0\big\}, \tag{1.27}$$

Ainsi, pour tout $P \in \mathfrak{F}$, nous avons

$$\begin{aligned}
\mathrm{FWER}_P(R) &= \mathbb{P}_{X\sim P}\big(\exists i \in \mathcal{H}_0(P) : i \in R(X)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists i \in [\![1, m]\!] : \hat{V}(\{i\}) = 0 \text{ et } V(\{i\}) = 1\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : V(S) > \hat{V}(S)\big) \\
&\leq \alpha,
\end{aligned}$$

où la dernière inégalité est due à (1.25). Remarquons ici que la procédure de tests multiples R définie dans (1.27) partage la même logique que la procédure de tests multiples R_{ψ_α} définie dans (1.11), qui est la procédure de tests multiples associée à la procédure de closed testing. Cette observation, combinée à (2.10), établit un lien fort entre le closed testing et les enveloppes de confiance. En effet, Goeman and Solari (2011) ont observé que les procédures de tests fermées apportent plus d'informations que la simple procédure de tests multiples R_{φ_α} à travers (2.10), et ont défini des enveloppes de confiance sur le nombre de fausses découvertes en utilisant le cadre du closed testing :

Proposition 1.3.8 (Goeman and Solari, 2011)**.** *Soit φ une famille de tests locaux telle que, pour tout $P \in \mathfrak{F}$, $\varphi_{\mathcal{H}_0(P)}$ soit un test de niveau α. Rappelons la définition de la famille de tests locaux effectifs ψ (2.9), et soit $\hat{V}_\varphi^{\mathrm{CT}}$ la fonction aléatoire définie comme suit :*

$$\forall S \subseteq [\![1, m]\!], \hat{V}_\varphi^{\mathrm{CT}}(S) = \max_{U\subseteq S} \big\{|U| : \psi_U(X) = 0\big\}. \tag{1.28}$$

Alors, pour tout $\alpha \in (0, 1]$, $\hat{V}_\varphi^{\mathrm{CT}}$ est une enveloppe de confiance sur le nombre de fausses découvertes au niveau α.

La démonstration de cette proposition découle immédiatement de (2.10) et de la définition de $\hat{V}_\varphi^{\mathrm{CT}}$:

Proof. Soit $P \in \mathfrak{F}$, nous avons

$$\begin{aligned}
&\mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : V(S) > \hat{V}_\varphi^{\mathrm{CT}}(S)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : V(S) > \max_{U\subseteq S}\{|U| : \psi_U(X) = 0\}\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : |S \cap \mathcal{H}_0(P)| > \max_{U\subseteq S}\{|U| : \psi_U(X) = 0\}\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : \psi_{S\cap\mathcal{H}_0(P)}(X) = 1\big) && (1.29) \\
&\leq \alpha, && (1.30)
\end{aligned}$$

où (1.29) provient du fait que si $\psi_{S\cap\mathcal{H}_0(P)}(X) = 0$, alors

$$\max_{U\subseteq S} \big\{|U| : \psi_U(X) = 0\big\} \geq |S \cap \mathcal{H}_0(P)|,$$

et (1.30) provient de (1.10). □

L'inconvénient de cette approche est que le calcul de $\hat{V}_\varphi^{\mathrm{CT}}$ est infaisable en pratique, car elle nécessite de calculer tous les tests locaux effectifs ψ_U pour tout $U \subseteq [\![1, m]\!]$, ce qui est exponentiel en m. Cependant, dans certains cas, le calcul de $\hat{V}_\varphi^{\mathrm{CT}}$ peut être simplifié par ce que l'on appelle des *shortcuts*, qui sont spécifiques à la famille de tests locaux φ, et permettent de calculer $\hat{V}_\varphi^{\mathrm{CT}}$ ou une borne supérieure de celle-ci en temps polynomial. Quelques exemples de tels shortcuts sont donnés dans la Section 3 avec la Proposition 3.3.12, ou la Proposition 3.4.2. Tout comme pour les méthodes de closed testing contrôlant le FWER, une notion de cohérence a été définie pour les enveloppes de confiance par Goeman et al. (2021), qui stipule que toute enveloppe de confiance sur le nombre de fausses découvertes cohérente peut être exprimée sous la forme d'une méthode de closed testing. Cependant, cette affirmation est remise en question par nos résultats de la Section 6.

Enfin, notons qu'il existe d'autres méthodes pour construire des enveloppes de confiance sur le nombre de fausses découvertes, telles que la procédure d'inversion (Genovese and Wasserman, 2006) reposant sur une famille de tests locaux (tout comme la procédure de closed testing), mais sans définir de tests locaux effectifs (voir la Proposition 3.3.10 pour une définition formelle), des méthodes reposant sur des chemins (Katsevich and Ramdas, 2020; Meah et al., 2024) ou encore des méthodes reposant sur le Joint Error Rate (JER) (Blanchard et al., 2020a; Durand et al., 2020) — voir la Section 6.2.1 et la Section 3.3.3. Notons que toutes ces méthodes, à l'exception de celle reposant sur les chemins, sont équivalentes, et les méthodes par chemins peuvent être améliorées en utilisant l'une de ces trois méthodes équivalentes.

1.3.5 Prédiction conforme et tests multiples

La prédiction conforme est un cadre général pour construire des ensembles de prédiction avec des garanties de couverture à échantillon fini à partir de garanties minimales, c'est-à-dire sans aucune hypothèse sur la distribution des données. Cette aspiration à des "garanties minimales" est motivée par l'utilisation croissante de modèles "boîte noire", tels que les réseaux de neurones. Considérons un exemple de régression, c'est-à-dire un problème statistique où nous supposons que deux variables $X \in \mathcal{X}$ et $Y \in \mathbb{R}$ vérifient $Y = f(X) + \varepsilon$ où ε est un terme de bruit. Supposons qu'une boîte noire ait produit un prédicteur $\hat{f}$, que nous ayons accès à un échantillon de taille $n \in \mathbb{N}$ de (X, Y) et que nous souhaitions appliquer notre prédicteur $\hat{f}$ à un *point de test* X_{n+1}. Alors, la méthode de prédiction conforme va construire n scores de non-conformité $S_1, \dots, S_n$, disons $S_i = \left|Y_i - \hat{f}(X_i)\right|$ pour tout $i \in \{1, \dots, n\}$, ainsi qu'un score $S_{n+1}(y) = \left|y - \hat{f}(X_{n+1})\right|$. Ensuite, pour un niveau $\alpha \in (0, 1]$ fixé au préalable, la méthode de prédiction conforme va construire un ensemble de prédiction

$$\hat{C}_\alpha(X_{n+1}) = \left\{y \in \mathbb{R} : S_{n+1}(y) \leq S_{(k)}\right\},$$

où k_α est égal à $\lceil (1-\alpha)(n+1) \rceil$. Sous une hypothèse d'*échangeabilité* sur les scores, Y_{n+1} appartient à $\hat{C}_\alpha(X_{n+1})$ avec une probabilité d'au moins $1 - \alpha$.

Le cadre de la prédiction conforme a été introduit par Gammerman et al. (1998), et pleinement formalisé par Vovk et al. (2005). Depuis lors, il a été largement étudié et appliqué dans divers domaines, tels que la classification (Ding et al., 2023; Gazin et al., 2025b) ou en inférence sélective (Jin and Candès, 2023; Gazin et al., 2025a). Un problème central de la prédiction conforme est de déterminer les scores de non-conformité à partir des observations, mais nous ne nous concentrerons pas sur ce problème ici et nous

supposerons que les scores sont donnés. Nous allons plutôt explorer certains liens entre la prédiction conforme et la théorie des tests multiples. Soient $n, m \in \mathbb{N}$, $S_1, \dots, S_n$ des scores de "référence" (ou scores de calibration) et $S_{n+1}, \dots, S_{n+m}$ des scores de "test", tous dans $\mathbb{R}$. L'hypothèse clé de la prédiction conforme est que les scores sont échangeables, ce qui est une hypothèse plus faible que celle d'indépendance et d'identique distribution.

Définition 1.3.9 (Échangeabilité)**.** *Soient $X_1, \dots, X_n$ $n \in \mathbb{N}$ variables aléatoires. Alors $X_1, \dots, X_n$ sont échangeables si pour toute permutation π de $\{1, \dots, n\}$,*

$$(X_1, \dots, X_n) \sim (X_{\pi(1)}, \dots, X_{\pi(n)}),$$

c'est-à-dire que la loi conjointe de $X_1, \dots, X_n$ est invariante par permutation.

Notons que l'hypothèse d'échangeabilité est satisfaite si les variables sont indépendantes et identiquement distribuées. De plus, elle implique que les variables ont la même loi marginale. Par conséquent, nous noterons par $S = (S_1, \dots, S_{n+m})$ le vecteur des scores, et nous supposerons que la loi P_0 de S appartient à un modèle $\mathfrak{F}$. Le problème de test d'hypothèse simple en prédiction conforme consiste à tester l'hypothèse nulle H_i : " $(S_1, \dots, S_n, S_{n+i})$ est échangeable" pour un certain $i \in [\![1, m]\!]$. Cela dérive une p-valeur p_i, appelée p-valeur conforme, pour chaque hypothèse nulle H_i, qui est définie comme suit :

$$p_i = \frac{1}{n+1}\left(\sum_{j=1}^{n} \mathbb{1}\{S_j \geq S_{n+i}\} + 1\right). \tag{1.31}$$

Cette définition des p-valeurs conformes définit bien une p-valeur valide sous l'hypothèse nulle H_i (Romano and Wolf, 2005, Lemme 1 ; Arlot et al., 2010, Lemme 5.2) :

Proposition 1.3.10 (Lemme du quantile)**.** *Soient $i \in \{1, \dots, m\}$ et $P \in H_i$. Alors, pour tout $\alpha \in [0, 1]$,*

$$\mathbb{P}_{S\sim P}\big(p_i \leq \alpha\big) \leq \alpha.$$

Notons que les p-valeurs conformes sont des p-valeurs *discrètes*, et que leur support est $\mathcal{A} = \left\{\frac{j}{n+1}, j \in [\![1, n+1]\!]\right\}$. De plus, pour tout $j \in [\![1, n+1]\!]$ et $P \in H_i$, nous avons $\mathbb{P}_{S\sim P}\big(p_i \leq \frac{j}{n+1}\big) = \frac{j}{n+1}$. On peut également définir des e-valeurs conformes. Un exemple est donné par Balinsky and Balinsky (2024), qui a proposé la e-valeur suivante pour tester l'hypothèse nulle H_i, pour un certain $i \in [\![1, m]\!]$:

$$e_i = \frac{(n+1)S_{n+i}}{\sum_{j=1}^{n} S_j + S_{n+i}}. \tag{1.32}$$

Revenons maintenant à notre problème de tests multiples. En prédiction conforme, afin de tester les hypothèses nulles $H_1, \dots, H_m$ simultanément, les théorèmes que nous allons présenter nécessitent des hypothèses supplémentaires sur les scores que nous définissons ci-dessous. La première hypothèse est simple :

Hypothèse 1.3. *Les scores $S_{k+1}, \dots, S_{n+m}$ ne sont pas égaux deux à deux presque sûrement.*

Si le modèle ne permet pas de distributions ayant des atomes, cette hypothèse est automatiquement satisfaite. La seconde hypothèse est l'échangeabilité des scores sous les hypothèses nulles avec les scores de calibration.

Hypothèse 1.4. *Pour tout $P \in \mathfrak{F}$, les scores $\{S_j : j \in [\![1,n]\!] \cup \{n+i, i \in \mathcal{H}_0(P)\}\}$ sont échangeables conditionnellement à $(S_{n+i})_{i\in\mathcal{H}_1}$.*

Notons que, pour $P \in \mathfrak{F}$ tel que $|\mathcal{H}_0(P)| > 1$, la propriété sous-jacente selon laquelle les scores $S_1, \ldots, S_n, S_{n+i}$ sont échangeables pour un $i \in \mathcal{H}_0(P)$ donné n'implique pas que les scores $\{S_j : j \in [\![1,n]\!] \cup \{n+i, i \in \mathcal{H}_0(P)\}\}$ sont échangeables. Nous illustrons ceci dans l'exemple suivant : soient X, Y, Z trois variables aléatoires de Bernoulli telles que :

$$\begin{cases} \mathbb{P}\big((X,Y,Z) = (1,1,0)\big) & = 0,5, \\ \mathbb{P}\big((X,Y,Z) = (0,0,1)\big) & = 0,5, \\ \mathbb{P}\big((X,Y,Z) = (a,b,c)\big) & = 0 \text{ sinon.} \end{cases}$$

Alors, les scores X, Y sont échangeables et les scores X, Z sont échangeables, mais les scores X, Y, Z ne sont pas échangeables. En effet, nous avons pour tous $a, b \in \{0,1\}$, $\mathbb{P}\big((X,Y) = (a,b)\big) = \mathbb{P}\big((Y,X) = (a,b)\big)$ et $\mathbb{P}\big((X,Z) = (a,b)\big) = \mathbb{P}\big((Z,X) = (a,b)\big)$, mais

$$\mathbb{P}\big((X,Y,Z) = (1,1,0)\big) = 0,5 \neq 0 = \mathbb{P}\big((X,Z,Y) = (1,1,0)\big).$$

À partir de l'Hypothèse 1.3 et de l'Hypothèse 1.4, Marandon et al. (2024) ont prouvé que les p-valeurs conformes sont PRDS, et par conséquent wPRDS. Remarquons que nos hypothèses nulles sont différentes de celles considérées dans Marandon et al. (2024). En effet, ils considèrent l'hypothèse nulle H_i : " $S_{n+i} \sim P_0$ " où P_0 est la distribution des scores de calibration. Cependant, sous l'Hypothèse 1.4, ces deux hypothèses nulles sont égales sur le sous-modèle correspondant $\mathfrak{F}'$ tel que pour tout $P \in \mathfrak{F}'$, P satisfait l'Hypothèse 1.4. Nos hypothèses nulles H_i ont l'avantage de conserver la propriété selon laquelle les p-valeurs conformes p_i sont sur-uniformes sous chacune des H_i, sans aucune hypothèse supplémentaire sur l'échangeabilité du score de non-conformité correspondant avec les scores de calibration, puisque cette hypothèse est déjà incluse dans la définition de H_i.

Cela garantit que la procédure BH contrôle le FDR au niveau α pour les p-valeurs conformes. De plus, en considérant l'Hypothèse 1.3 et l'Hypothèse 1.4 sous l'hypothèse nulle globale — c'est-à-dire en considérant le modèle

$$\mathfrak{F}_0 = \big\{P \in \mathfrak{F} : \mathcal{H}_0(P) = [\![1,m]\!] \text{ et } P \text{ satisfait l'Hypothèse 1.4}\big\},$$

Gazin et al. (2024) ont prouvé une inégalité de type DKW pour les p-valeurs conformes, ce qui permet de construire une enveloppe de confiance sur le nombre de fausses découvertes pour les p-valeurs conformes avec une hypothèse un petit peu plus faible que l'Hypothèse 1.4. Cette hypothèse plus faible ne requiert seulement que $\{S_j : j \in [\![1,n]\!] \cup \{n+i, i \in \mathcal{H}_0(P)\}\}$ soient échangeables pour tout $P \in \mathfrak{F}$, de façon similaire à la version affaiblie de l'hypothèse wPRDS requise par la procédure de Hommel pour contrôler le FWER dans la Section 2.3.2.

1.4 Contributions

Cette section présente les contributions de cette thèse, qui sont détaillées dans les chapitres suivants.

1.4.1 Enveloppes de confiance pour les fausses découvertes avec des données hétérogènes

Le Chapitre 3 présente un article soumis disponible sur arXiv, (Périer et al., 2026), qui est un travail conjoint avec Gilles Blanchard (Université Paris-Saclay), Sebastian Döhler (Université de Darmstadt), Guillermo Durand (Université Paris-Saclay), et Etienne Roquain (Sorbonne Université) [1]. Il introduit de nouvelles méthodes pour construire des enveloppes de confiance sur le nombre de fausses découvertes dans un cadre de tests multiples hétérogènes, c'est-à-dire lorsque les p-valeurs ne partagent pas la même loi marginale sous l'hypothèse nulle. Ces méthodes reposent sur divers outils, qui peuvent être classés en deux catégories. La première catégorie est composée d'outils probabilistes conçus sur-mesure pour le cadre hétérogène. Cette catégorie inclut en particulier l'inégalité de Bretagnolle (Shorack and Wellner, 1986, Chapitre 25, Inégalité 2) qui est une version hétérogène de l'inégalité DKWM (Dvoretzky et al., 1956; Massart, 1990), et la version hétérogène de l'inégalité de Simes que nous prouvons dans ce chapitre. Ce dernier résultat découle d'une procédure de tests multiples de contrôle du FDR développée dans Döhler et al. (2018), de la même manière que l'inégalité de Simes peut être obtenue via la procédure de BH. La seconde catégorie d'outils que nous utilisons dans ce chapitre est constituée de mécanismes algorithmiques qui optimisent l'information probabiliste fournie par la première catégorie, tout en restant à faible complexité algorithmique. Plus précisément, nous avons utilisé le cadre fourni par la littérature JER (Blanchard et al., 2020a) et la méthode de la procédure d'inversion (Genovese and Wasserman, 2006) pour construire des enveloppes de confiance sur le nombre de fausses découvertes à partir des outils probabilistes hétérogènes. Ce chapitre développe notamment de nouveaux shortcuts pour la procédure d'inversion, qui permettent de calculer une borne supérieure de l'enveloppe de confiance en temps polynomial, et d'améliorer d'anciennes méthodes conçues pour le cadre homogène. Nous énumérons ci-dessous les principales contributions de ce chapitre :

- nous avons fourni de nouveaux shortcuts génériques, $\hat{V}^{\mathrm{SC1}}$ et $\hat{V}^{\mathrm{SC2}}$, pour la procédure d'inversion qui permettent de calculer une borne supérieure de l'enveloppe de confiance de la procédure d'inversion en temps polynomial,

- nous avons appliqué ces nouveaux shortcuts génériques au cas homogène et les avons couplés avec des méthodes basées sur le JER, ce qui a permis de prouver l'exactitude — c'est-à-dire l'égalité avec la procédure de closed testing issue des mêmes outils probabilistes — d'anciennes méthodes conçues pour le cadre homogène,

- nous avons fourni de nouvelles enveloppes adaptées à l'hétérogénéité pour la fonction de répartition empirique des p-valeurs sous l'hypothèse nulle, et les avons appliquées pour construire de nouvelles enveloppes de confiance grâce à la procédure d'inversion et aux méthodes basées sur le JER,

- nous avons fourni des versions adaptatives des nouvelles enveloppes de confiance dans lesquelles le nombre de vraies hypothèses nulles est estimé via nos shortcuts pour la procédure d'inversion puis réinjecté dans les méthodes JER.

[1] Je certifie ici qu'en tant que premier auteur de l'article, j'ai contribué à une grande majorité du contenu de l'article dans tous les domaines : théorie, expériences et rédaction.

1.4.2 Contrôle conforme du FDR et bornes conformes sur le FDP pour la prédiction d'arêtes dans un graphe

Dans le Chapitre 4, nous présentons un article soumis disponible sur arXiv (Blanchard et al., 2024), qui est un travail conjoint avec Gilles Blanchard (Université Paris-Saclay), Guillermo Durand (Université Paris-Saclay), et Ariane Marandon (University College London). Il fournit formellement une preuve des résultats énoncés dans Marandon (2024) grâce aux outils abordés dans la Section 2.3.5. Marandon (2024) construit une procédure de contrôle du FDR dans un cadre de prédiction d'arêtes dans un graphe. Pour ce faire, l'autrice construit des p-valeurs conformes en échantillonnant des arêtes dans un graphe masqué selon un schéma d'échantillonnage spécifique, puis elle applique la procédure de BH à ces p-valeurs conformes. Cependant, l'article mentionné ne contient pas la preuve que les scores résultants satisfont effectivement l'Hypothèse 1.4. Nous fournissons une preuve formelle de ce résultat, et nous fournissons également une enveloppe de confiance sur le FDP en utilisant les résultats de Gazin et al. (2024).

1.4.3 p-valeurs composées hétérogènes et closure principle

Le Chapitre 5 présente un travail en cours, qui est un travail conjoint avec Gilles Blanchard (Université Paris-Saclay), et Guillermo Durand (Université Paris-Saclay). Il vise à fournir une généralisation du principe de e-closure pour le FDR (Xu et al., 2025) au cadre hétérogène. Notamment, ce chapitre introduit de nouvelles p-valeurs composées hétérogènes. Cette découverte apporte un nouvel éclairage sur la littérature des tests multiples hétérogènes, en fournissant une nouvelle preuve de la procédure DBY introduite dans Döhler (2018) et en étendant sa validité, ainsi qu'en fournissant une correction à la procédure de Heyse Heyse (2011). Ces résultats reposent sur des résultats de la théorie des p-valeurs composées (Barber and Samworth, 2025; Armstrong, 2026). Enfin, nous fournissons de nouvelles procédures de tests multiples contrôlant le FDR via le e-closure principle. Ces nouvelles procédures sont meilleures que la procédure DBY et que l'amélioration de la procédure de BY obtenue par l'application du e-closure principle par Xu et al. (2025), respectivement.

1.4.4 Sur les enveloppes de confiance fortement cohérentes pour les fausses découvertes

Le Chapitre 6 présente un travail en cours qui est un travail conjoint avec Gilles Blanchard (Université Paris-Saclay), et Guillermo Durand (Université Paris-Saclay). Ce travail discute des différentes notions d'interpolation issues de la littérature sur les enveloppes de confiance. Il montre notamment que la notion de cohérence définie dans Goeman et al. (2021) ne caractérise pas entièrement la classe des procédures de closed testing, et qu'une notion de cohérence plus forte est nécessaire pour cela. Il fournit également de nouvelles procédures pour construire des enveloppes de confiance sur le nombre de fausses découvertes, à partir d'une famille de régions de référence, en considérant des enveloppes de confiance locales au sein de ces régions de référence. Ces nouvelles procédures, sous certaines hypothèses sur la structure des régions de référence, sont calculables en pratique, et sont toujours au moins aussi performantes que celles de Durand et al. (2020) et Xu et al. (2024a).

1.4.5 Contributions additionnelles

En plus des contributions présentées ci-dessus, je fournis également deux autres contributions. La première est l'Appendice C et montre que la procédure $\mathrm{BH}+$ introduite dans Chen (2020) peut être réduite à la procédure de BH appliquée à des p-valeurs modifiées, et que la validité de la procédure $\mathrm{BH}+$ peut être dérivée de la validité de la procédure de BH, puisque nous montrons que la propriété wPRDS est préservée par transformation croissante des p-valeurs. La seconde contribution est une note sur la fission de variables aléatoires de Bernoulli, qui constitue l'Appendice D, dans le même esprit que Leiner et al. (2025). Il pourrait être particulièrement intéressant d'utiliser ce résultat dans le cadre du Chapitre 4 lors de travaux futurs.

Chapter 2

Introduction (English)

Contents

2.1 Motivations 26
2.2 Simple hypothesis testing 26
2.2.1 Formal framework 26
2.2.2 p-values and e-values 27
2.2.3 Discrete hypothesis testing and Fisher's exact test 29
2.3 Multiple testing framework 31
2.3.1 Formal framework 31
2.3.2 Family-wise error rate (FWER) and closed testing procedures . . . 33
2.3.3 False discovery rate (FDR) and BH procedure 37
2.3.4 Confidence envelopes on the number of false discoveries 39
2.3.5 Conformal Prediction and multiple testing 42
2.4 Contributions 44
2.4.1 Confidence envelopes for the false discoveries with heterogeneous data 44
2.4.2 FDR control and FDP bounds for conformal link prediction 45
2.4.3 Heterogeneous compound p-values and closure principle 45
2.4.4 On strongly coherent confidence envelopes for the false discoveries 45
2.4.5 Additional contributions 46

This chapter aims at providing a self-contained introduction to the multiple testing framework, and describes most of the basic concepts and tools that will be used in the rest of this thesis. It is organized as follows: Section 2.1 gives some motivations for multiple testing, Section 2.2 describes the simple hypothesis testing framework and the notions of p-values and e-values. Section 2.3 introduces the multiple testing framework and the basic notions of error rates from the multiple testing literature. It also provides a brief overview of the classical multiple testing procedures. This section finally introduces two more recent frameworks, the construction of confidence envelopes on the false discovery proportion, and the links between multiple testing and conformal prediction. This chapter is concluded by Section 2.4 which provides an overview of the subsequent contributions in each chapters of this thesis.

2.1 Motivations

Multiple testing arises in many scientific fields, such as genomics, clinical trials, neuroscience, astrophysics or social sciences. These fields often involve the simultaneous testing of a large number of hypotheses, which can lead to an increased risk of false discoveries. For instance, if one wants to determine which substances are responsible for a particular disease, one may test the effect of a large number of substances and, *statistically*, some of them will appear to be significant by chance, even if they have no real effect. We briefly describe below some of the afore mentioned applications.

Genomics A typical example of multiple testing in genomics is the genome-wide association studies (GWAS), which which have been a very active field of the bioinformatics literature since the 2000s, following the development of genotyping technologies. They aim to identify genetic variants (SNPs, for single nucleotide polymorphisms) associated with a particular trait or disease (*e.g.* diabetes, auto-immune diseases or schizophrenia). In a GWAS study, since no a priori hypothesis is specified, researchers test hundreds of thousands or even millions of SNPs for association with the trait of interest (Dudbridge and Gusnanto, 2008; Visscher et al., 2017).

Clinical trials Clinical trials are research studies that aim to evaluate the safety and efficacy of new medical treatments or interventions on human subjects. For instance, Cui et al. (2021), Chapter 11, describe a clinical trial that aims to evaluate the effect of a new drug with respect to the dose of the drug. The trial involves testing multiple hypotheses corresponding to different doses of the drug, and the goal is to identify the doses that are efficient while controlling the risk of false discoveries.

Replication crisis Multiple testing has been underused or misused in various fields with practices like *p-hacking* or *HARKing* (Hypothesizing After the Results are Known), leading partially to the replication crisis (Ioannidis, 2005a; Camerer et al., 2018), that is, the inability to reproduce results from published studies. Liu and Shiraito (2023) pointed out that most of studies in political science did not correct for multiple testing when they should have, and suggested methodologies to correct for multiple testing in this field.

2.2 Simple hypothesis testing

This section gives an introduction to simple hypothesis theory and two fundamental statistical objects in this framework: the p-value and the e-value.

2.2.1 Formal framework

We consider a statistical model $(\mathfrak{F}, \mathcal{X})$ where $\mathfrak{F}$ is a family of probability measures on a measurable space $(\mathcal{X}, \Xi)$. We observe a random variable X, defined on an abstract probability space, $(\Omega, \Sigma, \mathbb{P})$, following a distribution P_0 which belongs to the family of distributions $\mathfrak{F}$. Hypothesis testing aims at deciding if a prior assumption on the distribution P_0 is reasonable. More formally, a test is a 0/1 decision $\varphi(X)$ based on the observation X. It returns 1 if the prior assumption on the data distribution, that is

$P_0 \in H_0 \subseteq \mathfrak{F}$, is rejected and 0 otherwise. The set H_0 is called the null hypothesis and we define another subset of distributions, H_1, called the alternative hypothesis. Classically, two types of errors are considered in this framework: the type I error, which is the supremum of the probabilities to reject the null hypothesis whereas it is true, and the type II error, which is the supremum of the probabilities to accept the null hypothesis whereas it is false. Formally, the type I error of a test φ is defined as $\sup_{P\in H_0} \mathbb{E}_{X\sim P}\big[\varphi(X)\big] = \sup_{P\in H_0} \mathbb{P}_{X\sim P}\big(\varphi(X) = 1\big)$ and the type II error is defined as $\sup_{P\in H_1} \mathbb{E}_{X\sim P}\big[1-\varphi(X)\big] = \sup_{P\in H_1} \mathbb{P}_{X\sim P}\big(\varphi(X) = 0\big)$. The test is said to be a α-level test if its type I error is at most α.

In the one-sided Gaussian case, $\mathfrak{F}$ is the set of all Gaussian distributions with variance 1, that is $\mathfrak{F} = \big\{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\big\}$. A common testing problem in this case is to test H_0 : "$\mu \leq 0$ " against H_1 : "$\mu > 0$ ", that is to test if the mean of the distribution P_0 is positive or not. If we define $\mathbf{\Phi}$ as the cumulative distribution function *(c.d.f.)* of the standard normal distribution $\mathcal{N}(0, 1)$, then the test $\varphi = \mathbb{1}\big\{X > \mathbf{\Phi}^{-1}(1-\alpha)\big\}$, where $\mathbb{1}\{\cdot\}$ is the indicator function, is a level α test for this testing problem.

2.2.2 p-values and e-values

Defining a test for a level $\alpha \in [0, 1]$ may be too restrictive in practice, because it requires specifying the significance level a priori. Instead, it is common to define a p-value $p(X)$, which is a random variable with value in $[0, 1]$ that is *superuniform* under the null hypothesis.

Definition 2.2.1 (Superuniformity)**.** *Let T be a random variable. Then we say that T is superuniform if for all $\alpha \in [0, 1]$, $\mathbb{P}(T \leq \alpha) \leq \alpha$.*

We define formally a p-value as follows:

Definition 2.2.2 (p-value)**.** *Let $p : \mathcal{X} \to [0, 1]$ be a statistic. We say that p is a p-value for the null hypothesis H_0 if for all $P \in H_0$, $X \sim P$ implies that $p(X)$ is superuniform.*

Hence, a p-value defines a family of α-level tests $\varphi_\alpha = \mathbb{1}\big\{p(X) \leq \alpha\big\}$ for $\alpha \in [0, 1]$. Classically, if H_0 is a single point hypothesis — that is, $H_0 = \{P\}$ for $P \in \mathfrak{F}$ — and that X takes values in $\mathbb{R}$, then we can define a p-value as $p(X) = F(X)$, where F is the *c.d.f.* of P. The *c.d.f.* F of X under P is defined as $F(x) = \mathbb{P}_{X\sim P}(X \leq x)$ for all $x \in \mathbb{R}$. This p-value is of great interest because it is, in some sense, the best· p-value that we can define for this testing problem. Indeed, let G be a nondecreasing function from $\mathbb{R}$ to $[0, 1]$ such that $G(X)$ is superuniform under P. Then, for all $x \in \mathbb{R}$, we have $\mathbb{P}_{X\sim P}(X \leq x) \leq \mathbb{P}_{X\sim P}\big(G(X) \leq G(x)\big)$ because G is nondecreasing. Since $G(X)$ is superuniform under P, we have $\mathbb{P}_{X\sim P}\big(G(X) \leq G(x)\big) \leq G(x)$. Hence, combining these two inequalities, we have $F(x) \leq G(x)$ for all $x \in \mathbb{R}$. Hence, any other p-value $G(X)$ built from a nondecreasing function G is larger than $F(X)$.

Similarly, if G is nonincreasing, and $G(X)$ is superuniform under P, then $G(x) \geq 1 - F(x)$ for all $x \in \mathbb{R}$. Indeed, for all $x \in \mathbb{R}$, we have $1 - F(x) \leq \mathbb{P}_{X\sim P}(X \geq x) \leq \mathbb{P}_{X\sim P}\big(G(X) \leq G(x)\big) \leq G(x)$.

Jump back to the one-sided Gaussian case, a p-value for the testing problem H_0 : "$\mu \leq 0$ " against H_1 : "$\mu > 0$ " is given by $p(X) = 1 - \mathbf{\Phi}(X)$. Note that $p(X)$ is a p-value

because for all $\alpha \in [0,1]$,

$$\begin{aligned}
\sup_{P \in H_0} \mathbb{P}_{X\sim P}\big(p(X) \leq \alpha\big) &= \sup_{\mu \leq 0} \mathbb{P}_{X\sim \mathcal{N}(\mu,1)}\big(1 - \mathbf{\Phi}(X) \leq \alpha\big) \\
&= \sup_{\mu \leq 0} \mathbb{P}_{X\sim \mathcal{N}(\mu,1)}\big(X \geq \mathbf{\Phi}^{-1}(1-\alpha)\big) \\
&= \sup_{\mu \leq 0} \mathbb{P}_{X\sim \mathcal{N}(\mu,1)}\big(X - \mu \geq \mathbf{\Phi}^{-1}(1-\alpha) - \mu\big) \\
&= \sup_{\mu \leq 0} 1 - \mathbf{\Phi}\big(\mathbf{\Phi}^{-1}(1-\alpha) - \mu\big) \\
&= 1 - \mathbf{\Phi}\big(\mathbf{\Phi}^{-1}(1-\alpha)\big) = \alpha,
\end{aligned}$$

hence for all $P \in H_0$, $\mathbb{P}_{X\sim P}\big(p(X) \leq \alpha\big) \leq \alpha$.

Another recent formalism to define a family of tests is to define an e-value $e(X)$, which is a non-negative random variable such that $\sup_{P\in H_0} \mathbb{E}_{X\sim P}\big[e(X)\big] \leq 1$. Hence, an e-value defines a family of tests through $\varphi_\alpha = \mathbb{1}\big\{e(X) \geq 1/\alpha\big\}$ for $\alpha \in [0,1]$ using Markov's inequality. Note that for an e-value $e(X)$, the random variable $1/e(X) \wedge 1$ is a p-value. However, it is not true in general that for a p-value $p(X)$, the random variable $1/p(X)$ is an e-value — for example when $p(X) \sim \mathcal{U}\big([0,1]\big)$, since in this case $1/p(X)$ has an infinite expectation. A classical e-value, when testing a simple null hypothesis $H_0 = \{P\}$ against a simple alternative hypothesis $H_1 = \{Q\}$ such that $Q \ll P$, is the likelihood ratio $e(X) = \frac{\mathrm{d}Q}{\mathrm{d}P}(X)$, which is the Radon-Nikodym derivative of Q with respect to P. Indeed, we have $\mathbb{E}_{X\sim P}\big[e(X)\big] = \mathbb{E}_{X\sim P}\big[\frac{\mathrm{d}Q}{\mathrm{d}P}(X)\big] = \int \frac{\mathrm{d}Q}{\mathrm{d}P}(x)\mathrm{d}P(x) = Q(\mathcal{X}) = 1$. We can conceptualize from this simple example an e-value for the one-sided Gaussian case, given by $e(X) = \exp(\mu X - \mu^2/2)$, which is the likelihood ratio between the standard normal distribution and the normal distribution with mean $\mu > 0$ and variance 1. Indeed, for all $\nu \leq 0$, we have

$$\begin{aligned}
\mathbb{E}_{X\sim\mathcal{N}(\nu,1)}\big[e(X)\big] &= \mathbb{E}_{X\sim\mathcal{N}(\nu,1)}\Big[\exp\big(\mu X - \mu^2/2\big)\Big] \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\big(\mu x - \mu^2/2\big) \exp\big(-(x-\nu)^2/2\big) dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\big(\mu x - \mu^2/2 - x^2/2 + \nu x - \nu^2/2\big) dx \\
&= \frac{\exp(\mu\nu)}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\Big(-\big(x - (\mu+\nu)\big)^2/2\Big) dx \\
&= \exp(\mu\nu) \leq 1.
\end{aligned}$$

Remark that the e-value $e(X)$ is a non-negative random variable, and that the p-value $1/e(X) \wedge 1$ is strictly superuniform when $\nu \leq 0$. This is a general phenomenon: as explained before, if $p(X) \sim \mathcal{U}\big([0,1]\big)$, then $1/p(X)$ is not an e-value, and thus, for an e-value $e(X)$, the p-value $1/e(X) \wedge 1$ is strictly superuniform, otherwise $e(X)$ would not be an e-value. Therefore, the family of tests defined by the e-value $e(X)$, $\varphi_\alpha = \mathbb{1}\big\{e(X) \geq 1/\alpha\big\}$, is such that for all $\alpha \in [0,1]$, φ_α is a level α test, but its type I error is strictly smaller than α. As a result, this family of tests is more conservative than the family of tests defined by a p-value, since these statistical objects are allowed to be exactly uniform under the null hypothesis. At this point, the justification of e-values for testing a single null hypothesis may be questionable, but they have some practical properties that make them interesting, in particular in the multiple testing framework. One of these properties is that the set of e-values is equivalent to the set of post-hoc p-values (Koning, 2025) through the bijection $e \mapsto 1/e \wedge 1$. This subset of p-values has the property

that for all random variable $\hat{\alpha}$ such that $\hat{\alpha} \in (0, 1]$, we have

$$\mathbb{E}_{X\sim P}\left[\frac{\mathbb{1}\{p(X) \leq \hat{\alpha}\}}{\hat{\alpha}}\right] \leq 1, \quad \forall P \in H_0, \tag{2.1}$$

or equivalently,

$$\mathbb{E}_{X\sim P}\left[\sup_{\alpha\in(0,1]} \frac{\mathbb{1}\{p(X) \leq \alpha\}}{\alpha}\right] \leq 1, \quad \forall P \in H_0. \tag{2.2}$$

We will see in Section 2.3.3 that this property extends to the multiple testing framework, especially for the FDR control methods based on e-values.

2.2.3 Discrete hypothesis testing and Fisher's exact test

Statistical methods for discrete data are used when the response variable consists of counts or categorical outcomes rather than continuous measurements. Typical examples include binary outcomes (success/failure), contingency tables, or count data. In these settings, classical tests based on normality assumptions are often inappropriate, and methods specifically designed for discrete distributions are preferred. Hence, the p-values in this context are often discrete, which can lead to conservative tests.

We will illustrate this fact with one of the most widely used procedures, *Fisher's exact test* (Fisher, 1934), which assesses the association between two categorical variables in a 2×2 contingency table.

We first introduce the formal framework of Fisher's exact test, and then we discuss the issue of discreteness in this context. We follow here the formalism of Agresti (2013) in a *multinomial setting*. Let X and Y be two categorical variables, where X possesses $I \in \mathbb{N}$ categories and Y possesses $J \in \mathbb{N}$ categories. Hence X is a random variable such that $X \in \{1, \dots, I\}$ and Y is a random variable such that $Y \in \{1, \dots, J\}$. The observations are a set of n pairs $\left(X^\ell, Y^\ell\right)_{\ell\in\{1,\dots,n\}}$, or equivalently a set of n identical binary matrices $Z^1, \dots, Z^n \sim P$; for all $\ell \in \{1, \dots, n\}$ $Z^\ell = \left(Z^\ell_{ij}\right)_{\substack{i\in\{1,\dots,I\}\\ j\in\{1,\dots,J\}}}$ and we have for all $i \in \{1, \dots, I\}$ and $j \in \{1, \dots, J\}$,

$$Z^\ell_{ij} = \begin{cases} 1 \text{ if } (X_\ell, Y_\ell) = (i, j) \\ 0 \text{ otherwise} \end{cases}.$$

Usually, the quantity that is observed is the contingency matrix $T = \sum_{\ell=1}^n Z^\ell$. We define the following quantities for all $i \in \{1, \dots, I\}$ and $j \in \{1, \dots, J\}$:

$$\pi_{ij} = \mathbb{P}(Y = j, X = i),\ \pi_{i+} = \mathbb{P}(X = i) = \sum_{j=1}^J \pi_{ij},\ \pi_{+j} = \mathbb{P}(Y = j) = \sum_{i=1}^I \pi_{ij},$$

$$n_{ij} = \sum_{\ell=1}^n Z^\ell_{ij},\ n_{i+} = \sum_{j=1}^J n_{ij},\ n_{+j} = \sum_{i=1}^I n_{ij},\ n = \sum_{i=1}^I \sum_{j=1}^J n_{ij}.$$

The variables n_{ij} represent the counts of observations in each cell of the contingency table, and the variables n_{i+} and n_{+j} represent the row and column totals respectively — also called marginal counts. This is represented in the following contingency table:

	$Y = 1$	$Y = 2$	Total
$X = 1$	n_{11}	n_{12}	n_{1+}
$X = 2$	n_{21}	n_{22}	n_{2+}
Total	n_{+1}	n_{+2}	n

For the sake of simplicity, we assume here that $I = 2 = J$ since it is the framework of Fisher's exact test, but the results can be extended to the general case.

We assume that the data follows an *independent binomial sampling within rows*, that is, for all $i \in \{1, 2\}$, n_{i+} is fixed, and the couple (n_{i1}, n_{i2}) follows a multinomial distribution with parameters n_{i+} and $(\pi_{i1}/\pi_{i+}, \pi_{i2}/\pi_{i+})$, which is equivalent to say that $n_{i1} \sim \mathcal{B}(n_{i+}, \pi_{i1}/\pi_{i+})$ and $n_{i2} = n_{i+} - n_{i1}$, where $\mathcal{B}(n, p)$ is the binomial distribution with parameters n and p. Further, this assumption implies that n_{11} and n_{21} are independent. Then, we want to test the null hypothesis that X and Y are independent, that is, for all $i \in \{1, 2\}$ and $j \in \{1, 2\}$, $\pi_{ij} = \pi_{i+}\pi_{+j}$, against the alternative hypothesis that X and Y are not independent.

Let $P \in H_0$, we are reasoning conditionally on the counts n_{+1}. Note that $n_{+1} + n_{+2} = n$, and that the distribution of n_{11} is sufficient to determine the distribution of Z under the null hypothesis, since it fixes the values of n_{12}, n_{21}, and n_{22} because n_{1+}, n_{2+}, and n_{+1} are fixed. Let $k \in \{0, \dots, n_{1+}\}$, $a \in \{0, \dots, n\}$, the probability of observing $n_{11} = k$ under the null hypothesis is:

$$\begin{aligned}
\mathbb{P}(n_{11} = k \mid n_{+1} = a) =& \frac{\mathbb{P}(n_{11} = k, n_{+1} = a)}{\mathbb{P}(n_{+1} = a)} \\
=& \frac{\mathbb{P}(n_{11} = k, n_{21} = a - k)}{\mathbb{P}(n_{+1} = a)} = \frac{\mathbb{P}(n_{11} = k)\mathbb{P}(n_{21} = a - k)}{\mathbb{P}(n_{+1} = a)} \\
=& \frac{\binom{n_{1+}}{k}\left(\frac{\pi_{11}}{\pi_{1+}}\right)^k\left(1 - \frac{\pi_{11}}{\pi_{1+}}\right)^{n_{1+}-k}\binom{n_{2+}}{a-k}\left(\frac{\pi_{21}}{\pi_{2+}}\right)^{a-k}\left(1 - \frac{\pi_{21}}{\pi_{2+}}\right)^{n_{2+}-(a-k)}}{\binom{n}{a}(\pi_{+1})^a(1 - \pi_{+1})^{n-a}} \qquad (2.3) \\
=& \frac{\binom{n_{1+}}{k}\binom{n_{2+}}{a-k}}{\binom{n}{a}} = \frac{\binom{n_{1+}}{k}\binom{n-n_{1+}}{a-k}}{\binom{n}{a}},
\end{aligned}$$

where (2.3) occurs because $n_{+1} = n_{11} + n_{12}$ follows a binomial distribution of parameters n and π_{+1} hence, conditionally to $(n_{+1} = a)$, the probability of observing the table

	$Y = 1$	$Y = 2$	Total
$X = 1$	k	$n_{1+} - k$	n_{1+}
$X = 2$	$a - k$	$n_{2+} - (a - k)$	n_{2+}
Total	a	$n - a$	n

under the null hypothesis, is given by the hypergeometric distribution with parameters n, n_{1+}, and a. Note that under other assumptions on the distribution of the contingency data, such as *multinomial sampling* or *Poisson sampling*, the distribution of n_{11} conditionally to the marginals is still hypergeometric (see Section 3.5.1 of Agresti, 2013).

The one-sided p-value is then obtained by summing the probabilities of all tables that are at least as extreme as the observed one — let us say $k \in \{0, 1, \dots, n\}$ — that is, $p = \mathbb{P}(n_{11} \leq k \mid n_{1+}, n_{2+}, n)$. To obtain the two sided p-value, we sum the probabilities of all tables that are at least as extreme as the observed one in both directions, that is,

$$p(k) = \sum_{\substack{i \in \{0,\dots,n\} \text{such that} \\ \mathbb{P}(n_{11}=i|n_{+1}=a) \leq \mathbb{P}(n_{11}=k|n_{+1}=a)}} \mathbb{P}(n_{11} = i \mid n_{+1} = a).$$

Finally, remark that the p-values obtained from Fisher's exact test are discrete, and that their support may change depending on the values of the marginals. This observation is important in the context of multiple testing, where we may want to apply a

multiple testing procedure to a set of p-values obtained from Fisher's exact test with different marginals. This particular setting, where the marginals of the p-values vary across under the null hypothesis, is called *heterogeneous multiple testing*. Fisher's exact test will be used as a simulation example in Chapter 3 and Chapter 6.

The discreteness of the data can lead to conservative p-values, which may reduce the power of the test, especially when the level α of the test does not belong to the support of the p-value. To address this issue, various methods have been proposed to improve the power of discrete tests, such as randomization techniques. We give an example of a randomization technique in Appendix A.4. However, these methods may be inappropriate in some contexts, especially when the practitioner wants reproducible results. This fact motivates the consideration of the heterogeneous multiple testing framework since it allows for more powerful testing procedures without relying on external randomization.

2.3 Multiple testing framework

This section gives an introduction to the classical theory of multiple testing.

2.3.1 Formal framework

In the multiple testing framework, we consider the same statistical model $(\mathfrak{F}, \mathcal{X})$ as in Section 2.2.1, but we are interested in testing $m \in \mathbb{N}$ null hypotheses $H_1, \dots, H_m \subseteq \mathfrak{F}$. For all $P \in \mathfrak{F}$, we denote by $\mathcal{H}_0(P) = \{i \in [\![1, m]\!] : P \in H_i\}$ the set of true null hypotheses and by $\mathcal{H}_1(P) = \mathcal{H}_0^c$ the set of false null hypotheses, where S^c denotes the complement of S in $[\![1, m]\!]$ for all $S \subseteq [\![1, m]\!]$. We also define the notion of *intersection hypothesis* H_S as

$$H_S = \bigcap_{i \in S} H_i. \tag{2.4}$$

A special case of intersection hypothesis is the *global null hypothesis* $H_{[\![1,m]\!]} = \bigcap_{i \in [\![1,m]\!]} H_i$, and we say that we are in the *global null setting*, or *under the full null,* if we only consider $P \in H_{[\![1,m]\!]}$. Note that $H_{[\![1,m]\!]}$ may be empty, but in most practical situations, it is not the case. For example, in the multivariate Gaussian case, $X = (X_1, \dots, X_m)$ is a Gaussian vector of dimension m and we want to test the null hypotheses H_i : "$X_i \sim \mathcal{N}(\mu_i, 1), \mu_i \leq 0$ " for all $i \in [\![1, m]\!]$. In this case, the global null hypothesis satisfies

$$\left\{ \bigotimes_{1 \leq i \leq m} \mathcal{N}(\mu_i, 1) : \mu_1, \dots, \mu_m \leq 0 \right\} \subseteq H_{[\![1,m]\!]}.$$

A multiple testing procedure R is a function that takes as input the observation X and returns a subset of $[\![1, m]\!]$, that is the set of rejected null hypotheses. Hence, we are interested in controlling the number of false discoveries, that is the number of true null hypotheses that are falsely rejected. For a fixed distribution P, we define the number of false discoveries of a procedure R as

$$V\big(R(X)\big) = |\mathcal{H}_0(P) \cap R(X)|, \tag{2.5}$$

hence the function V depends on P through the set of true null hypotheses $\mathcal{H}_0(P)$.

The core goal of multiple testing is to control the number of false discoveries due to the multiplicity of the tests; as we illustrate with the following example: if we test

m true null hypotheses at level α, independently, the probability to reject at least one true null hypothesis is $1-(1-\alpha)^m$, which is much larger than α when m is large since $1-(1-\alpha)^m \to 1$ as $m \to \infty$. Therefore, there is a need for alternative error metrics to control the number of false discoveries.

In the following, depending on the context, we will consider either p-values or e-values for each null hypothesis. That is, we will consider m p-values $p_1(X), \dots, p_m(X)$ such that for all $\alpha \in [0,1]$,

$$\sup_{P \in H_i} \mathbb{P}_{X \sim P}\big(p_i(X) \leq \alpha\big) \leq \alpha, \quad i \in [\![1, m]\!], \tag{2.6}$$

or m e-values $e_1(X), \dots, e_m(X)$ such that

$$\sup_{P \in H_i} \mathbb{E}_{X \sim P}\big[e_i(X)\big] \leq 1, \quad i \in [\![1, m]\!]. \tag{2.7}$$

We will consider some dependence structures between the family of p-values or e-values. We begin with two classical dependence assumptions on the p-values, which are the independence assumption and the weak positive regression dependence on a subset (wPRDS) assumption. The independence assumption states that the p-values corresponding to true null hypotheses — that is, the p-values $p_i(X)$ for $i \in \mathcal{H}_0(P)$ — are independent random variables, and that these p-values are independent of the p-values corresponding to false null hypotheses:

Assumption 2.1 (Independence)**.** *For all $P \in \mathfrak{F}$, $(p_i)_{i \in \mathcal{H}_0(P)}$ is a vector of independent random variables and is independent of $(p_i)_{i \in \mathcal{H}_1(P)}$.*

This assumption is very strong and may not be satisfied in practice. Another classical assumption is the wPRDS assumption (Finner et al., 2009; Roquain, 2011), which states that the p-values are positively dependent in a certain sense.

Assumption 2.2 (wPRDS)**.** *Let $D \subseteq [0,1]^m$. We say that D is a nondecreasing set if for all $u \in D$, for all $v \in [0,1]^m$ such that $v \geq u$ componentwise, we have $v \in D$.*
We say that the p-values $(p_i)_{i \in [\![1,m]\!]}$ are wPRDS if they satisfy the following condition: for all $P \in \mathfrak{F}$, for all $i \in \mathcal{H}_0(P)$, for all nondecreasing measurable set $D \subseteq [0,1]^m$, the function

$$u \mapsto \mathbb{P}_{X \sim P}\big((p_j)_{j \in [\![1,m]\!]} \in D \mid p_i \leq u\big)$$

is nondecreasing on the set $\big\{u : 0 \leq u \leq 1, \mathbb{P}_{X \sim P}(p_i \leq u) > 0\big\}$.

This last assumption is weaker than the independence assumption, meaning that if the p-values satisfy Assumption 2.1, then they also satisfy Assumption 2.2, but the converse is not true in general. For example, in the multivariate Gaussian case, if $X = (X_1, \dots, X_m) \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma_{i,j} \geq 0$ for all $i, j \in [\![1, m]\!]$, then the p-values $p_i(X) = 1 - \mathbf{\Phi}\big(X_i/\sqrt{\Sigma_{i,i}}\big)$ satisfy Assumption 2.2 but not necessarily Assumption 2.1 (see Giraud, 2015, Exercise 10.6.3). Note that, historically, another assumption, called positive regression dependence on a subset (PRDS), was introduced by Benjamini and Yekutieli (2001), where the conditionning "$p_i \leq u$" is replaced by "$p_i = u$" in Assumption 2.2. However, while the PRDS assumption is stronger than the wPRDS assumption (see the proof of Proposition 3.6 in Blanchard and Roquain, 2008 for instance), the wPRDS assumption is a sufficient condition for the FDR control of the BH procedure, as we will see in Section 2.3.3. Assuming that one of these two assumptions holds means that we restrict the family of distributions $\mathfrak{F}$ to a submodel $\mathfrak{F}' \subseteq \mathfrak{F}$ such that for

all $P \in \mathfrak{F}'$, the p-values satisfy Assumption 2.1 or Assumption 2.2. In the case of the multivariate one-sided Gaussian model, assuming Assumption 2.1 implies that

$$H_{[\![1,m]\!]} = \left\{ \bigotimes_{1 \leq i \leq m} \mathcal{N}(\mu_i, 1) : \mu_1, \dots, \mu_m \leq 0 \right\},$$

for instance. Contrary to the simple hypothesis testing framework, where the type I error is the natural error metric to control, in the multiple testing framework there is no natural error metric to control. Should we control the probability to reject at least one true null hypothesis, or the expected proportion of true null hypotheses that are rejected, or something else?

It will be sometimes convenient to sort p-values in increasing order, to determine which null hypotheses to reject. To this end, we denote the order statistics of the p-values $p_1, \dots, p_m$ as $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$.

In the following, we introduce the most classical error metrics to control in the multiple testing framework, and we give some examples of procedures controlling these error metrics. Then we will introduce the notion of confidence envelope, which is a generalization of FWER-controling procedures, and we will give some examples of confidence envelopes. Finally, we will introduce the conformal prediction and provide some links with the multiple testing theory

2.3.2 Family-wise error rate (FWER) and closed testing procedures

The family-wise error rate (FWER) is the most classical error metric to control in the multiple testing framework. Its formal definition can be traced back to Tukey (1953) (Benjamini and Braun, 2002), but the issue of performing multiple tests simultaneously was already spotted by Bonferroni (1936) for example.

We define the family-wise error rate (FWER) of a procedure R as the probability to reject at least one true null hypothesis, that is , for all $P \in \mathfrak{F}$,

$$\mathrm{FWER}_P(R) = \mathbb{P}_{X \sim P}\big(\exists i \in \mathcal{H}_0 : i \in R(X)\big) = \mathbb{P}_{X \sim P}\Big(V\big(R(X)\big) > 0\Big). \tag{2.8}$$

We say that a procedure R achieves *strong control* of the FWER at level $\alpha \in [0,1]$ if for all $P \in \mathfrak{F}$, $\mathrm{FWER}_P(R) \leq \alpha$, and that R achieves *weak control* of the FWER at level α if $\mathrm{FWER}_P(R) \leq \alpha$ for all $P \in H_{[\![1,m]\!]}$, or, equivalently, $\mathbb{P}_{X \sim P}\Big(R(X) \neq \varnothing\Big) \leq \alpha$ for all $P \in H_{[\![1,m]\!]}$. There exist several procedures controlling the FWER; the most classical one is the Bonferroni procedure, which rejects all null hypotheses H_i such that the corresponding p-value $p_i(X)$ is smaller than α/m for a given significance level α. The Bonferroni procedure controls the FWER at level α, without any assumptions on the dependence structure of the p-values. Nevertheless, it is known that the Bonferroni procedure is conservative, and that it can be improved by using the closed testing procedures introduced by Marcus et al. (1976), and widely studied in the multiple testing literature (Sonnemann and Finner, 1988; Sonnemann, 1982; Sonnemann, 2008; Romano et al., 2011). To define the closure principle, we need to introduce the notion of *local tests*, which is a family of tests $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$ for all intersection hypotheses H_S. Then, the closure principle defines a new family of tests $\psi = (\psi_i)_{i \in [\![1,m]\!]}$, called *effective local tests*, as follows:

$$\forall S \subseteq [\![1,m]\!], \psi_S(X) = \min_{S \subseteq T \subseteq [\![1,m]\!]} \varphi_T(X). \tag{2.9}$$

Marcus et al. (1976) showed that, if the family of tests φ is such that for all $P \in \mathfrak{F}$, $\varphi_{\mathcal{H}_0(P)}$ is an α-level test, then the family of tests ψ satisfies the following property:

$$\forall P \in \mathfrak{F}, \mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big) \leq \alpha. \tag{2.10}$$

Note that (2.10) holds when φ is a family of α-level local tests — that is, for all $S \subseteq [\![1, m]\!]$, φ_S is an α-level test for the intersection hypothesis H_S — since it implies that for all $P \in \mathfrak{F}$, $\mathbb{P}_{X\sim P}\big(\varphi_{\mathcal{H}_0(P)}(X) = 1\big) \leq \alpha$. The proof of (2.10) is quite straightforward: if there exists $S \subseteq \mathcal{H}_0(P)$ such that $\psi_S(X) = 1$, then $\varphi_{\mathcal{H}_0(P)}(X) = 1$, otherwise it would contradict that $\psi_S(X) = 1$. Hence, we have, for all $P \in \mathfrak{F}$, $\mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big) \leq \mathbb{P}_{X\sim P}\big(\varphi_{\mathcal{H}_0(P)}(X) = 1\big) \leq \alpha$. Note that here, we already considered the formalism adapted to the construction of confidence envelopes on the number of false discoveries described in Goeman et al. (2021) and first introduced in Goeman and Solari (2011), but the closed testing procedure for the FWER control is a special case of this construction, where we only consider the local tests ψ_S for $S \subseteq [\![1, m]\!]$ such that $|S| = 1$. Indeed if we consider the multiple testing procedure

$$R_\varphi(X) = \big\{i \in [\![1, m]\!] : \psi_{\{i\}}(X) = 1\big\}, \tag{2.11}$$

we have

$$\begin{aligned}
\sup_{P\in\mathfrak{F}} \mathrm{FWER}_P\left(R_\varphi(X)\right) &= \sup_{P\in\mathfrak{F}} \mathbb{P}_{X\sim P}\big(\exists i \in \mathcal{H}_0(P) : \psi_{\{i\}}(X) = 1\big) \\
&\leq \sup_{P\in\mathfrak{F}} \mathbb{P}_{X\sim P}\big(\exists S \subseteq \mathcal{H}_0(P) : \psi_S(X) = 1\big) \\
&\leq \alpha,
\end{aligned}$$

where the last inequality comes from (2.10).

Let $\alpha \in (0, 1)$, the closed testing procedure allows to improve the Bonferroni procedure at level α by using the local tests φ_α defined as:

$$\forall S \subseteq [\![1, m]\!], \varphi_{\alpha,S} = \mathbb{1}\left\{\min_{i\in S} p_i \leq \frac{\alpha}{|S|}\right\}. \tag{2.12}$$

Hence, φ_α is an α-level local tests family since for $S \subseteq [\![1, m]\!]$, for all $P \in H_S$,

$$\begin{aligned}
\mathbb{P}_{X\sim P}(\varphi_{\alpha,S}(X) = 1) &= \mathbb{P}_{X\sim P}\left(\inf_{i\in S} p_i(X) \leq \frac{\alpha}{|S|}\right) \\
&= \mathbb{P}_{X\sim P}\left(\exists i \in S : p_i(X) \leq \frac{\alpha}{|S|}\right) \\
&\leq \sum_{i\in S} \mathbb{P}_{X\sim P}\left(p_i(X) \leq \frac{\alpha}{|S|}\right) \qquad (2.13) \\
&\leq \sum_{i\in S} \frac{\alpha}{|S|} = \alpha, \qquad (2.14)
\end{aligned}$$

where (2.13) comes from the union bound and (2.14) holds because $S \subseteq \mathcal{H}_0(P)$ and the p-values under the null hypothesis are superuniform. Then, the closed testing procedure define effective local tests $\psi_\alpha = (\psi_{\alpha,S})_{S\subseteq[\![1,m]\!]}$ by (2.9) with $\varphi = (\varphi_{\alpha,S})_{S\subseteq[\![1,m]\!]}$. It remains to make explicit the multiple testing procedure

$$R_{\varphi_\alpha}(X) = \big\{i \in [\![1, m]\!] : \psi_{\alpha,\{i\}}(X) = 1\big\}$$

for this choice of local tests. It is stated in the next proposition that the procedure R_{φ_α} is the Holm procedure (Holm, 1979):

Proposition 2.3.1. *The multiple testing procedure R_{φ_α} is the Holm procedure at level α, that is*

$$R_{\varphi_\alpha}(X) = R_\alpha^{\mathrm{Holm}} := \left\{ i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha}{m - k^* + 1} \right\}, \tag{2.15}$$

where $k^* = \max\left\{ k \in [\![1, m]\!] : \forall i \leq k, p_{(i)}(X) \leq \frac{\alpha}{m-i+1} \right\} \cup \{0\}$.

We show below the proof of this proposition.

Proof. Let $\tilde{k} = |R_{\varphi_\alpha}(X)|$. We first show that $\tilde{R}_\alpha^{\mathrm{Holm}}(X) = R_{\varphi_\alpha}(X)$, where $\tilde{R}_\alpha^{\mathrm{Holm}}(X)$ is defined with $\tilde{k}$ instead of k^* in the definition of $R_\alpha^{\mathrm{Holm}}(X)$. Let $i \in R_{\varphi_\alpha}^c$, we have $\psi_{\alpha,\{i\}}(X) = 0$, hence there exists $T \subseteq [\![1, m]\!]$ such that $i \in T$ and $\varphi_{\alpha,T}(X) = 0$. Furthermore, if $j \in R_{\varphi_\alpha}(X)$, then for all $T' \subseteq [\![1, m]\!]$ such that $j \in T'$, we have $\varphi_{\alpha,T'}(X) = 1$. Hence, $T \subseteq R_{\varphi_\alpha}^c$, and we have $\varphi_{\alpha,T}(X) = 0$, thus $\inf_{j \in T} p_j(X) > \frac{\alpha}{|T|}$, in particular with $j = i$, we have $p_i(X) > \frac{\alpha}{|T|} \geq \frac{\alpha}{|R_{\varphi_\alpha}^c|} = \frac{\alpha}{m-\tilde{k}} > \frac{\alpha}{m-\tilde{k}+1}$, and it holds that

$$R_{\varphi_\alpha}^c \subseteq \left\{ i \in [\![1, m]\!] : p_i(X) \geq \frac{\alpha}{m - \tilde{k}} \right\}.$$

Hence, we have $\tilde{R}_\alpha^{\mathrm{Holm}}(X) \subseteq R_{\varphi_\alpha}(X)$.

Conversely, if $i \in R_{\varphi_\alpha}(X)$, then $\psi_{\alpha,\{i\}}(X) = 1$, hence $\varphi_{\alpha,\{i\} \cup R_{\varphi_\alpha}^c}(X) = 1$, consequently

$$\inf_{j \in \{i\} \cup R_{\varphi_\alpha}^c} p_j(X) \leq \frac{\alpha}{m - \tilde{k} + 1}.$$

It implies $p_i(X) \leq \frac{\alpha}{m-\tilde{k}+1}$, because for all $j \in R_{\varphi_\alpha}^c$, we have $p_j(X) > \frac{\alpha}{m-\tilde{k}+1}$ from the previous paragraph. Finally, we have $R_{\varphi_\alpha}(X) \subseteq \tilde{R}_\alpha^{\mathrm{Holm}}(X)$, and we conclude that

$$R_{\varphi_\alpha}(X) = \tilde{R}_\alpha^{\mathrm{Holm}}(X).$$

It remains to show that $\tilde{k} = k^*$. Note that $p_{(\tilde{k}+1)}(X) > \frac{\alpha}{m-\tilde{k}}$ from the proof above because R_{φ_α} is the set of hypotheses indices corresponding to the $\tilde{k}$ smallest p-values, hence the hypothesis corresponding to the p-value $p_{(\tilde{k}+1)}(X)$ belongs to $R_{\varphi_\alpha}^c$ or else it would contradict the definition of $\tilde{k}$. Therefore, $k^* \leq \tilde{k}$, because k^* is such that $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$ for all $i \leq k^*$.
Conversely, we show that if $i \leq \tilde{k}$, then $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$.
Assume for the purpose of contradiction that $p_{(i)}(X) > \frac{\alpha}{m-i+1}$, and denote by $j \in [\![1, m]\!]$ the hypothesis index corresponding to the p-value $p_{(i)}(X)$. Then the set T defined as $T = \left\{ \ell \in [\![1, m]\!] : p_\ell(X) > \frac{\alpha}{m-i+1} \right\}$ satisfies $|T| \geq m - i + 1$ because $p_{(i)}(X) > \frac{\alpha}{m-i+1}$. Hence $\varphi_{\alpha,T}(X) = 0$ because $\min_{\ell \in T} p_\ell(X) > \alpha/(m - i + 1) \geq \alpha/|T|$, implying that $\psi_{\alpha,\{j\}}(X) = 0$, which contradicts the fact that $j \in R_{\varphi_\alpha}(X) = \left\{ \ell \in [\![1, m]\!] : \psi_{\alpha,\{\ell\}}(X) = 1 \right\}$. Hence $k^* \geq \tilde{k}$ by definition of k^* as the largest integer satisfying that for all $i \leq k^*$, $p_{(i)}(X) \leq \frac{\alpha}{m-i+1}$.

Therefore, $\tilde{k} = k^*$. □

Like the Bonferroni procedure, the Holm procedure controls strongly the FWER at level α without any assumptions on the dependence structure of the p-values, and is more powerful than the Bonferroni procedure, because the Bonferroni procedure is a subset of the Holm procedure. The Holm procedure is a special case of a more general class of procedures called step-down procedures, which are defined as follows:

Definition 2.3.2 (Step-down procedure)**.** *A step-down procedure is a multiple testing procedure R such that there exists a sequence of thresholds $\tau_1 \leq \tau_2 \leq \ldots \leq \tau_m$ such that*

$$R(X) = \mathrm{SD}(\tau) = \left\{i \in [\![1, m]\!] : p_i(X) \leq \tau_{k^*}\right\}, \tag{2.16}$$

where $k^* = \max\left\{k \in [\![1, m]\!] : \forall i \leq k, p_{(i)}(X) \leq \tau_i\right\} \cup \{0\}$.

The sequence of thresholds $\tau = (\tau_1, \ldots, \tau_m)$ is called the critical values of the step-down procedure. The Holm procedure is a step-down procedure with critical values $\tau_i = \alpha/(m - i + 1)$ for $i \in [\![1, m]\!]$. Note that the historical proof by Holm (1979) of the FWER control of the Holm procedure did not use the closure principle. Closed testing procedures plays a central role among FWER controlling procedures, because any *coherent* FWER controlling procedure can be expressed as a closed testing procedure (Sonnemann, 1982). Gabriel (1969) historically defined the notion of coherent procedures as procedures such that if a hypothesis H_i is not rejected with $i \in [\![1, m]\!]$, then any hypothesis H_j, $j \in [\![1, m]\!]$ with $H_i \subset H_j$ must also not be rejected, where $\subset$ denotes the strict subset relation. Note that this notion of coherence does not restrict the space of multiple testing procedures if there are no nested hypotheses.

Finally, we expose a last classical FWER controlling procedure, which is the Simes procedure (Simes, 1986). This procedure achieves weak control of the FWER, under the PRDS assumption on the p-values — although historically, Simes (1986) proved its validity under Assumption 2.1. It is defined as follows for a pre-specified significance level α:

$$R_\alpha^{\mathrm{Sim}}(X) = \left\{i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha k^*}{m}\right\}, \tag{2.17}$$

where $k^* = \max\left\{k \in [\![1, m]\!] : p_{(k)}(X) \leq \frac{\alpha k}{m}\right\} \cup \{0\}$. The Simes procedure is a special case of a step-up procedure with critical values $\tau = \left(\frac{k\alpha}{m}\right)_{1 \leq k \leq m}$, which is defined as follows:

Definition 2.3.3 (Step-up procedure)**.** *A step-up procedure is a multiple testing procedure R such that there exists a family of critical values $\tau = (\tau_1, \ldots, \tau_m)$ such that*

$$R(X) = \mathrm{SU}(\tau) = \left\{i \in [\![1, m]\!] : p_i(X) \leq \tau_{k^*}\right\}, \tag{2.18}$$

where $k^* = \max\left\{k \in [\![1, m]\!] : p_{(k)}(X) \leq \tau_k\right\} \cup \{0\}$.

Hence the step-up procedure differs from the step-down procedure in the definition of k^*, and we always have $\mathrm{SD}(\tau) \subseteq \mathrm{SU}(\tau)$.

The Simes procedure validity under the full null is a consequence of the Simes inequality (Seeger, 1968; Simes, 1986), which states that, for all $P \in H_{[\![1,m]\!]}$ such that the p-values are under Assumption 2.2 — although it was historically proved under Assumption 2.1,

$$\mathbb{P}_{X \sim P}\left(\exists k \in [\![1, m]\!] : p_{(k)}(X) \leq \frac{\alpha k}{m}\right) \leq \alpha. \tag{2.19}$$

This inequality has been a cornerstone to construct confidence envelopes on the number of false discoveries, as we will see in Section 2.3.4. Note that generalisations of this inequality exist, see e. g. Sarkar (2008). While this procedure does not have a practical interest — because it is valid only under $H_{[\![1,m]\!]}$, and, in practice, $\mathcal{H}_0(P_0)$ is unknown — it is a building block for the construction of the Hommel procedure (Hommel, 1988), which is the closed testing procedure associated with the Simes local tests. The Hommel procedure consists in rejecting the following hypotheses, for a pre-specified significance level α:

$$R_\alpha^{\mathrm{Hom}}(X) = \left\{i \in [\![1, m]\!] : p_i(X) \leq \frac{\alpha}{\hat{m}_0}\right\}, \tag{2.20}$$

where $\hat{m}_0 = \max\{k \in [\![1,m]\!] : \forall i \leq k, p_{(n-k+i)} > \frac{i\alpha}{k}\} \cup \{0\}$ and $\frac{\alpha}{0} = +\infty$ by convention. The Hommel procedure controls the FWER at level α under a weaker version of the wPRDS assumption on the p-values; it only needs that, for all $P \in \mathfrak{F}$, for any nondecreasing set $D \subseteq [0,1]^{\mathcal{H}_0(P)}$, $u \mapsto \mathbb{P}_{X\sim P}\big((p_j)_{j\in\mathcal{H}_0(P)} \in D \mid p_i \leq u\big)$ is nondecreasing for all $i \in \mathcal{H}_0(P)$ on the set $\{u : 0 \leq u \leq 1, \mathbb{P}_{X\sim P}(p_i \leq u) > 0\}$. The Hommel procedure can be considered as an adaptive version of the Bonferroni procedure in the following sense: if we knew the number of true null hypotheses m_0, we could use the Bonferroni procedure with critical values α/m_0 instead of α/m, which would be more powerful. Here, instead of m_0 which is unknown, we use the random integer $\hat{m}_0$ which is an *overestimator* of m_0 with probability at least $1-\alpha$. This observation is of particular importance in the construction of confidence envelopes on the number of false discoveries, see Proposition 3.3.12 for a generalization of this overestimator in this context, where the definition of $\hat{m}_0$ in Proposition 3.3.12 is exactly the same as the one we obtained above with $\ell_{i:n} = \frac{\alpha i}{n}$.

2.3.3 False discovery rate (FDR) and BH procedure

The false discovery rate (FDR) was popularized by the landmark paper Benjamini and Hochberg (1995) as an alternative error metric to the FWER that is less conservative and more powerful in practice. It is defined as the expected proportion of true null hypotheses that are rejected, that is, for any multiple testing procedure R, the FDR of R is defined, for all $P \in \mathfrak{F}$, as

$$\mathrm{FDR}_P(R) = \mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\right]. \tag{2.21}$$

Note that the idea of considering the ratio of false discoveries to the total number of rejections was already present in the works of Eklund and Seeger (1965) and Seeger (1966) — although not in expectation. We say that a procedure R achieves control of the FDR at level $\alpha \in [0,1]$ if for all $P \in \mathfrak{F}$, $\mathrm{FDR}_P(R) \leq \alpha$. This error metric can be viewed as a generalization of the FWER under the full null, that is:

$$\forall P \in H_{[\![1,m]\!]}, \mathrm{FDR}_P(R) = \mathrm{FWER}_P(R). \tag{2.22}$$

We prove this statement below:

Proof. Let $P \in H_{[\![1,m]\!]}$, we have $V\big(R(X)\big) = |R(X)|$, hence

$$\begin{aligned}
&\mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\right] \\
&= \mathbb{E}_{X\sim P}\left[\frac{V\big(R(X)\big)}{|R(X)| \vee 1}\mathbb{1}\{|R(X)| > 0\}\right] + \mathbb{E}_{X\sim P}\left[\frac{|R(X)|}{|R(X)| \vee 1}\mathbb{1}\{|R(X)| = 0\}\right] \\
&= \mathbb{P}_{X\sim P}\big(V\big(R(X)\big) > 0\big) + 0 = \mathbb{P}_{X\sim P}\big(V\big(R(X)\big) > 0\big).
\end{aligned}$$

□

We first introduce procedures using p-values that control the FDR. The most classical procedure controlling the FDR is the Benjamini-Hochberg (BH) procedure (Benjamini and Hochberg, 1995), which is a step-up procedure defined by the critical values $\tau = \left(\frac{k\alpha}{m}\right)_{1\leq k\leq m}$ for a prespecified level α. We denote this procedure by R_α^{BH}, and we have the following result:

Proposition 2.3.4. *Assume that Assumption 2.2 holds. Let $\alpha \in (0,1]$ and $P \in \mathfrak{F}$. Then the* BH *procedure R_α^{BH} controls FDR_P at level $\frac{m_0(P)}{m}\alpha$.*

Remark that the BH procedure is exactly the same as the Simes procedure, but the seminal paper Benjamini and Hochberg (1995) proved that it controls the FDR at level $\frac{m_0}{m}\alpha$ under Assumption 2.1, and then Benjamini and Yekutieli (2001) extended this result under the PRDS assumption. Finally, one can trace back the proof of the FDR control of the BH procedure under the wPRDS assumption to Blanchard and Roquain (2008), Lemma 3.2 although the authors did not explicitly state the wPRDS assumption. Note that some generalizations of the BH procedure exist, for example the Heyse procedure (Heyse, 2011), which is a step up procedure adapted to *heterogeneous* p-values, that is, p-values that do not share the same marginal distribution under the null hypothesis. However, the Heyse procedure does not control the FDR in general, as shown by Heller and Gur (2014) or Döhler et al. (2018). We give a formal definition of Heyse procedure in Section 5.2.2 and tackle this issue in Section 5.3.2.

The second procedure controlling the FDR that we introduce is the Benjamini-Yekutieli (BY) procedure (Benjamini and Yekutieli, 2001), which is a step-up procedure defined by the critical values $\tau = \left(\frac{k\alpha}{mh_m}\right)_{1\leq k\leq m}$ for a prespecified level α, where $h_m = \sum_{i=1}^m 1/i$. We denote this procedure by R_α^{BY}, and we have the following result:

Proposition 2.3.5. *Let $\alpha \in (0,1]$ and $P \in \mathfrak{F}$. Then the* BY *procedure, R_α^{BY}, controls FDR_P at level $\frac{m_0(P)}{m}\alpha$.*

Remark that the BY procedure controls the FDR at level $\frac{m_0(P)}{m}\alpha$ without any assumptions on the dependence structure of the p-values, but at the cost of a loss of power compared to the BH procedure. Indeed, $R_\alpha^{\mathrm{BY}} = R_{\alpha/h_m}^{\mathrm{BH}}$, and $h_m \sim \log(m)$, hence the critical values of the BY procedure correspond to the ones of the BH procedure divided by a logarithmic factor, which is significant when m is large. Note that the BY procedure is a special case of a more general class of procedures from Blanchard and Roquain (2008) designed to control the FDR under arbitrary dependence structure of the p-values. The BY procedure, as the BH procedure with the Heyse procedure, has been generalized to the case of heterogeneous p-values by the DBY procedure (Döhler, 2018). We give a formal definition of the DBY procedure in Section 5.2.2 and extend the validity of the DBY procedure in Section 5.3.2.

The last (non-adaptive) procedure controlling the FDR that we introduce is the e-BH procedure (Wang and Ramdas, 2022), which is designed to control the FDR using e-values instead of p-values. It is, as the BH procedure, a procedure defined by the following:

$$R_\alpha^{e\text{-BH}}(X) = \left\{ i \in [\![1,m]\!] : e_i(X) \geq \frac{m}{\alpha k^*} \right\}, \tag{2.23}$$

where $k^* = \max\left\{k \in [\![1,m]\!] : e_{[k]}(X) \geq \frac{m}{\alpha k}\right\} \cup \{0\}$, $e_{[k]}(X)$ is the k-th largest e-value, and we set $\frac{m}{\alpha k^*} = \infty$ if $k^* = 0$. Note that the e-BH procedure is equivalent to applying the BH procedure to the p-values $p_i(X) = (1/e_i(X)) \wedge 1$, for all $i \in [\![1,m]\!]$. Hence, the difference between the BH procedure and the e-BH procedure comes from the marginal distribution of the p-values under the null hypothesis; in case of e-BH we assume that the p-values used are not only super-uniformly distributed, but are also derived from e-values, which is a stronger assumption (see Section 2.2.2). We have the following result:

Proposition 2.3.6. *Let $\alpha \in (0,1]$ and $P \in \mathfrak{F}$. Then the e-BH procedure $R_\alpha^{e\text{-BH}}$ controls FDR_P at level $\frac{m_0(P)}{m}\alpha$. Furthermore, $R^{e\text{-BH}}$ controls the FDR in a post-hoc fashion, that is,*

$$\forall P \in \mathfrak{F}, \mathbb{E}_{X\sim P}\left[\sup_{\alpha\in(0,1]} \frac{V\left(R_\alpha^{e\text{-BH}}\right)}{\alpha|R_\alpha^{e\text{-BH}}| \vee 1}\right] \leq \frac{m_0(P)}{m}. \tag{2.24}$$

Hence the e-BH procedure satisfies an analog of the property (2.2), but in a multiple testing framework fashion. Note that the e-BH procedure can be used with *compound* e-values (see Definition 5.2.5 for a formal definition), instead of e-values, and the control over FDR_P, and the post hoc control, is then 1 instead of $\frac{m_0(P)}{m}$. Some new FDR controlling procedures combine e-values with a closure principle similar to the one presented in subsection 2.3.2, as in Xu et al. (2025) or Goeman (2026) to get a practical improvement over BY and BH procedures. We detail the Closure Principle for FDR control in Section 5.2.5.

Finally, we consider the class of adaptive procedures introduced in Storey (2002) (with results then formally proven in Storey et al., 2004). Storey (2002) proposed to estimate the number of true null hypotheses m_0 by $\hat{m}_0 = \frac{|\{i\in[\![1,m]\!]:p_i(X)>\lambda\}|+1}{1-\lambda}$ for a prespecified $\lambda \in (0,1)$, and then to reject the step-up procedure with critical values $\frac{k\alpha}{\hat{m}_0} \wedge \lambda$ for $k \in [\![1,m]\!]$, that we will denote by R_α^{St}. Storey et al. (2004) stated the following result:

Proposition 2.3.7. *Let $P \in \mathfrak{F}$. Assume that $(p_i(X))_{i\in\mathcal{H}_0(P)}$ are independent and that for all $i \in \mathcal{H}_0(P)$, $p_i(X) \sim U(0,1)$. Let $\alpha \in (0,1]$. Then the Storey procedure R_α^{St} controls FDR_P at level $\left(1-\lambda^{m_0(P)}\right)\alpha \leq \alpha$.*

Remark that the independence assumption on the p-values under the null hypothesis is slightly different from Assumption 2.1, because the proof relies on martingale properties of $\sum_{i\in\mathcal{H}_0(P)} \mathbb{1}\{p_i(X) \leq t\}$. Following this work, several papers have been devoted to handling dependency structures, in particular Blanchard and Roquain (2009), which expanded adaptive FDR theory to various dependency structures, and under super-uniformity of the p-values under the null hypothesis instead of uniformity. It especially showed that a slight modification of the Storey procedure, the step up procedure with critical values $\frac{k\alpha}{\hat{m}_0}$ for $k \in [\![1,m]\!]$ — hence without the minimization with λ, controls the FDR at level α under the assumption that all the p-values are independent, which is a mandatory assumption for the leave-one-out argument used in the proof. Recently, some articles tackled the problem of adaptive FDR control considering discrete data, as Döhler and Meah (2026), to gain power over the initial procedures, or considering compound e-values, as Ignatiadis et al. (2026), to get a uniform improvement over various adaptive procedures.

2.3.4 Confidence envelopes on the number of false discoveries

Confidence envelopes on the number of false discoveries were popularized by Goeman and Solari (2011), but construction of such envelopes can be traced back to Genovese and Wasserman (2004). In comparison to classical multiple testing procedures, that we illustrated through the FWER and FDR control, confidence envelopes on the number of false discoveries provide a more flexible framework to control the number of false discoveries, allowing to select any subset of hypotheses after observing the data, and still guarantee a valid upper bound on the number of false discoveries. It notably allows to select the hypotheses to reject in a post hoc fashion, that is, after observing the data. That property especially allows researchers to explore the data and select any subset

of hypotheses dynamically, while bringing an answer to some inapropriate practices in the scientific community, such as *p-hacking*, that is, the practice of selecting hypotheses to reject after observing the data, and then reporting only the selected hypotheses as significant.

A confidence envelope on the number of false discoveries is a random function $\hat{V} : \mathcal{P}(\llbracket 1, m \rrbracket) \to \llbracket 1, m \rrbracket$ — where $\mathcal{P}(\llbracket 1, m \rrbracket)$ is the power set of $\llbracket 1, m \rrbracket$ — such that for a pre-specified level $\alpha \in (0, 1]$,

$$\forall P \in \mathfrak{F}, \mathbb{P}_{X\sim P}\big(\forall S \subseteq \llbracket 1, m \rrbracket : V(S) \leq \hat{V}(S)\big) \geq 1 - \alpha. \tag{2.25}$$

Here, we already see why the notion of confidence envelope on the number of false discoveries is more flexible than classical multiple testing procedures: these methods do not prescribe a specific subset of hypotheses to reject, but rather provide a valid upper bound on the number of false discoveries for any subset of hypotheses, allowing the user to explore the data and select any subset of hypotheses to reject.

Note that these kind of procedures can be considered as generalizations of FWER control methods; if we consider a multiple testing procedure R that strongly controls the FWER at level α, we can define a confidence envelope $\hat{V}$ on the number of false discoveries as follows:

$$\forall S \subseteq \llbracket 1, m \rrbracket, \hat{V}(S) = |R(X)^c \cap S|, \tag{2.26}$$

which is a confidence envelope on the number of false discoveries at level α. Indeed, for all $P \in \mathfrak{F}$, we have

$$\begin{aligned}
&\mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : V(S) > \hat{V}(S)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : |R(X) \cap \mathcal{H}_0(P) \cap S| > |R(X)^c \cap \mathcal{H}_0(P)^c \cap S|\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : |R(X) \cap \mathcal{H}_0(P) \cap S| > 0\big) \\
&= \mathbb{P}_{X\sim P}\big(|R(X) \cap \mathcal{H}_0(P)| > 0\big) = \mathrm{FWER}_P\left(R(X)\right) \leq \alpha
\end{aligned}$$

Hence, any multiple testing procedure strongly controlling the FWER at level α can be used to construct a confidence envelope on the number of false discoveries at level α. Conversely, a confidence envelope on the number of false discoveries provides a multiple testing procedure which achieves strong control of the FWER at level α. Indeed, a confidence envelope on the number of false discoveries $\hat{V}$ at level α can be used to construct a multiple testing procedure R controlling the FWER at level α as follows:

$$R(X) = \Big\{i \in \llbracket 1, m \rrbracket : \hat{V}\big(\{i\}\big) = 0\Big\}, \tag{2.27}$$

Hence, for all $P \in \mathfrak{F}$, we have

$$\begin{aligned}
\mathrm{FWER}_P(R) &= \mathbb{P}_{X\sim P}\big(\exists i \in \mathcal{H}_0(P) : i \in R(X)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists i \in \llbracket 1, m \rrbracket : \hat{V}(\{i\}) = 0 \text{ and } V(\{i\}) = 1\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq \llbracket 1, m \rrbracket : V(S) > \hat{V}(S)\big) \\
&\leq \alpha,
\end{aligned}$$

where the last inequality comes from (2.25). Remark here that the multiple testing procedure R defined in (2.27) shares the same spirit as the multiple testing procedure R_{ψ_α} defined in (2.11), which is the multiple testing procedure associated with the closed testing procedure. This observation in addition to (2.10) brings a strong connection between

the closure principle and confidence envelopes. Indeed, Goeman and Solari (2011) observed that closed testing procedures bring more information than just the multiple testing procedure R_{φ_α} through (2.10), and defined confidence envelopes on the number of false discoveries using the closed testing framework:

Proposition 2.3.8 (Goeman and Solari, 2011)**.** *Let φ be a family of local tests such that, for all $P \in \mathfrak{F}$, $\varphi_{\mathcal{H}_0(P)}$ is an α-level test. Recall the definition of the family of effective local tests ψ (2.9), and let $\hat{V}_\varphi^{\mathrm{CT}}$ be the random function defined as follows:*

$$\forall S \subseteq [\![1, m]\!], \hat{V}_\varphi^{\mathrm{CT}}(S) = \max_{U \subseteq S} \big\{|U| : \psi_U(X) = 0\big\}. \tag{2.28}$$

Then, for all $\alpha \in (0, 1]$, $\hat{V}_\varphi^{\mathrm{CT}}$ is a confidence envelope on the number of false discoveries at level α.

The proof of this proposition immediately follows from (2.10) and the definition of $\hat{V}_\varphi^{\mathrm{CT}}$:

Proof. Let $P \in \mathfrak{F}$, we have

$$\begin{aligned}
&\mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : V(S) > \hat{V}_\varphi^{\mathrm{CT}}(S)\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : V(S) > \max_{U\subseteq S}\{|U| : \psi_U(X) = 0\}\big) \\
&= \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : |S \cap \mathcal{H}_0(P)| > \max_{U\subseteq S}\{|U| : \psi_U(X) = 0\}\big) \\
&\leq \mathbb{P}_{X\sim P}\big(\exists S \subseteq [\![1, m]\!] : \psi_{S\cap\mathcal{H}_0(P)}(X) = 1\big) && (2.29) \\
&\leq \alpha, && (2.30)
\end{aligned}$$

where (2.29) comes because if $\psi_{S\cap\mathcal{H}_0(P)}(X) = 0$, then

$$\max_{U\subseteq S} \big\{|U| : \psi_U(X) = 0\big\} \geq |S \cap \mathcal{H}_0(P)|,$$

and (2.30) comes from (2.10). □

The drawback of this approach is that the computation of $\hat{V}_\varphi^{\mathrm{CT}}$ is intractable in practice, because it requires to compute all the effective local tests ψ_U for all $U \subseteq [\![1, m]\!]$, which is exponential in m. However, in some cases, the computation of $\hat{V}_\varphi^{\mathrm{CT}}$ can be simplified by so-called *shortcuts*, which are specific to the family of local tests φ, and allow to compute $\hat{V}_\varphi^{\mathrm{CT}}$ or an upper bound of it in polynomial time. Some examples of such shortcuts are given in Section 3 with Proposition 3.3.12, or Proposition 3.4.2. As with the FWER control closed testing procedures, a notion of coherence has been defined for confidence envelopes by Goeman et al. (2021), which states that any confidence envelope on the number of false discoveries that is coherent can be expressed as a closed testing procedure. However, this statement is called into question by our findings in Section 6.

Finally, note that other methods to construct confidence envelopes on the number of false discoveries exist, such as the inversion procedure (Genovese and Wasserman, 2006) relying on a family of local tests as the closed testing procedure but without defining effective local tests (see Proposition 3.3.10 for a formal definition), methods relying on paths (Katsevich and Ramdas, 2020; Meah et al., 2024) or again methods relying on the Joint Error Rate (JER) (Blanchard et al., 2020a; Durand et al., 2020) — see Section 6.2.1 and Section 3.3.3. Note that all these methods, with an exception to the one relying on paths, are equivalent, and paths methods can be improved by using one of these three equivalent methods.

2.3.5 Conformal Prediction and multiple testing

Conformal prediction is a general framework to construct prediction sets with finite-sample coverage guarantees from minimal guarantees, that is with no assumptions on the distribution of the of the data. This "minimal guarantees" aspiration is motivated by the increasing use of black-box models, such as neural networks. Let us consider a regression example, that is, a statistical problem where we assume that two variables $X \in \mathcal{X}$ and $Y \in \mathbb{R}$ verify $Y = f(X) + \varepsilon$ where ε is a noise term. Assume that some black-box model produced a predictor $\hat{f}$, that we have access to a sample of length $n \in \mathbb{N}$ of (X, Y) and that we want to apply our predictor $\hat{f}$ to a *test point* X_{n+1}. Then, the conformal prediction method will construct n non-conformity scores $S_1, \dots, S_n$, let us say $S_i = \left|Y_i - \hat{f}(X_i)\right|$ for all $i \in \{1, \dots, n\}$, and a score $S_{n+1}(y) = \left|y - \hat{f}(X_{n+1})\right|$. Then, for a pre-specified level $\alpha \in (0, 1]$, the conformal prediction method will construct a prediction set

$$\hat{C}_\alpha(X_{n+1}) = \left\{y \in \mathbb{R} : S_{n+1}(y) \leq S_{(k)}\right\},$$

where k_α is equal to $\lceil (1-\alpha)(n+1) \rceil$. Then, under an *exchangeablility* assumption on the scores, Y_{n+1} belongs to $\hat{C}_\alpha(X_{n+1})$ with probability at least $1 - \alpha$.

The conformal prediction framework has been introduced by Gammerman et al. (1998), and fully formalized in Vovk et al. (2005). Since then it has been widely studied and applied in various fields, such as classification (Ding et al., 2023; Gazin et al., 2025b) or selective inference (Jin and Candès, 2023; Gazin et al., 2025a). A core problem in conformal prediction is to determine non-conformity scores from observations, but we will not focus on this problem here, and we will assume that the scores are given. Rather, we will expose some links between conformal prediction and multiple testing. Let $n, m \in \mathbb{N}$, $S_1, \dots, S_n$ be "reference" scores (or calibration scores) and $S_{n+1}, \dots, S_{n+m}$ be "test" scores, all in $\mathbb{R}$. The key assumption of conformal prediction is that the scores are exchangeable, which is a weaker assumption than independence and identical distribution.

Definition 2.3.9 (Exchangeability)**.** *Let $X_1, \dots, X_n$ be $n \in \mathbb{N}$ random variables. We say that $X_1, \dots, X_n$ are exchangeable if for any permutation π of $\{1, \dots, n\}$,*

$$(X_1, \dots, X_n) \sim (X_{\pi(1)}, \dots, X_{\pi(n)}),$$

that is, the joint distribution of $X_1, \dots, X_n$ is invariant under permutations.

Note that the exchangeability assumption is satisfied if the variables are independent and identically distributed. Furthermore, it implies that the variables have the same marginal distribution. Hence, we will denote by $S = (S_1, \dots, S_{n+m})$ the vector of scores, we will assume that the distribution P_0 of S belongs to a model $\mathfrak{F}$. The simple hypothesis testing problem in conformal prediction is to test the null hypothesis H_i : "$(S_1, \dots, S_n, S_{n+i})$ is exchangeable" for some $i \in [\![1, m]\!]$. It derives a p-value p_i, called conformal p-value, for each null hypothesis H_i, which is defined as follows:

$$p_i = \frac{1}{n+1}\left(\sum_{j=1}^{n} \mathbb{1}\{S_j \geq S_{n+i}\} + 1\right). \tag{2.31}$$

This definition of conformal p-values indeed defines a valid p-value under the null hypothesis H_i (Romano and Wolf, 2005, Lemma 1; Arlot et al., 2010, Lemma 5.2):

Proposition 2.3.10 (Quantile Lemma). *Let $i \in \{1, \dots, m\}$ and $P \in H_i$. Then, for all $\alpha \in [0, 1]$,*

$$\mathbb{P}_{S\sim P}\big(p_i \leq \alpha\big) \leq \alpha.$$

Note that the conformal p-values are *discrete* p-values, and that their support is $\mathcal{A} = \big\{\frac{j}{n+1}, j \in [\![1, n+1]\!]\big\}$. Furthermore, for all $j \in [\![1, n+1]\!]$ and $P \in H_i$, we have $\mathbb{P}_{S\sim P}\big(p_i \leq \frac{j}{n+1}\big) = \frac{j}{n+1}$. Note that one can also define conformal e-values. An example is given by Balinsky and Balinsky (2024), who proposed the following e-value to test the null hypothesis H_i, for some $i \in [\![1, m]\!]$:

$$e_i = \frac{(n+1)S_{n+i}}{\sum_{j=1}^{n} S_j + S_{n+i}}. \tag{2.32}$$

Let us now go back to our multiple testing problem. In conformal prediction, in order to test the null hypotheses $H_1, \dots, H_m$ simultaneously, the theorems that we will present need supplementary assumptions on the scores that we define below. The first assumption is simple:

Assumption 2.3. *The scores $S_{k+1}, \dots, S_{n+m}$ have no ties, almost surely.*

If the model does not allow distributions with atoms, this assumption is satisfied. The second assumption is the exchangeability of the scores under the null hypotheses with the calibration scores.

Assumption 2.4. *For all $P \in \mathfrak{F}$, the scores $\big\{S_j : j \in [\![1, n]\!] \cup \{n + i, i \in \mathcal{H}_0(P)\}\big\}$ are exchangeable conditionally on all $S_{n+i}, i \in \mathcal{H}_1$.*

Note that, for $P \in \mathfrak{F}$ such that $|\mathcal{H}_0(P)| > 1$, the underlying property that the scores $S_1, \dots, S_n, S_{n+i}$ are exchangeable for a given $i \in \mathcal{H}_0(P)$ does not imply that the scores $\big\{S_j : j \in [\![1, n]\!] \cup \{n + i, i \in \mathcal{H}_0(P)\}\big\}$ are exchangeable. We illustrate this point in the following example: let X, Y, Z be three Bernoulli random variables such that:

$$\begin{cases} \mathbb{P}\big((X, Y, Z) = (1, 1, 0)\big) & = 0.5, \\ \mathbb{P}\big((X, Y, Z) = (0, 0, 1)\big) & = 0.5, \\ \mathbb{P}\big((X, Y, Z) = (a, b, c)\big) & = 0 \text{ otherwise.} \end{cases}$$

Then, the scores X, Y are exchangeable and the scores X, Z are exchangeable, but the scores X, Y, Z are not exchangeable. Indeed, for all $a, b \in \{0, 1\}$, $\mathbb{P}\big((X, Y) = (a, b)\big) = \mathbb{P}\big((Y, X) = (a, b)\big)$ and $\mathbb{P}\big((X, Z) = (a, b)\big) = \mathbb{P}\big((Z, X) = (a, b)\big)$, but

$$\mathbb{P}\big((X, Y, Z) = (1, 1, 0)\big) = 0.5 \neq 0 = \mathbb{P}\big((X, Z, Y) = (1, 1, 0)\big).$$

From Assumption 2.3 and Assumption 2.4, Marandon et al. (2024) proved that the conformal p-values are PRDS, and hence wPRDS. Remark that our null hypotheses are different from the one considered in Marandon et al. (2024). Indeed, they consider the null hypothesis H_i : "$S_{n+i} \sim P_0$ " where P_0 is the distribution of the calibration scores. However, under Assumption 2.4, these two null hypotheses are equal on the correspondent submodel $\mathfrak{F}'$ such that for all $P \in \mathfrak{F}'$, P satisfies Assumption 2.4. Our null hypotheses H_i have the advantage to keep the property that the conformal p-value p_i are superuniform under each of the H_i, without any additional assumption on the exchangeability of the corresponding non-conformity score with the calibration scores, since this assumption is already included in the definition of H_i.

It guarantees that the BH procedure controls the FDR at level α over the conformal p-values. Furthermore, considering Assumption 2.3 and Assumption 2.4 under the full null— that is, considering the model

$$\mathfrak{F}_0 = \{P \in \mathfrak{F} : \mathcal{H}_0(P) = [\![1, m]\!] \text{ and } P \text{ satisfies Assumption 2.4}\},$$

Gazin et al. (2024) proved a DKW-type inequality for the conformal p-values, which allows to construct a confidence envelope on the number of false discoveries over the conformal p-values with a weaker assumption than Assumption 2.4, which only requires that $\left\{S_j : j \in [\![1, n]\!] \cup \{n + i, i \in \mathcal{H}_0(P)\}\right\}$ are exchangeable for all $P \in \mathfrak{F}$, in the same flavour as the weaker version of the wPRDS assumption required by the Hommel procedure to control the FWER in Section 2.3.2.

2.4 Contributions

This section presents the contributions of this thesis, which are detailed in the following chapters.

2.4.1 Confidence envelopes for the false discoveries with heterogeneous data

Chapter 3 presents a submitted paper available on arXiv, (Périer et al., 2026), which is a joint work with Gilles Blanchard (Université Paris-Saclay), Sebastian Döhler (Darmstadt University), Guillermo Durand (Université Paris-Saclay), and Etienne Roquain (Sorbonne Université)[1]. It introduces new methods to construct confidence envelopes on the number of false discoveries in a heterogeneous multiple testing framework, that is, when the p-values do not share the same marginal distribution under the null hypothesis. These methods rely on various tools, which can be classified into two categories. The first category is composed of the probabilistic tools tailored for the heterogeneous setting. This category in particular includes the Bretagnolle inequality (Shorack and Wellner, 1986, Chapter 25, Inequality 2) which is a heterogeneous version of the DKWM inequality (Dvoretzky et al., 1956; Massart, 1990), and the heterogeneous version of the Simes inequality that we prove in this chapter. This last result derives from a FDR control multiple testing procedure developed in Döhler et al. (2018) in the same way than the Simes inequality can be obtained via the BH procedure. The second category of tools that we use in this chapter is the low-complexity algorithmic devices that optimize the probabilistic information provided by the first category. Namely, we used the JER framework (Blanchard et al., 2020a) and the inversion procedure method (Genovese and Wasserman, 2006) to construct confidence envelopes on the number of false discoveries from the the heterogeneous probabilistic devices. This chapter especially develops new shortcuts for the inversion procedure, which allow to compute an upper bound of the confidence envelope in polynomial time, and to improve former methods designed for the homogeneous setting. We enumerate below the main contributions of this chapter:

- we provided new generic shortcuts, $\hat{V}^{\mathrm{SC1}}$ and $\hat{V}^{\mathrm{SC2}}$, for the inversion procedure that allow to compute an upper bound of the inversion procedure confidence envelope in polynomial time,

[1] I certify here that as a first author of the paper, I have contributed a large majority of the contents of the paper in all areas: theory, experiments and writing.

- we applied these new generic shortcuts to the homogeneous case and coupled them with JER-based methods, resulting in proving exactness — that is equality with the closed testing procedure resulting from the same probabilistic tools — of former methods designed for the homogeneous setting,
- we provided new heterogeneity-designed envelopes for the empirical cumulative distribution function of the p-values under the null hypothesis, and applied them to construct new confidence envelopes thanks to the inversion procedure and JER-based methods,
- we provided adaptive versions of the new confidence envelopes where the number of true null hypotheses is estimated via our shortcuts for the inversion procedure and plugged into JER methods.

2.4.2 FDR control and FDP bounds for conformal link prediction

In Chapter 4, we present a submitted paper available on arXiv (Blanchard et al., 2024), which is a joint work with Gilles Blanchard (Université Paris-Saclay), Guillermo Durand (Université Paris-Saclay), and Ariane Marandon (University College London). It formally provides a proof of the results stated in Marandon (2024) thanks to the tools discussed in Section 2.3.5. Marandon (2024) constructs a FDR controlling procedure in a link prediction framework. To achieve this, they construct conformal p-values by sampling edges in a masked graph following a specific sampling scheme and then they apply the BH procedure to these conformal p-values. However, they did not provide that the resulting scores indeed satisfy Assumption 2.4. We provide a formal proof of this result, and we also provide a confidence envelope on the FDP using results of Gazin et al. (2024).

2.4.3 Heterogeneous compound p-values and closure principle

Chapter 5 presents an ongoing work, which is a joint work with Gilles Blanchard (Université Paris-Saclay), and Guillermo Durand (Université Paris-Saclay). It aims to provide a generalization of the e-closure principle for the FDR (Xu et al., 2025) to the heterogeneous setting. Namely, this chapter introduces new heterogeneous compound p-values. This discovery brings new insights on the heterogeneous multiple testing literature, by providing a new proof to the procedure DBY introduced in Döhler (2018) and extending its validity, and by providing a correction to the Heyse procedure Heyse (2011). These findings rely on results from the theory of compound p-values (Barber and Samworth, 2025; Armstrong, 2026). Finally, we provide new multiple testing procedures controlling the FDR via the closure principle for e-values. These new procedures are better than DBY and the refinement of BY procedure obtained by application of the e-closure principle by Xu et al. (2025), respectively.

2.4.4 On strongly coherent confidence envelopes for the false discoveries

Chapter 6 presents an ongoing work which is a joint work with Gilles Blanchard (Université Paris-Saclay), and Guillermo Durand (Université Paris-Saclay). This work discusses the different notions of interpolation from the confidence envelopes literature. It notably shows that the notion of coherence defined in Goeman et al. (2021) does not fully

characterize the class of closed testing procedures, and that a stronger notion of coherence is required to do so. It also provides new procedures to construct confidence envelopes on the number of false discoveries, from a family of reference regions, by considering local confidence envelopes within these reference regions. These new procedures, under some assumptions, on the structure of the reference regions, are computationally tractable, and are always better than the ones from Durand et al. (2020) and Xu et al. (2024a).

2.4.5 Additional contributions

In addition to the contributions presented above, I also provide two other contributions. The first one is Appendix C and shows that the $\mathrm{BH}+$ procedure introduced in Chen (2020) can be reduced to the BH procedure applied to modified p-values, and that the validity of the $\mathrm{BH}+$ procedure can be derived from the validity of the BH procedure, since we show that the wPRDS property is preserved by the transformation of the p-values. The second contribution is a note on the fission of Bernoulli random variables, which is Appendix D, in the same spirit as Leiner et al. (2025). It could be particularly interesting to use this result in the setting of Chapter 4 in future works.

Chapter 3

Confidence envelopes for the false discoveries with heterogeneous data

In the context of selective inference, confidence envelopes for the false discoveries allow the user to select any subset of null hypotheses while having a statistical guarantee on the number of false discoveries in the selected set. Many constructions of such envelopes have been proposed recently, using local test families (Genovese and Wasserman, 2006; Goeman and Solari, 2011), paths (Katsevich and Ramdas, 2020) or interpolation (Blanchard et al., 2020a). All those methods have in common that they have been well-studied for the homogeneous case where all p-values under the null have a uniform distribution over $[0, 1]$. However, in many applications the data are heterogeneous and discrete, hence the p-values have heterogeneous, discrete distributions, and the previous constructions may incur a loss of power, in the sense that they over-estimate the number of false discoveries. In this chapter, we first review previous constructions, and introduce new tools (in particular shortcuts) that can be applied in both the homogeneous and heterogeneous settings, and allow to in particular to bridge previous constructions and extend existing theory in the homogeneous setting. Secondly, we also apply these tools to propose several confidence envelopes based on inequalities tailored for heterogeneous data, like the Bretagnolle inequality, or a new heterogeneous variant of the Simes inequality. We compare these new envelopes to their homogeneous counterparts on simulated data.

Contents

3.1 Introduction . . . 48
3.1.1 Overview of goals and motivations . . . 48
3.1.2 Confidence envelopes in the context of selective inference . . . 50
3.1.3 Contributions and chapter organization . . . 51
3.2 Multiple testing and confidence bound formalism . . . 52
3.2.1 Mathematical notation . . . 52
3.2.2 Classical multiple testing formalism . . . 52
3.2.3 Confidence bounds . . . 54
3.2.4 Heterogeneity . . . 55
3.3 Review and extension of existing theory . . . 56
3.3.1 Homogeneous concentration inequalities . . . 57
3.3.2 Confidence envelopes by the inversion procedure . . . 59
3.3.3 Reference families, JER and interpolation . . . 60

3.4 General shortcuts for the inversion procedure . . . 65
3.4.1 Shortcuts for the inversion procedure . . . 66
3.4.2 Applications to homogeneous inequalities . . . 67
3.5 Heterogeneous probabilistic inequalities and associated local test confidence envelopes . . . 70
3.5.1 Bretagnolle inequality and applications . . . 71
3.5.2 Heterogeneous Simes-like inequality . . . 73
3.6 Heterogeneous confidence envelopes using reference families . . . 77
3.6.1 Top-k setting . . . 77
3.6.2 Simultaneous k-FWER . . . 79
3.6.3 Deterministic setting . . . 81
3.7 Numerical experiments . . . 81
3.7.1 Simulation setting . . . 81
3.7.2 Analyses for top-k setting . . . 82
3.7.3 Analyses for k-FWER setting . . . 83
3.7.4 Comparison with other homogeneous bounds . . . 84
3.8 Conclusion and perspectives . . . 85

3.1 Introduction

3.1.1 Overview of goals and motivations

Consider a family of null statistical hypotheses to test, denoted $(H_i)_{i\in[\![1,m]\!]}$; each null hypothesis corresponds to a subset of possible probability distributions driving the observed data. (The objects used will be more formally defined in Section 3.2 but should be familiar to a reader already acquainted with the multiple testing literature.) For a data distribution μ, we say that " μ satisfies the null hypothesis H_i " if $\mu \in H_i$, and we denote $\mathcal{H}_0(\mu)$, immediately abbreviated to $\mathcal{H}_0$, the set of true nulls, i.e., indices i of null hypotheses H_i satisfied by μ. Based on the data, the user outputs a subset of indices of selected hypotheses $\widehat{S} \subseteq [\![1,m]\!]$; any selected index $i \in \widehat{S}$ which is actually in the null set $\mathcal{H}_0$ is called a false discovery (FD). A common goal in multiple testing is to return an upper bound $\widehat{V}(\widehat{S})$ on the number of false discoveries $|\widehat{S} \cap \mathcal{H}_0|$. In the present chapter, we focus on bounds holding with high probability, whatever the true μ (belonging to some overarching statistical model $\mathfrak{F}$).

Observe that $\widehat{S}$ is a random set whose dependence on the data can be complex and not fully known, especially if the user departs from a precisely prescribed algorithm to engage in various forms of data snooping. For this reason, an approach considered as more statistically sound than assuming a specific form for $\widehat{S}$ is to establish a family of upper bounds $(\widehat{V}(S))_{S\subseteq[\![1,m]\!]}$ holding (with guaranteed large probability) simultaneously for all potential selection sets $S \subseteq [\![1,m]\!]$. Such a family of bounds will be called a *FD confidence envelope* in the rest of the present chapter. (It is also sometimes called a *post hoc* multiple testing bound since it entails a guarantee in particular for any data-dependent $\widehat{S}$.)

Several approaches for constructing FD confidence envelopes have been proposed in the literature. A particularly important device to do so, also playing a central role in the present work, is the *inversion procedure* introduced by Genovese and Wasserman

(2006). For any subset $A \subseteq [\![1,m]\!]$, we denote H_A the set of distributions simultaneously satisfying $H_i, i \in A$; also called intersection hypothesis, since $H_A = \bigcap_{i\in A} H_i$. Assume that for any subset A, there exists a *local test* $\varphi_A \in \{0,1\}$ (the value 1 indicating rejection) such that if μ satisfies H_A, then the probability that the local test falsely rejects (local type I error) is bounded by some prescribed constant δ. Then, $V_\varphi(S) = \max_{A:\varphi_A=0} |A \cap S|$ is a FD confidence envelope with probability $1-\delta$. (An equivalent bound based on the notion of closed testing associated to the family of local tests was proposed by Goeman and Solari, 2011.)

This result shifts the problem to two questions: how to construct appropriate local tests and, for given local tests and rejection set S, how to compute efficiently the associated inversion bound $V_\varphi(S)$. Note that computing φ_A for all subsets $A \subseteq [\![1,m]\!]$ is exponential in m and typically infeasible. Sometimes one may be contented with an efficiently computable upper bound on $V_\varphi(S)$, also called *shortcut*.

As is common, we assume that the data has already been processed in such a way that for each hypothesis H_i, an observable p-value p_i has been constructed. Recall that the fundamental property of a p-value associated to a null hypothesis H is that, for any data distribution satisfying H, the random variable p has a *super-uniform* distribution (i.e., stochastically larger or equal to a uniform random variable on $[0,1]$). If we furthermore assume that p-values p_i for $i \in \mathcal{H}_0$ are independent and exactly uniform (also a common assumption), it is possible to construct a local test φ_A derived from an envelope on the empirical cumulative distribution function (ecdf) of i.i.d. uniform random variables. Namely, assume that $\mathcal{B}_n(t)$ is a $(1-\delta)$-upper confidence envelope on the ecdf $\widehat{F}_n(t) = \frac{1}{n}\sum_i \mathbb{1}\{U_i \le t\}$, if $(U_i)_{i\in[\![n]\!]}$ are i.i.d. uniform $[0,1]$, that is, it holds $\widehat{F}_n(t) \le \mathcal{B}_n(t)$ for all $t \in [0,1]$, with probability at least $(1-\delta)$. Then the test φ_A which rejects if $\widehat{F}_A(t) = \frac{1}{|A|}\sum_{i\in A} \mathbb{1}\{p_i \le t\} > \mathcal{B}_{|A|}(t)$ for any t is a local test for H_A. This principle of (i.i.d.) uniformity testing based on detecting violations of confidence envelopes on the ecdf is long-standing in statistics, from the Kolmogorov-Smirnov test (having asymptotic validity) to the more recent higher criticism (HC) tests in Donoho and Jin (2004) (also with asymptotic validity; for nonasymptotic and computationally feasible versions see e.g. Barnett and Lin, 2014). Various developments on this idea, based on more refined envelopes, can be found in Meah et al. (2024).

The approach described above, constructing local tests from upper envelopes for ecdf of uniform random variables, also applies to super-uniform null p-values because of a simple coupling device described in Section 3.3.1.

However, in many situations, such as when p-values are discrete, the exact cdf F_i of p-value p_i under the null is exactly known; in such a scenario, using "default" confidence envelopes holding for uniform variables may result in a loss of power of the associated local test. This is the starting motivation of this work: we first concentrate on deriving sharp confidence envelopes for ecdf of such variables (independent, but with different known distributions F_i, a setting we call *heterogeneous*), and then turn to constructing FD bounds that can be derived from them. A point of particular attention is to address the need for computable bounds (or shortcuts thereof), an issue which becomes thornier in the heterogeneous case. Namely, while homogeneous local ecdf confidence envelopes $\mathcal{B}_{|A|}(t)$ only depend on the cardinality $|A|$ of the considered subset of hypotheses, their heterogeneous counterpart typically depend on the whole set A. Driven by this need of computable bounds, we will also explore the alternative approach of Blanchard et al. (2020a) to construct FD bounds based on the control of a Joint Error Rate (JER) on a specific reference family of subsets of hypotheses. This approach has, until now, come in 3 different flavors depending on the reference family,

namely: k-FWER, top-k, and with deterministic regions. They will all be re-introduced in Section 3.3.3 and used in our contributions later.

3.1.2 Confidence envelopes in the context of selective inference

As mentioned before, data snooping (also called p-hacking, or significance chasing) plays a critical role in modern day applications and is considered one of the main causes of the replication crisis (see Wasserstein and Lazar, 2016, "Principle 4", and Ioannidis, 2005b). Indeed, classical statistical guarantees generally do not hold when the dataset has undergone a first selection step based on the observation of the data itself, but the classical method that has been used (whichever it is) does not take into account this random selection step. For instance, think of the Benjamini-Hochberg (BH) procedure applied to 10^6 tests, but where only the 10^3 smallest p-values have been selected by the data analyst as input to the procedure. Contrary to the plain BH procedure, there is no guarantee on the False Discovery Rate control of such a two-step procedure. While the flaw is obvious in this example, various, less blatant flavors of data-dependent selection often make their way into statistical practice while seeming innocuous to the practitioner, for example selection based on a different statistic which is not completely independent of the test statistic.

Selective inference is the branch of statistics that aims to answer this concern by providing methods and procedures with proven statistical guarantees when a first selection step occurred. Selective inference is generally tackled in two possible fashions. One consists in assuming that the selection procedure is known (for example, coming from the LASSO estimator) and to design valid inference by conditioning on the selection event, see for example Taylor and Tibshirani (2015); Lee et al. (2016); Fithian et al. (2017).

The other approach adopts a "simultaneous" point of view, where we seek to achieve guarantees for all possible selection subsets. This has been investigated for inference in, for instance, Scheffé (1953); Berk et al. (2013); Bachoc et al. (2019). For multiple testing purposes, this fashion takes the form of the construction of the aforementioned confidence envelopes $\left(\widehat{V}(S)\right)_{S\subseteq[\![1,m]\!]}$, first introduced by Genovese and Wasserman (2006). Since then, many works have focused on such constructions, with, notably, a prolific series of work focused on closed testing (Goeman and Solari, 2011; Meijer et al., 2015; Hemerik et al., 2019; Goeman et al., 2021; Vesely et al., 2023), and another series based on the JER approach (Blanchard et al., 2020a; Durand et al., 2020; Enjalbert-Courrech and Neuvial, 2022; Blain et al., 2022; Blain et al., 2023; Meah et al., 2024). The inversion procedure and the closed testing approaches, both based on local tests, are recalled in Section 3.3.2, while the JER approach is recalled in Section 3.3.3.

Local tests-based FD confidence envelopes have been considered in combination with knockoff local tests by Li et al. (2024). Confidence envelopes have notably been applied to gene expression analysis (Goeman and Solari, 2011; Hemerik and Goeman, 2018; Enjalbert-Courrech and Neuvial, 2022) while software coded with the `R` language (R Core Team, 2024) is already available for such analyses (Enjalbert Courrech and Neuvial, 2025) in a user-friendly `shiny` application (Chang et al., 2025). Confidence envelopes have also been applied extensively in functional Magnetic Resonance Imaging (fMRI) studies (Blain et al., 2022; Blain et al., 2023; Vesely et al., 2023).

3.1.3 Contributions and chapter organization

We describe the organization and main contributions of the chapter below. The contributions of the main body are also summarized in Table 3.1.

In Section 3.2, we give the formalism, notation and assumptions used in the remainder of the chapter.

In Section 3.3, we revisit past works on confidence bounds construction while consolidating this theory with several contributions, including:

- refined versions of known concentration inequalities,
- extension of past works in the top-k setting to super-uniformity (see Assumption 3.1),
- a new dynamic programming algorithm for confidence bound computation in the top-k setting.

In Section 3.4, we first provide new generic shortcuts for the inversion procedure, which are relevant in particular when local tests detect violations of uniform confidence envelopes. These shortcuts are applied in the homogeneous case to construct (over-)estimates of m_0. Such estimates can be plugged into the JER methods, leading to so-called adaptive versions, often leading to more powerful procedures that adapt to to the true (but unknown) number of null hypotheses. Finally, we establish a novel formal equivalence in certain settings between the inversion procedure bound and this adaptive JER approach.

In Section 3.5, we provide two ecdf confidence envelopes valid under heterogeneity. The first one is a slight refinement of a known one, the second one is completely new. Using those as local tests, we apply to each of them the generic shortcuts from Section 3.4; this allows us in turn to get estimates of m_0 under heterogeneity.

In Section 3.6, we demonstrate how to use the two previous ecdf confidence envelopes from Section 3.5 to construct new confidence bounds that use the JER formalism and that take the heterogeneity into consideration. In particular, we give constructions for the three JER settings: k-FWER, top-k, and deterministic regions. We also develop adaptive versions of our basic confidence bounds, using the estimators of m_0 constructed in Section 3.5.

In Section 3.7, we provide numerical experiments that demonstrate the power gain of our new constructions, with respect to their analogous past methods that do not take the heterogeneity into account. We do so with numerical simulations.

In Section 3.8, we give some concluding remarks and perspectives.

Some complementary results are given in the appendix. In Appendix A.1 we give a (previously known) third concentration inequality valid under heterogeneity and the related confidence bounds, but they behave rather poorly in practice so they were excluded from the main body. In Appendix A.2 we discuss and close some technical gaps in previous literature. In Appendix A.3, we give complements about the top-k setting, including links with another JER approach. In Appendix A.4, we give a randomization technique to construct uniform random variables with coupling properties, which covers the classical "randomized p-value" construction. Finally, in Appendix A.5, we discuss slight refinements about past confidence envelopes in the top-k setting, and Appendix A.6 is dedicated to the proofs.

Main contributions of this chapter, by section	
Section 3.3	• extension of known concentration inequalities and confidence envelopes to super-uniformity • dynamic programming algorithm for confidence bound computation in the top-k setting
Section 3.4	• generic shortcuts for the inversion procedure: SC1 & SC2 • estimator of m_0 using SC1 • recover the inversion procedure in the homogeneous setting by combining JER approach and plug-in of the above estimator
Section 3.5	• two concentration inequalities for heterogeneous p-values • derivation of local tests from these inequalities • derivation of SC1 & SC2 in a tractable way for these tests
Section 3.6	• derivation of confidence envelopes for heterogeneous p-values using JER (in the three JER sub-frameworks) and Section 3.5
Section 3.7	• validation of new confidence bounds on simulated data • comparison to known analog bounds not using the heterogeneity

Table 3.1: Summary of main original contributions, by section

3.2 Multiple testing and confidence bound formalism

3.2.1 Mathematical notation

In the remainder, $m \geq 1$ and $N \geq 1$ are two positive integers. Specifically, m will be used to denote the number of hypotheses tested, while N is used when we need some generic integer. For any two $a, b \in \mathbb{N}$ with $a \leq b$, $[\![a, b]\!]$ is the integer interval $\{a, a+1, \ldots, b-1, b\}$.

The binary operator $\wedge$ denotes the minimum of two terms, and $\vee$ denotes the maximum of two terms.

For a vector $(x_1, \ldots, x_N) \in \mathbb{R}^N$, the notation $x_{(1)}, \ldots, x_{(N)}$ denotes the order statistics of $(x_1, \ldots, x_N)$, such that $x_{(1)} \leq \cdots \leq x_{(N)}$, and the notation $x_{[1]}, \ldots, x_{[N]}$ denotes the reverse-order statistics of $(x_1, \ldots, x_N)$, such that $x_{[1]} \geq \cdots \geq x_{[N]}$. In case of ties, any permutation that orders can be used. Finally, for $A \subseteq [\![1, N]\!]$, the notation $x_{(1:A)}, \ldots, x_{(|A|:A)}$ denotes the order statistics of the elements $(x_i)_{i \in A}$ and, likewise, the notation $x_{[1:A]}, \ldots, x_{[|A|:A]}$ denotes the reverse-order statistics of the elements $(x_i)_{i \in A}$.

3.2.2 Classical multiple testing formalism

We let $\mathfrak{X} = (\mathcal{X}, \Xi)$ a measurable space and $\mathfrak{F}$ a model on $\mathfrak{X}$, that is a set of probability measures on $\mathfrak{X}$. We observe data belonging to $\mathcal{X}$ distributed according to a distribution that belongs to $\mathfrak{F}$. Formally, this data is represented by a random variable often

denoted X, defined on an abstract probability space $(\Omega, \Sigma, \mathbb{P})$, such that the pushforward measure $X_{\sharp\mathbb{P}}$ is an element of $\mathfrak{F}$. For a given $\mu \in \mathfrak{F}$, we say that "the law of X is μ " or that " X is distributed according to μ ", and we denote $X \sim \mu$, if $X_{\sharp\mathbb{P}} = \mu$.

For a given $\mu \in \mathfrak{F}$, a statistic $T : \mathfrak{X} \to \big(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N)\big)$, and a Borelian set $A \in \mathcal{B}\big(\mathbb{R}^N\big)$, the notations $\mathbb{P}_\mu(T \in A)$ and $\mathbb{P}_\mu(T(X) \in A)$ are short for " $\mathbb{P}\big(T(X) \in A\big)$ for some X with $X \sim \mu$ ". The notation $\mathbb{E}_\mu$ is used in the same way for expectations.

We consider $m \geq 1$ null hypotheses $H_1, \dots, H_m$ to test, which formally are submodels of $\mathfrak{F}$, that is, $H_i \subseteq \mathfrak{F}$ for all $i \in [\![1, m]\!]$. We denote by $\mathcal{H}_0 = \mathcal{H}_0(\mu)$ (the dependence in μ will be dropped when there is no ambiguity) the set of indices of null hypotheses that are true, that is $\mathcal{H}_0(\mu) = \{i \in [\![1, m]\!] : \mu \in H_i\}$. In other words, H_i is true if and only if $i \in \mathcal{H}_0$. The cardinal of $\mathcal{H}_0$ is always denoted $m_0 = m_0(\mu) = |\mathcal{H}_0(\mu)|$. We denote the complementary of $\mathcal{H}_0$ in $[\![1, m]\!]$ by $\mathcal{H}_1 = \mathcal{H}_1(\mu)$. Thus $\mathcal{H}_1$ indexes the true signal that we would be able to ideally detect.

To test each $H_i, i \in [\![1, m]\!]$, we assume that have at hand a statistic which takes its values in $[0, 1]$, named a p-value, and denoted p_i. When there is no ambiguity, we will identify the statistic p_i and the random variable $p_i(X)$. We make the following two assumptions on the p-values. The first one is the classical requirement for a statistic to be considered a p-value (and induce a test with type-I error control). The second one is an independence assumption required for the application of our concentration inequalities.

Assumption 3.1 (Super-uniformity under the null)**.** *For each $\mu \in \mathfrak{F}$, for each $i \in \mathcal{H}_0(\mu)$, for $X \sim \mu$, the distribution of $p_i(X)$ is super-uniform, which is often denoted by $p_i(X) \succeq \mathcal{U}\big([0, 1]\big)$. This means that the cdf of $p_i(X)$ is upper bounded by the cdf of $\mathcal{U}\big([0, 1]\big)$. Mathematically,*

$$\forall \mu \in \mathfrak{F}, \forall i \in \mathcal{H}_0(\mu), \forall x \in \mathbb{R},\ \mathbb{P}_\mu(p_i \leq x) \leq 0 \vee (x \wedge 1). \tag{3.1}$$

Assumption 3.2 (Independence under the null)**.** *For each $\mu \in \mathfrak{F}$, the p values $p_i(X)$, $i \in \mathcal{H}_0(\mu)$ are mutually independent.*

Remark 3.2.1. *Assumption 3.2 is rather classical in multiple testing, albeit not totally realistic. In the literature, it is sometimes required that the family of p-values $\big(p_i(X)\big)_{i \in \mathcal{H}_0}$ is also independent from the family $(p_i(X))_{i \in \mathcal{H}_1}$, for example to get FDR bounds in Döhler et al. (2018). In this chapter, however, classical multiple testing results and inequalities are only ever applied to $\big(p_i(X)\big)_{i \in \mathcal{H}_0}$, making this additional requirement unnecessary.*

The following is a simple toy example making use of the introduced notation.

Example 3.2.2 (Gaussian one-sided under independence)**.** *In this example we assume that $X = (X_1, \dots, X_m)$ is a vector of independent Gaussians and the null hypotheses refer to the nullity of their means in contrast to their positivity (we assume that they share the same variance for simplicity). That is, formally, $(\mathcal{X}, \Xi) = \big(\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m)\big)$, $\mathfrak{F} = \big\{\mathcal{N}(\boldsymbol{\nu}, \sigma^2 \mathrm{Id}) : \forall j \in [\![1, m]\!], \nu_j \geq 0, \sigma > 0\big\}$, and for each $i \in [\![1, m]\!]$, $H_i = \big\{\mathcal{N}(\boldsymbol{\nu}, \sigma^2 \mathrm{Id}) \in \mathfrak{F} : \nu_i = 0\big\}$. Then we can construct p-values by letting $p_i(X) = 1 - \Phi(X_i/\sigma)$, where Φ denotes the cdf of $\mathcal{N}(0, 1)$. Such p-values satisfy Assumptions 3.1 and 3.2. They are even exactly uniform under the null.*

For every subset of hypotheses $S \subseteq [\![1, m]\!]$, let $V(S) = |S \cap \mathcal{H}_0|$. If we think of S as a selection set of hypotheses deemed significant, $V(S)$ is then the number of false discoveries in S. The function $V(\cdot)$ is the main object of interest in multiple testing, with

past works proposing rejection procedures $R = R(X)$ with a given statistical control on $V(R)$. Popular error criteria include the Family-Wise Error Rate (FWER), defined as

$$\mathrm{FWER}(R) = \mathbb{P}(V(R) > 0),$$

the False Discovery Rate (FDR, Benjamini and Hochberg, 1995), defined as the mean of the False Discovery Proportion (FDP):

$$\mathrm{FDP}(R) = \frac{V(R)}{|R| \vee 1}, \quad \mathrm{FDR}(R) = \mathbb{E}[\mathrm{FDP}(R)],$$

and the False Discovery Exceedance (FDX, Genovese and Wasserman, 2006) for some level $\gamma \in (0, 1)$:

$$\mathrm{FDX}(R) = \mathbb{P}\big(\,\mathrm{FDP}(R) > \gamma\big).$$

However, in this work and in the confidence bounds literature, one does not want to prescribe a single rejection set R with proven control on $V(R)$ (and only $V(R)$), but to provide control on the function $V(\cdot)$ as a whole.

3.2.3 Confidence bounds

With the formalism introduced in last Section, a simultaneous confidence upper bound is a functional

$$\widehat{V} : \left\{ \begin{array}{ccc} \mathcal{X} \times (0,1) & \longrightarrow & \Big(\mathcal{P}\big(\llbracket 1, m \rrbracket\big) \to \llbracket 0, m \rrbracket\Big) \\ (x, \alpha) & \longmapsto & \widehat{V}_{(x,\alpha)}(\cdot) \end{array} \right. ,$$

such that the following inequality is true:

$$\forall \mu \in \mathfrak{F}, \forall \alpha \in (0,1),\ \mathbb{P}_\mu\Big(\forall S \subseteq \llbracket 1, m \rrbracket, V(S) \leq \widehat{V}_{(X,\alpha)}(S)\Big) \geq 1 - \alpha. \tag{3.2}$$

In the remainder or this work, the dependence in (X, α) will be dropped when there is no ambiguity and $\widehat{V}_{(X,\alpha)}$ will simply be written $\widehat{V}$.

Equation (3.2) means that, with high probability, we are able to bound the number of false discoveries in any selection set S. Note that the " $\forall S \subseteq \llbracket 1, m \rrbracket$ " being inside the probability is what makes the confidence bound "simultaneous".

From a simultaneous confidence upper bound on the false discoveries, we immediately get a simultaneous confidence lower bound on the trues discoveries, by letting

$$\widehat{D}(S) = |S| - \widehat{V}(S). \tag{3.3}$$

Sometimes, confidence bounds are reported into this form, as in Goeman and Solari (2011); Goeman et al. (2021), or in Section 3.7 here when we report the numerical results of our experiments.

Also note that, by a simple division by $|S| \vee 1$, Equation (3.2) can be rewritten as an equation giving a simultaneous upper bound on the FDP:

$$\forall \mu \in \mathfrak{F}, \forall \alpha \in (0,1),\ \mathbb{P}_\mu\left(\forall S \subseteq \llbracket 1, m \rrbracket, \mathrm{FDP}(S) \leq \frac{\widehat{V}_{(X,\alpha)}(S)}{|S| \vee 1}\right) \geq 1 - \alpha. \tag{3.4}$$

This, in turn, allows to use simultaneous upper bounds to construct procedures that control the FDX. Note that controlling the FDX is generally more desirable than controlling the FDR, given the nature of the latter as an expected value. See for example

Bogdan et al. (2015) (Figure 4) for a credible example where the FDR is controlled but the FDP has a highly undesirable behavior (either 0 because no discoveries are made, or higher than the target level). The construction for FDX control is the following: one can compute the largest S such that $\frac{\widehat{V}(S)}{|S|\vee 1}$ is less than or equal to a nominal level $\gamma \in (0,1)$, and (3.4) ensures that, with high probability, the FDP of S is upper bounded by γ, so that the FDX is bounded by α.

3.2.4 Heterogeneity

In this work, we are interested in the case where the distributions of p-values under the null are heterogeneous, departing from the classical assumption that they are all distributed as $\mathcal{U}([0,1])$ under the null (as in Meah et al., 2024, for instance), that we can call the *homogeneous uniform* case.

Also note that, with respect to the super-uniformity of Assumption 3.1, the homogeneous uniform case appears as a "worst case", in that many multiple testing procedures, designed to provide statistical control of the type-I error for that case, become more conservative under super-uniformity. This is easy to see with classical rejection procedures that often take the form of p-value sublevel sets of a statistical threshold, that is, $R(X) = \{i \in [\![1,m]\!] : p_i \leq \hat{t}(X)\}$. Given that super-uniform p-values are stochastically larger than uniforms, the threshold $\hat{t}(X)$, although ensuring statistical validity, becomes badly calibrated and "too small", leading to fewer rejections.

In order to be able to leverage heterogeneity and super-uniformity and create confidence bounds that are more powerful than those designed to guarantee control in the homogeneous uniform case, we need some knowledge about said heterogeneity, that is, to have access to the distribution of any p_i under the null, or at least to have a stochastic lower bound on that distribution. This motivates the following assumption:

Assumption 3.3 (Stochastic heterogeneous lower bound)**.** *For each $i \in [\![1,m]\!]$, there exists a known nondecreasing function F_i such that, if $\mu \in H_i$, the cdf of p_i is upper bounded (as a function) by F_i. Mathematically,*

$$\forall \mu \in \mathfrak{F}, \forall i \in \mathcal{H}_0(\mu), \forall x \in \mathbb{R},\ \mathbb{P}\big(p_i(X) \leq x\big) \leq F_i(x).$$

If Assumption 3.1 holds, then Assumption 3.3 is always satisfied by $F_i(x) = x \mapsto 0 \vee (x \wedge 1)$, but with Assumption 3.3 we hope to take advantage of tighter stochastic lower bounds if they are known, and we allow those to be heterogeneous. Assumption 3.3 also allows to generalize beyond the super-uniform case (i.e. if Assumption 3.1 does not hold), as is the case with weighted p-values (see Döhler et al., 2018).

Assumption 3.3 is satisfied in particular if, for all $\mu \in \mathfrak{F}$, $i \in \mathcal{H}_0(\mu)$, the cdf of p_i does not depend on μ and is known. This is typically the case with discrete tests, where heterogeneity arises naturally, as we can see in Example 3.2.4 below. More generally, Assumption 3.3 is also always satisfied by $x \mapsto \sup_{\mu \in H_i} \mathbb{P}_\mu(p_i \leq x)$. The latter choice of F_i is the one made in Döhler et al. (2018).

Definition 3.2.3 (Support)**.** *For $i \in [\![1,m]\!]$, we define the support of F_i, by analogy with the support of a measure, by*

$$\mathcal{A}_i = \{x \in \mathbb{R} : \forall \varepsilon > 0,\ F_i(x+\varepsilon) - F_i(x-\varepsilon) > 0\}.$$

Example 3.2.4 (Independent binomials)**.** *In this example we assume that $X = (X_1, \ldots, X_m)$ is a vector of independent binomials of respective numbers of trials $n_1, \ldots, n_m$ (known) and*

of unknown probabilities of success $\pi_1, \dots, \pi_m$. We test H_i: " $\pi_i = \frac{1}{2}$ " against " $\pi_i \neq \frac{1}{2}$ ". That is, formally, $(\mathcal{X}, \Xi) = \big(\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m)\big)$, $\mathfrak{F} = \big\{ \bigotimes_{j=1}^m \mathcal{B}(n_j, \pi_j) : \forall j \in [\![1, m]\!], \pi_j \in [0, 1] \big\}$, and for each $i \in [\![1, m]\!]$, $H_i = \big\{ \bigotimes_{j=1}^m \mathcal{B}(n_j, \pi_j) \in \mathfrak{F} : \pi_i = \frac{1}{2} \big\}$. Then we can construct two-sided p-values by letting $p_i(\cdot) : x \mapsto \mathbb{P}\big(f_i(N_i) \leq f_i(x_i)\big)$, where: $N_i \sim \mathcal{B}\big(n_i, \frac{1}{2}\big)$ and f_i denotes the probability mass function of $\mathcal{B}\big(n_i, \frac{1}{2}\big)$. Such p-values satisfy Assumptions 3.1, 3.2, and 3.3, and their distributions under the null are heterogeneous as soon as not all n_i are equal. In particular, we know the exact cdf of p_i under the null and we can let F_i be this cdf. Indeed, notice that $p_i(x) = \sum_{k=0}^{n_i} f_i(k) \mathbb{1}_{f_i(k) \leq f_i(x_i)}$ so, under the null, the support of p_i is $\mathcal{A}_i = \big\{ \sum_{k=0}^{n_i} f_i(k) \mathbb{1}_{f_i(k) \leq f_i(K)} : K \in [\![0, n_i]\!] \big\}$, and its cdf is the step function which has only discontinuities at the elements of $\mathcal{A}_i$ and which is equal to the identity on $\mathcal{A}_i$. For $n_i \in \{5, 15, 30\}$, the graphs of F_i, and the graph of the cdf of $\mathcal{U}\big([0, 1]\big)$, are represented in Figure 3.1.

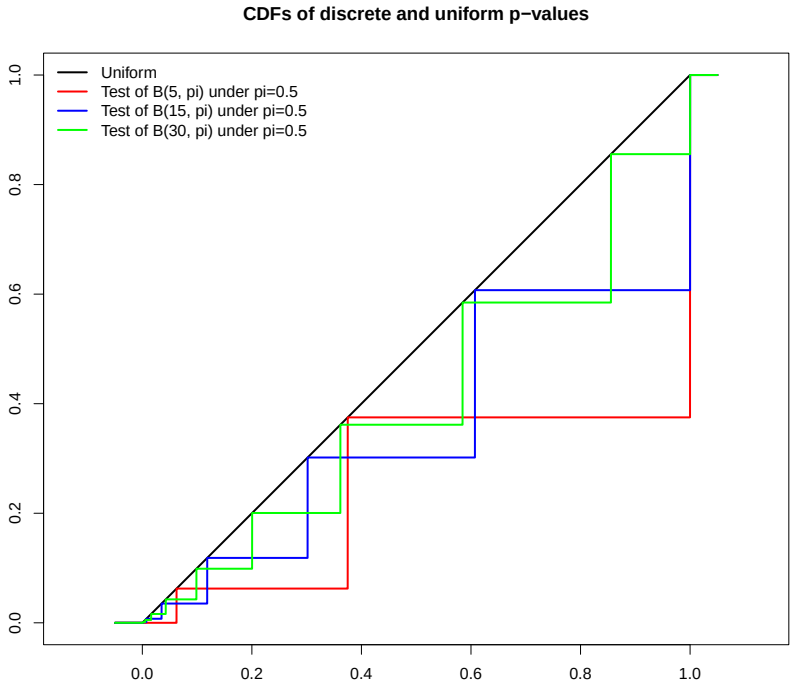


Figure 3.1: Graphs of the cdf of $\mathcal{U}([0, 1])$ (black) and the cdf of the p-value under the null testing if a binomial with 5, 15 or 30 trials is of parameter $\frac{1}{2}$ or not (other colors).

Heterogeneity has already been investigated as a way to correct the conservativeness induced by discrete tests, for the FWER control (Tarone, 1990), for the FDR control (Gilbert, 2005; Heller and Gur, 2014; Habiger, 2015; Döhler et al., 2018) and for the FDX control (Döhler and Roquain, 2020). To our knowledge, this manuscript is the first work tackling the exploitation of heterogeneity in the context of confidence envelopes.

3.3 Review and extension of existing theory

In this section, we revisit past works on confidence envelopes and tools used to build them, while introducing some new contributions to consolidate those works. Namely,

in Section 3.3.1, we refine some known concentration inequalities and extend them to the super-uniform case thanks to a coupling device.

Then, in Section 3.3.2, we recall the inversion method of Genovese and Wasserman (2006) (which is equivalent to the closed testing approach of Goeman and Solari, 2011, as proven by Blanchard et al., 2020b, Appendix S-1.3), and a practical way to compute the derived confidence envelope in favorable cases.

Finally, in Section 3.3.3, we recall the JER approach of Blanchard et al. (2020a) and its three usual settings: k-FWER, top-k, and with deterministic regions. As new contributions, we explain how past works on the top-k setting are valid under Assumption 3.1 with adequate modifications and how some bounds can be computed efficiently in the top-k setting with a new algorithm.

Other contributions are deferred to Appendices A.2 and A.3: the bridging of the theoretical gaps in some past proofs, the irrelevance of the slight difference between our definition of the top-k setting and the one of Meah et al. (2024), and the links between the k-FWER approach and the top-k approach.

3.3.1 Homogeneous concentration inequalities

We start with recalling classical concentration inequalities that have been used to construct confidence bounds. While they are usually stated to be true for uniform random variables (as they are in Meah et al., 2024), which imply that they are suitable to be used in the homogeneous uniform case (hence the name of this section), here we generalize them to super-uniform variables, which implies that they are also suitable under Assumption 3.1.

The underlying argument follows from coupling between super-uniform and uniform variables, we isolate this (standard) device first in the following result:

Lemma 3.3.1 (Super-uniform/uniform coupling)**.** *Let $(U_i)_{i\in[\![1,N]\!]}$ be N independent random variables that are super-uniform. Then there exist $\left(\widetilde{U}_i\right)_{i\in[\![1,N]\!]}$ independent random variables that are $\mathcal{U}[0,1]$ (possibly defined on another probability space) and $(V_i)_{i\in[\![1,N]\!]}$ random variables defined on the same space as $\left(\widetilde{U}_i\right)_{i\in[\![1,N]\!]}$, such that $(V_i)_{i\in[\![1,N]\!]}$ has the same distribution as $(U_i)_{i\in[\![1,N]\!]}$ and $V_i \geq \widetilde{U}_i$ for all $i \in [\![1,N]\!]$.*

The proof of this device is given in Appendix A.6.1.

The first inequality we depict here is the well-known one-sided DKW inequality (Dvoretzky et al., 1956), with the optimal constant of Massart (1990) and the validity domain extended by Reeve (2024) (to all nonnegative scalars λ).

Proposition 3.3.2 (DKW inequality - Reeve's update)**.** *Let $U_1, \ldots, U_N$ be N independent random variables that are super-uniform.*

Then, for all $\delta \in (0,1]$, with $\lambda_\delta = \sqrt{\frac{\log(1/\delta)}{2}}$,

$$\mathbb{P}\left(\sqrt{N}\sup_{t\in[0,1]}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\} - t\right) \leq \lambda_\delta\right) \geq 1-\delta. \tag{3.5}$$

This concentration inequality has been used in the past in the JER setting of deterministic regions (Durand et al., 2020) and in the top-k setting (Meah et al., 2024), see Section 3.3.3, although there was a gap in a proof in Durand et al. (2020), see Appendix A.2.

The second inequality is derived from Wellner's inequality and provided by Meah et al. (2024).

Proposition 3.3.3 (Uniform Wellner's inequality)**.** *Let $U_1, \dots, U_N$ be N independent random variables that are super-uniform, and let $\kappa = \frac{\pi^2}{6}$. For all $\delta \in (0, 1]$,*

$$\mathbb{P}\left(\forall t \in (0,1], \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{U_i \leq t\} \leq t h^{-1}\left(\frac{2\log(\kappa/\delta) + 4\log\left(1+\log_2(1/t)\right)}{Nt}\right)\right) \geq 1-\delta,$$

where h is the function defined by $h(x) = x(\log x - 1) + 1$ for all $x > 1$.

This concentration inequality has been used in the top-k setting, see Section 3.3.3. The proof of the two last inequalities to the super-uniform case is deferred to Appendix A.6.2. The third inequality, Simes inequality (Simes, 1986), has been extensively used in the literature (Goeman and Solari, 2011; Blanchard et al., 2020a).

Proposition 3.3.4 (Simes inequality)**.** *Let $U_1, \dots, U_N$ be N independent random variables that are super-uniform.*

Then for all $\delta \in [0, 1]$, $\mathbb{P}\left(\forall k \in [\![1, N]\!] : U_{(k)} > \frac{k\delta}{N}\right) \geq 1 - \delta$.

We recall its simple proof using the classical Benjamini-Hochberg (BH, Benjamini and Hochberg, 1995) procedure, because a similar argument will be used to prove the new heterogeneous analog to this inequality in Section 3.5.2.

Proof. We apply the BH procedure to $U_1, \dots, U_N$, seen as N p-values under the null, at level δ. It controls the FDR at level δ (even with super-uniformity instead of uniformity under the null), and given that all U_i are under the null, its FDR is equal to its FWER, which in turn is exactly $\mathbb{P}\left(\exists k \in [\![1, N]\!] : U_{(k)} \leq \frac{\delta k}{N}\right)$. □

Remark 3.3.5. *Given that the FDR control of the Benjamini-Hochberg procedure is valid under a more general dependence setting than independence, namely, the wPRDS condition (Benjamini and Yekutieli, 2001), such is also the case for the Simes inequality. For more information about the wPRDS condition, see Blanchard and Roquain (2008) and Giraud (2015).*

In Meah et al. (2024), Simes inequality is given under another form, closer to the formulation of the inequalities of Propositions 3.3.2 and 3.3.3.

Proposition 3.3.6 (Simes inequality - stochastic process version)**.** *Let $U_1, \dots, U_N$ be N independent random variables that are super-uniform. For all $\delta \in (0, 1]$,*

$$\mathbb{P}\left(\forall t > 0, \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{U_i \leq t\} < \frac{t}{\delta}\right) \geq 1 - \delta.$$

For completeness, we give the proof of the last proposition in Appendix A.6.3.

Remark 3.3.7. *Notice that in our statement, a "$<$" sign is used, contrary to the "$\leq$" used in Meah et al. (2024). It didn't matter in Meah et al. (2024) because the authors worked with uniform random variables, but in the general context of Assumption 3.1, our version can lead to less conservative bounds (see Section 3.3.3).*

Finally, we stress that these inequalities are not designed to use known information about heterogeneity of the data and are instead valid under the "worst case" that is the homogeneous uniform case. Heterogeneous analogues will be introduced in Section 3.5.

3.3.2 Confidence envelopes by the inversion procedure

In this section, we recall the inversion procedure introduced by Genovese and Wasserman (2006), which uses the notion of local test. We introduce, for $t \in \mathbb{R}$ and $A \subset [\![1, m]\!]$, a notation for the number of elements in $A \cap R(t)$, where $R(t) = \{i \in [\![1, m]\!] : p_i \leq t\}$:

$$i_A(t) = \sum_{i \in A} \mathbb{1}\{p_i \leq t\}. \tag{3.6}$$

When $A = [\![1, m]\!]$, we simplify the notation to $i(t)$.

Definition 3.3.8 (Local test). *Let $\alpha \in [0, 1]$ and $A \subseteq [\![1, m]\!]$. Then we call $H_A = \bigcap_{i \in A} H_i$ the intersection hypothesis for the subset A, and we say that a statistic φ_A is a local test for A at level $\alpha \in [0, 1]$ if, for all $\mu \in H_A, \mathbb{P}_\mu(\varphi_A = 1) \leq \alpha$.*
A family $(\varphi_A)_{A \subseteq [\![1,m]\!]}$ of local tests has level $\alpha \in [0, 1]$ if for all $A \subseteq [\![1, m]\!]$, φ_A is a local test for A at level α.

We can easily define local tests using concentration inequalities like the ones studied in previous section, as in the following example.

Example 3.3.9. *Let $\alpha \in (0, 1]$ and assume Assumption 3.1 and Assumption 3.2. Then for $A \subseteq [\![1, m]\!]$, a local test of level α derived from Proposition 3.3.6 is expressed by*

$$\varphi_A = \sup_{t \in (0,1]} \mathbb{1}\left\{ \frac{1}{|A|} i_A(t) \geq \frac{t}{\alpha} \right\}.$$

Denote $\mathcal{N}_\varphi(X) = \left\{A \subseteq [\![1, m]\!] : \varphi_A(X) = 0\right\}$ the set of potential candidates for $\mathcal{H}_0(\mu)$ according to our local tests (notice that it is a $(1 - \alpha)$-confidence set for $\mathcal{H}_0$).

Proposition 3.3.10 (Inversion procedure, Genovese and Wasserman, 2006). *Let $\alpha \in [0, 1]$. Assume that $(\varphi_A)_{A \subseteq [\![1,m]\!]}$ is a family of local tests at level α. Then, the functional $\hat{V}^{\mathrm{IP}}_\varphi$ defined by $\hat{V}^{\mathrm{IP}}_\varphi(S) = \max_{A \in \mathcal{N}_\varphi} |A \cap S|$ is a confidence bound at level α.*

We say that a confidence envelope is an *exact shortcut* (for the local test family φ) if it is equal to $\hat{V}^{\mathrm{IP}}_\varphi$, and we say that this confidence envelope is a *shortcut* if it is an upper bound of $\hat{V}^{\mathrm{IP}}_\varphi$.

Note that the computation of the inversion procedure envelope is NP-hard in general, hence a challenging goal is to express $\hat{V}^{\mathrm{IP}}_\varphi$ as an exact shortcut computable in polynomial time. Structural assumptions on the local tests can lead to such exact shortcuts. One of the most usual class of local tests is the following, where the rejection thresholds of a local test depend only on the cardinality of the matching subset of hypotheses.

Definition 3.3.11 (Homogeneous Simes-like local tests). *Let $\Phi = (\varphi_A)_{A \subseteq [\![1,m]\!]}$ be a family of local tests. We say that Φ is a homogeneous Simes-like local test family if, for all $A \subseteq [\![1, m]\!]$,*

$$\varphi_A = \max_{i \in [\![1, |A|]\!]} \mathbb{1}\left\{p_{(i:A)} \leq \ell_{i:|A|}\right\}, \tag{3.7}$$

with $(\ell_{i:n})_{\substack{0 \leq n \leq m \\ 1 \leq i \leq m}}$ such that $\ell_{i:n_1} \geq \ell_{i:n_2}$ for all $n_1 \leq n_2$ and $i \in [\![1, m]\!]$.

Note that the homogeneous term will be dropped when there is no ambiguity on the working case. We call *vanilla Simes local test family* the previous local test family for the specific choice $\ell_{i:n} = \frac{i\alpha}{n}$.

Proposition 3.3.12 (Goeman et al., 2021, Lemma 6)**.** *Assume that, for all $A \subseteq [\![1, m]\!]$, φ_A is of the form (3.7), then*

$$\forall S \subseteq [\![1, m]\!], \hat{V}_{\varphi}^{\mathrm{IP}}(S) = \min_{1 \leq u \leq |S|} \left\{ \left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{GHS}}} \right\} \right| + u - 1 \right\}, \tag{3.8}$$

where

$$\hat{m}_0^{\mathrm{GHS}} = \max\left\{ n \in [\![0, m]\!] : \forall i \in [\![1, n]\!], p_{[i]} > \ell_{(n-i+1):n} \right\}. \tag{3.9}$$

The same result holds true if the $\leq$ inequality is replaced by $<$ in (3.7), and $>$ by $\geq$ in (3.8) and (3.9).

The proof of this proposition in Goeman et al. (2021) relies on the closed testing framework first used by Goeman and Solari (2011) as an equivalent point of view to the inversion procedure confidence envelope. In Section 3.4.2, we present a new proof relying instead on JER tools (see Section 3.3.3). The above proposition offers computation in quadratic time when the statistical framework leads to homogeneous Simes-like local tests. Note that, in the above, $\hat{m}_0^{\mathrm{GHS}}$ acts as an over-estimator of m_0.

Remark 3.3.13. *Proposition 3.3.12 is well adapted to the homogeneous case, but the local tests derived from heterogeneous inequalities that will be introduced in Section 3.5 will fail to satisfy the conditions of Proposition 3.3.12. Namely, while they satisfy a form similar to the one of homogeneous Simes-like local tests, they are typically expressed by*

$$\varphi_A = \max_{i \in [\![1, |A|]\!]} \mathbb{1}\left\{ p_{(i:A)} \leq \ell_{i:A} \right\},$$

where, for all $i \in [\![1, m]\!]$, $(\ell_{i:A})_{A \subseteq [\![1,m]\!]}$ is nonincreasing with respect to the inclusion partial order, and, for all $A \subseteq [\![1, m]\!]$, $(\ell_{i:A})_{i \in [\![1,m]\!]}$ is nondecreasing.
Thus these local tests will, in general, fully depend on A rather than only $|A|$, which is the reason why the shortcut from Proposition 3.3.12 cannot be applied when the data heterogeneity is considered to construct local tests.

3.3.3 Reference families, JER and interpolation

JER methods

We present here the alternative approach to constructing confidence bounds originally introduced in Blanchard et al. (2020a), based on the two concepts of reference families and Joint Error Rate (JER) that we start by defining. A reference family is a finite family $\mathfrak{R}_\alpha(X) = (R_k, \zeta_k)_{k \in \mathcal{K}}$ with $R_k \subseteq [\![1, m]\!]$, $\zeta_k \in \mathbb{N}$, where everything (that is, $\mathcal{K}$ and all the R_k and ζ_k) depends on (X, α), but the dependency is not explicitly written. Likewise, when there is no ambiguity, the reference family will simply be denoted $\mathfrak{R}_\alpha$ or even $\mathfrak{R}$. The JER of a reference family is an error criterion that takes the following form:

$$\mathrm{JER}(\mathfrak{R}) = \mathbb{P}(\exists k \in \mathcal{K}, |R_k \cap \mathcal{H}_0| > \zeta_k) = \mathbb{P}(\exists k \in \mathcal{K}, V(R_k) > \zeta_k). \tag{3.10}$$

Think of each ζ_k as an over-estimator of the number of false discoveries in R_k. Then the JER is the probability that this upper bound fails for at least one k. If the JER is small, then, with high probability, the ζ_k's are simultaneous upper bounds of the false discoveries in all the regions R_k's.

We are then interested in constructing reference families that control the JER at any target level, the inequation that we want to see satisfied is then:

$$\forall \mu \in \mathfrak{F}, \forall \alpha \in (0,1),\ \mathrm{JER}_\mu(\mathfrak{R}_\alpha) \leq \alpha,$$

or, in other words, at any given $\alpha \in (0,1)$,

$$\forall \mu \in \mathfrak{F},\ \mathbb{P}_\mu(\forall k \in \mathcal{K}, V(R_k) \leq \zeta_k) \geq 1-\alpha. \tag{3.11}$$

Note that Equation (3.11) is really similar to Equation (3.2) except that the uniform guarantee, instead of being over all $S \subseteq [\![1,m]\!]$, is only over all the $R_k \subseteq [\![1,m]\!], k \in \mathcal{K}$, with $\mathcal{K}$ having cardinality potentially much smaller than 2^m. A simultaneous confidence bound over all subsets of hypotheses is then derived from a JER-controlling reference family by interpolation. Let

$$\mathfrak{A}(\mathfrak{R}) = \{A \subseteq [\![1,m]\!] : \forall k \in \mathcal{K}, |R_k \cap A| \leq \zeta_k\}. \tag{3.12}$$

The JER control informs us that, with high probability, $\mathcal{H}_0 \in \mathfrak{A}(\mathfrak{R})$. We leverage this information with the following confidence bound construction:

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \max_{A \in \mathfrak{A}(\mathfrak{R})} |S \cap A|, \tag{3.13}$$

which optimally uses the information provided by the JER control of the reference family, as proven by Proposition 2.1 of Blanchard et al. (2020a). Note that there exists an equivalence between constructions of confidence envelopes based on reference families and on the inversion procedure based on local tests. The links between the two approaches are discussed in Blanchard et al. (2020b) (Appendix S-1.3).

Because of the combinatorial maximum operation $\max_{A \in \mathfrak{A}(\mathfrak{R})}$, the computation of $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)$ is intractable in general (see Proposition 2.2 of Blanchard et al., 2020a), but for specific structures of reference families introduced in the next paragraph, a polynomial computation can be achieved.

First, if the set $\{R_k : k \in \mathcal{K}\}$ is totally ordered for the inclusion relation, that is, the R_k's are nested, then, for all $S \subseteq [\![1,m]\!]$,

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \min_{k \in \mathcal{K}} \left(\zeta_k + |S \setminus R_k|\right) \wedge |S|, \tag{3.14}$$

by Proposition 2.5 of Blanchard et al. (2020a). Second, if the reference family satisfies the forest structure constraint introduced by Durand et al. (2020) and defined by:

$$\forall k, k' \in \mathcal{K},\ R_k \cap R_{k'} \in \{R_k, R_{k'}, \varnothing\},$$

then fast algorithms also exist to compute $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)$ for a single $S \subseteq [\![1,m]\!]$ (Durand et al., 2020, Algorithm 1) or, even better, to compute $\left(\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_t)\right)_{1 \leq t \leq m}$ for a path of nested hypotheses subsets $S_1 \subsetneq \cdots \subsetneq S_m$ (Durand, 2025, Algorithm 4). Note that the forest structure condition encompasses the nested condition.

We can now recall the three settings that have been studied for the construction of reference families that control the JER. They will be studied again in the context of heterogeneity in Section 3.6.

k-FWER setting

The first setting is the " k-FWER" setting, in which we set $\mathcal{K} = [\![1, K]\!]$ for some $K \leq m$, $\zeta_k = k - 1$ and $R_k = \{i \in [\![1, m]\!] : p_i \leq \tau_k\}$, with a threshold family $(\tau_k)_{1 \leq k \leq K}$ that is nondecreasing and such that the JER control holds. By monotonicity of $(\tau_k)_{1 \leq k \leq K}$, such reference family satisfies the nested condition so Equation (3.14) holds and an algorithm given by Enjalbert-Courrech and Neuvial (2022) can be applied for fast computation of $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S)$ for any given S. The name of the setting comes from the fact that the JER control in that case is equivalent to the fact that, simultaneously on k, each R_k controls the k-FWER. An example of templates that yields JER control under our Assumptions is the Simes template introduced by Blanchard et al. (2020b) (Section S-2), defined by $\tau_k = \frac{\alpha k}{m}$:

Theorem 3.3.14 (Simes template yields JER control)**.** *Under Assumptions 3.1 and 3.2, for all $\mu \in \mathfrak{F}$, for all $\alpha \in (0, 1)$, the reference family $\mathfrak{R} = (R_k, k-1)_{1 \leq k \leq m}$, with $R_k = \left\{i \in [\![1, m]\!] : p_i \leq \frac{\alpha k}{m}\right\}$, controls the JER at level α.*

We defer the proof of the last theorem to Appendix A.6.4

Remark 3.3.15. *The same proof scheme proves that Hommel's inequality (Blanchard et al., 2020b) also induces a k-FWER reference family that controls the JER, given by*

$$R_k = \left\{ i \in [\![1, m]\!] : p_i \leq \frac{\alpha k}{m} \sum_{i=1}^{m} \frac{1}{i} \right\},$$

without needing Assumption 3.2.

Top-k setting

We move on to second setting, namely the "top-k " setting that appeared in Katsevich and Ramdas (2020). Let σ a (random) permutation that orders the p-values, i.e., $p_{\sigma(k)} = p_{(k)}$ for all $k \in [\![1, m]\!]$. Because we do not assume that the p-values have continuous distributions, there may be ties and σ may not be unique. The top-k setting consists in fixing

$$R_k = \{\sigma(1), \ldots, \sigma(k)\} \tag{3.15}$$

for all $k \in [\![1, m]\!]$, and to find appropriate ζ_k's such that the JER control holds.

Remark 3.3.16. *In past works assuming that the p-values have continuous distributions, R_k was rather defined as $R_k^\dagger = \left\{i \in [\![1, m]\!] : p_i \leq p_{(k)}\right\}$, and it is a.s. equal to our definition under such assumption. However, our definition ensures that $|R_k| = k$ even in the case of ties, which is practical for computational purposes, while we can only say that $\left|R_k^\dagger\right| \geq k$ under Assumption 3.1. Thankfully, the choice of R_k or $R_k^\dagger$ is inconsequential with respect to the confidence envelope, as is proven in Appendix A.3.1.*

Remark 3.3.17. *We can always assume that the sequence $(\zeta_k)_{1 \leq k \leq m}$ is nondecreasing, up to replacing ζ_k by $\min(\zeta_k, \zeta_{k+1}, \ldots, \zeta_m)$. Indeed, if the event inside the probability of Equation (3.11) is realized, for $k^* \geq k$ such that $\zeta_{k^*} = \min(\zeta_k, \zeta_{k+1}, \ldots, \zeta_m)$, we have $V(R_k) \leq V(R_{k^*}) \leq \zeta_{k^*}$.*

Many constructions are proposed by Meah et al. (2024), under the assumptions that the p-values are uniform under the nulls. These constructions can be extended to the case of Assumption 3.1 thanks to the following proposition.

Proposition 3.3.18 (Inequalities for the top-k setting, extended to super-uniformity)**.** *Assume that Assumptions 3.1 and 3.2 hold.*

Then, for all $\delta \in (0,1]$, and for all $\mu \in \mathfrak{F}$,

$$\mathbb{P}_\mu\left(\forall t \in [0,1], i_{\mathcal{H}_0(\mu)}(t) \leq mt + \sqrt{m}\sqrt{\frac{1}{2}\log\frac{1}{\delta}}\right) \geq 1-\delta, \tag{3.16}$$

$$\mathbb{P}_\mu\left(\forall t \in (0,1], i_{\mathcal{H}_0(\mu)}(t) \leq mth^{-1}\left(\frac{2\log(\kappa/\delta) + 4\log\left(1+\log_2(1/t)\right)}{mt}\right)\right) \geq 1-\delta, \tag{3.17}$$

$$\mathbb{P}_\mu\left(\forall t > 0, i_{\mathcal{H}_0(\mu)}(t) < \frac{mt}{\delta}\right) \geq 1-\delta. \tag{3.18}$$

The proof of the previous proposition is given in Appendix A.6.5. This proposition allows to deduce the following JER-controlling families, as defined in the following theorem.

Theorem 3.3.19 (Reference families for the top-k setting, extended to super-uniformity)**.** *Assume that Assumptions 3.1 and 3.2 hold. Let $\alpha \in (0,1)$, $\mu \in \mathfrak{F}$, and $X \sim \mu$. The following three reference families control the JER at level α.*

1. *the family $(R_k, \zeta_k)_{1\leq k\leq m}$ with, for all $k \in [\![1,m]\!]$, R_k as in (3.15) and*

$$\zeta_k = \mathbb{1}\{p_{(k)} > 0\}\left\lfloor mp_{(k)} + \sqrt{\frac{m}{2}\log\frac{1}{\alpha}}\right\rfloor \wedge k,$$

2. *the family $(R_k, \zeta_k)_{1\leq k\leq m}$ with, for all $k \in [\![1,m]\!]$, R_k as in (3.15) and*

$$\zeta_k = \mathbb{1}\{p_{(k)} > 0\}\left\lfloor mp_{(k)}h^{-1}\left(\frac{2\log(\kappa/\alpha) + 4\log\left(1+\log_2\left(1/p_{(k)}\right)\right)}{mp_{(k)}}\right)\right\rfloor \wedge k,$$

3. *the family $(R_k, \zeta_k)_{1\leq k\leq m}$ with, for all $k \in [\![1,m]\!]$, R_k as in (3.15) and*

$$\zeta_k = \mathbb{1}\{p_{(k)} > 0\}\left\lfloor \frac{mp_{(k)}{}^-}{\alpha}\right\rfloor \wedge k,$$

where $\lfloor\cdot\rfloor$ denotes the floor function and $\lfloor\cdot^-\rfloor$ is its left-limit.

Remark 3.3.20. *Those bounds are similar to the Equations (9), (10) and (15) of Meah et al. (2024), except that they are bounds for the false discoveries and not the FDP, that they are valid under super-uniformity under the null instead of just uniformity, that they use the floor functions to better exploit the fact that the number of false discoveries is an integer, and that they use a $\mathbb{1}\{p_{(k)} > 0\}$ term that was not useful in Meah et al. (2024) because the p-values were assumed to have continuous distributions, whereas here, in the context of Assumption 3.1, it becomes necessary.*

Remark 3.3.21. *Note that each definition of ζ_k in Theorem 3.3.19 is of the form $f\left(p_{(k)}\right) \wedge k$ for some measurable function f that only takes integer values. Also note that we can use the same argument than in Remark 3.3.17: up to replacing $f(\cdot)$ by $t \mapsto \min_{t\leq t'\leq 1} f(t')$ (the minimum is well-defined because f only takes integer values), we can furthermore assume that f is nondecreasing.*

Proof. First note that, almost surely, $V(R_k) \leq \sum_{i \in \mathcal{H}_0} \mathbb{1}\{p_i \leq p_{(k)}\}$ because, if $i \in R_k$, then $p_i \leq p_{\sigma(k)} = p_{(k)}$. Then note that, by super-uniformity, $p_i > 0$ a.s. if $i \in \mathcal{H}_0$, hence $V(R_k) = 0$ if $p_{(k)} = 0$.

Then, like in Meah et al. (2024), we just apply inequalities (3.16), (3.17) and (3.18) by setting $t = p_{(k)}$ if $p_{(k)} > 0$. The floor function can be applied because $V(R_k)$ is an integer. The minimum with k appears from $V(R_k) \leq |R_k| = k$. Finally, in the case of Equation (3.18), the left-limit of the floor function can be applied instead of the floor function thanks to the strict inequality in (3.18). □

Remark 3.3.22. *For computational purposes, note that, for all $t \in \mathbb{R}$, $\lfloor t^- \rfloor = \lceil t \rceil - 1$, where $\lceil \cdot \rceil$ denotes the ceiling function. See Example A.3.8 for an example where using the left-limit or not does matter.*

A top-k reference family satisfies the nested condition, so Equation (3.14) can also be applied for fast computation of $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S)$ for a given S. In particular, it can be applied to the reference top-k sets themselves, giving rise to a possibly improved bound $\tilde{\zeta}_k = \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k) \leq \zeta_k, k \in [\![1, m]\!]$ (see Blanchard et al., 2020a for a discussion). To compute the whole family $(\tilde{\zeta}_k)_{1 \leq k \leq m}$ more efficiently than by repeatedly applying (3.14), we introduce the following new Algorithm 1, which can spare the (possibly costly) computation of some of the original bounds ζ_k.

Algorithm 1: Computation of $\left(\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k)\right)_{1 \leq k \leq m}$

Data: p-values $(p_k)_{1 \leq k \leq m}$, measurable nondecreasing function f taking integer values
Result: $\left(\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k)\right)_{1 \leq k \leq m}$ on top-k path

$k \longleftarrow 1$;
$\tilde{\zeta}_0 \longleftarrow 0$;
while $k \leq m$ **do**
  $\zeta_k \longleftarrow f\left(p_{(k)}\right) \wedge k$;
  if $\zeta_k > \tilde{\zeta}_{k-1}$ **then**
    $j \longleftarrow \left(\zeta_k - \tilde{\zeta}_{k-1}\right) \wedge (m - k + 1)$;
    for $i = 1$ *to* j **do**
      $\tilde{\zeta}_{k+i-1} \longleftarrow \tilde{\zeta}_{k-1} + i$;
    $k \longleftarrow k + j$;
  else
    $\tilde{\zeta}_k \longleftarrow \tilde{\zeta}_{k-1}$;
    $k \longleftarrow k + 1$;
return $\left(\tilde{\zeta}_k\right)_{1 \leq k \leq m}$.

Proposition 3.3.23. *In the setting of a top-k reference family (see (3.15)), assume that $\zeta_k = f\left(p_{(k)}\right) \wedge k$ for some measurable, nondecreasing function f that only takes integer values (see Remark 3.3.21). Then, Algorithm 1 indeed returns the improved bound $\left(\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k)\right)_{1 \leq k \leq m} = \left(\tilde{\zeta}_k\right)_{1 \leq k \leq m}$ for the reference family $\mathfrak{R} = \left(R_k, \zeta_k\right)_{1 \leq k \leq m}$.*

We defer the proof of the previous proposition to Appendix A.6.6.

Remark 3.3.24. *Equation (A.4) provides a simpler algorithm than Algorithm 1, that could be implemented with a simple* `for` *loop. However, this would imply to compute each* ζ_k*, which can be costly, depending on the function* f*. On the contrary, Algorithm 1 leverages the fact that not all* ζ_k *need to be computed:* ζ_k *is computed only at the steps* k *visited by the* `while` *loop.*

We end this overview of the top-k setting by remarking that there exists a direct correspondence between this setting and the k-FWER setting. Namely, from a reference family built according to one of these settings, we can build another one in the other setting, and they yield the exact same confidence envelopes. Furthermore, the two reference families using Simes inequality that we encountered in Theorems 3.3.14 and 3.3.19 also yield the exact same confidence envelopes. This point is formalized precisely in Appendix A.3.2.

Deterministic setting

We finally move to the third setting relevant for applying the JER point of view, called "deterministic setting". In that setting, the regions R_k are deterministic, fixed in advance (possibly reflecting some prior information), and we build appropriate ζ_k's such that the JER control holds. Any generic method that can upper bound the number of false discoveries in a deterministic set of hypotheses at any confidence level (including other confidence envelopes) can be applied to R_k with the confidence level set to $\frac{\alpha}{K}$, where $K = |\mathcal{K}|$, and JER control ensues by a simple union bound argument. We tend to choose our regions such that the forest structure conditions hold, so that Durand (2025) (Algorithm 4) can be applied.

One such construction has been done in Durand et al. (2020), based on the DKW inequality (see Proposition 3.3.2).

Proposition 3.3.25 (Proposition 1 of Durand et al., 2020)**.** *The family* $(R_k, \zeta_k)_{k\in\mathcal{K}}$ *with* $\mathcal{K}$ *a deterministic set and, for all* $k \in \mathcal{K}$*,* R_k *deterministic, and*

$$\zeta_k = \min_{t\in[0,1)} \left\lfloor \left(\frac{\lambda_{\alpha/K}}{2(1-t)} + \sqrt{\frac{\lambda_{\alpha/K}^2}{4(1-t)^2} + \frac{|R_k| - i_{R_k}(t)}{1-t}} \right)^2 \right\rfloor \wedge |R_k|, \tag{3.19}$$

where $\lambda_{\alpha/K} = \sqrt{\frac{1}{2}\log\frac{K}{\alpha}}$*, controls the JER at level* α*.*

3.4 General shortcuts for the inversion procedure

This section introduces new shortcuts for local tests having a specific structure. Namely, we assume that our local tests, for all $A \subseteq [\![1, m]\!]$, are expressed as

$$\varphi_A = \max_{t\in[0,1]} \varphi_{A,t}. \tag{3.20}$$

While any local test can obviously be written under this form through the trivial choice $\varphi_{A,t} = \varphi_A$ for all t, the underlying guiding idea for considering this expression is that, for particular representations taking this form, we are able to compute efficiently — that is, in polynomial time — the minimum of $\varphi_{A,t}$ for a fixed t, over sets A of a fixed cardinality. Therefore, our aim in this section is to derive shortcuts relying first on taking such minima for fixed t.

In the remainder of this section, we assume that we have at hand a family of local tests expressed in the form (3.20). Section 3.4.1 focuses on defining and establishing validity of the new shortcuts. In Section 3.4.2 we provide a methodology to derive confidence bounds, which turns out to be an exact shortcut for the inversion procedure in the special case of homogeneous Simes-like local tests.

Local tests derived from ecdf envelopes, like the DKW and Wellner inequalities, have a natural expression in the form (3.20). That holds for Simes-like local tests as well, as stated by the next lemma. Furthermore, the heterogeneous local test families that will be introduced in Section 3.5 are expressed as (3.20) as well.

Lemma 3.4.1 (Alternative form of Simes-like tests)**.** *Let $A \subseteq [\![1, m]\!]$, and assume that $(\varphi_A)_{A\subseteq[\![1,m]\!]}$ is a homogeneous Simes-like local test family with the sequence $(\ell_{i:n})_{1\leq i\leq m}$ non-decreasing for all $n \in [\![1, m]\!]$. Then, with the convention $\ell_{0:n} = -1$ for all $n \in [\![1, m]\!]$, for all $A \subseteq [\![1, m]\!]$:*

$$\varphi_A = \max_{t\in[0,1]} \mathbb{1}\big\{t \leq \ell_{i_A(t):|A|}\big\},$$

where i_A is defined by (3.6).

We give the proof of the previous lemma in Appendix A.6.7.

3.4.1 Shortcuts for the inversion procedure

This section introduces new shortcuts for the inversion procedure. These shortcuts rely on the fundamental statistic $\varphi_{S,n,t}$ that is defined for $S \subseteq [\![1, m]\!]$, $t \in [0, 1]$ and $n \in [\![0, |S|]\!]$ by

$$\varphi_{S,n,t} = \min_{\substack{A\subseteq[\![1,m]\!]\\ |A\cap S|=n}} \varphi_{A,t}. \tag{3.21}$$

For fixed S and t, this statistic is computable in polynomial time (in m, independently of n) for all the local test families considered in the present article (including heterogeneous settings). Furthermore, as will be seen shortly (in Proposition 3.4.3), taking the minimum or maximum over $t \in [0, 1]$ can in fact be restricted to finite subsets of $[0, 1]$, thus preserving the polynomial complexity for such operations.

Proposition 3.4.2. *The following confidence envelopes, defined for all $S \subseteq [\![1, m]\!]$ by*

$$\hat{V}_\varphi^{\mathrm{SC1}}(S) = \max\left\{n \in [\![0, |S|]\!] : \max_{t\in[0,1]} \varphi_{S,n,t} = 0\right\}, \tag{3.22}$$

and

$$\hat{V}_\varphi^{\mathrm{SC2}}(S) = \min_{t\in[0,1]} \big\{ \max\{n \in [\![0, |S|]\!] : \varphi_{S,n,t} = 0\}\big\}, \tag{3.23}$$

satisfy

$$\forall S \subseteq [\![1, m]\!], \hat{V}_\varphi^{\mathrm{IP}}(S) \leq \hat{V}_\varphi^{\mathrm{SC1}}(S) \leq \hat{V}_\varphi^{\mathrm{SC2}}(S).$$

The proof of Proposition 3.4.2 is provided in Appendix A.6.8.

To fully benefit from the efficient computations of $\varphi_{S,n,t}$, one needs to be able to reduce the " $\max_{t\in[0,1]}$ " and the " $\min_{t\in[0,1]}$ " in Proposition 3.4.2 into a " $\max_{t\in\mathcal{A}}$ " and a " $\min_{t\in\mathcal{A}}$ " with $\mathcal{A}$ the set of realized p-values. The next proposition provides a monotonicity assumption on $\varphi_{A,\cdot}$, for $A \subseteq [\![1, m]\!]$, to this end.

Proposition 3.4.3. *Suppose, defining $p_{(0:A)} = p_0 = 0$ for all $A \subseteq [\![1, m]\!]$, that for all $A \subseteq [\![1, m]\!]$ $\varphi_{A,\cdot}$ is nonincreasing on $[p_{(i:A)}, p_{((i+1):A)})$ for all $i \in [\![0, |A| - 1]\!]$.*

Then the three following equalities occur for all $A \subseteq [\![1, m]\!]$, $S \subseteq [\![1, m]\!]$ and $n \in [\![0, m]\!]$:

$$\varphi_A = \max_{i \in A \cup \{0\}} \varphi_{A,p_i},$$

$$\max_{t \in [0,1]} \varphi_{S,n,t} = \max_{i \in [\![0,m]\!]} \varphi_{S,n,p_i},$$

$$\hat{V}_{\varphi}^{\mathrm{SC2}}(S) = \min_{i \in [\![0,m]\!]} \{\max\{n \in [\![0, |S|]\!] : \varphi_{S,n,p_i} = 0\}\}.$$

Proposition 3.4.3 is proved in Appendix A.6.9.

The monotonicity assumption from the previous proposition will cover all the local tests provided by concentration inequalities considered in the present article. Thus, while we keep using the notation $\max_{t \in [0,1]}$ for notational convenience in the rest of the chapter, in practice only evaluations at t equal to one of the p-values (or $t = 0$) are required.

Finally, we give a sufficient monotonicity condition on $\varphi_{S,\cdot,t}$ for our two shortcuts for the inversion procedure to be equal:

Proposition 3.4.4. *Assume that, for $S \subseteq [\![1, m]\!]$, for all $t \in [0, 1]$, $\varphi_{S,\cdot,t}$ is nondecreasing, that is, for all $n \in [\![0, |S|]\!]$:*

$$\varphi_{S,n,t} = 1 \Rightarrow \forall k \geq n, \varphi_{S,k,t} = 1.$$

Then $\hat{V}_{\varphi}^{\mathrm{SC1}}(S) = \hat{V}_{\varphi}^{\mathrm{SC2}}(S)$.

The proof of Proposition 3.4.4 is deferred to Appendix A.6.10. Note that under the assumptions of Proposition 3.4.3, we only need that $\varphi_{S,\cdot,p}$ is nondecreasing for all $p \in \{0\} \cup \{p_i, i \in [\![1, m]\!]\}$.

3.4.2 Applications to homogeneous inequalities

We now study the shortcuts defined previously when the local tests are based on homogeneous inequalities, namely homogeneous Simes-like local tests, and DKW inequality based local tests. In both cases, we first give an explicit expression for the shortcuts, then investigate their use to obtain an over-estimator $\hat{m}_0^{\mathrm{SC1}} = \hat{V}_{\varphi}^{\mathrm{SC1}}([\![1, m]\!])$ for $m_0 = |\mathcal{H}_0|$, which is then combined with the JER point of view, in order to obtain "adaptive JER" envelopes.

In both cases, we establish that this adaptive JER envelope coincides with the inversion procedure. This shows as a by-product that the adaptive Simes procedure and the adaptive DKW-based top-k procedure introduced by Meah et al. (2024) are both exact shortcuts for their respective associated local test families.

The exactness of the proposed shortcuts in the homogeneous case provides strong motivation for using a similar approach in the heterogeneous setting in Section 3.6.

Case of homogeneous Simes-like local tests

In this section, we assume that the local tests φ_A are homogeneous Simes-like local tests, as by Definition 3.3.11. We define $\ell_{0:n} = -1$ for all $n \in [\![1, m]\!]$ to handle the case when p-values are equal to 0. Recall that, from Lemma 3.4.1, homogeneous Simes-like local tests can be reformulated as $\varphi_A = \max_{t \in [0,1]} \varphi_{A,t}$ with $\varphi_{A,t} = \mathbb{1}\{t \leq \ell_{i_A(t):|A|}\}$.

Based on this representation, we start with a lemma giving an expression of $\varphi_{S,n,t}$, defined by Equation (3.21), which will allow us to compute our shortcuts.

Lemma 3.4.5. *Let $S \subseteq [\![1,m]\!]$ and $n \in [\![0,|S|]\!]$. Then, for all $t \in [0,1]$,*

$$\varphi_{S,n,t} = \min_{0 \le k \le i_{S^c}(t)} \mathbb{1}\left\{t \le \ell_{((n+i_S(t)-|S|)\vee 0+k):(n+m-|S|-i_{S^c}(t)+k)}\right\}.$$

In particular,

$$\varphi_{[\![1,m]\!],n,t} = \mathbb{1}\left\{t \le \ell_{(n+i(t)-m):n}\right\}.$$

The proof of Lemma 3.4.5 is given in Appendix A.6.11.

The following example gives an explicit expression for $\varphi_{[\![1,m]\!],n,t}$ in the specific case of vanilla Simes local tests, as a by-product of Lemma 3.4.5.

Example 3.4.6. *Consider the vanilla Simes local tests at level $\alpha \in (0,1]$, that is such that for all $n \in [\![1,m]\!]$ and $i \in [\![1,n]\!]$, $\ell_{i:n} = \frac{i}{m}\alpha$ in the setting of Definition 3.3.11. Then for all $t \in (0,1]$ and $n \in [\![1,m]\!]$,*

$$\varphi_{[\![1,m]\!],n,t} = \mathbb{1}\left\{t \le \frac{n+i(t)-m}{n}\alpha\right\}.$$

We now propose the following "adaptive JER" methodology based on this shortcut applied to $S = [\![1,m]\!]$:

- Define $\hat{m}_0^{\mathrm{SC1}} = \hat{V}_\varphi^{\mathrm{SC1}}([\![1,m]\!])$.
- Introduce the reference family $\mathfrak{R} = (R_k, k-1)_{k \in [\![1,\hat{m}_0^{\mathrm{SC1}}+1]\!]}$ with

$$\begin{cases} R_k & = \left\{i \in [\![1,m]\!] : p_i \le \ell_{k:\hat{m}_0^{\mathrm{SC1}}}\right\}, \text{ for } k \in [\![1,\hat{m}_0^{\mathrm{SC1}}]\!]; \\ R_{\hat{m}_0^{\mathrm{SC1}}+1} & = [\![1,m]\!]. \end{cases} \tag{3.24}$$

- Use the confidence envelope based on the JER bound using this reference family.

The main result of this section is that the adaptive JER confidence envelope coincides with the inversion procedure confidence envelope using the initial local test family.

The following lemma establishes that the $\hat{m}_0^{\mathrm{SC1}}$ coincides with the one from Proposition 3.3.12, Equation (3.9).

Lemma 3.4.7. *Defining $\hat{m}_0^{\mathrm{SC1}} = \hat{V}_\varphi^{\mathrm{SC1}}([\![1,m]\!])$, we have*

$$\hat{m}_0^{\mathrm{SC1}} = \max\left\{n \in [\![0,m]\!] : \forall t \in [0,1], t > \ell_{(n+i(t)-m):n}\right\}.$$

Furthermore, if for all $n \in [\![1,m]\!]$ and $i \in [\![1,n]\!]$, $\ell_{i+1:n+1} \ge \ell_{i:n}$, then

$$\hat{m}_0^{\mathrm{SC1}} = \hat{V}_\varphi^{\mathrm{SC2}}([\![1,m]\!]) = \min_{t \in [0,1]} \left\{ \max\left\{n \in [\![0,m]\!] : t > \ell_{(n+i(t)-m):n}\right\}\right\}.$$

Proof. The expression from Lemma 3.4.5 for $S = [\![1,m]\!]$ is nondecreasing with respect to n when $\ell_{i+1:n'+1} \ge \ell_{i:n'}$ for all i and n', thus applying Proposition 3.4.4 leads to the conclusion. □

Note that the assumption " $\ell_{i+1:n+1} \ge \ell_{i:n}$ " was already considered in Goeman et al. (2021) as a sufficient condition for using bisection to compute $\hat{m}_0^{\mathrm{GHS}}$.

Corollary 3.4.8. *In the same framework as Example 3.4.6,*

$$\hat{m}_0^{\mathrm{SC1}} = \inf_{t\in(0,\alpha)} \left(\left\lceil \frac{m - i(t)}{1 - t/\alpha} \right\rceil - 1 \right) \wedge \big(m - i(0)\big). \tag{3.25}$$

The proof of the previous corollary is deferred in Appendix A.6.12.

A similar bound was already considered for Simes top-k adaptive bound in Meah et al. (2024), but it did not consider cases with p-values equal to 0 or to one of Simes thresholds. Denoting $\hat{m}_0^{\mathrm{MBR}}$ the bound from Meah et al. (2024), we have here $\hat{m}_0^{\mathrm{SC1}} = \big(\lceil \hat{m}_0^{\mathrm{MBR}} \rceil - 1\big) \wedge \big(m - i(0)\big)$ in the framework of vanilla Simes local tests.

The following proposition states that our methodology produces an exact shortcut.

Proposition 3.4.9. *For a homogeneous Simes-like local test family, consider the adaptive reference family given by (3.24). Then $\mathrm{JER}(\mathfrak{R}) \leq \alpha$ and $\hat{V}_\varphi^{\mathrm{IP}} = \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$.*

For completeness, we provide a proof of this proposition in Appendix A.6.13

Combining Proposition 3.4.9 and Corollary 3.4.8, the adaptive interpolated Simes bound from Meah et al. (2024) applied with our over-estimator defined by (3.25) is an exact shortcut, computable in time $\mathcal{O}\big(m \log(m)\big)$.

Remark 3.4.10. *Note that, for the family $\mathfrak{R}$ defined in Proposition 3.4.9, the expression of $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ does not match exactly the right-hand side of Equation (3.8) from Proposition 3.3.12. The latter can be written $\hat{V}_{\mathfrak{R}_S}^{\mathrm{JER}}$ with the pseudo-reference family $\mathfrak{R}_S$ (which depends on S) defined by $\mathfrak{R}_S = (R_k, k-1)_{k\in[\![1,|S|]\!]}$, with $R_k = \Big\{ i \in [\![1,m]\!] : p_i \leq \ell_{k:\hat{m}_0^{\mathrm{SC1}}} \Big\}$ for $k \in [\![1,|S|]\!]$. Finally it can be checked that the proof of Proposition 3.4.9 also implies that $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}} = \hat{V}_{\mathfrak{R}_S}^{\mathrm{JER}}$, thus we obtain a new proof of Proposition 3.3.12, without using the closed testing formalism.*

Exactness of the adaptive top-k procedure for the DKW inequality

We first introduce an alternative expression of Proposition 3.3.12 suited to the top-k setting, as it allows the user to avoid translating their top-k designed local test into homogeneous Simes-like designed local test (see Remark A.6.2). It can be useful when the functional f_n is computationally hard to invert and it is convenient to prove theoretical results on top-k bounds.

Proposition 3.4.11. *Assume that there exists a family of nonnegative valued function (f_n) such that, for all $A \subseteq [\![1,m]\!]$,*

$$\varphi_A = \max_{t\in[0,1]} \mathbb{1}\big\{ i_A(t) > f_{|A|}(t) \big\}. \tag{3.26}$$

If, for $A \subseteq [\![1,m]\!]$, $f_{|A|}$ is nondecreasing, $\varphi_A = \max_{j\in A} \mathbb{1}\big\{ i_A(p_j) > f_{|A|}(p_j) \big\}$.

If, furthermore, the function f_n is right continuous for all $n \in [\![1,m]\!]$ and the function sequence (f_n) is nondecreasing,

$$\hat{V}_\varphi^{\mathrm{IP}}(S) = \min_{0\leq k\leq |S|} \Big(\Big\lfloor f_{\hat{m}_0^{\mathrm{SC1}}}\big(p_{(k:S)}\big) \Big\rfloor + \big| S \setminus R\big(p_{(k:S)}\big) \big| \Big) \tag{3.27}$$

where

$$\hat{m}_0^{\mathrm{SC1}} = \max \Big\{ n \in [\![0,m]\!] : \forall t \in [0,1], f_n(t) \geq n + i(t) - m \Big\}. \tag{3.28}$$

The proof of this proposition is given in Appendix A.6.14.

In the remainder of this section, the local tests considered are the ones derived from Proposition 3.3.2, that is, we let $\alpha \in (0,1]$ and $\lambda_\alpha = \sqrt{\frac{\log(1/\alpha)}{2}}$, and for all $A \subseteq [\![1,m]\!]$, a local test is defined by $\varphi_A = \max_{t\in[0,1]} \varphi_{A,t}$, where, for all $t \in [0,1]$ and all $n \in [\![0,m]\!]$,

$$f_n(t) = nt + \sqrt{n}\lambda_\alpha,$$

and

$$\varphi_{A,t} = \mathbb{1}\{i_A(t) > f_{|A|}(t)\}. \tag{3.29}$$

Lemma 3.4.12. *Considering the DKW local tests, $\hat{m}_0^{\mathrm{SC1}}$ has the following closed form:*

$$\hat{m}_0^{\mathrm{SC1}} = \min_{t\in[0,1)} \left\lfloor \left(\frac{\lambda_\alpha}{2(1-t)} + \sqrt{\frac{\lambda_\alpha^2}{4(1-t)^2} + \frac{m-i(t)}{1-t}} \right)^2 \right\rfloor \wedge m. \tag{3.30}$$

This expression already appeared in Durand et al. (2020) and was used as a plug-in estimator for the adaptive top-k procedure derived from DKW inequality in Meah et al. (2024). Note that the principle of deriving an over-estimator for m_0 (or equivalently the proportion $\pi_0 = m_0/m$ of null hypotheses) from cdf confidence envelopes has been proposed and analyzed by Meinshausen and Rice (2006),Genovese and Wasserman (2004); but the above bound is sharper than those of these earlier references when specialized to the DKW inequality.

Remark 3.4.13. *Applying Lemma 3.4.12 and Equation (3.30) to a deterministic subset $R_k \subseteq [\![1,m]\!]$ and at level α/K, instead of $[\![1,m]\!]$ and α, exactly recovers the formula of ζ_k given in Equation (3.19).*

The next proposition proves that the adaptive top-k DKW method described in Meah et al. (2024) is an exact shortcut for the inversion procedure. Furthermore, this confidence bound is computable in time $\mathcal{O}\big(m\log(m)\big)$.

Proposition 3.4.14. *Considering DKW local test family, $\hat{V}_\varphi^{\mathrm{IP}} = \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ with $\mathfrak{R} = (R_k, \zeta_k)$ where for all $k \in [\![1,m]\!]$, $\zeta_k = k \wedge \left\lfloor \hat{m}_0^{\mathrm{SC1}} p_{(k)} + \sqrt{\hat{m}_0^{\mathrm{SC1}}}\lambda_\alpha \right\rfloor$ with $\hat{m}_0^{\mathrm{SC1}}$ defined by Equation (3.30), and $R_k = \{\sigma(i), 1 \le i \le k\}$ with σ a (stochastic) permutation ordering p-values by increasing order.*

We prove Lemma 3.4.12 and Proposition 3.4.14 in Appendix A.6.15.

3.5 Heterogeneous probabilistic inequalities and associated local test confidence envelopes

In this section, we prove new probabilistic inequalities adapted to Assumption 3.2 and Assumption 3.3. Section 3.5.1 revisits the so-called Bretagnolle inequality depicted in Shorack and Wellner (1986), an analogue of DKW inequality relying on a known data heterogeneity, and updates it thanks to a recent work (Reeve, 2024). Then, Section 3.5.2 introduces a new k-FWER inequality, using a FDR control method from Döhler et al. (2018) and extracts an m_0-adaptive version of this inequality. From all those new inequalities, we construct local tests. Note that a third approach using an inequality due to van Zuijlen (1978) is deferred to Appendix A.1 because of its lack of practical power.

3.5.1 Bretagnolle inequality and applications

The first inequality that we present is a slight refinement of an inequality found in Shorack and Wellner (1986) and that is attributed to Bretagnolle. It can be thought as a heterogeneous analogue of DKW inequality (Proposition 3.3.2).

Proposition 3.5.1 (Bretagnolle inequality)**.** *Let $X_1, \dots, X_N$ be N independent random variables, with respective cdf denoted by $J_1, \dots, J_N$. Assume that there exist some functions $K_1, \dots, K_N$ such that $J_i(x) \leq K_i(x)$ for all $i \in [\![1, N]\!], x \in \mathbb{R}$, and let $\bar{K} : x \mapsto \frac{1}{N}\sum_{i=1}^{N} K_i(x)$. Then, for all $\lambda \geq 0$,*

$$\mathbb{P}\left(\sqrt{N}\sup_{t\in\mathbb{R}}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{X_i \leq t\} - \bar{K}(t)\right) > \lambda\right) \leq e \cdot \exp\left(-2\lambda^2\right). \tag{3.31}$$

We defer a proof based on the one from Shorack and Wellner (1986) updated by the results from Reeve (2024) in Appendix A.6.16.

If the random variables $X_1, \dots, X_N$ are super-uniform and we use the cdf of the uniform distribution as $K_1, \dots, K_N$, then (3.31) becomes (3.5), that is DKW inequality, except that the right-hand side has a multiplicative factor of e, that we can think of as a penalty term due to the heterogeneity. In that sense, Bretagnolle inequality is indeed a heterogeneous analog of DKW inequality.

In the remainder of the section, we fix $\alpha \in (0, 1]$, define $\tilde{\lambda}_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$, the random vector $\mathbb{G}$ and the non-random vector $\mathbf{F}$ under Assumption 3.3:

$$\forall t \in [0, 1], \mathbb{G}(t) = \left(\mathbb{1}\{p_i \leq t\} - F_i(t)\right)_{1\leq i\leq m}, \tag{3.32}$$

$$\mathbf{F}(t) = \left(F_i(t)\right)_{1\leq i\leq m}. \tag{3.33}$$

For $A \subseteq [\![1, m]\!]$, we will consider the local test deduced from Bretagnolle inequality (in the meaning of Definition 3.3.8), $\varphi_A = \max_{t\in[0,1]} \varphi_{A,t}$, where for all $t \in [0, 1]$,

$$\varphi_{A,t} = \mathbb{1}\left\{\sum_{i\in A}\mathbb{G}_i(t) > \tilde{\lambda}_\alpha\sqrt{|A|}\right\}. \tag{3.34}$$

Corollary 3.5.2. *Assume Assumption 3.2 and Assumption 3.3 hold. Then, for all $A \subset [\![1, m]\!]$, φ_A is a local test at level α.*

The proof of the previous corollary is given in Appendix A.6.17.

Remark 3.5.3. *Note that $\tilde{\lambda}_\alpha = \lambda_{\tilde{\alpha}}$ with $\tilde{\alpha} = e^{-1}\alpha$. Thus when Assumption 3.3 holds with for all $i \in [\![1, m]\!]$, $F_i = F$, the Bretagnolle inequality reduces to the DKW inequality at level $e^{-1}\alpha$.*

Similarly to the local tests considered in the homogeneous setting, we get an expression of $\varphi_{S,n,t}$ (3.21) for the local test family aforementioned.

Lemma 3.5.4. *For all $S \subseteq [\![1, m]\!]$, $n \in [\![0, |S|]\!]$ and $i \in [\![1, m]\!]$,*

$$\varphi_{S,n,t} = \min_{0\leq j\leq m-|S|} \mathbb{1}\left\{\sum_{1\leq k\leq n}\left(\mathbb{G}(t)\right)_{(k:S)} + \sum_{1\leq k\leq j}\left(\mathbb{G}(t)\right)_{(k:S^c)} > \tilde{\lambda}_\alpha\sqrt{n+j}\right\}.$$

We defer the proof to Appendix A.6.18. This lemma allows us to compute a first confidence envelope thanks to Proposition 3.4.2 in the following proposition. This envelope is a shortcut computable in polynomial time for the inversion procedure applied to Bretagnolle local test.

Proposition 3.5.5. *Assume that Assumption 3.2 and Assumption 3.3 hold. Then,*

$$\hat{V}_{\varphi}^{\mathrm{SC1}}(S) = \max\left\{ n \in [\![0, |S|]\!] : \forall t \in [0,1], \exists j \leq m - |S|, \sum_{1\leq k\leq n} \big(\mathbb{G}(t)\big)_{(k:S)} + \sum_{1\leq k\leq j} \big(\mathbb{G}(t)\big)_{(k:S^c)} \leq \tilde{\lambda}_\alpha \sqrt{n+j} \right\}. \tag{3.35}$$

In particular, when $S = [\![1, m]\!]$, with probability $1-\alpha$,

$$\hat{m}_0^{\mathrm{SC1}} = \max\left\{ n \in [\![0, m]\!] : \forall t \in [0,1], \sum_{1\leq j\leq n} \big(\mathbb{G}(t)\big)_{(j)} \leq \tilde{\lambda}_\alpha \sqrt{n} \right\}. \tag{3.36}$$

Equation (3.36) allows the user to approximate the upper bound on m_0 given by the inversion procedure applied to φ_A, which may not be computable in practice, by a statistic computable in polynomial time. Indeed applying Proposition 3.4.3 to compute (3.36), we first need for all $t \in \{p_i, i \in [\![1, m]\!]\}$ to sort $\big(F_j(t)\big)_{j\in R(t)}$ which is in time $\mathcal{O}\big(\log(m)m^2\big)$. Then the following identity for $n > m - i(t)$:

$$\sum_{1\leq j\leq n} \big(\mathbb{G}(t)\big)_{(j)} = n - \big(m - i(t)\big) - \sum_{j\in R(t)^c} F_j(t) - \sum_{1\leq j\leq n-(m-i(t))} \big(\mathbf{F}(t)\big)_{[j:R(t)]}$$

allows to compute the matrix $\left(\sum_{1\leq j\leq n} \big(\mathbb{G}(p_i)\big)_{(j)}\right)_{(n,i)\in[\![1,m]\!]^2}$ in time $\mathcal{O}\big(m^2\big)$, and finally we can find the maximum in (3.36) in time $\mathcal{O}\big(m^2\big)$. Hence, $\hat{m}_0^{\mathrm{SC1}}$ is computable in time $\mathcal{O}\big(\log(m)m^2\big)$ for Bretagnolle local test family.

By the same type of reasoning, $\hat{V}_{\varphi}^{\mathrm{SC1}}(S)$ can be computed in time $\mathcal{O}\big(m^3\big)$.

Similar to previous adaptive procedures that plug in an over-estimator of m_0 (*e.g.* Proposition 3.4.14), we are interested in using an over-estimator $\hat{m}_0$, such as the one defined by (3.36), in combination with Bretagnolle inequality. The following proposition states an adaptive version of the inequality by plug-in, suited to deriving such an adaptive procedure, which will be done in Section 3.6.1.

Proposition 3.5.6. *Assume Assumption 3.2 and Assumption 3.3. Let $\hat{m}_0$ such that*

$$\left\{ \sup_{t\in[0,1]} \sum_{k\in\mathcal{H}_0} \mathbb{G}_k(t) \leq \sqrt{m_0}\tilde{\lambda}_\alpha \right\} \subseteq \{m_0 \leq \hat{m}_0\},$$

then, with $\mathbf{F}$ defined by (3.33),

$$\mathbb{P}\left(\exists t \in [0,1], i_{\mathcal{H}_0}(t) > \sum_{1\leq k\leq \hat{m}_0} \big(\mathbf{F}(t)\big)_{[k]} + \sqrt{\hat{m}_0}\tilde{\lambda}_\alpha \right) \leq \alpha.$$

The proof is given in Appendix A.6.19.

3.5.2 Heterogeneous Simes-like inequality

In this section, we turn to a different type of concentration inequality, which we derive from an existing step-up FDR control procedure designed for the heterogeneous setting (Döhler et al., 2018). This is similar to the way one can recover Simes inequality from the Benjamini-Hochberg procedure in the homogeneous setting.

To this end we define, for a subset of hypotheses $U \subseteq [\![1,m]\!]$ and a threshold family $\tau = (\tau_k)_{1\leq k\leq m}$ (*i.e.* a nondecreasing family of m elements of $\mathbb{R}_+$), the step-up procedure on U with respect to τ; $SU(\tau,U) = \{i \in U : p_i \leq \tau_{\hat{k}}\}$, where $\hat{k} = \max\{k \in [\![0,|U|]\!] : p_{(k:U)} \leq \tau_k\}$, with, by convention, $p_{(0:U)} = 0$ and $\tau_0 = \tau_1$. We will consider, for $X \in \mathbb{R}^d$ $(d \in \mathbb{N})$, that $X_{[d+1]} = +\infty$. This last convention is merely considered to get aesthetic expressions in Corollary 3.5.15, similarly to the convention $\ell_{0:n} = -1$ in Section 3.4.2.

If Assumption 3.3 holds, we denote, for all $(s,t) \in (\mathbb{R}_+)^2$ such that $F_i(s) < 1$,

$$\mathbf{H}(s,t) = \left(\frac{F_i(t)}{1-F_i(s)}\right)_{1\leq i\leq m}.$$

We introduce a condition involving the vector $\mathbf{H}$ on a threshold family.

Definition 3.5.7. *We say that the threshold family τ satisfies (C1) at level $\alpha \in [0,1]$ with respect to $U \subseteq [\![1,m]\!]$ if $F_i(\tau_{|U|}) < 1$ for all $i \in U$ and*

$$\max_{1\leq k\leq |U|}\left\{\max_{\substack{A\subseteq U\\ |A|=|U|-k+1}}\left\{\frac{1}{k}\sum_{i\in A}\mathbf{H}_i(\tau_{|U|},\tau_k)\right\}\right\} \leq \alpha. \tag{C1}$$

The condition (C1) on the threshold family τ is relevant to get a control on the FDR at level α in addition to the the result of Döhler et al. (2018) that we recall:

Theorem 3.5.8 (Döhler et al., 2018, Theorem 1)**.** *Assume Assumption 3.2 and Assumption 3.3.*

Let $U \subseteq [\![1,m]\!]$ and $\alpha \in [0,1]$. Furthermore, assume that the two families $(p_i)_{i\in U\cap\mathcal{H}_0}$ and $(p_i)_{i\in U\cap\mathcal{H}_1}$ are independent.

Let τ be a threshold family satisfying (C1) at level α with respect to U. Then,

$$\mathrm{FDR}\left(SU(\tau,U)\right) \leq \alpha.$$

The contribution of $\tau_{|U|}$ in (C1) is different from the contributions of the other thresholds. This asymmetry naturally results in letting $\tau_{|U|}$ parametrize a sequence of thresholds, like it is designed in the following example.

Example 3.5.9. *Let $\alpha \in (0,1]$ and*

$$\lambda \in \left\{t \in \mathcal{A} : \forall i \in [\![1,m]\!], F_i(t) < 1, \text{ and } \left(\mathbf{H}(t,t)\right)_{[1]} \leq m\alpha\right\}, \tag{3.37}$$

and, for all $k \in [\![1,m-1]\!]$, let

$$\tau_k = \max\left\{t \in \mathcal{A}, t \leq \lambda : \left(\mathbf{H}(\lambda,t)\right)_{[1]} + \cdots + \left(\mathbf{H}(\lambda,t)\right)_{[m-k+1]} \leq k\alpha\right\}. \tag{3.38}$$

Then $\tau = (\tau_1,\ldots,\tau_{m-1},\lambda)$ satisfies (C1) at level α with respect to $[\![1,m]\!]$.
Thus $\mathrm{FDR}\left(SU\left(\tau,[\![1,m]\!]\right)\right) \leq \alpha$.

Remark 3.5.10. *In comparison to Döhler et al. (2018), we introduce an additional flexibility in the threshold family via the parameter λ. We can recover the threshold sequence of the [AHSU] procedure of Döhler et al. (2018) with the particular choice*

$$\lambda = \max\left\{ t \in \mathcal{A} : \sum_{i=1}^{m} \mathbf{H}_i(t,t) \leq m\alpha \right\}$$

in Example 3.5.9. The role of the parameter λ is analogous to the parameter of the adaptive one-stage step-up procedure of Blanchard and Roquain (2009).

An other fundamental example is the following.

Example 3.5.11. *Let $\alpha \in (0,1]$, $U \subseteq [\![1,m]\!]$ and*

$$\lambda \in \left\{ t \in \mathcal{A} : \forall i \in U, F_i(t) < 1, \text{ and } \big(\mathbf{H}(t,t)\big)_{[1:U]} \leq |U|\alpha \right\}, \tag{3.39}$$

and, for all $k \in [\![1, |U|-1]\!]$, let

$$\tilde{\tau}_k = \max\left\{ t \in \mathcal{A}, t \leq \lambda : \big(\mathbf{H}(\lambda,t)\big)_{[1:U]} + \cdots + \big(\mathbf{H}(\lambda,t)\big)_{[|U|-k+1:U]} \leq k\alpha \right\}. \tag{3.40}$$

Then $\tilde{\tau} = \big(\tau_1, \ldots, \tau_{|U|-1}, \lambda, \bar{\mathcal{A}}, \ldots, \bar{\mathcal{A}}\big)$ where $\bar{\mathcal{A}} = \sup \mathcal{A}$, satisfies (C1) at level α with respect to U.

This last example will be particularly useful with $U = \mathcal{H}_0$, in combination with the following lemma which states a practical monotonicity property of the condition (C1).

Lemma 3.5.12. *Let τ be a threshold family from $\mathcal{A}$, $V \subseteq [\![1,m]\!]$ and $\alpha \in (0,1]$. Assume that τ satisfies (C1) at level α with respect to V.*
Then for all $U \subseteq V$, for $\tilde{\tau}$ defined by (3.40) with $\lambda \in \{\tau_{|U|}, \tau_{|V|}\}$, $\tau_k \leq \tilde{\tau}_k$ holds for all $k \in [\![1,m]\!]$.
In particular, τ satisfies (C1) at level α with respect to all $U \subseteq V$.

We prove Lemma 3.5.12 in Appendix A.6.20.

In particular, this last lemma states with $U = V$ that Example 3.5.11 gives an optimal threshold family satisfying (C1) with respect to V, up to the choice of $\tau_{|V|}$. Furthermore, it states that a threshold family satisfying (C1) at level α with respect to $[\![1,m]\!]$ also satisfies (C1) at level α with respect to $\mathcal{H}_0$, especially the thresholds defined in Example 3.5.9.

From the FDR control of a general step-up procedure with threshold family τ satisfying (C1), we deduce an analogue of Simes inequality (Proposition 3.3.4), using τ instead of the family $\left(\frac{\alpha k}{m}\right)_{1\leq k\leq m}$.

Theorem 3.5.13 (Heterogeneous Simes-like inequality)**.** *Let $U \subseteq [\![1,m]\!]$ such that $\mu \in H_U$. Let τ be a threshold family satisfying (C1) at level $\alpha \in [0,1]$ with respect to U, and assume Assumption 3.2 and Assumption 3.3. Then, the following inequality holds:*

$$\mathbb{P}\big(\exists i \in [\![1,|U|]\!],\, p_{(i:U)} \leq \tau_i\big) \leq \alpha.$$

Proof.

$$\begin{aligned}
&\mathbb{P}\big(\exists i \in [\![1,|U|]\!],\, p_{(i:U)} \leq \tau_i\big) = \mathrm{FWER}\left(SU\big((\tau_k)_{1\leq k\leq m_0}, U\big)\right) \\
&= \mathrm{FDR}\left(SU\big((\tau_k)_{1\leq k\leq m_0}, U\big)\right) \text{ since each null hypothesis in } U \text{ is true,} \\
&\leq \alpha,
\end{aligned}$$

where the inequality comes from Theorem 3.5.8 applied to U.

□

Remark that, in opposition to Theorem 3.5.8, this last theorem does not require any assumptions on $\mathcal{H}_1$, therefore the assumptions on the null hypotheses are the same as the ones mandatory in Section 3.5.1. Theorem 3.5.13, in addition to Lemma 3.5.12, entails the following corollary:

Corollary 3.5.14. *Let τ be a threshold family satisfying (C1) at level $\alpha \in [0,1]$ with respect to $[\![1,m]\!]$, and assume Assumption 3.2 and Assumption 3.3. Then, the following inequality holds:*

$$\mathbb{P}\big(\exists i \in [\![1,m_0]\!],\, p_{(i:\mathcal{H}_0)} \leq \tau_i\big) \leq \alpha.$$

The first fundamental application of Theorem 3.5.13 is to define local tests in the heterogeneous setting.

Corollary 3.5.15 (Simes-like heterogeneous local test)**.** *Assume Assumption 3.2 and Assumption 3.3. Let $\alpha \in (0,1]$ and*

$$\lambda \in \Big\{ t \in \mathcal{A} : \forall i \in [\![1,m]\!], F_i(t) < 1, \text{ and } \big(\mathbf{H}(t,t)\big)_{[1]} \leq \alpha \Big\}. \tag{3.41}$$

Let $A \subseteq [\![1,m]\!]$ and define $k = |A|$. Then,

$$\varphi_A = \max_{i \in [\![1,k]\!]} \mathbb{1}\left\{ \sum_{\ell=1}^{k-i+1} \Big(\mathbf{H}\big(\lambda, p_{(i:A)}\big)\Big)_{[\ell:A]} \leq i\alpha \right\} \mathbb{1}\big\{p_{(i:A)} \leq \lambda\big\} \tag{3.42}$$

$$= \max_{t \in [0,\lambda]} \mathbb{1}\left\{ \sum_{\ell=1}^{k-i_A(t)+1} \big(\mathbf{H}(\lambda,t)\big)_{[\ell:A]} \leq i_A(t)\alpha \right\}, \tag{3.43}$$

where $i_A(t)$ is defined by (3.6), is a local test of level α.

The proof of the validity of these local tests is given in Appendix A.6.21.

Remark 3.5.16. *Here, our local tests are defined with a "$\max_{t\in[0,\lambda]}$" instead of a "$\max_{t\in[0,1]}$". All the propositions from Section 3.4.1 can easily be adapted to this case.*

This local test family, thanks to the mild assumption on λ (3.41), derives a computable expression of $\varphi_{S,n,t}$ (3.21) stated in the next lemma.

Lemma 3.5.17. *Assume Assumption 3.2 and Assumption 3.3. Let $\alpha \in [0,1]$, $S \subseteq [\![1,m]\!]$, $n \in [\![0,|S|]\!]$, λ defined by (3.41), $t \in [0,\lambda]$ and consider the local tests from Corollary 3.5.15. Define· $k_S(t) = (n - |R(t)^c \cap S|) \vee 0$ and $S(t) = R(t) \cap S$, then,*

$$\varphi_{S,n,t} = \mathbb{1}\left\{ \sum_{k=1}^{m-i(t)+1} \Big(\mathbf{H}_{R(t)^c}(\lambda,t), \big(\mathbf{H}(\lambda,t)\big)_{[1:S(t)]}, \ldots, \big(\mathbf{H}(\lambda,t)\big)_{[k_S(t):S(t)]}\Big)_{[k]} \leq k_S(t)\alpha \right\}.$$

This previous lemma is proven in Appendix A.6.22. It allows us to easily express the confidence envelopes achieved by the combination of the local tests described in Corollary 3.5.15 and Proposition 3.4.2.

Proposition 3.5.18. *Assume Assumption 3.2 and Assumption 3.3. Let $\alpha \in [0,1]$, $S \subseteq [\![1,m]\!]$, $n \in [\![0,|S|]\!]$, λ defined by (3.41) and consider the local tests from Corollary 3.5.15. Define for all $t \in [0,\lambda]$, $k_S(t) = (n - |R(t)^c \cap S|) \vee 0$ and $S(t) = R(t) \cap S$, then:*

$$\hat{V}_{\varphi}^{\mathrm{SC1}}(S) = \max\left\{ n \in [\![0,|S|]\!] : \forall t \in [0,\lambda],\ \sum_{k=1}^{m-i(t)+1} \Big(\mathbf{H}_{R(t)^c}(\lambda,t), \big(\mathbf{H}(\lambda,t)\big)_{[1:S(t)]}, \ldots, \big(\mathbf{H}(\lambda,t)\big)_{[k_S(t):S(t)]}\Big)_{[k]} > k_S(t)\alpha \right\}. \tag{3.44}$$

In particular, when $S = [\![1, m]\!]$, with probability $1 - \alpha$,

$$\hat{m}_0^{\mathrm{SC1}} = \max\left\{ n \in [\![0, m]\!] : \forall t \in [0, \lambda], \sum_{k=1}^{m-i(t)+1} \left(\mathbf{H}_{R(t)^c}(\lambda, t), \left(\mathbf{H}(\lambda, t)\right)_{[1:R(t)]}, \ldots, \left(\mathbf{H}(\lambda, t)\right)_{[k(t):R(t)]} \right)_{[k]} > k(t)\alpha \right\}. \tag{3.45}$$

Note that the computation of $\hat{m}_0^{\mathrm{SC1}}$ is in time $\mathcal{O}\big(\log(m)m^2\big)$ similarly to the framework of Bretagnolle's local test family.

The second fundamental application of Theorem 3.5.13 is to design adaptive, data-dependent threshold families. The idea is that we can replace theoretical thresholds satisfying condition (C1) depending on the (unknown) $\mathcal{H}_0$ by data-dependent thresholds that are smaller than the theoretical ones on a suitable event.

Lemma 3.5.19 (Data-dependent threshold family)**.** *Let $\alpha \in [0, 1]$ and $\tilde{\tau}$ be a threshold family satisfying (C1) at level α with respect to $\mathcal{H}_0$. Suppose Assumption 3.2 and Assumption 3.3 hold. Then, for any (possibly data-dependent) threshold family τ such that $\tau_k \leq \tilde{\tau}_k$ for all $k \in [\![1, m_0]\!]$ whenever the event $\left\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\}$ is realized, the following inequality holds:*

$$\mathbb{P}\big(\exists i \in [\![1, m_0]\!], p_{(i:\mathcal{H}_0)} \leq \tau_i\big) \leq \alpha.$$

Proof. Define the events $N = \{\forall k \in [\![1, m_0]\!], \tau_k \leq \tilde{\tau}_k\}$, $\tilde{B} = \left\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\}$ and $B = \left\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tau_i\right\}$. The assumption on τ translates to $\tilde{B} \subseteq N$, hence $N \cap \tilde{B} = \tilde{B}$. Furthermore $N \cap \tilde{B} \subseteq B$. Therefore $\tilde{B} \subseteq B$, and $\mathbb{P}\big(\tilde{B}\big) \geq 1 - \alpha$ by Theorem 3.5.13 with $U = \mathcal{H}_0$, thus $\mathbb{P}(B) \geq 1 - \alpha$. □

Lemma 3.5.19 will be useful to define a m_0-adaptive form of the first threshold family from Example 3.5.9 in the following example:

Proposition 3.5.20. *Let $\alpha \in (0, 1]$, $\tilde{\tau}$ be defined by Example 3.5.11 with $U = \mathcal{H}_0$ and $\hat{m}_0$ be a statistic such that*

$$\left\{\forall i \in [\![1, m_0]\!], p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\} \subseteq \{m_0 \leq \hat{m}_0\}.$$

Furthermore, assume Assumption 3.2 and Assumption 3.3. Define $\tau_{\hat{m}_0} = \tilde{\tau}_{m_0}$ and

$$\forall k \in [\![1, \hat{m}_0 - 1]\!], \tau_k = \max\left\{ t \in \mathcal{A}, t \leq \tilde{\tau}_{\hat{m}_0} : \left(\mathbf{H}(\tau_{\hat{m}_0}, t)\right)_{[1]} + \cdots + \left(\mathbf{H}(\tau_{\hat{m}_0}, t)\right)_{[\hat{m}_0 - k + 1]} \leq k\alpha \right\}. \tag{3.46}$$

Then $\tau = \big(\tau_1, \ldots, \tau_{\hat{m}_0}, \bar{\mathcal{A}}, \ldots, \bar{\mathcal{A}}\big)$ satisfies the inequality

$$\mathbb{P}\big(\exists i \in [\![1, m_0]\!], p_{(i:\mathcal{H}_0)} \leq \tau_i\big) \leq \alpha.$$

Proof. We know that under the event $\left\{\forall i \in [\![1, m_0]\!], p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\right\}$, the event $\{m_0 \leq \hat{m}_0\}$ occurs, thus, for all $U \supseteq \mathcal{H}_0$ such that $|U| = \hat{m}_0$, τ satisfies condition (C1) at level α with respect to U by Equation (3.46). Therefore, by Lemma 3.5.12, $\tau_k \leq \tilde{\tau}_k$, hence Lemma 3.5.19 indeed applies. □

Proposition 3.5.20 allows us to derive adaptive procedures in Section 3.6.2 using two different upper bounds on m_0.

Remark 3.5.21. *In the statement of Proposition 3.5.20, $\tilde{\tau}$ can be considered as an "oracle" threshold family for the parameter $\lambda = \tilde{\tau}_{m_0}$, because it relies on the knowledge of $\mathcal{H}_0$. However, defining λ by (3.38) with $k = m_0$ or by (3.41) implies in particular that λ verifies (3.39) with $U = \mathcal{H}_0$, thus the knowledge of $\mathcal{H}_0$ is not mandatory to define adaptive thresholds.*

3.6 Heterogeneous confidence envelopes using reference families

This section formally introduces our new heterogeneous confidence envelopes derived from the heterogeneous inequalities depicted in Section 3.5, following the methodology introduced in Section 3.4.2.

Note that these inequalities have already been used to build confidence envelopes in Section 3.5 thanks to the local test formalism — derived from inexact shortcuts for the inversion procedure. Nevertheless, these confidence envelopes have a complexity in $\mathcal{O}(m^3)$ in general, where the one we will exhibit here are in time $\mathcal{O}\big(m^2 \log(m)\big)$.

Finally, recall that for a nested reference family $\mathfrak{R} = (R_k, \zeta_k)_{1 \leq k \leq m}$, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is defined by:

$$\forall S \subseteq [\![1, m]\!], \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S) = \min_{k \in [\![1,m]\!]} (\zeta_k + |S \setminus R_k|) \wedge |S|.$$

3.6.1 Top-k setting

This section will introduce our new bounds relying on the Bretagnolle inequality from Section 3.5.1. We will first introduce the non-adaptive bound which is a consequence of our new adaptive bound on Section 3.6.1. Recall that, in the top-k setting, $R_k = \{\sigma(i), 1 \leq i \leq k\}$ with σ a (stochastic) permutation ordering p-values by increasing order.

Non adaptive bound

The first bound that we introduce here is the nonadaptive one.

Corollary 3.6.1. *Let $\alpha \in]0, 1[$. Assume Assumption 3.2 and Assumption 3.3. Define*

$$\tilde{\lambda}_\alpha = \sqrt{\frac{1 + \log(1/\alpha)}{2}} \text{ and } \mathfrak{R} = (R_k, \zeta_k)_{1 \leq k \leq m},$$

where for all $k \in [\![1, m]\!]$,

$$\zeta_k = k \wedge \left\lfloor m\bar{F}\big(p_{(k)}\big) + \sqrt{m}\tilde{\lambda}_\alpha \right\rfloor.$$

Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is a confidence bound of level α.

Proof. It is a direct application of Proposition 3.5.6 with $\hat{m}_0 = m$ knowing that

$$|\mathcal{H}_0 \cap R_k| \leq i_{\mathcal{H}_0}\big(p_{(k)}\big) \text{ for all } k \in [\![1, m]\!]$$

. □

Note that the previous confidence bound is computed in time $\mathcal{O}\big(m \log(m)\big)$.

Compared to the DKW calibration, the Bretagnolle calibration does not use an upper bound F on the F_i but the mean $\bar{F}$ of the individual F_i, which is always smaller than F, and $\tilde{\lambda}_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$ which is always greater than the constant $\sqrt{\frac{\log(1/\alpha)}{2}}$ used in the case of the DKW calibration. Note that this slight change on $\tilde{\lambda}_\alpha$ has consequences as it is discussed in Section 3.7.

This envelope has the advantage, like all other nonadaptive envelope, to upper bound $|R(t) \cap \mathcal{H}_0|$ deterministically, and thus gives the experimenter the opportunity to decide which bound to use according to the domain of p-values on which they want the better envelope (without looking at the p-values).

Adaptive bound

The following bounds is the adaptive version of the Bretagnolle calibration.

Theorem 3.6.2. *Let $\alpha \in (0,1]$. Assume Assumption 3.2 and Assumption 3.3. Define $\hat{m}_0^{\mathrm{SC1}}$ by (3.36), $\tilde{\lambda}_\alpha = \sqrt{\frac{1+\log(1/\alpha)}{2}}$ and $\mathfrak{R} = (R_k, \zeta_k)_{1\le k\le m}$ where for all $k \in [\![1,m]\!]$,*

$$\zeta_k = k \wedge \hat{m}_0^{\mathrm{SC1}} \wedge \left\lfloor \sum_{1\le i\le \hat{m}_0^{\mathrm{SC1}}} \Big(\mathbf{F}\big(p_{(k)}\big)\Big)_{[i]} + \sqrt{\hat{m}_0^{\mathrm{SC1}}}\tilde{\lambda}_\alpha \right\rfloor. \tag{3.47}$$

Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is a confidence bound of level α.

Proof. By Proposition 3.5.6, $\mathbb{P}\Big(\exists t \in [0,1], i_{\mathcal{H}_0}(t) > \sum_{1\le k\le \hat{m}_0^{\mathrm{SC1}}} \big(\mathbf{F}(t)\big)_{[k]} + \sqrt{\hat{m}_0^{\mathrm{SC1}}}\tilde{\lambda}_\alpha\Big) \le \alpha$ because $\left\{ \sup_{t\in[0,1]} \sum_{k\in\mathcal{H}_0} \mathbb{G}_k(t) \le \sqrt{m_0}\tilde{\lambda}_\alpha \right\} = \{\varphi_{\mathcal{H}_0} = 0\} \subseteq \left\{m_0 \le \hat{m}_0^{\mathrm{SC1}}\right\}$ with $\varphi_{\mathcal{H}_0}$ defined by (3.34). The conclusion follows from the inequality $|\mathcal{H}_0 \cap R_k| \le i_{\mathcal{H}_0}\big(p_{(k)}\big)$ for all $k \in [\![1,m]\!]$. □

Note that the previous confidence bound is computed in time $\mathcal{O}\big(m^2\log(m)\big)$, because the family $(\zeta_k)_{k\in[\![1,m]\!]}$ is computed in time $\mathcal{O}\big(m^2\log(m)\big)$. Then, the effective computation of $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S)$ is in time $\mathcal{O}(|S|)$, which is convenient for exploratory purposes.

This adaptive top-k procedure reduces to the one derived from the DKW inequality described in Proposition 3.4.14 when $F_k = F_1$ for all $k \in [\![1,m]\!]$ with $\tilde{\lambda}_\alpha$ instead of λ_α. Thus, when no heterogeneity occurs in the data, the adaptive top-k confidence bound, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ with $\mathfrak{R}$ defined in Theorem 3.6.2 is equal to the bound $\hat{V}_{\varphi}^{\mathrm{IP}}$. Hence, under this assumption, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}} \le \hat{V}_{\varphi}^{\mathrm{SC1}}$. Nevertheless, under heterogeneity, the inequality $\hat{V}_{\varphi}^{\mathrm{SC1}}(S) < \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S)$ can occur, as it is illustrated by the next example.

Example 3.6.3. *Let $\alpha > \exp\big(-\frac{5}{3}\big)$ and $\frac{2-\tilde{\lambda}_\alpha\sqrt{3}}{2} > \varepsilon > 0$. Define $a_1 = \frac{2-\tilde{\lambda}_\alpha\sqrt{3}}{2} - \varepsilon_1$ and $a_2 = \frac{2-\tilde{\lambda}_\alpha\sqrt{3}}{2}$. Assume that $p_1 = a_1$, $p_2 = a_2$, $p_3 = a_2$ and $p_4 = 1$ are such that under their respective null hypotheses,*

- $\mu_{p_1} = a_1\delta_{a_1} + (1-a_1)\delta_1$,
- $\mu_{p_2} = a_2\delta_{a_2} + (1-a_2)\delta_1 = \mu_{p_3}$,
- $\mu_{p_4} = \delta_1$.

Then, $\hat{V}_{\varphi}^{\mathrm{SC1}}\big(\{1,2,4\}\big) < \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}\big(\{1,2,4\}\big)$ with $\mathfrak{R}$ defined in Theorem 3.6.2.

In Appendix A.6.23, we show the statement of Example 3.6.3.

Remark 3.6.4. *Since the validity of $\hat{V}_{\varphi}^{\mathrm{SC1}}$ and $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ rely on the same event that holds with high probability·, we define a confidence bound with the following: $\hat{V} = \hat{V}_{\varphi}^{\mathrm{SC1}} \wedge \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$. This confidence bound inherits of the time complexity of the two bounds, but is valid and uniformly more powerful than each other.*

Proposition 3.6.5. *The adaptive top-k procedure described in Theorem 3.6.2 is a uniform improvement of the inversion procedure applied to the homogenized Bretagnolle local test family:*

$$\forall A \subseteq [\![1,m]\!], \varphi_A^{hom} = \max_{t\in\mathbb{R}} \mathbb{1}\big\{i_A(t) > f_{|A|}(t)\big\},$$

where, for all $t \in \mathbb{R}$, and $n \in [\![1,m]\!]$, $f_n(t) = \sum_{1\le i\le n} \big(\mathbf{F}(t)\big)_{[i]} + \tilde{\lambda}_\alpha\sqrt{n}$.

The proof of Proposition 3.6.5 is deferred to Appendix A.6.24

3.6.2 Simultaneous k-FWER

In this section, recall that the k-FWER setting assume to have a Simes-type inequality, that is, equivalently, some family of local test of level α expressed as in (3.7). Then, the appropriate confidence envelope can be obtained as $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$, where

$$\mathfrak{R} = \big(\{i \in [\![1,m]\!] : p_i \leq \ell_{k:n}\}, k-1\big)_{1\leq k\leq m}$$

with n possibly data dependant, such that $\mathbb{P}\big(\exists k \in [\![1,m_0]\!] : p_{(k:\mathcal{H}_0)} \leq \ell_{k:n}\big) \leq \alpha$.

Non adaptive envelope

Applying Corollary 3.5.14 and Example 3.5.9, we easily derive a confidence envelope:

Theorem 3.6.6. *Let $\alpha \in [0,1]$ and assume Assumption 3.2 and Assumption 3.3. Define a threshold family τ by (3.38) and $\mathfrak{R} = \big(\{i \in [\![1,m]\!] : p_i \leq \tau_k\}, k-1\big)_{1\leq k\leq m}$.*
Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is a confidence bound of level α.

This confidence envelope gives a convenient over-estimator of m_0 to get an adaptive confidence envelope. Indeed,

$$\hat{m}_0^{\mathrm{JER}} := \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}\big([\![1,m]\!]\big) = \min_{1\leq k\leq m}\Big\{\big|\{i \in [\![1,m]\!] : p_i > \tau_k\}\big| + k - 1\Big\} \tag{3.48}$$

verifies the assumption $\big\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\big\} \subseteq \{m_0 \leq \hat{m}_0\}$ from Proposition 3.5.20, because $\big\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\big\} \subseteq \big\{\forall i \in \mathcal{H}_0, p_{(i:\mathcal{H}_0)} > \tau_i\big\}$. Thus it will allow us to define a family of adaptive thresholds thanks to Proposition 3.5.20.

Adaptive envelopes

Recall here, for $(s,t) \in \mathbb{R}^2$ such that for all $i \in [\![1,m]\!]$, $F_i(s) < 1$, $\mathbf{H}(s,t) = \left(\frac{F_i(t)}{1-F_i(s)}\right)_{i\in[\![1,m]\!]}$.
We provide here two adaptive envelopes derived from Section 3.5.2. The first one will be computed thanks to the expression of $\hat{m}_0^{\mathrm{JER}}$ defined by (3.48):

Theorem 3.6.7. *Let $\alpha \in [0,1]$ and assume Assumption 3.2 and Assumption 3.3. Define a threshold family $\tau^{(1)}$ by (3.46), with $\hat{m}_0^{\mathrm{JER}}$ expressed by (3.48), $\lambda = \tau_{\hat{m}_0^{\mathrm{JER}}}$ and*

$$\mathfrak{R} = \left(\left\{i \in [\![1,m]\!] : p_i \leq \tau_k^{(1)}\right\}, k-1\right)_{1\leq k\leq \hat{m}_0^{\mathrm{JER}}+1}.$$

Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is a confidence bound of level α.

Proof. Define $\tilde{\tau}$ by Example 3.5.11 with $\lambda = \tau_{m_0}$ and $U = \mathcal{H}_0$. Assume that the event $\big\{\forall i \in [\![1,m_0]\!], p_{(i:\mathcal{H}_0)} > \tilde{\tau}_i\big\}$ occurs. Then $m_0 \leq \hat{m}_0^{\mathrm{JER}}$ therefore $\tau_{\hat{m}_0^{\mathrm{JER}}} \geq \tau_{m_0}$, thus $\tau_k^{(1)} \leq \tilde{\tau}_k$ for all $k \leq \hat{m}_0^{\mathrm{JER}}$, and Lemma 3.5.19 concludes. □

Observe that we can re-enact this step of adaptivity to improve our bound again. Thus, a strategy would be to repeat this plug-in steps until $\hat{m}_0^{\mathrm{JER}}$ stops decreasing. This step-down strategy is classical in multiple testing literature and can be traced back to Holm (1979). It has also been used for confidence envelopes in Blanchard et al. (2020a).

The second envelope requires a more stringent assumption on λ.

Theorem 3.6.8. *Let $\alpha \in [0,1]$ and assume Assumption 3.2 and Assumption 3.3. Define a threshold family $\tau^{(2)}$ by (3.46) with λ verifying (3.41), that is*

$$\lambda \in \Big\{ t \in \mathcal{A} : \forall i \in [\![1,m]\!], F_i(t) < 1, \text{ and } \big(\mathbf{H}(t,t)\big)_{[1]} \leq \alpha \Big\},$$

and $\hat{m}_0^{\mathrm{SC1}}$ defined by (3.45) and

$$\mathfrak{R} = \left(\left\{ i \in [\![1,m]\!] : p_i \leq \tau_k^{(2)} \right\}, k-1 \right)_{1 \leq k \leq \hat{m}_0^{\mathrm{SC1}}+1}. \tag{3.49}$$

Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ is a confidence bound of level α.

Proof. Apply Proposition 3.5.20 with $\hat{m}_0 = \hat{m}_0^{\mathrm{SC1}}$.

□

This assumption on λ will generally be benign with respect to the thresholds range; indeed, in the case of the vanilla Simes inequality thresholds, the larger threshold is equal to α, and here, under the assumption that $F_i(t) = t$ for all $t \in \mathcal{A}_i$, $\lambda = \frac{\alpha}{1+\alpha}$ satisfies (3.41).

Proposition 3.6.9. *Define for $n \leq m$ and λ which verifies (3.41), and define for all $i \leq n$,*

$$\ell_{i:n} = \max \Big\{ t \in \mathcal{A}, t \leq \lambda : \big(\mathbf{H}(\lambda,t)\big)_{[1]} + \cdots + \big(\mathbf{H}(\lambda,t)\big)_{[n-i+1]} \leq i\alpha \Big\}.$$

The family $(\ell_{i:n})$ yields a homogeneous Simes-like local test family at level α:

$$\begin{aligned} \forall A \subseteq [\![1,m]\!], \varphi_A^{hom} &= \max_{t \in [0,\lambda]} \mathbb{1}\left\{ \sum_{\ell=1}^{|A|-i_A(t)+1} \big(\mathbf{H}(\lambda,t)\big)_{[\ell]} \leq i_A(t)\alpha \right\} \\ &= \max_{1 \leq i \leq |A|} \mathbb{1}\big\{ p_{(i:A)} \leq \ell_{i:|A|} \big\}. \end{aligned}$$

Then, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}} \leq \hat{V}_{\varphi^{hom}}^{\mathrm{SC1}}$ with $\mathfrak{R}$ defined by (3.49).

The proof of the previous proposition is given in Appendix A.6.25.

Remark 3.6.10. *The adaptive envelopes from Theorem 3.6.7 and Theorem 3.6.8 differ from the shortcut for the inversion procedure described by Proposition 3.3.12 applied to φ^{hom} by the over estimators of m_0 considered. The one provided by Proposition 3.3.12 is*

$$\hat{m}_0^{hom} = \inf_{t \in (0,\lambda]} \left(\left\lceil \frac{1}{\alpha} \sum_{k=1}^{m-i(t)+1} \big(\mathbf{H}(\lambda,t)\big)_{[k]} \right\rceil + m - (i(t)+1) \right) \wedge \big(m - i(0)\big),$$

and is lower bounded by $\hat{m}_0^{\mathrm{SC1}}$ expressed by (3.45), hence the adaptive envelope from Theorem 3.6.8 is uniformly better than the one provided by the inversion procedure applied to the homogenized local tests.

3.6.3 Deterministic setting

This section focuses on applying the Bretagnolle m_0 estimation on the framework from Durand et al. (2020), that we recall here: assume that the family $(R_k)_{k\in\mathcal{K}}$ is deterministic, that is $\mathcal{K}$ is deterministic and, for all $k \in \mathcal{K}$, R_k is deterministic.

Theorem 3.6.11. *Let $\alpha \in (0, 1]$. Assume Assumption 3.3 and that for all $k \in \mathcal{K}$, $(p_i)_{i\in R_k\cap\mathcal{H}_0}$ is a family of independent variables. For all $k \in \mathcal{K}$, define:*

$$\zeta_k = \max\left\{ n \in [\![0, |R_k|]\!] : \forall i \in R_k, \sum_{1\le j\le n} \big(\mathbb{G}(p_i)\big)_{(j:R_k)} \le \tilde{\lambda}_{\frac{\alpha}{|\mathcal{K}|}}\sqrt{n} \right\}.$$

Then, the reference family $\mathfrak{R} = (R_k, \zeta_k)_{k\in\mathcal{K}}$ has its JER *bounded by α.*

Proof. For all $k \in \mathcal{K}$, by Proposition 3.5.5 applied with R_k instead of $[\![1, m]\!]$ and $\frac{\alpha}{|\mathcal{K}|}$ instead of α, $\mathbb{P}\big(|R_k \cap \mathcal{H}_0| > \zeta_k\big) \le \frac{\alpha}{|\mathcal{K}|}$. It follows, by union bound, that

$$\mathrm{JER}(\mathfrak{R}) = \mathbb{P}\big(\exists k \in \mathcal{K} : |R_k \cap \mathcal{H}_0| > \zeta_k\big) \le |\mathcal{K}|\frac{\alpha}{|\mathcal{K}|} = \alpha.$$

□

3.7 Numerical experiments

The purpose of this section is to empirically visualize the gain of power on confidence envelopes by considering the heterogeneity of the data. We will compare our new procedures with the ones defined for the homogeneous setting, which we recall is the classical setting that was investigated in previous works. The code for the numerical experiments is available at https://github.com/RomainPerier/Heterogeneous_confidence_envelopes.

3.7.1 Simulation setting

We follow the simulation setting from Döhler et al. (2018) to generate our data. We recall here this setting: we simulate two groups of $N = 50$ subjects, and for each subject, we observe m independent binary responses (*e.g.* treatment effects, gene variants, ...). We denote by β_{1i} and β_{2i} the success probability, respectively for the first group and the second group. Then, we simultaneously test the null hypotheses H_i :"$\beta_{1i} = \beta_{2i}$" by the mean of two-sided Fisher's exact tests producing for each $i \in [\![1, m]\!]$ a p-value p_i with its *c.d.f.* F_i. We will then have three distinct sets of p-values. The first one has a cardinality of $m'_0 < m$ and the paired p-values is generated by underlying data such that $\beta_{1i} = \beta_{2i} = 0.01$. We denote by $\pi'_0 = m'_0/m_0$ the proportion of null hypotheses lying in this set of null p-values, and it will be set to 0.5 for our experiments. The second set of p-values has a cardinality of $m''_0 < m - m'_0$ and will be generated by underlying data such that $\beta_{1i} = \beta_{2i} = 0.1$. That is, $m_0 = m'_0 + m''_0$. Finally, the third set of p-values has a cardinality $m_1 = m - (m'_0 + m''_0)$, determined by the proportion of effects $1 - \pi_0$, where $\pi_0 = m_0/m$, and generated by underlying data such that $\beta_{1i} = 0.1$ and $\beta_{2i} = q$, where q represents the strength of the signal. For all of our simulations, π_0 and q will be specified and we will set $m = 2000$, $\pi'_0 = 0.5$. Furthermore, for each couple (π_0, q) we draw m contingency tables, compute the associated p-values and apply our methods to this unique set of p-values.

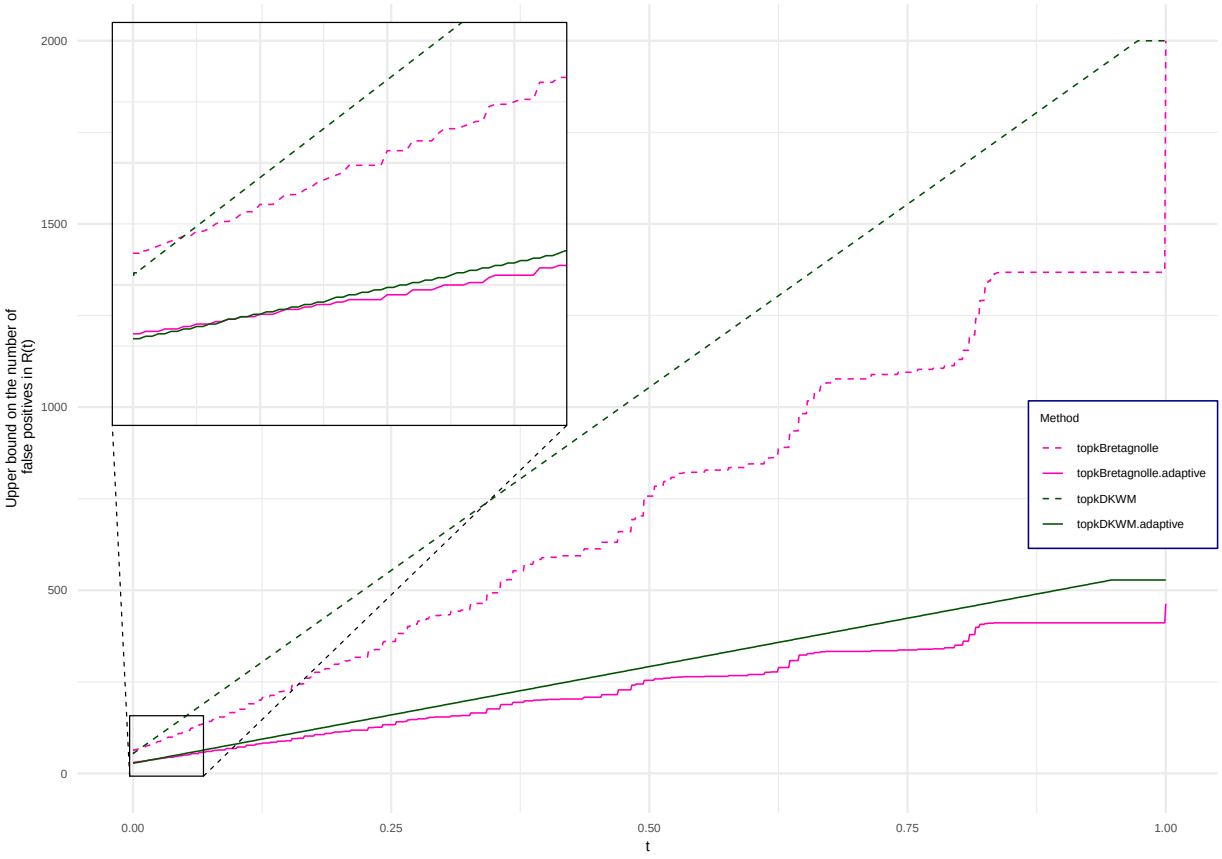


Figure 3.2: Upper bounds $\zeta(\cdot)$ provided by the DKW and Bretagnolle inequalities, with their respective adaptive bounds. We set $\pi_0 = 0.2$ and $q = 0.4$. Lower is better.

3.7.2 Analyses for top-k setting

This section will focus on our results in the top-k setting. We will compare our Bretagnolle bounds for heterogeneous data with the DKW bounds provided in the homogeneous case. Note that in the top-k setting, $\zeta_k = \zeta(p_{(k)})$ derives from a more general bound $\zeta(t)$ on $|R(t) \cap \mathcal{H}_0|$ which is deterministic when it derives from non-adaptive bounds. In this kind of simulation, an upper bound $\zeta(\cdot)$ on the number of false discoveries performs better if it has small values. We recall here the expression of $\zeta(\cdot)$ for the Bretagnolle and DKW inequalities Theorem 3.3.19:

$$\zeta^{DKW}(t) = \left\lfloor mt + \sqrt{m}\lambda_\alpha \right\rfloor, \zeta^{Bret}(t) = \left\lfloor m\bar{F}(t) + \sqrt{m}\tilde{\lambda}_\alpha \right\rfloor,$$

and their adaptive counterparts:

$$\zeta^{DKW}_{adap}(t) = \left\lfloor \hat{m}_{0,1}t + \sqrt{\hat{m}_{0,1}}\lambda_\alpha \right\rfloor, \zeta^{Bret}_{adap}(t) = \left\lfloor \sum_{1 \le i \le \hat{m}_{0,2}} \big(\mathbf{F}(t)\big)_{[i]} + \sqrt{\hat{m}_{0,2}}\tilde{\lambda}_\alpha \right\rfloor,$$

where $\hat{m}_{0,1}$ is defined by (3.30) and $\hat{m}_{0,2}$ is defined by (3.36).

Here, we expect the DKW bounds to be slightly better than the Bretagnolle bounds on smaller p-values — due to the difference between λ_α and $\tilde{\lambda}_\alpha$, see Remark 3.5.3 — if the number of p-values in the area near 0, where $\zeta^{DKW}(\cdot) < \zeta^{Bret}(\cdot)$, oversteps the $\sqrt{m}\lambda_\alpha$ or $\sqrt{\hat{m}_0}\lambda_\alpha$ term. In Figure 3.2, we plot the four functions $\zeta^{DKW}, \zeta^{Bret}, \zeta^{DKW}_{adap}, \zeta^{Bret}_{adap}$,

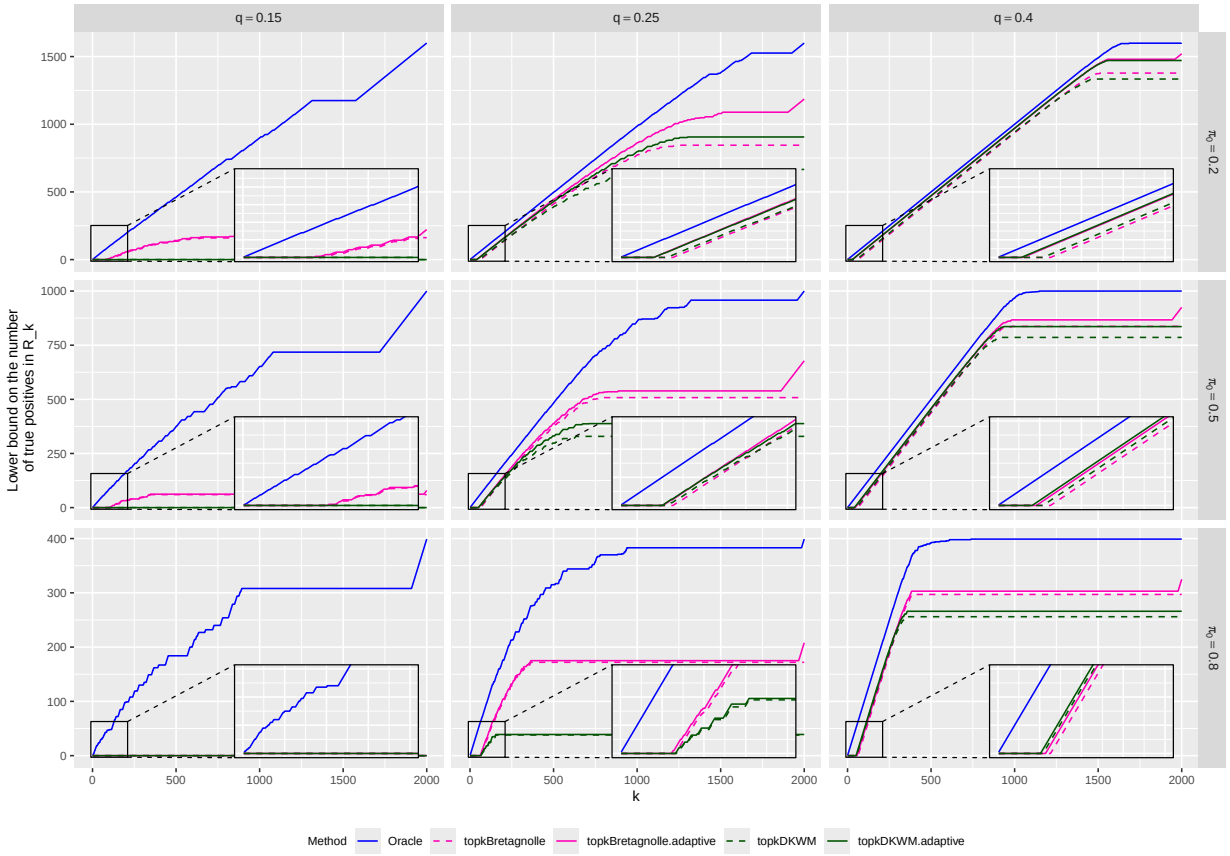


Figure 3.3: Lower bounds on number of true positives in R_k provided by DKW and Bretagnolle inequalities, with their respective adaptive bounds. Higher is better.

for one realization of $(\hat{m}_{0,1}, \hat{m}_{0,2})$. The lowest a function is at a given point t, the better the estimation of $|R(t) \cap \mathcal{H}_0|$. We see that, indeed, the functions based on the DKW inequality are better than their Bretagnolle counterparts for small values of t. We also remark that there is a huge gap between the adaptive and the vanilla methods, in favor of the adaptive ones, because the signal is strong and large (we used $\pi_0 = 0.2$ and $q = 0.4$), which is a favourable case for adaptive methods.

Figure 3.3 illustrates the previous analysis of for different values of q and π_0 by plugging the p-values to the bounds $\zeta(\cdot)$. We compare here the lower bound provided by the top-k methods, derived from the sequence of inequalities $|\mathcal{H}_1 \cap S| = |S| - |\mathcal{H}_0 \cap S| \geq |S| - \hat{V}(S)$. Note that here, the Oracle curve is the number of elements from R_k in $\mathcal{H}_1$. As we expected, in settings with strong signal or with moderately strong and moderate to high signal proportion, the Bretagnolle bound is more conservative than the DKW bound for R_k such that $p_{(k)}$ is small. This last phenomena is due to the corrective term from the Bretagnolle inequality to consider heterogeneity. However when the signal is weak, the Bretagnolle bound detects signal when the DKW inequality does not.

3.7.3 Analyses for k-FWER setting

Likewise to the top-k case, one can construct an upper bound $\zeta(t)$ on $|R(t) \cap \mathcal{H}_0|$ in the k-FWER setting. To obtain this bound we define, for a thresholds family τ and all $t \in [0, 1]$, $\zeta(t) = |\{i \in [\![1, m]\!] : \tau_i < t\}|$. Note that it can be checked that this construction is exactly the same as the one displayed in Section A.3.2. We can observe

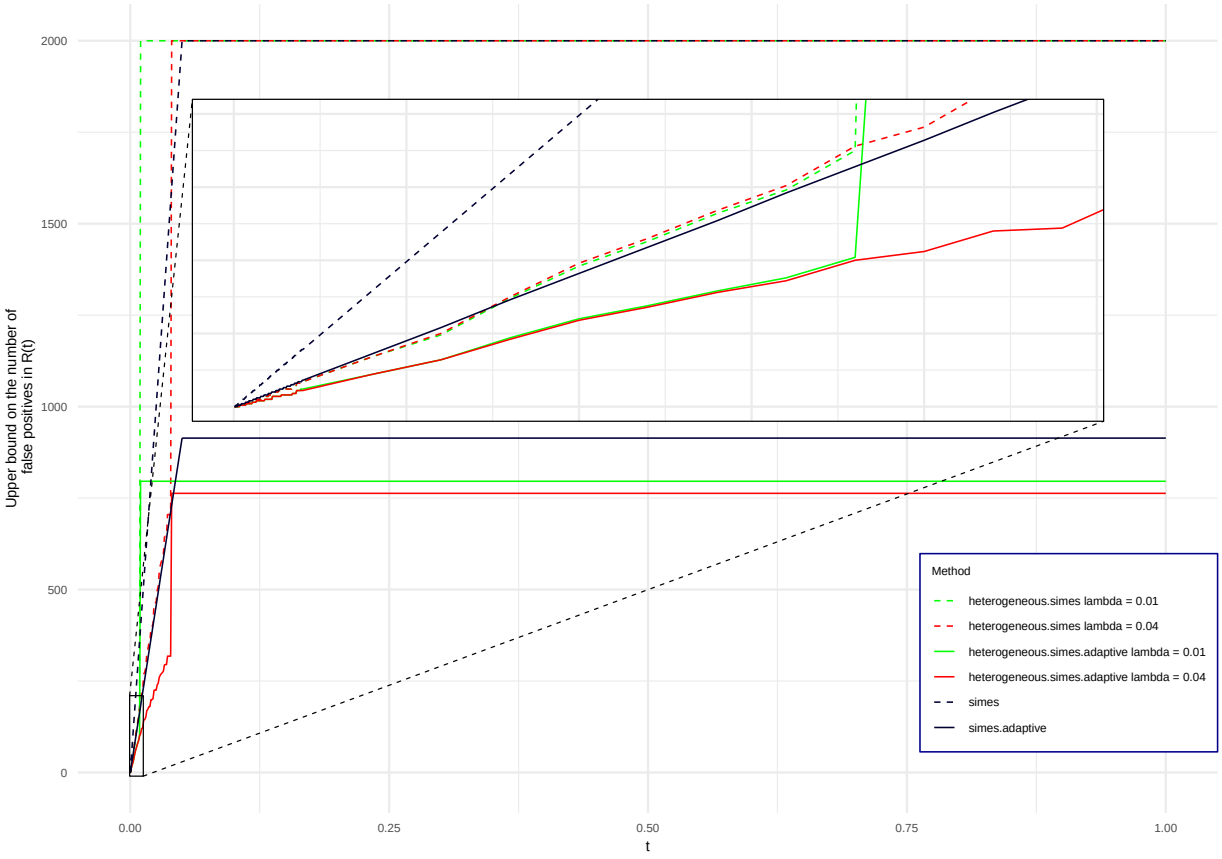


Figure 3.4: Upper bounds $\zeta(\cdot)$ provided by Simes and heterogeneous Simes inequalities, with their respective adaptive bounds. We set $\pi_0 = 0.2$, $\pi_0' = 0.5$ and $q = 0.4$. Lower is better.

two phenomena in Figure 3.4 ; when we reduce the parameter λ in the heterogeneous Simes-like procedures, we get bounds slightly better in the non-adaptive fashion. This phenomenon was expected, since we enlarge the denominator of the coordinates of $\mathbf{H}(\cdot, \cdot)$. However, it deteriorates the bound in the adaptive fashion. It is a consequence of limiting the computation of $\hat{m}_0$ to p-values smaller than λ.

Figure 3.5 shows efficiency of Simes-like confidence envelopes via lower bounds on the number of true positives in R_k. Likewise to the vanilla Simes method and accordingly to power studies from Döhler et al. (2018), this class of k-FWER methods does not detect the signal if it is weak. However, with moderate and strong signal, the heterogeneous Simes methods will detect the signal, and is less conservative than the vanilla Simes method. The improvement using adaptive methods is substantial when the signal is strong and wide, and the slight difference between the adaptive methods mentioned earlier in the analysis of Figure 3.4 is visible for the couple of parameters $(q, \pi_0) = (0.4, 0.2)$. However, these improvements are slight or inexistent in the other settings hence these heterogeneous bounds inherits of the vanilla Simes inequality.

3.7.4 Comparison with other homogeneous bounds

We compare our new envelopes for heterogeneous data with the bounds from Meah et al. (2024) and Katsevich and Ramdas (2020) provided in the homogeneous case. We only show here the adaptive versions of the bounds. As it is illustrated by Figure 3.6, there is not an uniformly better upper bound over $[0, 1]$. While the heterogeneous Simes-like

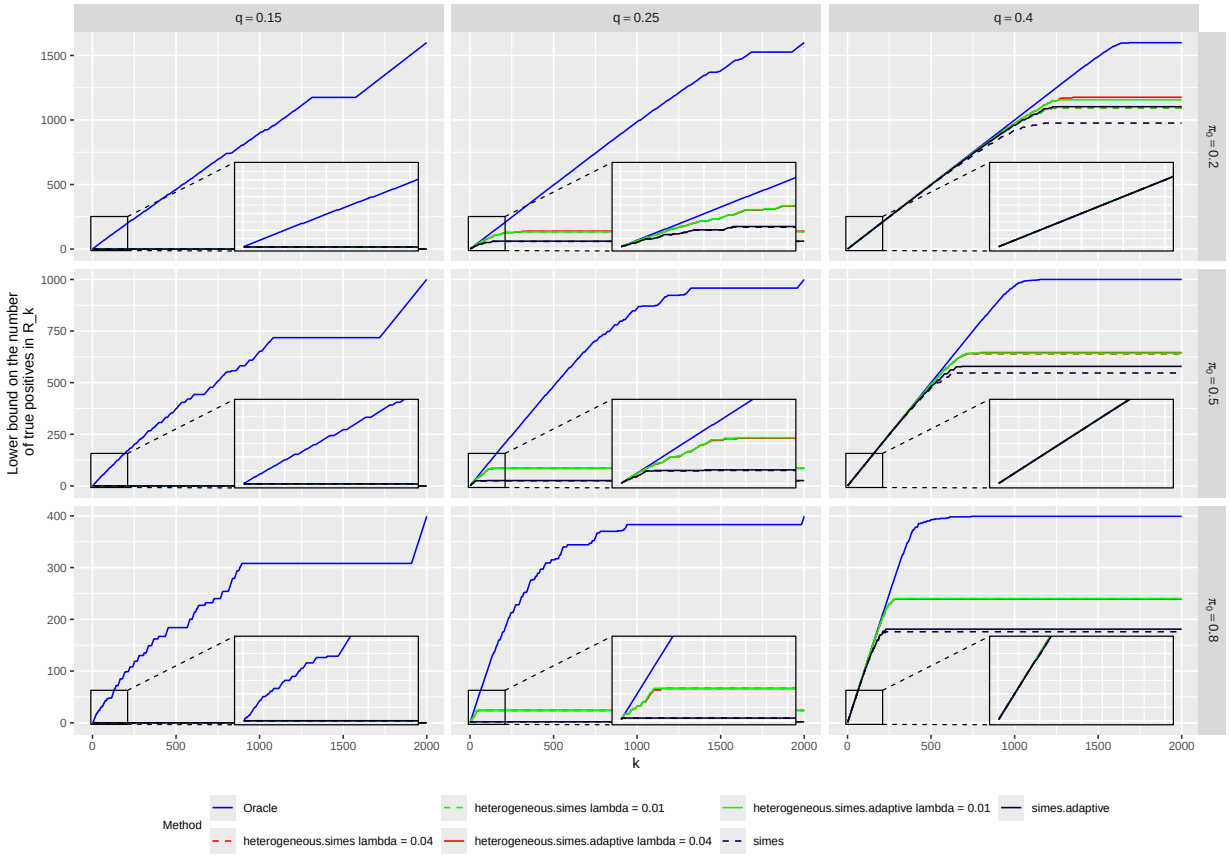


Figure 3.5: Lower bounds on number of true positives in R_k provided by Simes and heterogeneous Simes inequalities, with their respective adaptive bounds. Higher is better.

bounds are the least conservative for the smallest t, this trend reverses as t becomes less extreme — but still inferior to approximatively 0.05. For these values of t, the KR and Wellner confidence bounds are less conservative than the heterogeneous envelopes developed in the present article. However, for larger t, the less conservative bound is the Bretagnolle bound.

The previous analysis is corroborated by Figure 3.7 where we plugged the p-values in these adaptive bounds.

3.8 Conclusion and perspectives

In this chapter, we introduced new confidence envelopes for false discoveries by incorporating heterogeneous upper bounds on the cumulative distributive functions of null distribution. Our approach bridges key frameworks in the existing literature on confidence envelopes — specifically the JER approach (Blanchard et al., 2020a) and the inversion procedure using local tests (Genovese and Wasserman, 2006; Goeman and Solari, 2011).

We established generic novel shortcuts for the inversion procedure, providing significant new insights even in the standard homogeneous case. For instance, applied to the complete set of hypotheses, the new shortcuts are exact (they coincide with the theoretical inversion procedure) and thus provide optimal (over-)estimators for m_0. Furthermore, using these over-estimators as plug-ins into the JER framework yields a

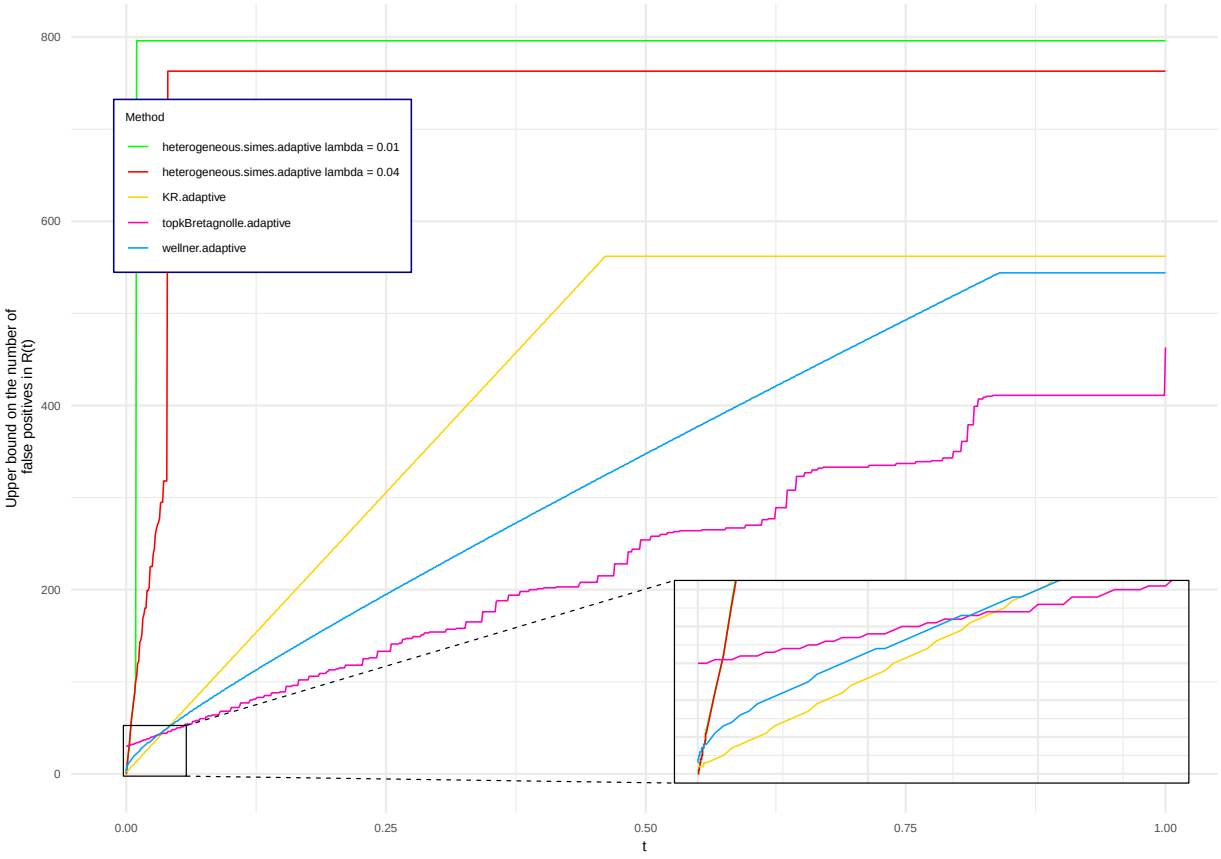


Figure 3.6: Upper bounds $\zeta(\cdot)$ provided by adaptive bounds from Meah et al. (2024), Katsevich and Ramdas (2020) and our adaptive heterogeneous bounds. We set $\pi_0 = 0.2$, $\pi_0' = 0.5$ and $q = 0.4$. Lower is better.

bound which coincides for all rejection sets with the inversion procedure in the homogeneous case, formally validating the optimality of several previous constructions from Meah et al. (2024).

The methodologies introduced here extend naturally to the heterogeneous setting. By combination with heterogeneous ecdf inequalities (some of which are newly introduced in this work and of independent interest), our novel shortcuts provide new confidence envelopes for heterogeneous tests and yield over-estimators for the number of null hypotheses. These bounds are less conservative than those obtained via a naive homogenization of the local test family. Consequently, when plugged in into the JER methodology, these estimators provide sharper, more efficient shortcuts. Additionally, our new shortcuts for the inversion procedure also produce alternative confidence envelopes that can be less conservative than their JER counterparts in specific cases.

We validated our methods with numerical experiments highlighting the power gain of using envelopes tailored to the heterogeneity of the data compared to previous approaches designed for the homogeneous case.

A promising direction for future research involves deriving heterogeneous versions of the inequalities proposed by Katsevich and Ramdas (2020) or the uniform Wellner inequality from Meah et al. (2024). Since these bounds outperform heterogeneous ones in specific p-value regions without requiring prior distributional information, getting heterogeneous versions could further enhance the power of these envelopes in heterogeneous context.

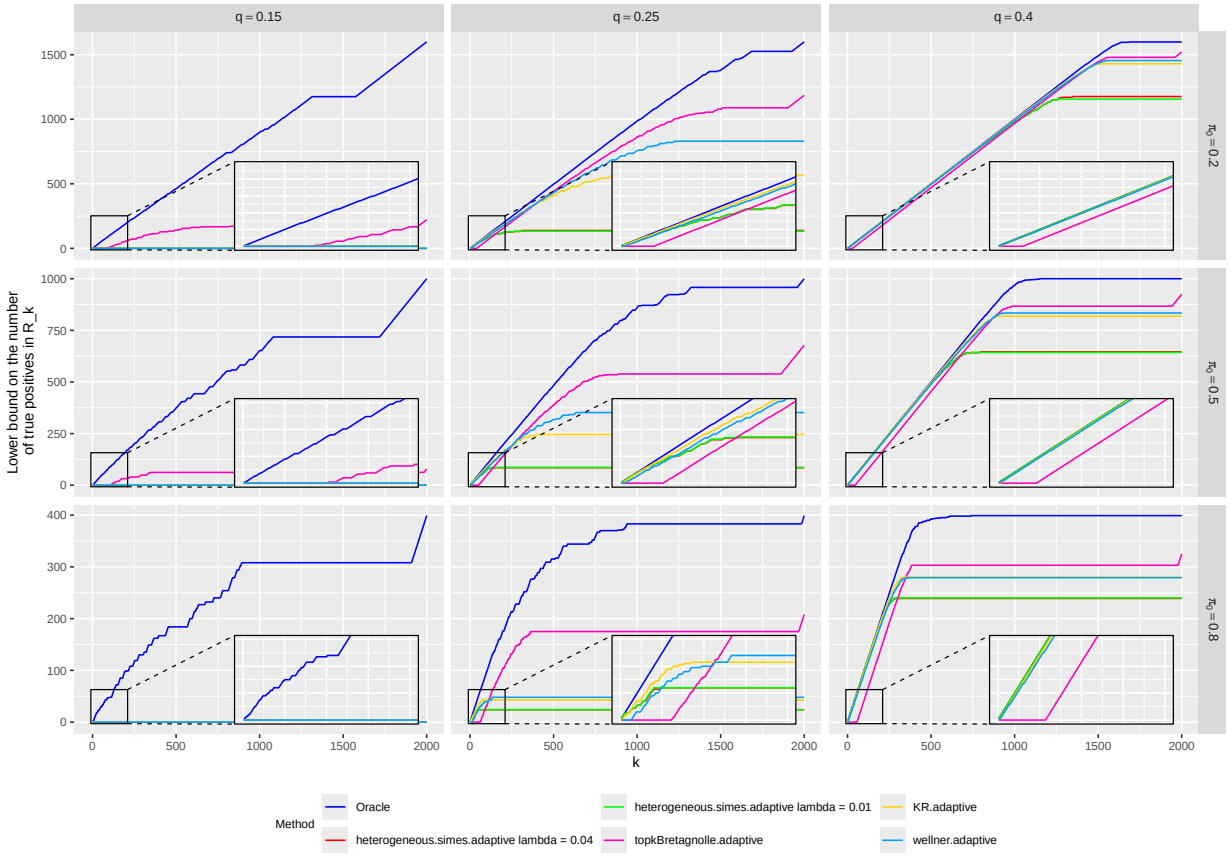


Figure 3.7: Lower bounds on number of true positives in R_k adaptive bounds from Meah et al. (2024), Katsevich and Ramdas (2020) and our adaptive heterogeneous bounds. Higher is better.

Chapter 4

FDR control and FDP bounds for conformal link prediction

In Marandon, 2024, the author introduces a procedure to detect true edges from a partially observed graph using a conformal prediction fashion: first computing scores from a trained function, deriving conformal p-values from them and finally applying a multiple testing procedure. In this chapter, we prove that the resulting procedure indeed controls the FDR, and we also derive uniform FDP bounds, thanks to an exchangeability argument and the previous work of Marandon et al., 2025.

Contents

4.1 Introduction 89
4.2 Conformal link prediction 90
4.2.1 Model and notation 90
4.3 Conformal link prediction procedure 91
4.4 FDR and FDP control guarantees 93
4.5 Discussion 97

4.1 Introduction

Consider a link prediction setting where we partially observe a graph, that is, the presence or absence of an edge (borrowing to multiple testing literature, we will use the terms "true" and "false" edge, respectively) between two nodes is either observed, either masked (Lu and Zhou, 2011). Furthermore, consider that the missing data mask is known. Hence, the absence of connection between two nodes is either an observed false edge, either a masked edge (and the state of the edge, true or false, is unobserved), and we know if we are in the first or the second case. The task at hand is to provide a label (true or false) to each unobserved edge. In Marandon (2024), the author introduced the first, to our knowledge, method which aims to control the False Discovery Rate (FDR) in the given context, using tools from the Conformal Inference (CI) literature, namely conformal p-values (Vovk et al., 2005). However, because of the graph structure inducing intricate dependence in the data, the setup is different from the usual case encountered in CI: typically, in CI, the data is assumed to be exchangeable. Hence, theoretical FDR control was not proven, but was observed empirically on simulated data.

In this work, we prove that the procedure of Marandon (2024) controls the FDR under a general missing data setup and without any distributional assumptions on the complete network. The core observation to make is that even if the data is not exchangeable, the scores are, under really mild assumptions (no ties among them, almost surely). This allows to use the results of Marandon et al. (2025) to achieve FDR control. Furthermore, the exchangeability of the scores also allows to use Gazin et al. (2024) to get a uniform False Discovery Proportion (FDP) bound for the path $(R(t))_{t\in[0,1]}$, where $R(t)$ is the set of unobserved edges with conformal p-value lesser or equal than t. This uniform bound in turn allows to construct confidence post hoc bounds à la Blanchard et al. (2020a).

For an extensive review of related works in link prediction, conformal inference, and multiple testing, we refer the reader to Marandon (2024) (Sections 1.3, 3.1, 3.3).

The rest of the chapter is organized as follows. In Section 4.2, we first define formally the model, the problem and the notation, before recalling the conformal link prediction procedure introduced in Marandon (2024). Then, in Section 4.4, we state and prove our two main theorems. We conclude in Section 4.5 with a short discussion of the relation to recent literature, in particular the work of Huang et al. (2023).

4.2 Conformal link prediction

4.2.1 Model and notation

Let $A^* = (A^*_{i,j})_{1\leq i,j\leq n}$ be the adjacency matrix of the true complete graph $\mathcal{G}$, $X \in \mathbb{R}^{n\times d}$ a matrix of node covariates (if available), and $\Omega = (\Omega_{i,j})_{1\leq i,j\leq n}$ the sampling matrix such that $\Omega_{i,j} = 1$ if the interaction status (true/false) of (i,j) is observed, and 0 otherwise. We denote by A the observed adjacency matrix with $A_{i,j} = \Omega_{i,j}A^*_{i,j}$. Thus, $A_{i,j} = 1$ indicates that there is an observed true edge between i and j, whereas $A_{i,j} = 0$ indicates either the observed lack of an edge or an unreported edge. The sampling matrix Ω is assumed to be observed, so that it is known which zero-entries $A_{i,j} = 0$ correspond to observed false edges and which ones correspond to missing information. We denote by P the joint distribution of $Z^* = (A^*, X, \Omega)$, Z the observation (A, X, Ω) and $\mathcal{Z}$ the observation space. P belongs to a model $\mathcal{P}$. We consider a general missing data setup in which the entries of Ω are independently generated as:

$$\Omega_{i,j}|A^*, X \sim \mathcal{B}\big(w_0\mathbb{1}\{A^*_{i,j} = 0\} + w_1\mathbb{1}\{A^*_{i,j} = 1\}\big), \quad 1 \leq i,j \leq n$$

for some unknown sampling rates w_0, w_1. This type of missing data setup is called double standard sampling (Tabouy et al., 2020; Sportisse et al., 2020) and is a generalization of the case where the entries of Ω are i.i.d, that is more relevant for practical applications (see Section 2 in Marandon (2024) and the references therein for a discussion of this assumption with regards to the literature). Let us introduce the following notations:

- We denote by $\mathcal{D}_{\text{test}}(Z) = \{(i,j) : \Omega_{i,j} = 0\}$ the set of non-sampled (or missing) node pairs and by $\mathcal{D}(Z) = \{(i,j) : \Omega_{i,j} = 1\}$ the set of sampled pairs, with $\mathcal{D}^0 = \{(i,j) \in \mathcal{D} : A^*_{i,j} = 0\}$ the set of observed false edges and $\mathcal{D}^1 = \{(i,j) \in \mathcal{D} : A^*_{i,j} = 1\}$ the set of observed true edges. We refer to $\mathcal{D}_{\text{test}}(Z)$ as the test set.
- We denote by $\mathcal{H}_0 = \{(i,j) : \Omega_{i,j} = 0, A^*_{i,j} = 0\}$ the (unobserved) set of false edges in the test set and $\mathcal{H}_1 = \{(i,j) : \Omega_{i,j} = 0, A^*_{i,j} = 1\}$ the (unobserved) set of true edges in the test set.

In link prediction, one is interested in classifying the unobserved node pairs of $\mathcal{D}_{\text{test}}$ into true edges and false edges, or in other words, selecting a set of unobserved node pairs to be declared as true edges, based on the observed graph structure. Define a selection procedure as a (measurable) function $R = R(Z)$ that returns a subset of $\mathcal{D}_{\text{test}}$ corresponding to the indices (i, j) where an edge is declared. For any such procedure R, the False Discovery Rate (FDR) of R is defined as the average of the False Discovery Proportion (FDP) of R under the model parameter $P \in \mathcal{P}$, that is,

$$\mathrm{FDR}(R) = \mathbb{E}_{Z^* \sim P}[\mathrm{FDP}(R)], \quad \mathrm{FDP}(R) = \frac{\sum_{i \in \mathcal{H}_0} \mathbb{1}\{i \in R\}}{1 \vee |R|}.$$

Similarly, the true discovery rate (TDR) is defined as the average of the true discovery proportion (TDP), that is,

$$\mathrm{TDR}(R) = \mathbb{E}_{Z^* \sim P}[\mathrm{TDP}(R)], \quad \mathrm{TDP}(R) = \frac{\sum_{i \in \mathcal{H}_1} \mathbb{1}\{i \in R\}}{1 \vee |\mathcal{H}_1|}.$$

The aim considered here is to build a procedure R that controls the FDR while having a TDR (measuring the *power* of the procedure) as large as possible.

Notably, a well-known procedure controlling the FDR is the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995) which, given m p-values $p_1, \dots, p_m$, sorts them: $0 = p_{(0)} \leq p_{(1)} \leq \cdots p_{(m)}$, defines $\hat{k} = \max\{k \in [\![0, m]\!] : p_{(k)} \leq \alpha \frac{k}{m}\}$, and rejects the p-values smaller than $\alpha \frac{\hat{k}}{m}$.

Let us emphasize that, in our setting, $\mathcal{H}_0$ is random, which is not entirely classical in the multiple testing litterature, but it is also not unusual, and can be traced back to, at least, the mixture model presented in Efron et al. (2001).

4.3 Conformal link prediction procedure

We briefly recall the procedure which is described in details in Marandon (2024) (Section 3.2). Assume that we have at hand a score estimator function $g : \mathcal{Z} \to \mathbb{R}^{n \times n}$ that takes as input an observation $z \in \mathcal{Z}$ and returns a score matrix $(S_{i,j})_{1 \leq i,j \leq n} \in \mathbb{R}^{n \times n}$ that scores the likeliness of a true edge between couples of nodes. The function g corresponds to a link prediction algorithm that would be trained on the "training" dataset z. Any off-the-shelf link prediction algorithm can be used, up to a specification detailed in the next paragraph. Our training set has to be different from a "calibration" set on which we will compute scores and compare them to the scores computed on the test set to build the conformal p-values. We build this calibration set $\mathcal{D}_{\text{cal}}$ of false edges by sampling uniformly $\ell = |\mathcal{D}_{\text{cal}}|$ edges without replacement from the set of observed false edges $\mathcal{D}^0$. The algorithm is then trained on $Z_{\text{train}} = (A, X, \Omega_{\text{train}})$ where $(\Omega_{\text{train}})_{i,j} = 0$ if $(i, j) \in \mathcal{D}_{\text{cal}}$ and $\Omega_{i,j}$ otherwise. That is, formally, we simply compute $g(Z_{\text{train}})$. The scores $S_{i,j} = g(Z_{\text{train}})_{i,j}$ are then computed for all $(i, j) \in \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{test}}$. Conformal p-values $(p_{i,j})_{(i,j) \in \mathcal{D}_{\text{test}}}$ are then computed following

$$p_{i,j} = \frac{1}{\ell + 1} \left(1 + \sum_{(u,v) \in \mathcal{D}_{\text{cal}}} \mathbb{1}\{S_{i,j} \leq S_{u,v}\} \right), \quad (i, j) \in \mathcal{D}_{\text{test}} \tag{4.1}$$

and finally the BH procedure is applied to them. Those steps are recapped in Algorithm 2.

Algorithm 2: Conformal link prediction

Input: Observation $Z = (A, X, \Omega)$: (Observed) adjacency matrix A, node covariate matrix X, sampling matrix Ω; LP algorithm g; sample size ℓ of the reference set

Define $\mathcal{D} = \{(i,j) : \Omega_{ij} = 1\}$; $\mathcal{D}_{\text{test}} = \{(i,j) : \Omega_{ij} = 0\}$; $\mathcal{D}^0 = \{(i,j) \in \mathcal{D} : A_{ij} = 0\}$

1. Sample $\mathcal{D}_{\text{cal}}$ of size ℓ uniformly without replacement from $\mathcal{D}^0$

2. Learn g on "masked" observation $Z_{\text{train}} = (A, X, \Omega_{\text{train}})$, with Ω_{train} given by

$$(\Omega_{\text{train}})_{i,j} = \begin{cases} 0 & \text{if } (i,j) \in \mathcal{D}_{\text{cal}}, \\ \Omega_{i,j} & \text{otherwise.} \end{cases}$$

3. For each $(i,j) \in \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{test}}$, compute the score $S_{i,j} = g(Z_{\text{train}})_{i,j}$

4. For each $(i,j) \in \mathcal{D}_{\text{test}}$, compute the conformal p-value $p_{i,j}$ given by (4.1)

5. Apply BH using $(p_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}}$ as input p-values, providing a rejection set $R(Z) \subset \mathcal{D}_{\text{test}}$.

Output: $R(Z)$

Although g can be any measurable function, we now discuss a general shape of g mild and commonly used. Let us introduce, for a given $K \in \{1, \dots, n\}$, and any adjacency matrix a,

$$w(a)_{i,j} = (a_{i,\bullet}, a_{j,\bullet}, a^2_{i,\bullet}, a^2_{j,\bullet}, \dots, a^K_{i,\bullet}, a^K_{j,\bullet}),$$

where $a^k_{i,\bullet} = (a^k_{i,u})_{1\le u\le n}$ for $1 \le k \le K$, that is, $a^k_{i,\bullet}$ is the i-th row of a to the power k. For $z = (a, x, \omega) \in \mathcal{Z}$, the commonly used form for $g(z)$ is then

$$g(z)_{i,j} = h\big((w(a)_{i,j}, x_{N(i,K),\bullet}, x_{N(j,K),\bullet}); \{(w(a)_{u,v}, x_{N(u,K),\bullet}, x_{N(v,K),\bullet}, a_{u,v}), \omega_{u,v} = 1\}\big) \tag{4.2}$$

with $h(\cdot;\cdot)$ some real-valued measurable function, $N(i,K)$ the K-hop neighborhood of node i, and $x_{S,\bullet}$ is the submatrix $(x_{u,l})_{\substack{u\in S \\ 1\le l\le d}}$ for any subset $S \in [\![1, n]\!]$. Note that the second argument of h means exactly that g is trained only on the information provided by the non-missing edges. Also note that because $(\Omega_{\text{train}})_{i,j} = 0$ for all $(i,j) \in \mathcal{D}_{\text{cal}}$, Ω_{train} treats the edges of $\mathcal{D}_{\text{cal}}$ as missing. Hence $g(Z_{\text{train}})$ indeed is trained on a set of observed edges that is disjoint from the edges of the calibration set, thus avoiding overfitting.

This formulation indeed is mild and encompasses numerous link prediction algorithms. For example the simple common neighbors heuristic (Lu and Zhou, 2011) given by $g(z)_{i,j} = a^\top_{i,\bullet} a_{j,\bullet}$. The larger class of Empirical Risk Minimization (ERM) methods (like in Bleakley et al. (2007) or Zhang and Chen (2018)) also generally fall into this form. Take a class $\mathcal{F}$ of measurable functions taking values in $[0,1]$, that can be thought of as binary classifiers. It could be a class of (graph) neural networks for example. Also let $\mathcal{L}$ a measurable loss function, for example the cross-entropy loss function. Then let $\hat{f} \in \arg\min_{f\in\mathcal{F}} \sum_{(u,v):\omega_{u,v}=1} \mathcal{L}(f(w(a)_{u,v}), a_{u,v})$ and finally $h\big(\cdot; \{(w(a)_{u,v}, a_{u,v}), \omega_{u,v} = 1\}\big) = \hat{f}(\cdot)$. This ERM framework indeed fits Equation (4.2). More examples are provided in Marandon (2024) (Section 3.3).

4.4 FDR and FDP control guarantees

In this section, we establish that the procedure controls the FDR in finite samples and the FDP with high probability. In the sequel, we use the following notation: $k_0 = |\mathcal{D}^0|$, $m = |\mathcal{D}_{\text{test}}|$, $\mathcal{D}^0_{\text{train}} = \mathcal{D}^0 \setminus \mathcal{D}_{\text{cal}}$.

We will consider the following assumptions:

Assumption 4.1. *The matrix score $g(Z_{train}) = (S_{i,j})_{1\leq i,j\leq n}$ has no ties a.s.*

Assumption 4.2. *The calibration set size ℓ is a function of $k_0 = |\mathcal{D}^0|$ and $\mathcal{D}^1$ only.*

The main result of this section is the following:

Theorem 4.4.1. *Consider the output R of the procedure proposed in Algorithm 2 with a scoring function g of the form (4.2), and let Assumptions 4.1 and 4.2 hold true. Then, it holds that $\mathrm{FDR}(R) \leq \alpha$.*

The proof relies on reformulating the proposed procedure as the one of Marandon et al. (2025) applied to a suitable ordering of the scores, which is shown to satisfy the sufficient exchangeability condition in Marandon et al. (2025) for FDR control, conditionally on a certain random variable W.

The following lemma states two key properties of the calibration scores $(S_{i,j})_{(i,j)\in\mathcal{D}_{\text{cal}}}$ and test scores $(S_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}}$.

Lemma 4.4.2. *Let W denote the r.v. given by $(k_0, A^*, \mathcal{D}^1, \mathcal{D}^0_{train}, X)$, and let Assumption 4.2 hold true. Then:*

(i) $(S_{i,j})_{1\leq i,j\leq n}$ is measurable with respect to W.

(ii) Denote, for any nonempty finite set C and $N \in \mathbb{N}^*$ with $N \leq |C|$, $\Xi(C, N)$ the distribution that corresponds to sampling uniformly and without replacement an N-sized subset of the set C. Then it holds:

$$\mathcal{D}_{cal}\,|W \sim \Xi(\mathcal{E}^0 \setminus \mathcal{D}^0_{train}, \ell), \tag{4.3}$$

$$\mathcal{H}_0 = (\mathcal{E}^0 \setminus \mathcal{D}^0_{train}) \setminus \mathcal{D}_{cal}, \tag{4.4}$$

$$\mathcal{D}_{test} = \left(\llbracket 1, n\rrbracket^2 \setminus (\mathcal{D}^1 \cup \mathcal{D}^0_{train})\right) \setminus \mathcal{D}_{cal}, \tag{4.5}$$

where $\mathcal{E}^0 = \{(i,j) \in \llbracket 1, n\rrbracket^2 : A^*_{i,j} = 0\}$.

Proof. Point (i) comes from $(S_{i,j})_{1\leq i,j\leq n} = g(A, X, \Omega_{\text{train}})$ as, first, g is measurable, second, the knowledge of $\mathcal{D}^1$ is equivalent to the knowledge of A, and third, Ω_{train} can be rewritten as

$$(\Omega_{\text{train}})_{i,j} = \begin{cases} 1 & \text{if } (i,j) \in \mathcal{D}^1 \cup \mathcal{D}^0_{\text{train}}, \\ 0 & \text{otherwise,} \end{cases}$$

which in turn entails that $(S_{i,j})_{1\leq i,j\leq n}$ is measurable with respect to $(\mathcal{D}^1, \mathcal{D}^0_{\text{train}}, X)$.

For (ii), we reason conditionally to $(A^*, k_0, \mathcal{D}^1)$ and assume those to be fixed. Because Ω are i.i.d. on $\mathcal{E}^0$ conditionally to (A^*, X), recalling $\mathcal{D}^0 = \{(i,j) \in \mathcal{E}^0 : \Omega_{i,j} = 1\}$

and recalling the definition of $\mathcal{D}_{\text{cal}}$ as a sub-sampling without replacement of ℓ elements from $\mathcal{D}^0$, it follows that

$$\begin{aligned}
\mathcal{D}^0|(A^*, k_0, \mathcal{D}^1, X) &\sim \Xi(\mathcal{E}^0, k_0);\\
\mathcal{D}_{\text{cal}}\,|(A^*, k_0, \mathcal{D}^1, \mathcal{D}^0, X) &\sim \Xi(\mathcal{D}^0, \ell);\\
\mathcal{D}^0_{\text{train}} &= \mathcal{D}^0 \setminus \mathcal{D}_{\text{cal}};\\
\mathcal{H}_0 &= \mathcal{E}^0 \setminus \mathcal{D}^0;\\
\mathcal{D}_{\text{test}} &= [\![1, n]\!]^2 \setminus (\mathcal{D}^0 \cup \mathcal{D}^1).
\end{aligned}$$

It is easy to check that the above joint distribution of $(\mathcal{D}^0, \mathcal{D}_{\text{cal}}, \mathcal{D}^0_{\text{train}}, \mathcal{D}_{\text{test}}, \mathcal{H}_0)$ conditional to $(A^*, k_0, \mathcal{D}^1)$ is equivalently represented as

$$\begin{aligned}
\mathcal{D}^0_{\text{train}}\,|(A^*, k_0, \mathcal{D}^1, X) &\sim \Xi(\mathcal{E}^0, k_0 - \ell);\\
\mathcal{D}_{\text{cal}}\,|(A^*, k_0, \mathcal{D}^1, \mathcal{D}^0_{\text{train}}, X) &\sim \Xi(\mathcal{E}^0 \setminus \mathcal{D}^0_{\text{train}}, \ell);\\
\mathcal{D}^0 &= \mathcal{D}_{\text{cal}} \cup \mathcal{D}^0_{\text{train}};\\
\mathcal{H}_0 &= (\mathcal{E}^0 \setminus \mathcal{D}^0_{\text{train}}) \setminus \mathcal{D}_{\text{cal}};\\
\mathcal{D}_{\text{test}} &= \left([\![1, n]\!]^2 \setminus (\mathcal{D}^1 \cup \mathcal{D}^0_{\text{train}})\right) \setminus \mathcal{D}_{\text{cal}}\,.
\end{aligned}$$

This implies the claim.

□

The next lemma introduces a suitable (random) ordering of the scores in order to formally grant the sufficient score exchangeability condition of Marandon et al. (2025).

Lemma 4.4.3. *Use the same notation and assumptions as in Lemma 4.4.2, denote* $m_0 = |\mathcal{H}_0|$.
For given W*, let* σ_0 *be a purely random ordering of* $\mathcal{E}^0 \setminus \mathcal{D}^0_{train}$ *(i.e. a random element drawn uniformly from* $\mathrm{Bij}(\{1, \dots, \ell + m_0\}, \mathcal{E}^0 \setminus \mathcal{D}^0_{train})$*) and* σ_1 *an arbitrary ordering of* $\mathcal{H}_1$ *(an element of* $\mathrm{Bij}(\{1, \dots, |\mathcal{H}_1|\}, \mathcal{H}_1)$*), e.g. lexical ordering. Let:*

$$\begin{aligned}
\mathcal{D}^{\sigma_0}_{cal} &= \{\sigma_0(i), i = 1, \dots, \ell\};\\
\mathcal{H}^{\sigma_0}_0 &= \{\sigma_0(i), i = \ell + 1, \dots, \ell + m_0\};\\
\mathcal{D}^{\sigma_0}_{test} &= \mathcal{H}^{\sigma_0}_0 \cup \{\sigma_1(i), i = 1, \dots, |\mathcal{H}_1|\};\\
S'_i &= S_{\sigma_0(i)}, i = 1, \dots, \ell + m_0;\\
S'_{\ell+m_0+j} &= S_{\sigma_1(j)}, j = 1, \dots, |\mathcal{H}_1|.
\end{aligned}$$

Then the distribution of $(\mathcal{D}^{\sigma_0}_{cal}, \mathcal{D}^{\sigma_0}_{test}, \mathcal{H}^{\sigma_0}_0)$ *conditionally to* W *is identical to that of* $(\mathcal{D}_{cal}, \mathcal{D}_{test}, \mathcal{H}_0)$ *given by (4.3)-(4.4), and the scores* $(S'_i)_{1 \le i \le \ell + m_0}$ *form an exchangeable family of random variables (conditionally to* W*, hence also unconditionally).*

Proof. Straightforward since a (set-valued) random variable $F \sim \Xi(C, N)$ can be represented by a set whose elements are N successive draws from C without replacement, in turn equivalent to (i) drawing uniformly an ordering of C; (ii) taking the N first elements according to that order. □

Proof of Theorem 4.4.1. Let us consider the same construction and notation as in Lemma 4.4.3 and define additionally

$$\widetilde{p}_{i,j} = \frac{1}{\ell + 1}\left(1 + \sum_{(u,v) \in \mathcal{D}^{\sigma_0}_{\text{cal}}} \mathbb{1}\{S_{i,j} \le S_{u,v}\}\right), \quad (i,j) \in \mathcal{D}^{\sigma_0}_{\text{test}};$$

observe that it is merely definition (4.1) with $\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$ replaced by $\mathcal{D}_{\text{cal}}^{\sigma_0}, \mathcal{D}_{\text{test}}^{\sigma_0}$.

We recall the definition of the FDP of a rejection set R:

$$\text{FDP}(R, \mathcal{H}_0) = \frac{|R \cap \mathcal{H}_0|}{|R| \vee 1},$$

where for precision of notation, we have specified explicitly the set of null hypotheses $\mathcal{H}_0$ on the left-hand side, because it is random in our observation model.

For a rejection procedure $R((p_{i,j}))$ only depending on the family of p-values, observe that we have

$$\text{FDP}\left(R\Big((p_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}}\Big), \mathcal{H}_0\right) \overset{\mathcal{L}}{=} \text{FDP}\left(R\Big((\widetilde{p}_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}^{\sigma_0}}\Big), \mathcal{H}_0^{\sigma_0}\right), \tag{4.6}$$

where $\overset{\mathcal{L}}{=}$ represents equality in distribution. Namely, by inspection of the definition of the family p-values (4.1), notice that conditionally to W those only depend on $(\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}})$ (since the scores are measurable with respect to W, see Lemma 4.4.2). As a consequence, the left-hand side of (4.6) only depends on $(\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}, \mathcal{H}_0)$, and the right-hand-side is obtained by replacing those by $(\mathcal{D}_{\text{cal}}^{\sigma_0}, \mathcal{D}_{\text{test}}^{\sigma_0}, \mathcal{H}_0^{\sigma_0})$, which according to Lemma 4.4.3 have the same joint distribution conditional to W. Hence (4.6) holds conditionally to W (hence also unconditionally). We therefore focus on the right-hand side of (4.6) from now on.

For any set of p-values $(p_i)_{i\in\mathcal{I}}$, denote by $R_{BH}((p_i)_{i\in\mathcal{I}}) \subseteq \mathcal{I}$ the rejection set of the BH procedure; it takes the form

$$R_{BH}((p_i)_{i\in\mathcal{I}}) = \Big\{i \in I : p_i \leq t_{BH}\big(\{p_i, i \in I\}\big)\Big\},$$

where t_{BH} is a threshold that depends only on the (multi)set of p-values $\{p_i, i \in I\}$. Let us introduce the set of r.v. $p'_1, \dots, p'_m$ such that

$$p'_a = \frac{1}{\ell+1}\left(1 + \sum_{b=1}^{\ell} \mathbb{1}\big\{S'_{\ell+a} \leq S'_b\big\}\right), \quad 1 \leq a \leq m, \tag{4.7}$$

with the r.v. $(S'_a)_{1\leq a\leq m}$ defined in Lemma 4.4.3.

It can be readily checked that

$$p'_a = \widetilde{p}_{\sigma_0(\ell+a)}, \quad 1 \leq a \leq m_0; \quad p'_{m_0+a} = \widetilde{p}_{\sigma_1(a)}, \quad 1 \leq a \leq |\mathcal{H}_1|. \tag{4.8}$$

As a consequence, we have that

$$\begin{aligned}
\text{FDP}\left(R_{BH}\Big((\widetilde{p}_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}^{\sigma_0}}\Big), \mathcal{H}_0^{\sigma_0}\right) &= \frac{\Big|\mathcal{H}_0^{\sigma_0} \cap R_{BH}\Big((\widetilde{p}_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}^{\sigma_0}}\Big)\Big|}{\Big|R_{BH}\Big((\widetilde{p}_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}^{\sigma_0}}\Big)\Big| \vee 1} \\
&= \frac{\sum_{(i,j)\in\mathcal{H}_0^{\sigma_0}} \mathbb{1}\big\{\widetilde{p}_{i,j} \leq t_{BH}\big(\{p_{i,j}, (i,j) \in \mathcal{D}_{\text{test}}^{\sigma_0}\}\big)\big\}}{\Big|R_{BH}\Big((\widetilde{p}_{i,j})_{(i,j)\in\mathcal{D}_{\text{test}}^{\sigma_0}}\Big)\Big| \vee 1} \\
&= \frac{\sum_{a=1}^{m_0} \mathbb{1}\big\{p'_a \leq t_{BH}\big(\{p'_b, 1 \leq b \leq m\}\big)\big\}}{\big|R_{BH}\big((p'_b)_{1\leq b\leq m}\big)\big| \vee 1} \qquad (4.9) \\
&= \text{FDP}\Big(R_{BH}\big((p'_a)_{1\leq a\leq m}\big), \{1, \dots, m_0\}\Big), \qquad (4.10)
\end{aligned}$$

where (4.9) is due to (4.8) and the definitions in Lemma 4.4.3.

Under Assumption 4.1, Theorem 3.3. of Marandon et al. (2025) entails that conditionally on W, the r.v. $(p'_a)_{1\leq a\leq m_0}$ are each marginally super-uniform and the r.v. $(p'_a)_{1\leq a\leq m}$ are PRDS on $\{1,\dots,m_0\}$. It follows from Benjamini and Yekutieli (2001) that the conditional expectation with respect to W of the right term in (4.10) is below α (thus this holds also for the unconditional expectation). This concludes the proof.

□

Furthermore, the same arguments as above (representing the distributions of null scores as a conditionally exchangeable distribution) also allow us to leverage recent results of Gazin et al. (2024) to get a control of the false discovery proportion (FDP) with high probability, and which hold uniformly over the choice of the rejection threshold.

Theorem 4.4.4. *Consider the same setting and assumptions as in Theorem 4.4.1, but instead of applying the BH procedure as the last step of Algorithm 2, denote $R(t)$ as the rejection set obtained as the set of edges having conformal p-value less than t, for $t\in[0,1]$.*
For any $\delta\in(0,1)$, with probability at least $1-\delta$ with respect to the sampling matrix Ω and the draw of $\mathcal{D}_{cal}$, it holds

$$P\left(\forall t\in[0,1]:\mathrm{FDP}(R(t))\leq\frac{m}{1\vee|R(t)|}\left(\frac{\lfloor(n+1)t\rfloor}{n+1}+\lambda(m,\ell,\delta)\right)\right)\geq 1-\delta, \tag{4.11}$$

where $\lambda(m,\ell,\delta)$ is the $1-\delta$ quantile of a certain universal probability distribution $P_{m,\ell}$, and satisfies

$$\lambda(m,\ell,\delta)\leq\left(\frac{\log\delta^{-1}+\log\left(1+2\sqrt{\pi}(m\wedge\ell)\right)}{m\wedge\ell}\right)^{\frac{1}{2}}. \tag{4.12}$$

Proof. Following the same argument as in the proof of Theorem 4.4.1, we use the fact that

$$\left((p_{i,j})_{j\in\mathcal{D}_{\text{test}}},\mathcal{H}_0\right)\overset{\mathcal{L}}{=}\left((\widetilde{p}_{i,j})_{j\in\mathcal{D}_{\text{test}}^{\sigma_0}},\mathcal{H}_0^{\sigma_0}\right).$$

Hence, since the event considered in the claim (4.11) only depends on $\left((p_{i,j})_{j\in\mathcal{D}_{\text{test}}},\mathcal{H}_0\right)$, its probability is unchanged when those are replaced by $\left((\widetilde{p}_{i,j})_{j\in\mathcal{D}_{\text{test}}^{\sigma_0}},\mathcal{H}_0^{\sigma_0}\right)$.

Consider now the p-values $(p'_i)_{1\leq i\leq m}$ given by explicit reindexing of $(\widetilde{p}_{ij})_{j\in\mathcal{D}_{\text{test}}^{\sigma_0}}$ via (4.7)-(4.8); again, since the event in (4.11) only depends on the p-values and on $\mathcal{H}_0$ as (multi)sets and not of the ordering of their indexation, we can use the reindexing given by $(p'_i)_{1\leq i\leq m}$ (and the corresponding $\mathcal{H}'_0=\{1,\dots,m_0\}$) without changing the probability. Since the reindexed p-values $(p'_i)_{1\leq i\leq m}$ can now be seen as conformal p-values based on exchangeable scores without ties $(S'_i)_{1\leq i\leq\ell+m}$, the claimed bound then holds, conditionally to W, as a direct application of Corollary 4.1 and Theorem 2.4 of Gazin et al. (2024) (with some straightforward estimates to simplify slightly the final expression). Thus the claimed probability bound also holds unconditionally. □

Remark 4.4.5. *Observe that the uniformity with respect to t allows for a post hoc (i.e. data-dependent) choice of the rejection threshold, rather than having to set a target level α for the FDR in advance, see Blanchard et al. (2020a). This is of particular interest since the output of a link prediction procedure is often presented to the user as a list of edges ordered by their scores rather than a fixed rejection set; the user can then monitor by themselves the guaranteed FDP bound in dependence of the threshold and decide of an adequate trade-off based on possibly additional criteria (e.g. concerning the geometry of rejected edges, or using additional information).*

From the results of Gazin et al. (2024), the "universal probability" $P_{m,\ell}$ appearing in Theorem 4.4.4 is that of the supremum deviation from identity of the empirical cdf for the color histogram of a Pólya urn scheme with ℓ draws from an urn initially containing $(m+1)$ balls of different colors. It can be easily approximated by Monte-Carlo simulation to get a sharper estimate. The explicit estimate (4.12) has the advantage of clearly showing an exponential concentration of order $\mathcal{O}\left(\sqrt{\log\left((m\wedge\ell)/\delta\right)/(m\wedge\ell)}\right)$. The bound of Gazin et al. (2024) also allows the replacement of m by a specific built-in estimate $\widehat{m}_0$ of m_0, and a slightly sharper estimate where the logarithmic dependence in $(n\wedge m)$ can be improved to an iterated logarithm. Both of these refinements are ignored here for simplicity.

4.5 Discussion

Comparison to Huang et al., 2023 : Our arguments on score exchangeability bears similarity to the reasoning of Huang et al., 2023, who considered conformal prediction intervals for label values on the nodes of a graph. A first (superficial) difference is that we predict on edges rather than nodes of the graph, and that we consider FDR and FDP for edge presence/absence rather than false coverage rate prediction intervals. We do not need an analogue of Assumption 1 of Huang et al., 2023, which assumes explicitly that the score function is invariant with respect to permutation of indices of the calibration+test set. Namely, this is in fact automatically satisfied in our setting, since the training function g applied to the masked observation data Z_{train} in Algorithm 2 does not have the information of which edges are in $\mathcal{D}_{\text{cal}}$ and which in $\mathcal{D}_{\text{test}}$.

More importantly, the nature of our results are different because, instead of fixing a threshold t and get guarantees for the FDR or FDP of $R(t)$ as in Huang et al., 2023, we consider either a (specific) data-dependant threshold t_{BH} which ensures a control of the FDR, or a bound uniform with respect to t on the FDP which allows to use any data-dependent threshold and still ensures valid bounds on the FDP. For a fixed beforehand threshold t, it is possible to follow the same line of argumentation we used to explicitly represent the scores as an exchangeable uple, and combine with the results of Marques F., 2023 instead of Marandon, 2024 or Gazin et al., 2024. In this case the same control as appearing in Huang et al., 2023 is recovered. See also Gazin et al., 2024 for the relation of the distribution $P_{m,\ell}$ to the results of Marques F., 2023; Huang et al., 2023.

Chapter 5

Heterogeneous compound p-values and closure principle

The closure principle applied to FDR control have been recently studied (Xu et al., 2025; Goeman, 2026) and has been shown to improve former FDR control methods, as the BY procedure in Xu et al. (2025) or the BH procedure in Goeman (2026). In this chapter, we show that the closure principle can be applied to heterogeneous compounds p-values, and therefore improve former FDR control methods designed for heterogeneous p-values, as the closed BY procedure in Xu et al. (2025). Hence considering heterogeneity within the closure principle derive less conservative methods. These heterogeneous compounds p-values allows to retrieve former FDR control methods and to extend their validity to more general settings.

Contents

5.1 Introduction . . . 100
5.1.1 Background . . . 100
5.1.2 Contributions . . . 101
5.2 Preliminaries . . . 101
5.2.1 Model and definitions . . . 101
5.2.2 DBY procedure (Döhler, 2018) and Heyse procedure (Heyse, 2011) 102
5.2.3 Compound p-values, compound e-values and calibrators . . . 102
5.2.4 Compound p-values and multiple testing framework . . . 103
5.2.5 The e-Closure Principle (Xu et al., 2025) for false discovery rate control 104
5.3 Heterogeneous compound p-values and Heyse and DBY procedures . . . 104
5.3.1 Heterogeneous locally compound p-values . . . 104
5.3.2 Applications to the heterogeneous multiple testing framework . . . 105
5.4 Heterogeneity and the e-Closure Principle . . . 106
5.4.1 Local heterogeneous e-values for the e-Closure Principle . . . 106
5.4.2 Closed DBY procedure . . . 107
5.5 Conclusion . . . 108

5.1 Introduction

5.1.1 Background

Closed testing procedures (Marcus et al., 1976) are a classical tool to control the family-wise error rate (FWER) in multiple hypothesis testing and is a useful device to construct confidence envelopes on the number of false discoveries (Goeman and Solari, 2011; Goeman et al., 2021). In a recent work, Xu et al. (2025) introduced the e-Closure Principle, a generalization of the closed testing procedure that allows to control any error rate coupled with a loss function using e-values, that is, nonnegative random variables with expectation less than or equal to 1 under the null hypothesis. In particular, they showed that the e-Closure Principle can be used to control the false discovery rate (FDR, Benjamini and Hochberg, 1995) and derived in particular a closed version of the Benjamini-Yekutieli (BY) procedure (Benjamini and Yekutieli, 2001). This closed version of the BY procedure has been shown to be less conservative than the BY procedure while still controlling the FDR under arbitrary dependence of the p-values.

Related to the work of Xu et al. (2025), the theory of compound p-values and e-values has been developed, notably in Ignatiadis et al. (2025) and Ramdas and Wang (2025). Thanks to this notion Barber and Samworth (2025) and Armstrong (2026) showed that the Benjamini-Hochberg (BH) procedure (Benjamini and Hochberg, 1995) is still valid for compound p-values up to a corrective factor on the target FDR level under various dependence structures, like independence, weak positive regression dependence on a subset (wPRDS Benjamini and Yekutieli, 2001) or unknown dependence. An interesting fact is that the corrective factor under unknown dependence is exactly the same as the one used in the BY procedure, despite using compound p-values instead of standard p-values (see Theorem 5.2.9).

This chapter aims to consider the case of heterogeneous p-values, that is p-values with different cumulative distribution functions (CDF), and uses tools from compound p-values, compound e-values and e-values multiple testing theory. This notion of heterogeneous p-values is particularly relevant in the context of multiple testing with discrete data. Heterogeneity in multiple testing has been investigated as a framework to correct the conservativeness induced by heterogeneous tests without external randomization since (Tarone, 1990) for the FWER control. It has been then considered for the FDR control (Gilbert, 2005; Heller and Gur, 2014; Habiger, 2015; Döhler et al., 2018), for the FDX control (Döhler and Roquain, 2020), and for confidence envelopes on the false discoveries (Périer et al., 2026). In addition to these works, Döhler (2018) introduced the DBY procedure, a generalization of the BY procedure to the heterogeneous framework, that allows to control the FDR under arbitrary dependence of the p-values, and Heyse (2011) introduced the Heyse procedure as a generalization of the BH procedure to the heterogeneous framework. These two procedures have in common that they respectively apply BY procedure and BH procedure to the same statistical object — that is, they consider the compound p-values from Proposition 5.3.1 — but the Heyse procedure does not control the FDR in general, as shown by Heller and Gur (2014) and Döhler (2018).

5.1.2 Contributions

Our results in this chapter rely on the core finding that we prove in this chapter: the statistical objects considered in Döhler (2018) and Heyse (2011) to describe DBY and Heyse procedure are compound p-values. This property notably allows to derive several findings of interest, in particular:

- Deduce a new and simple proof of the DBY procedure in Section 5.3.2,
- Bring a correction to the Heyse procedure to control the FDR in general in Section 5.3.2,
- Bring heterogeneity to the e-Closure Principle in Section 5.4.

5.2 Preliminaries

5.2.1 Model and definitions

Let us observe a random variable $X \sim P_0 \in \mathfrak{F}$ where $\mathfrak{F}$ is a family of probability distributions over some underlying space $\mathcal{X}$. We consider a family of $m \in \mathbb{N}$ null hypotheses, denoted $(H_i)_{i\in[\![1,m]\!]}$, that is a family of subsets of $\mathfrak{F}$. We say, for $P_0 \in \mathfrak{F}$ and $i \in [\![1,m]\!]$, that "P_0 satisfies H_i" (or that H_i is true) if $P_0 \in H_i$ and we denote $\mathcal{H}_0(P_0) = \{i \in [\![1,m]\!] : P_0 \in H_i\}$ the set of true null hypotheses under P_0. Furthermore, we say that a random variable $p_i(X)$ (the X dependency will be omitted in the remainder of the chapter) is a p-value for H_i if, for all $P \in H_i$, $\mathbb{P}_{X\sim P}(p_i \leq t) \leq t$ for all $t \in [0,1]$. Let $p_1, \dots, p_m$ be m p-values for the null hypotheses $H_1, \dots, H_m$ respectively, we recall here the main assumption on the marginal distributions of the p-values from the heterogeneous multiple testing framework:

Assumption 5.1 (Stochastic heterogeneous lower bound)**.** *For each $i \in [\![1,m]\!]$, there exists a known nondecreasing function F_i upper bounded by the identity function such that, if $P_0 \in H_i$, the cdf of p_i is upper bounded (as a function) by F_i. Mathematically,*

$$\forall P \in \mathfrak{F}, \forall i \in \mathcal{H}_0(P), \forall x \in \mathbb{R},\ \mathbb{P}\big(p_i(X) \leq x\big) \leq F_i(x) \leq x.$$

In this chapter, we are interested in controlling the FDR under Assumption 5.1, that is, for a multiple testing procedure $R(X) \subseteq [\![1,m]\!]$ (the X dependency will be omitted in the remainder of the chapter), the following constraint should hold for a prescribed level $\alpha \in (0,1)$:

$$\mathrm{FDR}(R) = \sup_{P\in\mathfrak{F}} \mathbb{E}\left[\frac{|R \cap \mathcal{H}_0(P)|}{|R| \vee 1}\right] \leq \alpha.$$

Classical multiple testing procedures, like the BH procedure (Benjamini and Hochberg, 1995) or the BY procedure (Benjamini and Yekutieli, 2001), are designed to control the FDR — under the wPRDS assumption for the BH procedure, and under arbitrary dependence for the BY procedure. The set rejected by the BH procedure at level $\alpha \in (0,1)$ is defined by

$$R_\alpha^{\mathrm{BH}} = \left\{i \in [\![1,m]\!] : p_i \leq \frac{\alpha k^*}{m}\right\},$$

where $k^* = \max\{k : p_{(k)} \leq \frac{\alpha k}{m}\} \cup \{0\}$. The set rejected by the BY procedure at level $\alpha \in (0,1)$ is defined by

$$R_\alpha^{\mathrm{BY}} = R_{\alpha/h_m}^{\mathrm{BH}},$$

where $h_m = \sum_{i=1}^m \frac{1}{i}$.

5.2.2 DBY procedure (Döhler, 2018) and Heyse procedure (Heyse, 2011)

This section introduces the DBY procedure and the Heyse procedure.

The first procedure, R_α^{DBY}, controls the FDR under arbitrary dependence of the p-values and is a generalization of the BY procedure. It consists in applying the BY procedure to $\left(\bar{J}(p_i)\right)_{i\in[\![1,m]\!]}$ where $\bar{J} = \frac{1}{m}\sum_{i=1}^m J_i$ where J_i is the *c.d.f.* of p_i when $X \sim P_0 \in H_i$.

Hence, this procedure requires that for all $P \in \mathfrak{F}$, for all $i \in \mathcal{H}_0(P)$, $p_i \sim J_i$ in opposition to Assumption 5.1 which allows to consider an upper bound F_i on the *c.d.f.* of the p-values under the null hypotheses — hence to consider composite null hypotheses.

The second procedure, the Heyse procedure R_α^H, is not theoretically true — even under independence of the p-values — and is a generalization of the BH procedure. It consists in applying the BH procedure to $\left(\bar{J}(p_i)\right)_{i\in[\![1,m]\!]}$.

5.2.3 Compound p-values, compound e-values and calibrators

This section introduces the notion of compound e-values (Ignatiadis et al., 2025; Ramdas and Wang, 2025) and calibrator Vovk and Wang (2021).

Definition 5.2.1 (p-to-e calibrator, Vovk and Wang, 2021)**.** *A nonincreasing function $g : [0,1] \to [0,+\infty]$ is a p-to-e calibrator if $\int_0^1 g \,\mathrm{d}\lambda \leq 1$, where λ is the Lebesgue measure.*

In the following, we call a p-to-e calibrator a *calibrator* since we only focus on converting p-values to e-values. Note that for any probability space, calibrator g and p-value p, $g(p)$ is an e-value.

A key example of calibrator, that will be used in the remainder of the chapter, is the Benjamini-Yekutieli calibrator:

Example 5.2.2 (BY calibrator, Xu et al., 2024b)**.** *Let $\alpha \in \mathbb{R}^*$, $s \in \mathbb{N}$. Define $h_s = \sum_{i=1}^s \frac{1}{i}$ and g_s^α, for all $x \in [0,1]$, by,*

$$g_s^\alpha(x) = \frac{s}{\alpha}\frac{\mathbb{1}\{h_s x \leq \alpha\}}{\lceil s h_s x/\alpha\rceil \vee 1}.$$

Then g_s^α is a calibrator.

This last example possesses the following property, that will be used in the remainder of the chapter:

Lemma 5.2.3. *Let $\alpha \in \mathbb{R}^*$, $s \in \mathbb{N}$, $x \in [0,1]$ and $c \in (0,1]$. Then $g_s^\alpha(cx) = \frac{1}{c} g_s^{\alpha/c}(x)$.*

We then define compound p-values. The idea behind compound p-values is that the individual constraints in distribution on the p-values are leveraged and replaced by a joint constraint.

Definition 5.2.4 (Compound p-values, Ignatiadis et al., 2025)**.** *Let $p_1,\dots,p_m$ be $[0,1]$-valued random variables. Then $p_1,\dots,p_m$ are compound p-values if the following holds :*

$$\forall t \in [0,1],\ \sup_{P\in\mathfrak{F}} \sum_{i\in\mathcal{H}_0(P)} \mathbb{P}_P(p_i \leq t) \leq mt.$$

In the same flavor as compound p-values, the individual constraints of compound e-values are replaced by a joint constraint.

Definition 5.2.5 (Compound e-values, Ignatiadis et al., 2025)**.** *Let $e_1, \dots, e_m$ be $[0, \infty]$-valued random variables. Then $e_1, \dots, e_m$ are compound e-values if the following holds :*

$$\sup_{P \in \mathfrak{F}} \sum_{i \in \mathcal{H}_0(P)} \mathbb{E}_P[e_i] \leq m.$$

Finally, Ignatiadis et al. (2025) proved the following link between compound p-values and compound e-values:

Proposition 5.2.6. *Let $p_1, \dots, p_m$ be compound p-values and g be a calibrator. Then $g(p_1), \dots, g(p_m)$ are compound e-values.*

However, to use these notions in the e-Closure Principle, we need to define a local version of compound p-values and compound e-values.

Definition 5.2.7 (S-locally compound p-values and e-values)**.** *Let $S \subseteq [\![1, m]\!]$. Let $(p_i)_{i \in S}$ be $[0, 1]$-valued random variables. Then $(p_i)_{i \in S}$ are S-locally compound p-values if the following holds :*

$$\forall t \in [0, 1], \ \sup_{P \in \mathfrak{F}} \sum_{i \in \mathcal{H}_0(P) \cap S} \mathbb{P}_P(p_i \leq t) \leq |S| t,$$

and $e_1, \dots, e_{|S|}$ are locally compound e-values if the following holds :

$$\sup_{P \in \mathfrak{F}} \sum_{i \in \mathcal{H}_0(P) \cap S} \mathbb{E}_P[e_i] \leq |S|.$$

Hence, S-locally compound p-values and e-values are compound p-values and e-values for the sub multiple testing model $(H_i)_{i \in S}$, that is the multiple testing model restricted to the hypotheses in S.

Obviously, Proposition 5.2.6 holds to convert locally compound p-values to locally compound e-values. Furthermore, S-locally compound e-values $e_1, \dots, e_{|S|}$ will define a local e-value $e_S = \frac{1}{|S|} \sum_{i=1}^{|S|} e_i$ for the intersection hypothesis H_S. See Definition 5.2.10 for the definition of local e-values.

5.2.4 Compound p-values and multiple testing framework

This section recalls the main results from Barber and Samworth (2025) and Armstrong (2026). The first result that we recall is that the BH procedure is still valid for compound p-values up to a corrective factor on the target FDR level under independence.

Theorem 5.2.8 (Barber and Samworth, 2025, Theorem 1)**.** *Let $p_1, \dots, p_m$ be independent compound p-values. Then, for all $\alpha \in (0, 1)$, the BH procedure at level α applied to $p_1, \dots, p_m$, R_α^{BH}, controls the FDR at level 1.93α, that is*

$$\mathrm{FDR}\big(R_\alpha^{\mathrm{BH}}\big) \leq 1.93\alpha.$$

Hence, from this result, we can deduce that the BH procedure applied to compound p-values at level $\alpha/1.93$, $R_{\alpha/1.93}^{\mathrm{BH}}$, controls the FDR at level α under independence. The second result that we recall is that the BY procedure is still valid for compound p-values, without corrective factor, under arbitrary dependence of the compound p-values.

Theorem 5.2.9 (Armstrong, 2026, Corollary 3.1)**.** *Let $p_1, \dots, p_m$ be compound p-values. Then, for all $\alpha \in (0, 1)$, the BY procedure at level α applied to $p_1, \dots, p_m$, R_α^{BY}, controls the FDR at level α, that is*

$$\mathrm{FDR}\big(R_\alpha^{\mathrm{BY}}\big) \leq \alpha.$$

5.2.5 The e-Closure Principle (Xu et al., 2025) for false discovery rate control

This section introduces the notion of e-Closure Principle considered by Xu et al. (2025). While this principle can be applied to any loss function $f_\cdot(\cdot) : \mathcal{P}(\llbracket 1, m \rrbracket) \times \mathcal{P}(\llbracket 1, m \rrbracket) \to \mathbb{R}_+$ (where $\mathcal{P}(\llbracket 1, m \rrbracket)$ is the power set of $\llbracket 1, m \rrbracket$) and its coupled error rate $\mathrm{ER}_f(\cdot) = \mathbb{E}[f_{\mathcal{H}_0}(\cdot)]$, we focus here on the special case from Xu et al. (2025) where for all $S, R \subseteq \llbracket 1, m \rrbracket$, $f_S(R) = \frac{|R \cap S|}{|R| \vee 1}$ and so $\mathrm{ER}_f = \mathrm{FDR}$.

We denote, for $S \subseteq \llbracket 1, m \rrbracket$, $H_S = \cap_{i \in S} H_i$ the intersection hypothesis for the subset S, and we state the folowing definition:

Definition 5.2.10. *A nonnegative variable e_S is a local e-value for S if, for all $P \in H_S$, $\mathbb{E}_P[e_S] \leq 1$.*

Finally, we say that $\mathbf{e} = (e_s)_{S \subseteq \llbracket 1, m \rrbracket}$ is an e-collection if for all $P \in \mathfrak{F}$, $\mathbb{E}_P\left[e_{\mathcal{H}_0(P)}\right] \leq 1$. Remark that a family of local e-values is an e-collection.

We finally state the main result from Xu et al. (2025): for any e-collection $\mathbf{e}$ and $\alpha \in (0, 1]$, denote

$$\mathcal{R}_\alpha(\mathbf{e}) = \left\{ R \subseteq \llbracket 1, m \rrbracket : \alpha e_S \geq \frac{|R \cap S|}{|R| \vee 1} \; \forall S \subseteq \llbracket 1, m \rrbracket \right\}. \tag{5.1}$$

Then, for all $P \in \mathfrak{F}$, $\mathbb{E}_P\left[\max_{R \in \mathcal{R}_\alpha(\mathbf{e})} \frac{|R \cap \mathcal{H}_0(P)|}{|R| \vee 1}\right] \leq \alpha$; hence for all random set $R \subseteq \llbracket 1, m \rrbracket$ such that $R \in \mathcal{R}_\alpha(\mathbf{e})$ *a.s.*, R has its FDR upper-bounded by α. We will say that $\mathcal{R}_\alpha(\mathbf{e})$ is a FDR *controlling procedure set at level* α. It notably allows to choose in a post hoc fashion the rejection set within $\mathcal{R}_\alpha(\mathbf{e})$.

This e-Closure Principle for FDR control has been used to derive the closed BY procedure (Xu et al., 2025) denoted $\mathcal{R}_\alpha^{\overline{\mathrm{BY}}}$. (Xu et al., 2025), from a set of p-values $p_1, \ldots, p_m$ with arbitrary dependence, defines, for all $S \subseteq \llbracket 1, m \rrbracket$, the local local e-values $e_S = \frac{1}{|S|} \sum_{i \in S} e_{S,i}^{\mathrm{BY}}$ where

$$e_{S,i}^{\mathrm{BY}} = \frac{|S|}{\alpha} \frac{\mathbb{1}\{h_{|S|} p_i \leq \alpha\}}{\lceil |S| h_{|S|} p_i / \alpha \rceil \vee 1} = g_{|S|}^\alpha(p_i). \tag{5.2}$$

This procedure has been shown to be a uniform improvement in simultaneity over the BY procedure R_α^{BY} (Benjamini and Yekutieli, 2001), in the sense that $R_\alpha^{\mathrm{BY}} \in \mathcal{R}_\alpha^{\overline{\mathrm{BY}}}$ *a.s.*.

5.3 Heterogeneous compound p-values and Heyse and DBY procedures

5.3.1 Heterogeneous locally compound p-values

This section introduces a family of locally compound p-values under heterogeneity assumption on the p-values.

Proposition 5.3.1. *Assume Assumption 5.1 holds and let $S \subseteq \llbracket 1, m \rrbracket$. Define*

$$\bar{F}^S = \frac{1}{|S|} \sum_{i \in S} F_i.$$

Then $\left(\bar{F}^S(p_i)\right)_{i \in S}$ are S-locally compound p-values.

Proof. Let $t \in [0,1]$ and $P \in \mathfrak{F}$. Denote, for $i \in [\![1,m]\!]$, by $F_{i,P}$ the *c.d.f.* of p_i and P_i the distribution of p_i when $X \sim P$. Furthermore, define $\bar{P}^T = \frac{1}{|T|}\sum_{i\in T} P_i$ and $\bar{F}_P^T = \frac{1}{|T|}\sum_{i\in T} F_{i,P}$ for any $T \subseteq [\![1,m]\!]$. Then, define $m_0^S = |\mathcal{H}_0 \cap S|$. If $m_0^S = 0$, the result is trivial. Otherwise, we have

$$\sum_{i\in\mathcal{H}_0\cap S} \mathbb{P}_P\big(\bar{F}^S(p_i) \le t\big) = \sum_{i\in\mathcal{H}_0\cap S} \mathbb{E}_P\big[\mathbb{1}\big\{\bar{F}^S(p_i) \le t\big\}\big] \tag{5.3}$$

$$= \sum_{i\in\mathcal{H}_0\cap S} \int \mathbb{1}\big\{\bar{F}^S(p) \le t\big\} \mathrm{d}P_i(p) \tag{5.4}$$

$$= m_0^S \int \mathbb{1}\big\{\bar{F}^S(p) \le t\big\} \mathrm{d}\bar{P}^{\mathcal{H}_0\cap S}(p) \tag{5.5}$$

$$\le m_0^S \int \mathbb{1}\left\{\frac{m_0^S}{|S|}\bar{F}^{\mathcal{H}_0\cap S}(p) \le t\right\} \mathrm{d}\bar{P}^{\mathcal{H}_0\cap S}(p) \tag{5.6}$$

$$= m_0^S \mathbb{P}_{p\sim\bar{P}^{\mathcal{H}_0\cap S}}\left(\bar{F}^{\mathcal{H}_0\cap S}(p) \le \frac{|S|}{m_0^S}t\right) \tag{5.7}$$

$$\le m_0^S \mathbb{P}_{p\sim\bar{P}^{\mathcal{H}_0\cap S}}\left(\bar{F}_P^{\mathcal{H}_0\cap S}(p) \le \frac{|S|}{m_0^S}t\right) \tag{5.8}$$

$$\le |S|t, \tag{5.9}$$

where (5.6) comes from the inequality $\frac{m_0^S}{|S|}\bar{F}^{\mathcal{H}_0\cap S} \le \bar{F}^S$, (5.8) occurs because $\bar{F}_P^{\mathcal{H}_0\cap S} \le \bar{F}^{\mathcal{H}_0\cap S}$ from Assumption 5.1, and (5.9) occurs because if p follows the distribution $\bar{P}^{\mathcal{H}_0\cap S}$, its *c.d.f.* is $\bar{F}_P^{\mathcal{H}_0\cap S}$ hence $\bar{F}_P^{\mathcal{H}_0\cap S}(p)$ is a superuniform random variable. □

5.3.2 Applications to the heterogeneous multiple testing framework

Application to the DBY procedure

This section is dedicated to the applications of Proposition 5.3.1 in the heterogeneous multiple testing litterature. The first application is that we provide another proof of the DBY procedure from Döhler (2018) and a relaxation of the hypotheses of Döhler (2018) to get a more general result. Indeed, Döhler (2018) only considered the case where the *c.d.f.* of the p-values under the null hypotheses are known, and the generalization of this proof to Assumption 5.1 is not straightforward. Here, we simply use **??** to show that the DBY procedure is valid under Assumption 5.1:

Theorem 5.3.2. *Assume Assumption 5.1 holds. Then, for all $\alpha \in (0,1]$, the DBY procedure at level α controls the FDR at level α, that is*

$$\mathrm{FDR}\big(R_\alpha^{\mathrm{DBY}}\big) \le \alpha.$$

Proof. Let $\alpha \in (0,1]$. From Proposition 5.3.1 with $S = [\![1,m]\!]$, $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ are compound p-values. Hence, from Theorem 5.2.9, the BY procedure applied to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ at level α controls the FDR at level α. Finally, the DBY procedure is exactly the BY procedure applied to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ so the result follows. □

Remark 5.3.3. *The result from Armstrong (2026), is more general than the one stated in Theorem 5.2.9 since it states that the general step-up procedures from Blanchard and Roquain (2008) — which include the particular case of the BY procedure — are still valid using compound p-values. Hence it allows to expand the result from Blanchard and Roquain (2008) to the heterogeneous multiple testing framework.*

Application to the Heyse procedure

This section is dedicated to the applications of Proposition 5.3.1 to the Heyse procedure from Heyse (2011). The Heyse procedure is a generalization of the BH procedure and consists in applying the BH procedure to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$. However, as shown by Heller and Gur (2014), the Heyse procedure does not control the FDR in general. Here, we show that a corrected version of the Heyse procedure controls the FDR under independence of the p-values and Assumption 5.1.

Theorem 5.3.4. *Assume Assumption 5.1 holds and that $p_1, \dots, p_m$ are independent. Let $\alpha \in (0,1]$ and denote by R_α^{c-H} the corrected Heyse procedure at level α, that is, the BH procedure applied to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ at level $\alpha/1.93$. Then, the c-Heyse procedure controls the FDR at level α, that is*

$$\mathrm{FDR}\big(R_\alpha^{c-H}\big) \leq \alpha.$$

Proof. Let $\alpha \in (0,1]$. From Proposition 5.3.1 with $S = [\![1,m]\!]$, $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ are compound p-values. Hence, from Theorem 5.2.8, the BH procedure applied to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ at level $\alpha/1.93$ controls the FDR at level α. Finally, the corrected Heyse procedure is exactly the BH procedure applied to $\big(\bar{F}(p_i)\big)_{i\in[\![1,m]\!]}$ at level $\alpha/1.93$ and the result follows. □

5.4 Heterogeneity and the e-Closure Principle

5.4.1 Local heterogeneous e-values for the e-Closure Principle

In this section, we derive from Proposition 5.3.1 a new procedure for heterogeneous data that improves the closed BY procedure from Xu et al. (2025). For $\alpha \in (0,1]$ fixed, we first need to define the following for all $S \subseteq [\![1,m]\!]$ and $i \in S$,

$$e_{S,i}^{\mathrm{HBY}} := \frac{|S|}{\alpha} \frac{\mathbb{1}\big\{h_{|S|}\bar{F}^S(p_i) \leq \alpha\big\}}{\big\lceil |S| h_{|S|} \bar{F}^S(p_i)/\alpha \big\rceil \vee 1} = g_{|S|}^\alpha\big(\bar{F}^S(p_i)\big), \tag{5.10}$$

where $g_{|S|}^\alpha$ is defined by Example 5.2.2. Hence the following property holds:

Proposition 5.4.1. *Assume Assumption 5.1 holds. Let $S \subseteq [\![1,m]\!]$ and $i \in S$. Then $\big(e_{S,i}^{\mathrm{HBY}}\big)_{i\in S}$ is a family of S-locally compound e-values. Therefore,*

$$e_S^{\mathrm{HBY}} := \frac{1}{|S|}\sum_{i\in S} e_{S,i}^{\mathrm{HBY}} \tag{5.11}$$

is a local e-value for S.

Proof. Using Proposition 5.3.1 for S and the fact that $g_{|S|}^\alpha$ is a calibrator, $\big(e_{S,i}^{\mathrm{HBY}}\big)_{i\in S}$ is a family of S-locally compound e-values by Proposition 5.2.6. Hence for all $P \in H_S \subseteq \mathfrak{F}$, $\mathbb{E}_P\big[e_S^{\mathrm{HBY}}\big] = \frac{1}{|S|}\sum_{i\in S}\mathbb{E}_P\big[e_{S,i}^{\mathrm{HBY}}\big] \leq 1$ and e_S^{HBY} is a local e-value for S. □

This last property allows us to define a family of local e-values $\mathbf{e}^{\mathrm{HBY}} = \big(e_S^{\mathrm{HBY}}\big)_{S\subseteq[\![1,m]\!]}$. This family, in addition to the framework described in Section 5.2.5, allows us to define the closed heterogeneous BY procedure, and to show that it is a FDR controlling procedure set at level α. Furthermore, we can show the following:

Theorem 5.4.2. *Assume Assumption 5.1 holds. Then, the closed heterogeneous* BY *procedure, denoted* $\mathcal{R}_\alpha^{\overline{\mathrm{HBY}}}$*, and define as* $\mathcal{R}_\alpha(\mathrm{e}^{\mathrm{HBY}})$*, is a uniform improvement in simultaneity over* $\mathcal{R}_\alpha^{\overline{\mathrm{BY}}}$*.*

Proof. This is a direct consequence of the inequality $e_{S,i}^{\mathrm{HBY}} = g_{|S|}^{\alpha}\big(\bar{F}^S(p_i)\big) \geq g_{|S|}^{\alpha}(p_i) = e_i^{\mathrm{BY}}$, for all $S \subseteq [\![1,m]\!]$ and $i \in S$ because $\bar{F}^S(p_i) \leq p_i$. Hence the inequality $e_S^{\mathrm{HBY}} \geq e_S^{\mathrm{BY}}$ on the average holds for all $S \subseteq [\![1,m]\!]$ and the result follows. □

5.4.2 Closed DBY procedure

This section is dedicated to derive from the e-closure principle a uniform improvement of the DBY procedure from Döhler (2018), that we recalled in Section 5.2.2.

For $\alpha \in (0,1]$ fixed, we define the following for all $S \subseteq [\![1,m]\!]$ and $i \in S$,

$$e_{S,i}^{\mathrm{DBY}} := \frac{|S|}{\alpha} \frac{\mathbb{1}\Big\{h_m \frac{|S|}{m} \bar{F}^S(p_i) \leq \alpha\Big\}}{\big\lceil |S| h_m \bar{F}^S(p_i)/\alpha \big\rceil \vee 1} = g_m^{\alpha m/|S|}\big(\bar{F}^S(p_i)\big), \tag{5.12}$$

where we recall that $\bar{F}^S = \frac{1}{|S|}\sum_{i\in S} F_i$.

The results from Section 5.3.1 allow us to show that the following property holds:

Proposition 5.4.3. *Assume Assumption 5.1 holds. Let* $S \subseteq [\![1,m]\!]$ *and* $i \in S$*. Then* $\big(e_{S,i}^{\mathrm{DBY}}\big)_{i\in S}$ *is a family of* S*-locally compound e-values. Therefore,*

$$e_S^{\mathrm{DBY}} := \frac{1}{|S|}\sum_{i\in S} e_{S,i}^{\mathrm{DBY}} \tag{5.13}$$

is a local e-value for S*.*

Proof. For all $i \in S$, $e_{S,i}^{\mathrm{DBY}} = g_m^{\alpha m/|S|}\big(\bar{F}^S(p_i)\big)$. Using Proposition 5.3.1 for S and the fact that $g_m^{\alpha m/|S|}$ is a calibrator, $\big(e_{S,i}^{\mathrm{DBY}}\big)_{i\in S}$ is a family of S-locally compound e-values. Hence, $e_S^{\mathrm{DBY}} = \frac{1}{|S|}\sum_{i\in S} e_{S,i}^{\mathrm{DBY}}$ is a local e-value for S. □

Proposition 5.4.3 allows us to define an FDR contolling procedure at level α, the closed DBY procedure, thanks to the tools presented in Section 5.2.5, as follows:

Theorem 5.4.4. *The closed DBY procedure, denoted* $\mathcal{R}_\alpha^{\overline{\mathrm{DBY}}} = \mathcal{R}_\alpha\Big(\big(e_S^{\mathrm{DBY}}\big)_{S\subseteq[\![1,m]\!]}\Big)$*, is a uniform improvement in simultaneity over* R_α^{DBY}*.*

Proof. Define, for all $i \in [\![1,m]\!]$,

$$\tilde{e}_i := g_m^{\alpha}\big(\bar{F}(p_i)\big) = \frac{m}{\alpha}\frac{\mathbb{1}\big\{h_m \bar{F}(p_i) \leq \alpha\big\}}{\big\lceil m h_m \bar{F}(p_i)/\alpha \big\rceil \vee 1},$$

where $\bar{F} = \frac{1}{m}\sum_{i=1}^m F_i$. It holds that $\tilde{e}_1, \ldots, \tilde{e}_m$ are compound e-values by Proposition 5.3.1 and Proposition 5.2.6. Thus these compound e-values define a local e-value for $S \subseteq [\![1,m]\!]$ as $\tilde{e}_S = \frac{1}{m}\sum_{i\in S}\tilde{e}_i$ and an e-collection with $\tilde{\mathbf{e}} = (\tilde{e}_S)_{S\subseteq[\![1,m]\!]}$.

For all $i \in S$, $\tilde{e}_i = g_m^\alpha\big(\bar{F}(p_i)\big) \leq g_m^\alpha\Big(\frac{|S|}{m}\bar{F}^S(p_i)\Big) = \frac{m}{|S|}e_{S,i}^{\mathrm{DBY}}$ by Lemma 5.2.3. Therefore

$$\begin{aligned}\tilde{e}_S = \frac{1}{m}\sum_{i\in S}\tilde{e}_i &\leq \frac{1}{m}\sum_{i\in S}\frac{m}{|S|}e_{S,i}^{\mathrm{DBY}} \\ &= \frac{1}{|S|}\sum_{i\in S}e_{S,i}^{\mathrm{DBY}} \\ &= e_S^{\mathrm{DBY}},\end{aligned}$$

and $\mathcal{R}_\alpha(\tilde{\mathbf{e}}) \subseteq \mathcal{R}_\alpha\big(\mathbf{e}^{\mathrm{DBY}}\big)$.

Now, we show that $R_\alpha^{DBY} \in \mathcal{R}_\alpha(\tilde{\mathbf{e}})$. Let $S \subseteq [\![1,m]\!]$. Then,

$$\begin{aligned}\alpha\tilde{e}_S &= \sum_{i\in S}\frac{\mathbb{1}\{h_m\bar{F}(p_i)\leq\alpha\}}{\lceil mh_m\bar{F}(p_i)/\alpha\rceil \vee 1} \\ &\geq \sum_{i\in S\cap R_\alpha^{DBY}}\frac{\mathbb{1}\{h_m\bar{F}(p_i)\leq\alpha\}}{\lceil mh_m\bar{F}(p_i)/\alpha\rceil \vee 1} \\ &= \sum_{i\in S\cap R_\alpha^{DBY}}\frac{1}{\lceil mh_m\bar{F}(p_i)/\alpha\rceil \vee 1} \\ &\geq \sum_{i\in S\cap R_\alpha^{DBY}}\frac{1}{|R_\alpha^{DBY}|} = \frac{\big|S\cap R_\alpha^{DBY}\big|}{|R_\alpha^{DBY}|},\end{aligned}$$

where the last inequality occurs because for all $i \in R_\alpha^{DBY}$, $\bar{F}(p_i) \leq \frac{|R_\alpha^{DBY}|\alpha}{mh_m}$.
Hence, $R_\alpha^{DBY} \in \mathcal{R}_\alpha(\tilde{\mathbf{e}})$. Remark that the last proof remains valid for all sets $R \subseteq [\![1,m]\!]$ such that, for all $i \in R$, $\bar{F}(p_i) \leq \frac{|R|\alpha}{mh_m}$, which can be considered as heterogeneous self consistent sets. For more details on self consistency and its key role in FDR control, see Blanchard and Roquain (2008).
Therefore, $R_\alpha^{DBY} \in \mathcal{R}_\alpha(\tilde{\mathbf{e}}) \subseteq \mathcal{R}_\alpha\big(\mathbf{e}^{\mathrm{DBY}}\big)$ and the result follows. □

Remark 5.4.5. *It is not straightforward whether the closed DBY procedure is a uniform improvement in simultaneity over the closed heterogeneous BY procedure. Indeed, the local e-values $e_{S,i}^{\mathrm{HBY}}$ and $e_{S,i}^{\mathrm{DBY}}$ are not comparable in general. This question is parallel to the question of the comparison of the different ways to derive e-values from p-values. Indeed, as shown by Vovk and Wang (2021), there is no uniformly best way to calibrate p-values into e-values, and, here, there is no uniformly better calibration of compound p-values into compound e-values.*

5.5 Conclusion

In this chapter, we have introduced the notion of heterogeneous locally compound p-values and e-values. We have shown that these notions allow to derive new procedures for heterogeneous multiple testing problems, that are uniform improvements in simultaneity over the existing procedures from Döhler (2018) and Xu et al. (2025). A by-product of our work is that we have provided a new proof of the DBY procedure from Döhler (2018) and a relaxation of the hypotheses of Döhler (2018) to get a more general result. Furthermore, we have brought a new perspective on the Heyse procedure from Heyse (2011) by providing a corrected version of the Heyse procedure that

controls the FDR under independence of the p-values and Assumption 5.1. Finally, we have derived a closed version of the DBY procedure from Döhler (2018) that is a uniform improvement in simultaneity over the DBY procedure. These two last results are of particular interest since they provide new procedures for heterogeneous multiple testing problems that control the FDR under independence or arbitrary dependence of the p-values, respectively. As it is usually the case when applying closed testing procedures, the computational cost of the procedures must be taken into account, and shortcuts to reduce the computational cost of this procedure are of particular interest. We leave this question for future work, but we note that some inexact shortcuts can be used to reduce the computational cost of these closed heterogeneous procedures, in the same spirit than the one provided by Périer et al. (2026).

Chapter 6

On strongly coherent confidence envelopes for the false discoveries

Coherence is a long term notion in the literature of multiple testing. From FWER control multiple testing theory to confidence envelopes on the number of false positives, coherence is a desirable property for multiple testing procedures. However, we show in this chapter that coherence is not a sufficient property to ensure that a multiple testing procedure is nonimprovable via an interpolation mechanism. We introduce a new notion, the notion of strong coherence, which is a stronger property than coherence and ensures that a multiple testing procedure is nonimprovable via the interpolation mechanisms considered in the literature. Finally, we provide a uniform improvement of the confidence envelope from Durand et al. (2020) that is computable in polynomial time and demonstrate the improvement on simulated data.

Contents

6.1 Introduction . . . 112
6.1.1 Motivations . . . 112
6.1.2 Model and notations . . . 112
6.2 Literature review . . . 113
6.2.1 Interpolation devices from the JER literature . . . 113
6.2.2 Interpolation device from Goeman et al. (2021) . . . 114
6.2.3 Discussion on Xu et al. (2024a) procedure . . . 115
6.3 Interpolation operators and strong coherence . . . 116
6.3.1 Interpolation operators from the JER literature . . . 116
6.3.2 Strong coherence . . . 118
6.4 Local confidence bounds: a uniform improvement of Durand et al. (2020) methods . . . 119
6.4.1 Considering local confidence bounds in Durand et al. (2020) methods 119
6.4.2 Simulations . . . 121
6.5 Conclusion . . . 123

6.1 Introduction

6.1.1 Motivations

Many modern statistical data analyses involve high-dimensional data, with a large number of variables and many questions of interest. Frequently, scientists are interested in identifying a subset of variables that are responsible of a particular phenomenon, hence, multiple hypothesis testing theory provides a natural framework to address this problem. While classical multiple hypothesis testing prescribes algorithms to control specific error metrics, scientists can be tempted to snoop into data to find a subset of variables that are more likely to be significant, and then report the results of the analysis on this subset. However, this practice can lead to a significant inflation of the error metrics and hence to misleading conclusions. In this work, we propose a new methodology relying on the one from Durand et al. (2020) to provide a valid uniform improvement of the confidence bounds from the latter article. We also provide a new theoretical notion, the notion of *strong coherence* for confidence envelopes. A confidence envelope is strongly coherent if it is non-improvable by the interpolation mechanism used in Blanchard et al. (2020a) in a sense defined in (6.8). This notion is stronger than the notion of coherence from Goeman et al. (2021), hence the name of strong coherence.

6.1.2 Model and notations

For a positive integer m, we denote by $[\![1, m]\!] = \{1, \dots, m\}$ and, for a set A, $\mathcal{P}(A)$ the power set of A. For two real numbers a and b, we denote by $a \wedge b$ the minimum of a and b.
We consider a statistical model $(\mathfrak{F}, \mathcal{X})$ where $\mathfrak{F}$ is a family of probability measures on a measurable space $(\mathcal{X}, \Xi)$. We denote by X an observed random variable taking values in $\mathcal{X}$ defined on an abstract probability space $(\Omega, \Sigma, \mathbb{P})$, such that the pushforward measure of X is in $\mathfrak{F}$. We will denote by $X \sim \mu$ or say that X follows the distribution μ when the pushforward measure of X is $\mu \in \mathfrak{F}$. The notations $\mathbb{E}_\mu[\cdot]$ and $\mathbb{P}_\mu(\cdot)$ denote the expectation and the probability when $X \sim \mu$. We consider $m \geq 1$ null hypotheses $H_1, \dots, H_m \subseteq \mathfrak{F}$. For all $\mu \in \mathfrak{F}$, we denote by $\mathcal{H}_0(\mu) = \{i \in [\![1, m]\!] : \mu \in H_i\}$ the set of true null hypotheses, and by $\mathcal{H}_1(\mu) = \mathcal{H}_0(\mu)^c$ the set of false null hypotheses and by $m_0(\mu) = |\mathcal{H}_0(\mu)|$ the number of true null hypotheses. The dependence in μ is omitted when it is clear from the context. Further, for all $S \subseteq [\![1, m]\!]$, we denote by $V(S) = |S \cap \mathcal{H}_0|$ the number of true null hypotheses in S. With the formalism introduced up to now, we define a simultaneous confidence bound for the number of false discoveries by the following functional:

$$\widehat{V} : \left\{ \begin{array}{ccc} \mathcal{X} \times (0,1) & \longrightarrow & \Big(\mathcal{P}\big([\![1, m]\!]\big) \to [\![0, m]\!]\Big) \\ (x, \alpha) & \longmapsto & \widehat{V}_{(x,\alpha)}(\cdot) \end{array} \right. ,$$

such that the following inequality is true:

$$\forall \mu \in \mathfrak{F}, \forall \alpha \in (0,1),\ \mathbb{P}_\mu\Big(\forall S \subseteq [\![1, m]\!], V(S) \leq \widehat{V}_{(X,\alpha)}(S)\Big) \geq 1 - \alpha. \tag{6.1}$$

The dependence in X and α is omitted when it is clear from the context, and $\widehat{V}_{(X,\alpha)}$ will be written as $\widehat{V}$. For simplicity, we assume that, for all $S \subseteq [\![1, m]\!]$, $\widehat{V}(S) \leq |S|$, as if $\widehat{V}$ is a valid confidence bound, then $\tilde{V}(\cdot) = \widehat{V}(\cdot) \wedge |\cdot|$ is also a valid confidence bound.

6.2 Literature review

6.2.1 Interpolation devices from the JER literature

The notion of interpolation to improve existing confidence envelopes was first introduced in Blanchard et al. (2020a) and Durand et al. (2020), although they were applied to particular confidence envelopes, as it will be specified in Section 6.3.1. In this section, we introduce these JER-based interpolation devices, and we give a simpler expression of the interpolation device from Durand et al. (2020). Let us recall, for a reference family $\mathfrak{R} = \big((R_k(X), \zeta_k(X))\big)_{k\in\mathcal{K}}$ where $\mathcal{K}$ is a finite set and for all $k \in \mathcal{K}$, $R_k \subseteq [\![1, m]\!]$ and $0 \leq \zeta_k \leq |R_k|$, the definition of the optimal bound $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ from Blanchard et al. (2020a) and Durand et al. (2020) — denoted $V^*_{\mathfrak{R}}$ in the aforementioned articles. $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ is defined, for all $S \subseteq [\![1, m]\!]$, by:

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \max_{A\in\mathcal{A}(\mathfrak{R})} |S \cap A|, \text{ where} \tag{6.2}$$

$$\mathcal{A}(\mathfrak{R}) = \{A \subseteq [\![1, m]\!] : \forall k \in \mathcal{K}, |A \cap R_k| \leq \zeta_k\}. \tag{6.3}$$

Another interpolation device was proposed by Durand et al. (2020). This interpolation device expression is given, for all $S \subseteq [\![1, m]\!]$, by

$$\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S) = \min_{Q\subseteq\mathcal{K}} \left(\sum_{k\in Q} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in Q} R_k \right| \right). \tag{6.4}$$

We can state an equivalent but simpler expression of $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$:

Lemma 6.2.1. *Let $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{k\in\mathcal{K}}$ be a reference family. Then, for all $S \subseteq [\![1, m]\!]$, we have*

$$\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S) = \min_{Q\subseteq\mathcal{K}} \left(\sum_{k\in Q} \zeta_k + \left| S \setminus \bigcup_{k\in Q} R_k \right| \right). \tag{6.5}$$

We prove below expression (6.5).

Proof. We show that there exists some $\bar{Q} \subseteq \mathcal{K}$ such that the inequality $|S \cap R_k| > \zeta_k$ holds for all $k \in \bar{Q}$ and $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S) = \sum_{k\in\bar{Q}} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in\bar{Q}} R_k \right|$.

Assume that $Q \subseteq \mathcal{K}$ is such that $\sum_{k\in Q} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in Q} R_k \right| = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S)$. Assume that for some $\ell \in Q$, $|R_\ell \cap S| \leq \zeta_\ell$ and consider $\tilde{Q} = Q \setminus \{\ell\}$. The inequality

$$\left| S \setminus \bigcup_{k\in Q} R_k \right| = \left| S \setminus \bigcup_{k\in\tilde{Q}} R_k \right| - \left| S \cap R_\ell \setminus \bigcup_{k\in\tilde{Q}} R_k \right| \geq \left| S \setminus \bigcup_{k\in\tilde{Q}} R_k \right| - |S \cap R_\ell|$$

holds. Hence, we have

$$\sum_{k\in Q} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in Q} R_k \right| \geq \sum_{k\in\tilde{Q}} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in\tilde{Q}} R_k \right|,$$

Hence, $\sum_{k\in\tilde{Q}} \zeta_k \wedge |S \cap R_k| + \left| S \setminus \bigcup_{k\in\tilde{Q}} R_k \right| = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S)$ and, by a reasoning by induction, we conclude that $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S)$ is attained for a $\bar{Q} \subseteq \mathcal{K}$ such that for all $k \in \bar{Q}$, $|S \cap R_k| > \zeta_k$. Hence, we have $\sum_{k\in\bar{Q}} \zeta_k \wedge |S \cap R_k| = \sum_{k\in\bar{Q}} \zeta_k$ and $\sum_{k\in\bar{Q}} \zeta_k + \left| S \setminus \bigcup_{k\in\bar{Q}} R_k \right| = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S)$, which concludes the proof.

□

This expression is useful in order to consider $Q \subseteq \mathcal{K}$ on which $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}(S)$ is attained. Note that Durand et al. (2020) have shown that $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}} \geq \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ for any reference family $\mathfrak{R}$, with an equality if $\mathfrak{R}$ has a forest structure, that is if for all $k, k' \in \mathcal{K}$, $R_k \cap R_{k'} \in \{\varnothing, R_k, R_{k'}\}$. Hence, the bound $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ performs better than $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ on the same information basis.
The idea behind these two bounds is that if we have a reference family $\mathfrak{R}$ such that its Joint Error Rate (JER) is controlled, that is

$$\mathrm{JER}(\mathfrak{R}) = \sup_{\mu \in \mathfrak{F}} \mathbb{P}_{X \sim \mu}(\exists k \in \mathcal{K}, |R_k \cap \mathcal{H}_0| > \zeta_k) \leq \alpha \tag{6.6}$$

for some prespecified $\alpha \in (0, 1)$, then $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ and $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ are valid confidence envelopes.

6.2.2 Interpolation device from Goeman et al. (2021)

This section is dedicated to introducing the interpolation device from Goeman et al. (2021), the notion of *coherence* defined in the aforementioned article and the notion of interpolation operator. We also show that the confidence envelopes $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ and $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ are coherent confidence envelopes.

Definition 6.2.2. *An interpolation operator is a deterministic mapping* $\mathrm{op} : \hat{V} \mapsto \mathrm{op}(\hat{V})$ *such that if* $\hat{V}$ *is a confidence envelope, then* $\mathrm{op}(\hat{V})$ *is also a confidence envelope.*

Goeman et al. (2021) proposed another notion of interpolation, dedicated to improve a given confidence envelope $\hat{V}$, that is an operator $\mathrm{op}_{\mathrm{GHS}} : \hat{V} \mapsto \mathrm{op}_{\mathrm{GHS}}(\hat{V})$ where, for all $S \subseteq [\![1, m]\!]$,

$$\mathrm{op}_{\mathrm{GHS}}(\hat{V})(S) = \min_{U \subseteq [\![1,m]\!]} \Big\{ \hat{V}(U) + \hat{V}(S \setminus U) \Big\}. \tag{6.7}$$

From this notion of interpolation, Goeman et al. (2021) defines a notion of *coherence* for confidence envelopes that we rephrase here in terms of interpolation operators:

Definition 6.2.3. *Let* $\hat{V}$ *be a confidence envelope. We say that* $\hat{V}$ *is coherent if* $\mathrm{op}_{\mathrm{GHS}}(\hat{V}) = \hat{V}$*, that is if for all* $S \subseteq [\![1, m]\!]$*,* $\hat{V}(S) = \min_{U \subseteq [\![1,m]\!]} \Big\{ \hat{V}(U) + \hat{V}(S \setminus U) \Big\}$.

Remark 6.2.4. *Xu et al. (2024a) and Goeman et al. (2021) results are expressed in a lower bound on the true discoveries fashion. However, it is easy to express them in term of confidence envelopes on the false discoveries since, if* $\hat{d}$ *is a confidence envelope on the true discoveries, then* $|\cdot| - \hat{d}$ *is a confidence envelope on the false discoveries since, with high probability, for all* $S \subseteq [\![1, m]\!]$*,* $|S \cap \mathcal{H}_0| = |S| - |S \cap \mathcal{H}_1| \leq |S| - \hat{d}(S)$*. This explains that* (6.7) *is expressed differently than in Goeman et al. (2021) and Xu et al. (2024a).*

We state below a core property on the confidence envelopes $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ and $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ from Durand et al. (2020) and Blanchard et al. (2020a).

Proposition 6.2.5. *The confidence envelopes* $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ *and* $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ *are coherent for any reference family* $\mathfrak{R}$*.*

The proof is provided in Appendix B.1.2. This property is important since it shows that the interpolation device from Goeman et al. (2021) will not improve further the confidence envelopes $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ and $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$.

6.2.3 Discussion on Xu et al. (2024a) procedure

This section discusses the framework from Xu et al. (2024a) and its relation with the JER-based interpolation devices. More precisely, we show that the confidence envelope $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ from Durand et al. (2020) is always better than the one from Xu et al. (2024a). Further, we show that the interpolation device from Goeman et al. (2021) to define the notion of coherence is not adequate because it is possible to build better confidence envelopes with the approach from the JER literature.
From its definition of a family of " focus sets" R_k and " local bounds" $\hat{V}_{R_k}$ for each $k \in \mathcal{K}$ — that is a family of sets $\left(R_k, k \in \mathcal{K}\right)$ and a family of functionals $\left(\hat{V}_{R_k}, k \in \mathcal{K}\right)$ such that for all $k \in \mathcal{K}$, $\hat{V}_{R_k}$ verifies Equation 6.1 with R_k instead of $[\![1, m]\!]$ — Xu et al. (2024a) have defined a confidence envelope $\hat{V}^{(0)}$ by $\hat{V}^{(0)}(R_k) = \hat{V}_{R_k}(R_k)$ for all $k \in \mathcal{K}$ and $\hat{V}^{(0)}(S) = |S|$ for all $S \subseteq [\![1, m]\!]$ such that $S \notin \{R_k, k \in \mathcal{K}\}$. Equivalently, they defined a reference family $\mathfrak{R} = \left(\left(R_k, \hat{V}_{R_k}(R_k)\right)\right)_{k \in \mathcal{K}}$, since no other information from the local bounds $\hat{V}_{R_k}, k \in \mathcal{K}$ is used than the one given on the R_k. Hence, the name " focus sets" is another name for the already existing reference family concept. Then, Xu et al. (2024a) defined by induction a sequence of confidence envelopes, for all $i \in [\![1, N]\!]$, for all $S \subseteq [\![1, m]\!]$, by $\hat{V}^{(i+1)}(S) = \min_{U \subseteq [\![1,m]\!]} \left\{\hat{V}^{(i)}(U) + \hat{V}^{(i)}(S \setminus U)\right\}$ — that is $\hat{V}^{(i+1)} = \mathrm{op}_{\mathrm{GHS}}(\hat{V}^{(i)})$ — and the confidence envelope of interest is $\hat{V}^{(r)}$ where r is such that $\hat{V}^{(r)} = \hat{V}^{(r+1)}$ — hence a fixed point for $\mathrm{op}_{\mathrm{GHS}}$, that is a *coherent* confidence envelope. Remark that the integer r is finite since the sequence is nonincreasing and bounded, and that r is stochastic.
From Proposition 6.2.5, we can show the following proposition, which states that the bound $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ is always as good as $\hat{V}^{(r)}$.

Proposition 6.2.6. *We have the following inequality :*

$$\hat{V}_{\mathfrak{R}}^{\mathrm{JER}} \leq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}} \leq \hat{V}^{(r)}.$$

Proof. We show that $\hat{V}^{(0)} \geq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$. Indeed, by definition of $\hat{V}^{(0)}$, we have $\hat{V}^{(0)}(R_k) = \zeta_k \geq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}(R_k)$ for all $k \in \mathcal{K}$ taking $Q = \{k\}$ in (6.5), and $\hat{V}^{(0)}(S) = |S| \geq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}(S)$ if $S \subseteq [\![1, m]\!]$ is not a reference region. As $\hat{V}^{(0)} \geq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ and the " $\geq$ " relation on confidence envelopes is stable by $\mathrm{op}_{\mathrm{GHS}}$, we have $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}} \leq \hat{V}^{(r)}$ because $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ is coherent by Proposition 6.2.5. □

Remark 6.2.7. *Xu et al. (2024a) used the Holm-Based Combination, introduced in Durand (2018), as a slight refinement of the Bonferroni-Based Combination, which is the tool used in Durand et al. (2020) to build the reference family based on DKWM inequality. However, this improvement is over the reference family — equivalently over $\hat{V}^{(0)}$ — but not over the interpolation operator.*

Remark 6.2.8. *Proposition 6.2.6 also shows that the interpolation device from Goeman et al. (2021) is not optimal in the sense that with the same information (i.e. the bound on the number of false positive in the reference regions, or " focus sets " in Xu et al. (2024a) terminology), it is possible to build a better confidence envelope from Durand et al. (2020) and Blanchard et al. (2020a) via the confidence envelope $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}$ because $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}} \leq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ and Durand et al. (2020) provide an example where $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S) < \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}(S)$ for some S.*

We provide here the aforementioned example for completeness.

Example 6.2.9. *Let* $[\![1,m]\!] = \{1,2,3,4\}$ *and* $\mathcal{K} = \{1,2,3\}$ *with* $R_1 = \{1,2,4\}$, $R_2 = \{2,3,4\}$ *and* $R_3 = \{1,3,4\}$. *Let* $\zeta_1 = \zeta_2 = \zeta_3 = 1$. *Then, for* $S = [\![1,m]\!]$, *we have* $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S) = 2$ *and* $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = 1$.

Hence there is a gap in the notion of coherent confidence envelope. It is possible, with the same amount of information, to construct a better confidence envelope with an other interpolation device than $\mathrm{op}_{\mathrm{GHS}}$. The next section is devoted to tackle this issue by introducing new interpolation operators relying on the JER literature. These new interpolation operators will be used to define a new notion of coherence, called *strong coherence*, which is stronger than the notion of coherence from Goeman et al. (2021).

6.3 Interpolation operators and strong coherence

6.3.1 Interpolation operators from the JER literature

This section is dedicated to introduce the interpolation operators derived from the confidence envelopes $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ and $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ from Blanchard et al. (2020a) and Durand et al. (2020). Let us first introduce, for a given confidence envelope $\hat{V}$, the reference family $\mathfrak{R}_{\hat{V}} = \Big(\big(S, \hat{V}(S)\big)\Big)_{S\subseteq[\![1,m]\!]}$ that we call the reference family associated to $\hat{V}$. We define then two new interpolation operators, $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$, by the following equations:

$$\mathrm{op}_{\mathrm{JER}}\big(\hat{V}\big)(S) = \max_{A\in\mathcal{A}(\mathfrak{R}_{\hat{V}})} |S\cap A| = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}}}(S), \tag{6.8}$$

$$\mathrm{op}_{\mathrm{DBNR}}\big(\hat{V}\big)(S) = \min_{\mathcal{B}\subseteq\mathcal{P}([\![1,m]\!])} \left(\sum_{T\in\mathcal{B}} \hat{V}(T) + \left|S\setminus\bigcup_{T\in\mathcal{B}} T\right|\right) = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}}}(S). \tag{6.9}$$

Note that the expression of $\mathrm{op}_{\mathrm{DBNR}}$ derives from (6.5), and that in this special case we can again simplify the expression of $\mathrm{op}_{\mathrm{DBNR}}$:

Lemma 6.3.1. *Let* $\hat{V}$ *be a confidence envelope. Then, for all* $S\subseteq[\![1,m]\!]$, *we have*

$$\mathrm{op}_{\mathrm{DBNR}}\big(\hat{V}\big)(S) = \min_{\substack{\mathcal{B}\subseteq\mathcal{P}([\![1,m]\!])\\ s.t.\ S\subseteq\bigcup_{T\in\mathcal{B}} T}} \sum_{T\in\mathcal{B}} \hat{V}(T). \tag{6.10}$$

We provide below the proof of this expression.

Proof. Let $\mathcal{B}\subseteq\mathcal{P}([\![1,m]\!])$. Then consider $\tilde{\mathcal{B}} = \mathcal{B}\cup\big\{S\setminus\bigcup_{T\in\mathcal{B}} T\big\}$ and we have

$$\begin{aligned}\sum_{T\in\tilde{\mathcal{B}}} \hat{V}(T) + \left|S\setminus\bigcup_{T\in\tilde{\mathcal{B}}} T\right| &\leq \sum_{T\in\mathcal{B}} \hat{V}(T) + \hat{V}\left(S\setminus\bigcup_{T\in\mathcal{B}} T\right)\\ &\leq \sum_{T\in\mathcal{B}} \hat{V}(T) + \left|S\setminus\bigcup_{T\in\mathcal{B}} T\right|\end{aligned}$$

where the inequality comes from the fact that $\hat{V}(T)\leq|T|$ for all $T\subseteq[\![1,m]\!]$. Hence, we can restrict the minimization to the sets $\tilde{\mathcal{B}}$ such that $S\setminus\bigcup_{T\in\tilde{\mathcal{B}}} T = \varnothing$, which gives the result. □

These two interpolation operators, in the same spirit as the interpolation operator $\mathrm{op}_{\mathrm{GHS}}$ from Goeman et al. (2021), improve a given confidence envelope $\hat{V}$. It is a consequence of the following proposition.

Proposition 6.3.2. *For all confidence envelope $\hat{V}$, we have*

$$\mathrm{op}_{\mathrm{JER}}(\hat{V}) \leq \mathrm{op}_{\mathrm{DBNR}}(\hat{V}) \leq \mathrm{op}_{\mathrm{GHS}}(\hat{V}) \leq \hat{V}.$$

Proof. The first inequality is a consequence of Lemma 1 from Durand et al. (2020) and the second one comes from Expression (6.10) with $\mathcal{B} = \{U, S \setminus U\}$. The last inequality is a consequence of the definition of $\mathrm{op}_{\mathrm{GHS}}$ (6.7) with $U = S$. □

Contrary to the interpolation operator $\mathrm{op}_{\mathrm{GHS}}$, the interpolation operators $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$ are idempotent operators. It means that if we apply the interpolation operator $\mathrm{op}_{\mathrm{JER}}$ or $\mathrm{op}_{\mathrm{DBNR}}$ to a confidence envelope $\hat{V}$, then we cannot improve further the confidence envelope $\hat{V}$ by applying again the same interpolation operator. The following proposition states that property.

Proposition 6.3.3. *The operators $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$ are idempotent operators, that is for all confidence envelope $\hat{V}$, $\mathrm{op}_{\mathrm{JER}}\Big(\mathrm{op}_{\mathrm{JER}}(\hat{V})\Big) = \mathrm{op}_{\mathrm{JER}}(\hat{V})$ and $\mathrm{op}_{\mathrm{DBNR}}\Big(\mathrm{op}_{\mathrm{DBNR}}(\hat{V})\Big) = \mathrm{op}_{\mathrm{DBNR}}(\hat{V})$.*

The proof is deferred to Appendix B.1.1.

Let us define the operator $\mathrm{op}_{\mathrm{XSG}}$ by a rule similar to the one from Xu et al. (2024a), that is, for all confidence envelope $\hat{V}$, we define $\mathrm{op}_{\mathrm{XSG}}(\hat{V}) = \lim_{r\to\infty} \mathrm{op}^r_{\mathrm{GHS}}(\hat{V})$. $\mathrm{op}_{\mathrm{XSG}}$ exists by monotonicity of $\mathrm{op}_{\mathrm{GHS}}$ (Proposition 6.3.2) and because the set of confidence envelopes is finite. Then, for all confidence envelope $\hat{V}$, we have $\mathrm{op}_{\mathrm{XSG}}(\hat{V}) \geq \mathrm{op}_{\mathrm{DBNR}}(\hat{V})$ by Proposition 6.3.2 which gives that $\mathrm{op}_{\mathrm{DBNR}}$ is nondecreasing and that $\mathrm{op}_{\mathrm{DBNR}} \leq \mathrm{op}_{\mathrm{GHS}}$, and the idempotency of $\mathrm{op}_{\mathrm{DBNR}}$ (Proposition 6.3.3).

Recall here the definition of the confidence envelope $\hat{V}^{(0)}_{\mathfrak{R}}$ from Xu et al. (2024a) for a reference family $\mathfrak{R} = \Big((R_k, \zeta_k)\Big)_{k\in\mathcal{K}}$, that is $\hat{V}^{(0)}_{\mathfrak{R}}(R_k) = \zeta_k$ for all $k \in \mathcal{K}$ and $\hat{V}^{(0)}_{\mathfrak{R}}(S) = |S|$ for all $S \subseteq [\![1, m]\!]$ such that $S \notin \{R_k, k \in \mathcal{K}\}$.

An other fundamental property is that the interpolation operators $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$ are a generalization of the confidence envelopes $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ and $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ from Blanchard et al. (2020a) and Durand et al. (2020).

Proposition 6.3.4. *For all reference family $\mathfrak{R} = \Big((R_k, \zeta_k)\Big)_{k\in\mathcal{K}}$, we have $\mathrm{op}_{\mathrm{JER}}(\hat{V}^{(0)}_{\mathfrak{R}}) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ and $\mathrm{op}_{\mathrm{DBNR}}(\hat{V}^{(0)}_{\mathfrak{R}}) = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$.*

The proof is provided in Appendix B.1.3.

From Proposition 6.3.4 and Example 6.2.9, we have an example of a confidence envelope $\hat{V}^{(0)}_{\mathfrak{R}}$ such that $\mathrm{op}_{\mathrm{JER}}(\hat{V}^{(0)}_{\mathfrak{R}}) \neq \mathrm{op}_{\mathrm{DBNR}}(\hat{V}^{(0)}_{\mathfrak{R}}) \leq \hat{V}^{(r)}$, hence the interpolation operators $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$ are different. Furthermore, it provides an example of a confidence envelope $\hat{V}^{(0)}_{\mathfrak{R}}$ such that $\mathrm{op}_{\mathrm{DBNR}}(\hat{V})$ is a coherent confidence envelope, but cannot be written as a closed testing procedure. This example is detailed in Appendix B.2. It implies that Corollary 1 from Goeman et al. (2021) is invalid, and that the notion of *coherence* is not enough to define a notion of admissibility. A detailed inspection needs to be performed on the collection of claims from Goeman et al. (2021). Still, we believe that the core message of Goeman et al. (2021), namely that only closed testing procedure are admissible, remains valid with the notion of strong coherence, that we introduce in the next section, instead of the notion of coherence.

Finally, we can show that the range of the interpolation operators $\mathrm{op}_{\mathrm{JER}}$ and $\mathrm{op}_{\mathrm{DBNR}}$ lies in the set of coherent confidence envelopes:

Proposition 6.3.5. *We have* $\mathrm{op}_{\mathrm{GHS}}(\mathrm{op}_{\mathrm{JER}}) = \mathrm{op}_{\mathrm{JER}}$ *and* $\mathrm{op}_{\mathrm{GHS}}(\mathrm{op}_{\mathrm{DBNR}}) = \mathrm{op}_{\mathrm{DBNR}}$.

Proof. The proof is a direct consequence of Proposition 6.2.5 and the fact that $\mathrm{op}_{\mathrm{JER}}(\hat{V}) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}}}$ and $\mathrm{op}_{\mathrm{DBNR}}(\hat{V}) = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}}}$. □

6.3.2 Strong coherence

We define here the notion of strong coherence for confidence envelopes. This notion is stronger than the one of *coherence* from Goeman et al. (2021), and provides a correction to Corollary 1 from Goeman et al. (2021).

Definition 6.3.6. *Let* $\hat{V}$ *be a confidence envelope. We say that* $\hat{V}$ *is strongly coherent if* $\mathrm{op}_{\mathrm{JER}}(\hat{V}) = \hat{V}$*, that is if for all* $S \subseteq [\![1, m]\!]$, $\hat{V}(S) = \max_{A \in \mathcal{A}(\mathfrak{R}_{\hat{V}})} |S \cap A|$.

The following proposition shows that the notion of strong coherence is indeed stronger than the notion of coherence from Goeman et al. (2021).

Proposition 6.3.7. *If a confidence envelope* $\hat{V}$ *is strongly coherent, then it is coherent.*

Proof. Let $\hat{V}$ be a strongly coherent confidence envelope. Then, $\hat{V} = \mathrm{op}_{\mathrm{JER}}(\hat{V}) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}}}$ hence by Proposition 6.2.5, $\hat{V}$ is coherent. □

Hence, every strongly coherent confidence envelope is coherent, but the converse is not true, as we have already seen in Example 6.2.9. Then, we show that a confidence envelope $\hat{V}$ is strongly coherent if and only if $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ for some reference family $\mathfrak{R}$.

Proposition 6.3.8. *Let* $\hat{V}$ *be a confidence envelope. Then,* $\hat{V}$ *is strongly coherent if and only if there exists a reference family* $\mathfrak{R}$ *such that* $\hat{V} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$.

Proof. Assume $\hat{V}$ is a strongly coherent confidence envelope. Then, $\hat{V} = \mathrm{op}_{\mathrm{JER}}(\hat{V}) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}}}$ hence by taking $\mathfrak{R} = \mathfrak{R}_{\hat{V}}$, we have $\hat{V} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$. Assume now that there exists a reference family $\mathfrak{R}$ such that $\hat{V} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$. Then, Lemma B.1.1 concludes that $\hat{V} = \mathrm{op}_{\mathrm{JER}}(\hat{V})$. □

This previous proposition allows us to show that the space of closed testing procedures is the same as the space of strongly coherent confidence envelopes. We recall here the closed testing procedure for the confidence envelope on false positives associated to a family of local tests $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$, as defined in Goeman et al. (2021).

Definition 6.3.9. *Let* $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$ *be a family of local tests. The closed testing procedure associated to* φ *is the confidence envelope* $\hat{V}^{\mathrm{CT}}_{\varphi}$ *defined by*

$$\hat{V}^{\mathrm{CT}}_{\varphi}(S) = \max_{A \subseteq S} \left\{ |A|, \inf_{U \supseteq A} \varphi_U = 0 \right\}. \tag{6.11}$$

Proposition 6.3.10. *Let* $\hat{V}$ *be a confidence envelope. Then,* $\hat{V}$ *is a closed testing procedure if and only if it is strongly coherent.*

Proof. Assume $\hat{V}$ is a closed testing procedure. Then, by Blanchard et al. (2020b), Section S-1.3, there exists a reference family $\mathfrak{R}$ such that $\hat{V} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$. By Proposition 6.3.8, we have that $\hat{V}$ is strongly coherent. Assume now that $\hat{V}$ is strongly coherent. Then, by Proposition 6.3.8, there exists a reference family $\mathfrak{R}$ such that $\hat{V} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$. By Blanchard et al. (2020b), Section S-1.3, we have that $\hat{V}$ is a closed testing procedure with the special choice of local tests $\varphi_S = 0$ if and only if $S \in \mathcal{A}(\mathfrak{R})$. □

Hence this notion of strong coherence allows us to correct Corollary 1 from Goeman et al. (2021) by introducing the notion of strong coherence instead of the notion of coherence. From this last proposition, we have an alternative proof to show that the bound given in Example 6.2.9 cannot be written as a closed testing procedure, since, in this case, $\hat{V}_{\mathfrak{R}}^{DBNR}$ is not strongly coherent.

6.4 Local confidence bounds: a uniform improvement of Durand et al. (2020) methods

6.4.1 Considering local confidence bounds in Durand et al. (2020) methods

In this section, we show that the confidence envelope $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ from Durand et al. (2020) can be improved by considering, for all $k \in \mathcal{K}$, a local confidence bound $\hat{V}_k$, on the reference region R_k instead of only the number of false positive in R_k as it was done in the original work Durand et al. (2020) and in Xu et al. (2024a). These two confidence envelopes, $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ and $\hat{V}^{(r)}$ only use the upper bound on the number of false positive $V(R_k)$ in the reference regions given by the local confidence bounds. We can improve upon this by using the local confidence bounds, with $S \subseteq R_k$, to bound $V(S \cap R_k)$.
The main idea is that for top-k and k-FWER confidence bounds methods, following the analysis from Périer et al. (2026) concerning adaptive top-k and k-FWER methods, the event used to compute the number of false positive in R_k — that is $\zeta_k = \hat{V}_k(R_k)$ — provides, for free, a local top-k structure within R_k. Hence, instead of only considering the number of false positive in R_k, we can consider the local confidence bounds $\hat{V}_k(S \cap R_k)$ for all $S \subseteq R_k$.
Moreover, the computation of those local confidence envelopes usually requires the same computational cost than the cost to compute ζ_k, since it is mandatory to prior compute ζ_k to get $\hat{V}_k(S \cap R_k)$ for some $S \subseteq R_k$.
The following proposition shows that if, for all $k \in \mathcal{K}$, we have local confidence bounds on the reference regions R_k at level α_k, such that $\sum_{k \in \mathcal{K}} \alpha_k$ is upper bounded by α, the informal construction above provides a confidence envelope $\tilde{V}$ that is an α-confidence envelope.

Proposition 6.4.1. *Let $(R_k)_{k \in \mathcal{K}}$ be reference regions, let $(\alpha_k)_{k \in \mathcal{K}}$ such that $\sum_{k \in \mathcal{K}} \alpha_k \leq \alpha$, and for all $k \in \mathcal{K}$, let $\hat{V}_k$ be a local α_k-confidence envelope over R_k, meaning that with probability $1 - \alpha_k$, for all $S \subseteq R_k$, $V(S) \leq \hat{V}_k(S)$.*
Then the bound $\tilde{V}$ defined by

$$\tilde{V}(S) = \min_{Q \subseteq \mathcal{K}} \left(\sum_{k \in Q} \hat{V}_k(S \cap R_k) + \left| S \setminus \bigcup_{k \in Q} R_k \right| \right) \tag{6.12}$$

is an α-confidence envelope.
Further, if for all $k \in \mathcal{K}$, $\hat{V}_k(S \cap R_k) \leq \hat{V}_k(R_k)$, then $\tilde{V}$ is a uniform improvement of the confidence envelope $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$, where $\mathfrak{R} = \left(\left(R_k, \hat{V}_k(R_k) \right) \right)_{k \in \mathcal{K}}$.

Proof. • Let $A = \bigcup_{k \in \mathcal{K}} \left\{ \exists S \subseteq R_k : V(S) > \hat{V}_k(S \cap R_k) \right\}$. By union bound, $\mathbb{P}(A) \leq \sum_{k \in \mathcal{K}} \alpha_k \leq \alpha$.

- Within the event A^c, for all $Q \subseteq \mathcal{K}$,

$$V(S) \leq \sum_{k \in Q} |R_k \cap S \cap \mathcal{H}_0| + \left| (S \cap \mathcal{H}_0) \setminus \bigcup_{k \in Q} R_k \right|$$
$$\leq \sum_{k \in Q} \hat{V}_k(S \cap R_k) + \left| S \setminus \bigcup_{k \in Q} R_k \right|.$$

  The first part of this proposition follows by considering the minimum over this inequality.

The second part of the proposition comes by comparing the definition of $\tilde{V}$ and $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$. □

This confidence envelope is naturally a uniform improvement of the confidence envelope $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ from Durand et al. (2020) since for all $k \in \mathcal{K}$, $\hat{V}_k(R_k) \geq \hat{V}_k(R_k \cap S)$ for all $S \subseteq [\![1, m]\!]$ when the local bounds are coherent by Lemma 3 from Goeman et al. (2021) — as are the top-k and k-FWER methods described in Périer et al. (2026) since they are strongly coherent hence coherent by Proposition 6.3.7. Thus considering these underlying structures will always be better than only bounding the number of false positive in R_k by $\hat{V}_k(R_k)$.

The following proposition shows that, when the local confidence envelopes are provided by the JER constructions, the confidence envelope $\tilde{V}$ from Proposition 6.4.1 is lower bounded by the strongly coherent associated JER envelope. However, this strongly coherent JER envelope is not computable in general, while $\tilde{V}$ is computable if the reference regions have a forest structure, see Algorithm 1 from Durand et al. (2020).

Proposition 6.4.2. *Let $(R_k)_{k \in \mathcal{K}}$ be reference regions, and, for all $k \in \mathcal{K}$, $\mathfrak{R}_k = (R_{k,i}, \zeta_{k,i})_{i \in \mathcal{K}_k}$ such that for all $i \in \mathcal{K}_k$, $R_{k,i} \subseteq R_k$. Then, defining $\tilde{\mathcal{K}} = \bigcup_{k \in \mathcal{K}} \{(k, i), i \in \mathcal{K}_k\}$ and $\tilde{\mathfrak{R}} = (R_{k,i}, \zeta_{k,i})_{(k,i) \in \tilde{\mathcal{K}}}$,*

$$\hat{V}^{\mathrm{JER}}_{\tilde{\mathfrak{R}}}(S) \leq \min_{Q \subseteq \mathcal{K}} \left(\sum_{k \in Q} \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_k}(S \cap R_k) + \left| S \setminus \bigcup_{k \in Q} R_k \right| \right). \tag{6.13}$$

Proof. Let $A \in \mathcal{A}(\tilde{\mathfrak{R}}) = \left\{ A \subseteq [\![1, m]\!] : \forall (k, i) \in \tilde{\mathcal{K}}, |A \cap R_{k,i}| \leq \zeta_{k,i} \right\}$ and $Q \subseteq \mathcal{K}$. Then, for all $k \in \mathcal{K}$, $A \cap R_k \in \left\{ B \subseteq R_k : \forall i \in \tilde{\mathcal{K}}, |A \cap R_{k,i}| \leq \zeta_{k,i} \right\}$, thus

$$|A \cap S| = \sum_{k \in Q} |S \cap R_k \cap A| + \left| S \cap A \setminus \bigcup_{k \in Q} R_k \right|$$
$$\leq \sum_{k \in Q} \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_k}(S \cap R_k) + \left| S \setminus \bigcup_{k \in Q} R_k \right|.$$

It implies $\hat{V}^{\mathrm{JER}}_{\tilde{\mathfrak{R}}}(S) \leq \sum_{k \in Q} \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_k}(S \cap R_k) + \left| S \setminus \bigcup_{k \in Q} R_k \right|$ and the inequality follows taking the minimum over Q. □

Remark 6.4.3. *As it is the case for $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$, if the reference regions $(R_k)_{k \in \mathcal{K}}$ have a complete forest structure — that is the union of the regions equals $[\![1, m]\!]$ and that each node from the forest structure is either a leaf or partitioned by its children (i.e., the sets $R_\ell \subsetneq R_k$) — then*

the minimum over $Q \subseteq \mathcal{K}$ in (6.12) and (6.13) can be restricted to a minimum over the sets $Q \subseteq \mathcal{K}$ such that $(R_k)_{k\in Q}$ is a partition of $[\![1, m]\!]$. Practically, for a specific set $S \subseteq [\![1, m]\!]$, we are applying the Algorithm 1 from Durand et al. (2020) with $\hat{V}_k(S \cap R_k)$ instead of ζ_k for all $k \in \mathcal{K}$.

6.4.2 Simulations

In this section, we provide simulations to illustrate the improvement of the confidence envelope $\tilde{V}$ from Proposition 6.4.1 over the confidence envelope $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ from Durand et al. (2020) — and the confidence envelope from Xu et al. (2024a), by extension. The code to reproduce the simulations is available at `https://github.com/RomainPerier/Local_CB`.

We consider the same simulation setting than in Périer et al. (2026). This setting is an *heterogeneous* setting, meaning that the p-values are not identically distributed, that we recall here for completeness.

We simulate two groups of $N = 50$ subjects, and for each subject, we observe m independent binary responses (*e.g.* treatment effects, gene variants, ...). We denote by β_{1i} and β_{2i} the success probability, respectively for the first group and the second group. Then, we simultaneously test the null hypotheses H_i :"$\beta_{1i} = \beta_{2i}$" by two-sided Fisher's exact tests — see Agresti (2013) for more details on these tests — producing for each $i \in [\![1, m]\!]$ a p-value p_i with its *c.d.f.* denoted F_i. We will then have three distinct sets of p-values. The first set of p-values is generated by underlying data such that $\beta_{1i} = \beta_{2i} = 0.01$. Hence it is a set of p-values such that the underlying null hypotheses are true. The second set of p-values is generated by underlying data such that $\beta_{1i} = \beta_{2i} = 0.1$ hence this is a set of p-values such that the underlying null hypotheses are true. Each of these two sets of p-values is equally represented in the simulations, meaning that 50% of the p-values under the null hypothesis belong to the first set and 50% to the second set. Finally, the third set of p-values is generated by underlying data such that $\beta_{1i} = 0.1$ and $\beta_{2i} = q$, where $q \in \{0.2, 0.3, 0.4\}$ represents the strength of the signal.

These p-values are localized in a binary tree structured reference family $\mathfrak{R}$. We give an example of such a structure in Figure 6.1, albeit simpler than the one we used. For each of the leaves of the tree — that is, for each reference region R_k, $k \in \mathcal{K}$, such that $R_k \cap R_{k'} \in \{\varnothing, R_k\}$ for all $k' \in \mathcal{K}$ — $d_k \in \{0, 0.2, 0.5, 0.8\}$ represents the proportion of p-values under the alternative hypothesis within the leaf. This proportion is common to all active leaves, that is, there exists a common value $d \in \{0.2, 0.5, 0.8\}$ such that for any leaf R_k, $k \in \mathcal{K}$, verifying $d_k \neq 0$, $d_k = d$.

Finally, for all of our simulations, the proportion d and the strength of the signal q are specified and we set the number of hypotheses to $m = 3200$. In addition, the number of active leaves is set to 8 and the number of hypotheses within each leaf is set to 100, therefore the forest structure possesses 32 leaves.

Furthermore, for each couple (d, q) we draw m contingency tables, compute the associated p-values and apply four methods to this unique set of p-values. For each of the p-values p_i, the cumulative distribution function *c.d.f* is kwown and denoted F_i. We describe below the four methods that we consider in our simulations, $\hat{V}^{(1)}, \hat{V}^{(2)}, \hat{V}^{(3)}, \hat{V}^{(4)}$, but before we introduce some notations. For a reference family $\mathfrak{R} = \big((R_k, \zeta_k)\big)_{k\in\mathcal{K}}$, we denote $K = |\mathcal{K}|$ and $m_k = |R_k|$ for all $k \in \mathcal{K}$. For a vector $x \in \mathbb{R}^m$ and $A \subseteq [\![1, m]\!]$, $x_{[i:A]}$ denotes the i-th order statistic of the vector $(x_j)_{j\in A}$ in descending order and $x_{(i:A)}$ denotes the i-th order statistic of the vector $(x_j)_{j\in A}$ in ascending order. The methods

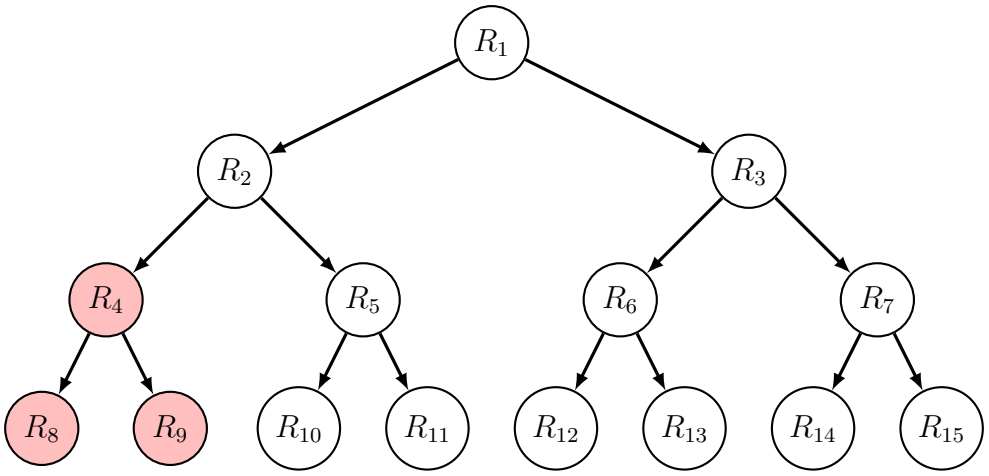


Figure 6.1: Perfect binary tree structures, here with a number of active leaves equal to 2. The pink nodes are those containing signal. This figure is one of the figures from Durand et al. (2020), but slightly modified.

$\hat{V}^{(1)}, \hat{V}^{(2)}$ are derived from the DKW inequality (Massart, 1990; Dvoretzky et al., 1956); $\hat{V}^{(1)}$ is $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ where $\mathfrak{R} = \Big((R_k, \zeta_k)\Big)_{k\in\mathcal{K}}$ with ζ_k defined by

$$\zeta_k = \min_{t\in[0,1)} \left\lfloor \left(\frac{\lambda_{\alpha/K}}{2(1-t)} + \sqrt{\frac{\lambda^2_{\alpha/K}}{4(1-t)^2} + \frac{|R_k| - i_{R_k}(t)}{1-t}} \right)^2 \right\rfloor,$$

where $\lambda_{\alpha/K} = \sqrt{\frac{1}{2}\log\frac{K}{\alpha}}$. This overestimator on the number of false discoveries in R_k has been considered for instance in Durand et al. (2020), Meah et al. (2024) and Périer et al. (2026). $\hat{V}^{(2)}$ is defined by (6.12) with, for all $k \in \mathcal{K}$, $\hat{V}_k$ defined by $\hat{V}_k = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}_k}$ where $\mathfrak{R}_k = \Big((R_{k,i}, \zeta_{k,i})\Big)_{i\in[\![1,m_k]\!]}$ is such that for all $i \in [\![1, m_k]\!]$, $R_{k,i} = \big\{j \in R_k : p_j \le p_{(i:R_k)}\big\}$ and $\zeta_{k,i} = \big\lfloor \zeta_k p_{(i:R_k)} + \sqrt{\zeta_k}\lambda_{\alpha/|\mathcal{K}|}\big\rfloor$ — hence the adaptive DKW procedure described in Périer et al. (2026), Proposition 4.14, but restricted to the set R_k.

The last two methods, $\hat{V}^{(3)}$ and $\hat{V}^{(4)}$, are methods derived from the Bretagnolle inequality (Shorack and Wellner, 1986); $\hat{V}^{(3)}$ is $\hat{V}^{\mathrm{DBNR}}_{\bar{\mathfrak{R}}}$ where $\bar{\mathfrak{R}} = \big((R_k, \bar{\zeta}_k)\big)_{k\in\mathcal{K}}$ with $\bar{\zeta}_k$ defined by

$$\bar{\zeta}_k = \max\left\{ n \in [\![0, m_k]\!] : \forall t \in [0,1],\ \sum_{1\le j\le n} \big(\mathbb{G}(t)\big)_{(j:R_k)} \le \lambda_{\alpha/(eK)}\sqrt{n} \right\},$$

where $e = \exp(1)$ and for all $t \in [0,1]$, $\mathbb{G}(t) = \big(\mathbb{1}\{p_i \le t\} - F_i(t)\big)_{1\le i\le m}$. This overestimator on the number of false discoveries in R_k has been considered in Périer et al. (2026). $\hat{V}^{(4)}$ is defined by (6.12) with, for all $k \in \mathcal{K}$, $\hat{V}_k = \bar{V}_k$ defined by $\bar{V}_k = \hat{V}^{\mathrm{JER}}_{\bar{\mathfrak{R}}_k}$ where $\bar{\mathfrak{R}}_k = \Big((R_{k,i}, \bar{\zeta}_{k,i})\Big)_{i\in[\![1,m_k]\!]}$ $\bar{\zeta}_{k,i} = \bar{\zeta}_k \wedge \Big\lfloor \sum_{1\le i\le\bar{\zeta}_k} \big(\mathbf{F}(p_{(i:R_k)})\big)_{[i:R_k]} + \sqrt{\bar{\zeta}_k}\lambda_{\alpha/(eK)} \Big\rfloor$. where for all $t \in [0,1]$, $\mathbf{F}(t) = \big(F_i(t)\big)_{1\le i\le m}$. Hence, for all $k \in \mathcal{K}$, the procedure $\bar{V}_k$ is the adaptive top-k procedure from Périer et al. (2026), Theorem 6.2, but restricted to R_k.

The results of these simulations are summerized by Figure 6.2. where we plot the lower bounds on the number of true positives for the top-k sets S_k with $k \in [\![1, m]\!]$, that is, the hypotheses corresponding to the k smallest p-values. Therefore, in Figure 6.2 at

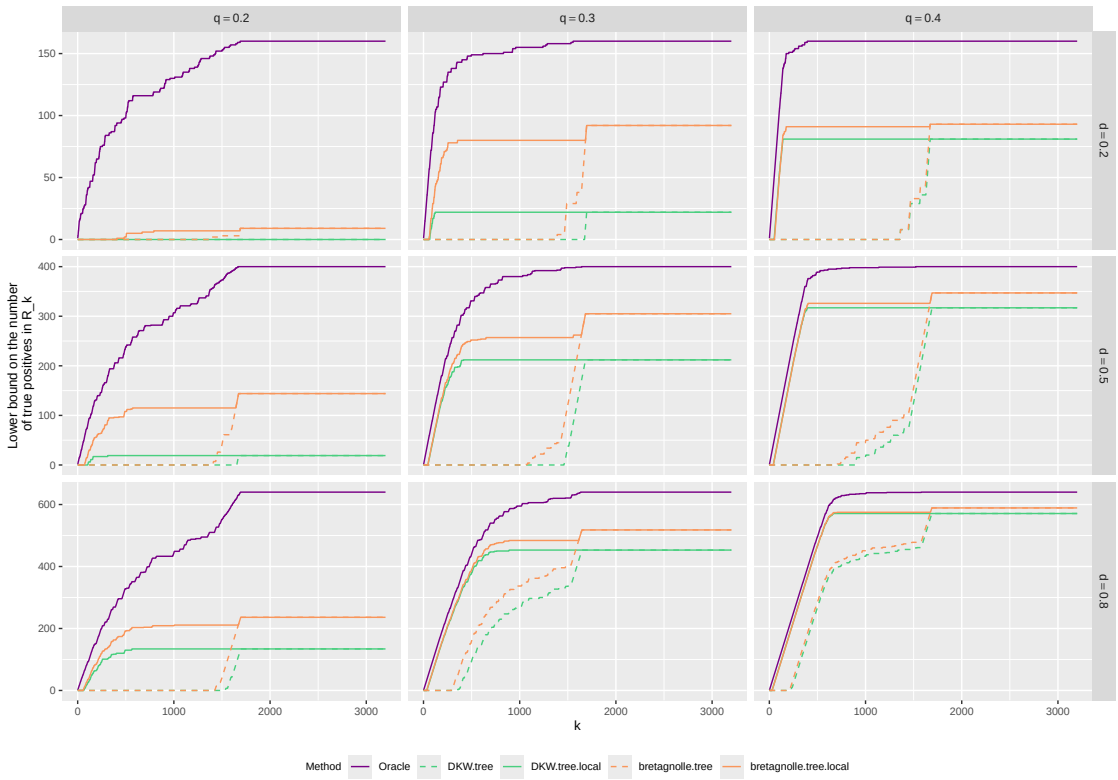


Figure 6.2: Lower bounds on number of true positives in the top-k sets provided by DKW and Bretagnolle inequalities based confidence envelopes. Higher is better.

abscissa k, we plot $|S_k| - \hat{V}^{(i)}(S_k)$, for all $i \in \{1, 2, 3, 4\}$. Each of the procedures based on $\hat{V}^{\mathrm{DBNR}}$ are highly outperformed by the corresponding procedures based on $\tilde{V}$, which is consistent with Proposition 6.4.1. In particular, a caveat of the $\hat{V}^{\mathrm{DBNR}}$ procedures is that it performs poorly when density of the signal is not high enough, like it is the case for our simulations, because the smallest p-values may not be in the same reference region.

6.5 Conclusion

In this chapter, we have shown that the notion of coherence from Goeman et al. (2021) is not sufficient ensure that a confidence envelope can be written as a closed testing procedure. We have introduced the notion of strong coherence, which is a stronger notion than coherence, and we have shown that a confidence envelope is strongly coherent if and only if it can be written as a JER confidence envelope. We have also provided a uniform improvement of the confidence envelope from Durand et al. (2020), and thus from Xu et al. (2024a), by considering local confidence bounds on the reference regions. Although this improvement is not equal to the optimal JER confidence envelope, it is computable in practice and provides a significant improvement over the original methods, as illustrated by our simulations, using methods derived from DKW and Bretagnolle inequalities. At some point, it would be interesting to investigate other inequalities to derive confidence envelopes based on local confidence bounds, and to compare their performance with the methods used without structure, as the confidence envelopes from Périer et al. (2026).

Chapter 7

Conclusion and perspectives

In this thesis, I mainly investigated the heterogeneity of the p-values and how to exploit it to improve former multiple testing procedures. One of the main questions that arise when considering heterogeneous p-values is how to handle computational issues, since the heterogeneity of the p-values can be very complex and can lead to intractable procedures. These results open new perspectives for future research, and we discuss some of them in the following.

Concentration inequalities The (heterogeneous) concentration inequalities presented in Chapter 3, designed for heterogeneous p-values, suffer of a few caveats. First, they assume independence of the p-values under the null hypotheses, which is a strong assumption in practice. Therefore, it would be interesting to develop new concentration inequalities for heterogeneous p-values under dependence assumptions. An easy way to do so would be to use the procedure DBY from Döhler (2018) and use it to derive new Simes-like inequalities for heterogeneous p-values under any dependence assumption. Second, the developed inequalities do not reduce to the classical analog inequalities in the homogeneous case. For instance, this is the case for the Bretagnolle inequality, which does not reduce to the DKW inequality in the homogeneous case due to a multiplicative constant equal to $\exp(1)$. Recent developments, like the ones in Reeve (2024) or Boudec (2026), bring new proofs of the DKW inequality, and these may be adapted to the heterogeneous case to reduce the constant in the Bretagnolle inequality. Finally, there is a need for heterogeneous versions of the K&R inequality from Katsevich and Ramdas (2020), or of the uniform Wellner inequality from Meah et al. (2024), since these inequalities perform better than heterogeneous inequalities under some conditions. Hence, developing new heterogeneous inequalities that reduce to the K&R or Wellner inequalities in the homogeneous case would be of great interest.

Multiple testing procedures The main issue adressed in this thesis regarding multiple testing procedures is the issue of *computational tractability*. Some questions that arise are the following:

- can we produce a better bound than $\tilde{V}$ in Chapter 6, Equation (6.12), that is still computationally tractable, since $\tilde{V}$ does not attain the bound $\hat{V}^{\mathrm{JER}}$ in general?

- how to produce a computationally tractable subset of FDR control procedures set that we developed in Chapter 5 that will not be too conservative, since the closure principle is intractable in general?

Furthermore, the developments in Chapter 5 will drive new research on the heterogeneous multiple testing procedures, since it is possible to apply the whole developments from the compound p-values and e-values literature to the heterogeneous case. For instance, it would be interesting to apply Theorem 4 from Barber and Samworth (2025), which states that the BH procedure at level α on compound p-values achieves weak control of the FWER at level $\alpha + \alpha^2$ under independence. This result would derive a new confidence envelope for the false discoveries that could be of interest even if it would not reduce to the Simes confidence envelope in the homogeneous case.

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Appendix A

Supplementary material for Chapter 3

Contents

A.1 Van Zuijlen inequality and confidence envelope . . . 134
A.2 Bridging some gaps in the literature . . . 135
A.3 Complements on the top-k setting . . . 135
A.3.1 Top-k with ties . . . 135
A.3.2 Links between the top-k and k-FWER settings . . . 136
A.4 Uniformization of super-uniform p-values . . . 139
A.5 Complements on Katsevich and Ramdas (2020) and Meah et al. (2024) confidence envelopes . . . 141
A.6 Proofs of Chapter 3 . . . 143
A.6.1 Proof of Lemma 3.3.1 . . . 143
A.6.2 Proofs of Proposition 3.3.2 and Proposition 3.3.3 . . . 143
A.6.3 Proof of Proposition 3.3.6 . . . 143
A.6.4 Proof of Theorem 3.3.14 . . . 143
A.6.5 Proof of Proposition 3.3.18 . . . 144
A.6.6 Proof of Proposition 3.3.23 . . . 144
A.6.7 Proof of Lemma 3.4.1 . . . 145
A.6.8 Proof of Proposition 3.4.2 . . . 145
A.6.9 Proof of Proposition 3.4.3 . . . 146
A.6.10 Proof of Proposition 3.4.4 . . . 146
A.6.11 Proof of Lemma 3.4.5 . . . 147
A.6.12 Proof of Corollary 3.4.8 . . . 147
A.6.13 Proof of Proposition 3.4.9 . . . 147
A.6.14 Proof of Proposition 3.4.11 . . . 149
A.6.15 Proofs of Lemma 3.4.12 and Proposition 3.4.14 . . . 150
A.6.16 Proof of Proposition 3.5.1 . . . 151
A.6.17 Proof of Corollary 3.5.2 . . . 151
A.6.18 Proof of Lemma 3.5.4 . . . 151
A.6.19 Proof of Proposition 3.5.6 . . . 152
A.6.20 Proof of Lemma 3.5.12 . . . 152
A.6.21 Proof of Corollary 3.5.15 . . . 152
A.6.22 Proof of Lemma 3.5.17 . . . 153
A.6.23 Proof of Example 3.6.3 . . . 154
A.6.24 Proof of Proposition 3.6.5 . . . 155
A.6.25 Proof of Proposition 3.6.9 and Remark 3.6.10 . . . 155

A.1 Van Zuijlen inequality and confidence envelope

In this appendix, we focus on using the concentration inequality provided by Shorack and Wellner (1986) as a heterogeneous form of Proposition 3.3.18. Unfortunately, this inequality turned out to not be sharp enough in our simulations, as it is outperformed by the heterogeneous Simes method.
Let us define the function

$$h : x \mapsto x \exp(x),$$

denote W_0 its inverse on $[-1, +\infty)$ (called principal real branch of the Lambert function) and W_{-1} its inverse on $(-\infty, -1]$.

Theorem A.1.1 (Van Zuijlen inequality)**.** *Let $X_1, \dots, X_n$ be n independent variables. Denote by F_i their respective cdf and $\bar{F}$ the mean of these cdf. Then, for $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\forall t \in \mathbb{R}, \frac{1}{n} \sum_{1 \le i \le n} \mathbb{1}_{X_i \le t} < \frac{\bar{F}(t)}{-W_0\left(-\frac{\delta}{e(1+\delta)}\right)}.$$

Proof. Let $\delta \in (0, 1)$. By Van Zuijlen inequality (see Shorack and Wellner (1986) chapter 25, from van Zuijlen (1978)), for all $\lambda > 1$,

$$\forall t \in \mathbb{R}, \frac{1}{n} \sum_{1 \le i \le n} \mathbb{1}_{X_i \le t} \ge \lambda \bar{F}(t)$$

with probability at most $\frac{\exp(1-1/\lambda)(1/\lambda)}{1-(1/\lambda)\exp(1-1/\lambda)}$. Solving $\frac{\exp(1-1/\lambda)(1/\lambda)}{1-(1/\lambda)\exp(1-1/\lambda)} = \delta$ yields

$$\lambda = \frac{1}{-W_{-1}\left(-\frac{\delta}{e(1+\delta)}\right)} \text{ or } \lambda = \frac{1}{-W_0\left(-\frac{\delta}{e(1+\delta)}\right)}.$$

The fact that $\lambda > 1$ then imposes that $\lambda = \frac{1}{-W_0\left(-\frac{\delta}{e(1+\delta)}\right)}$, as W_{-1} takes its values in the set $(-\infty, -1]$. □

Remark A.1.2. *There is no restriction on δ because*

$$-\frac{1}{2}\exp(-1/2) \le -\frac{1}{2}\exp(-1) \le -\frac{\delta}{1+\delta}\exp(-1).$$

It follows that, because W_0 is increasing, for all $\delta \in (0, 1)$,

$$-1/2 = W_0\left(-\frac{1}{2}\mathrm{e}^{-1/2}\right) \le W_0\left(-\frac{\delta}{e(1+\delta)}\right),$$

thus $\frac{1}{-W_0\left(-\frac{\delta}{e(1+\delta)}\right)} \ge 2 > 1$.

Theorem A.1.3. *Let $\alpha \in]0, 1[$. Assume that $(p_i)_{i \in \mathcal{H}_0}$ is an independent family of p-values. Define*

$$\zeta_k = k \wedge \frac{m}{-W_{-1}\left(-\frac{\delta}{e(1+\delta)}\right)} \bar{F}\left(p_{(k)}\right).$$

Then, the reference family $\mathfrak{R} = (R_k, \zeta_k)_{1 \le k \le m}$ has its JER *bounded by α, where the R_k are defined as the top-k sets, see Section 3.3.3 and Equation (3.15).*

Proof. Use Van Zuijlen's inequality with $(p_i)_{i \in \mathcal{H}_0}$, $n = m_0$, and upper bound $\sum_{i \in \mathcal{H}_0} \bar{F}\left(p_{(k)}\right)$ by $\sum_{1 \le i \le m} \bar{F}\left(p_{(k)}\right)$. □

A.2 Bridging some gaps in the literature

In Durand et al. (2020), in the proof of Proposition 1, the authors used the fact that $N_t(S)$ stochastically dominates $c = \sum_{i=1}^{\nu} \mathbb{1}\{U_i > t\}$, where the U_i are independent uniform variables (we use the notations of that paper) to conclude. But that domination was only true at a fixed $t \in [0,1)$: for $t \in [0,1)$, for all $x \in \mathbb{R}$, $\mathbb{P}(N_t(S) \leq x) \leq \mathbb{P}(c \leq x)$. Actually, we needed a much stronger result to be able to conclude, like the following: almost surely, $\tilde{N}_t(S) \geq c$ for all $t \in [0,1)$, for some process $\left(\tilde{N}_t(S)\right)_{t\in[0,1)}$ with the same distribution as $(N_t(S))_{t\in[0,1)}$. This kind of result can be obtained by coupling but, given our Proposition 3.3.2 (which is already proved by coupling), it is actually unnecessary given that now the U_i can be considered super-uniform. In other words, by using the version of the DKW inequality that we gave here in Proposition 3.3.2, the gap in the proof is closed.

Also note that the proof of Proposition 1 in Durand et al. (2020) required $\lambda < 1/2$, and so $\alpha/K < 1/2$, because the authors used the version of the DKW inequality given by Massart (1990). Thanks to Reeve (2024), Proposition 3.3.2 does not have a similar constraint for its validity domain and so there is no more constraint on α, as stated in Proposition 3.3.25.

Finally, in Döhler et al. (2018), there was a slight oversight regarding the definition of $\mathcal{A}$: 0 should be included in $\mathcal{A}$, that is, we should have $\mathcal{A} = \{0\} \cup \bigcup_{i=1}^{m} \mathcal{A}_i$ instead of just $\mathcal{A} = \bigcup_{i=1}^{m} \mathcal{A}_i$. Otherwise, the sets that appear in the definitions of τ_m and τ_k for the [HSU], [HSD], [AHSU], [AHSD] and [HBR-λ] procedures may be empty, and so τ_m and τ_k may not be defined. This oversight has been corrected in the implementations of these procedures in the `R` package `DiscreteFDR` (R Core Team, 2024; Döhler et al., 2024).

A.3 Complements on the top-k setting

A.3.1 Top-k with ties

Recall that in the top-k setting, we defined R_k with Equation (3.15), while past works like Meah et al. (2024) used $R_k^\dagger = R(p_{(k)}) = \left\{i \in [\![1,m]\!] : p_i \leq p_{(k)}\right\}$. Also recall Remark 3.3.21: we assume that ζ_k is of the form $f\left(p_{(k)}\right) \wedge k$ for some measurable function f. The next proposition shows that using R_k or $R_k^\dagger$ in that context is inconsequential, in the sense that it gives raise to the same confidence envelope.

Proposition A.3.1. *Let $\zeta_k = f\left(p_{(k)}\right) \wedge k$ for all $k \in [\![1,m]\!]$, where f is any measurable function with real values.*

Let the following reference families: $\mathfrak{R} = (R_k, \zeta_k)$, $\mathfrak{R}^\sim = \left(R_k, f\left(p_{(k)}\right)\right)$, and $\mathfrak{R}^\dagger = \left(R_k^\dagger, f\left(p_{(k)}\right)\right)$.

Then, for all $S \subseteq [\![1,m]\!]$,

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\sim}(S) \tag{A.1}$$

$$= \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\dagger}(S). \tag{A.2}$$

Note that the statement of Proposition A.3.1 is purely deterministic. Also note that the statement of Proposition A.3.1 implies that the choice of the permutation σ that orders the p-values is irrelevant, given that $\mathfrak{R}^\dagger$ does not depend on σ.

Proof. First, note that for any $A \subseteq [\![1,m]\!]$, $|A \cap R_k| \leq \zeta_k = f\big(p_{(k)}\big) \wedge k$ if and only if $|A \cap R_k| \leq f\big(p_{(k)}\big)$, because we always have $|A \cap R_k| \leq |R_k| = k$. This implies that $\mathfrak{A}(\mathfrak{R}) = \mathfrak{A}(\mathfrak{R}^{\sim})$ (recall Equation (3.12)). And so $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^{\sim}}(S)$ for all $S \subseteq [\![1,m]\!]$, by Equation (3.13). Hence (A.1) holds.

Note that $R_k \subseteq R_k^{\dagger}$ and so if $\left|A \cap R_k^{\dagger}\right| \leq f\big(p_{(k)}\big)$, then $|A \cap R_k| \leq f\big(p_{(k)}\big)$ also. Hence $\mathfrak{A}(\mathfrak{R}^{\dagger}) \subseteq \mathfrak{A}(\mathfrak{R}^{\sim})$.

Denote $g(k) = \max\big\{j : p_{(j)} = p_{(k)}\big\}$ and recall that $p_{(j)} = p_{\sigma(j)}$ for all $j \in [\![1,m]\!]$. First note that $f\big(p_{(g(k))}\big) = f\big(p_{(k)}\big)$ and $R^{\dagger}_{g(k)} = R^{\dagger}_k$ for all $k \in [\![1,m]\!]$. Second, note that if $p_i \leq p_{(k)}$ and $i \notin R_{g(k)}$, then $i = \sigma(\ell)$ with $\ell > g(k)$, so $p_i = p_{\sigma(\ell)} = p_{(\ell)} > p_{(g(k))} = p_{(k)}$ by definition of $g(k)$, which is a contradiction, and so $R_{g(k)} = R^{\dagger}_k$. Let $A \in \mathfrak{A}(\mathfrak{R}^{\sim})$. For all $k \in [\![1,m]\!]$,

$$\begin{aligned}\left|A \cap R_k^{\dagger}\right| &= \left|A \cap R_{g(k)}\right| \\ &\leq f\big(p_{(g(k))}\big) \\ &= f\big(p_{(k)}\big),\end{aligned}$$

so $A \in \mathfrak{A}(\mathfrak{R}^{\dagger})$ and $\mathfrak{A}(\mathfrak{R}^{\sim}) \subseteq \mathfrak{A}(\mathfrak{R}^{\dagger})$.

To conclude, $\mathfrak{A}(\mathfrak{R}^{\sim}) = \mathfrak{A}(\mathfrak{R}^{\dagger})$, and so $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}^{\dagger}}(S) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^{\sim}}(S)$ for all $S \subseteq [\![1,m]\!]$. Hence (A.2) holds. □

A.3.2 Links between the top-k and k-FWER settings

From k-FWER to top-k

In this Section, we show that from every reference family in the k-FWER setting, a top-k family can be built that leads to the same confidence envelope.

Let $\mathfrak{R} = (R_k, k-1)_{1 \leq k \leq K}$ a reference family in the k-FWER setting, which means that for all $k \leq K$, $R_k = \{i : p_i \leq t_k\}$ for some nondecreasing sequence $(t_k)_{1 \leq k \leq K}$. (This sequence may be data-dependent. The argument in this section is deterministic and holds for any fixed data.)

We introduce the following notation: for two integers $a \leq b$, $(\![a,b]\!] = [\![a,b]\!] \setminus \{a\}$. In particular, note that $(\![a,b]\!] = \varnothing$ if $a = b$.

The construction of the top-k family is the following. Let σ a (data-dependent) permutation that sorts the p-values in nondecreasing order, and $R^{\sigma}_{\ell} = \{\sigma(1), \dots, \sigma(\ell)\}$. We let $\ell_0 = 0$, $\ell_{K+1} = m$ and, for any $k \in [\![1,K]\!]$,

$$\ell_k = \max\Big\{\tilde{\ell} \in [\![1,m]\!] : p_{(\tilde{\ell})} \leq t_k\Big\}.$$

For any $k \in [\![1,K]\!]$ and any $\ell \in (\![\ell_{k-1}, \ell_k]\!]$, we let $\zeta^{\sigma}_{\ell} = k-1$. For any $\ell \in (\![\ell_K, \ell_{K+1}]\!]$, we let $\zeta^{\sigma}_{\ell} = m$. Finally, we let $\mathfrak{R}^{\sigma} = (R^{\sigma}_{\ell}, \zeta^{\sigma}_{\ell})_{1 \leq \ell \leq m}$, which is indeed built according to the top-k setting.

An example of such construction, that highlights some formal difficulties, is given below (Example A.3.4).

Proposition A.3.2. *For all $S \subseteq [\![1,m]\!]$, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^{\sigma}}(S)$.*

Like Proposition A.3.1, this Proposition is purely deterministic. The proof uses arguments similar to the proof of Proposition A.3.1, that is, the use of Equations (3.12) and (3.13).

Proof. First note that, for all $k \in [\![1, K]\!]$, $R_k = R^\sigma_{\ell_k}$. Indeed, if $p_i \leq t_k$, then $i = \sigma(\ell)$ with $\ell \leq \ell_k$ by definition of ℓ_k as a maximum, and conversely, for all $\ell \leq \ell_k$, $p_{\sigma(\ell)} \leq p_{\sigma(\ell_k)} \leq t_k$.

Second, note that for all $k \in [\![1, K]\!]$, $\zeta^\sigma_{\ell_k} \leq k - 1$. Indeed, let $k \in [\![1, K]\!]$. Given that

$$\bigcup_{k'=1}^{k}]\!]\ell_{k'-1}, \ell_{k'}]\!] = [\![1, \ell_k]\!],$$

there exists k', $1 \leq k' \leq k$, such that $\ell_k \in]\!]\ell_{k'-1}, \ell_{k'}]\!]$. Then, by definition, $\zeta^\sigma_{\ell_k} = k' - 1 \leq k - 1$.

Now, let $A \in \mathfrak{A}(\mathfrak{R})$. Let any $\ell \in [\![1, m]\!]$, there exists a unique $k \in [\![1, K+1]\!]$ such that $\ell \in]\!]\ell_{k-1}, \ell_k]\!]$. If $k = K + 1$, then $\zeta^\sigma_\ell = m$ and so obviously $|A \cap R^\sigma_\ell| \leq m = \zeta^\sigma_\ell$. If $k \leq K$, then $\zeta^\sigma_\ell = k - 1$. By the first remark, $R_k = R^\sigma_{\ell_k}$, and given that $R^\sigma_\ell \subseteq R^\sigma_{\ell_k}$, we have:

$$\begin{aligned} |A \cap R^\sigma_\ell| &\leq \left|A \cap R^\sigma_{\ell_k}\right| \\ &= |A \cap R_k| \\ &\leq k - 1 \text{ because } A \in \mathfrak{A}(\mathfrak{R}) \\ &= \zeta^\sigma_\ell. \end{aligned}$$

This proves that $\mathfrak{A}(\mathfrak{R}) \subseteq \mathfrak{A}(\mathfrak{R}^\sigma)$.

Let $A \in \mathfrak{A}(\mathfrak{R}^\sigma)$. Let any $k \in [\![1, K]\!]$. We have:

$$\begin{aligned} |A \cap R_k| &= \left|A \cap R^\sigma_{\ell_k}\right| \text{ by the first remark} \\ &\leq \zeta^\sigma_{\ell_k} \text{ because } A \in \mathfrak{A}(\mathfrak{R}^\sigma) \\ &\leq k - 1 \text{ by the second remark.} \end{aligned}$$

This proves that $\mathfrak{A}(\mathfrak{R}^\sigma) \subseteq \mathfrak{A}(\mathfrak{R})$.

In the end, $\mathfrak{A}(\mathfrak{R}) = \mathfrak{A}(\mathfrak{R}^\sigma)$ and so $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(\cdot) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\sigma}(\cdot)$. □

Remark A.3.3. *Given that each R_k is a particular $R^\sigma_{\ell_k}$, the previous construction just comes back to adding the missing R^σ_ℓ's to the family and to associate them a conservative upper bound for $V(R^\sigma_\ell)$.*

Example A.3.4. *Assume that $m = 4$ and that the threshold family is the Simes family as described in Theorem 3.3.14: $t_1 = \frac{\alpha}{4}, t_2 = \frac{2\alpha}{4}, t_3 = \frac{3\alpha}{4}, t_4 = \alpha$. Assume that $p_1 = \frac{\alpha}{4000}, p_2 = \frac{1.5\alpha}{4}, p_3 = \frac{1.9\alpha}{4}, p_4 = \frac{3.5\alpha}{4}$. Then $\ell_1 = 1$, $\ell_2 = \ell_3 = 3$, and $\ell_4 = 4$. Notably, in the proof of Proposition A.3.2, the k' such that $\ell_3 \in]\!]\ell_{k'-1}, \ell_{k'}]\!]$ is $k' = 2$, because $\ell_2 = \ell_3$ and so $]\!]\ell_2, \ell_3]\!] = \varnothing$.*

From top-k to k-FWER

Conversely, in this Section, we show that from every reference family in the top-k setting, a k-FWER family can be built that leads to the same confidence envelope.

Let $\mathfrak{R} = (R_k, \zeta_k)_{1 \leq k \leq K}$ a reference family in the top-k setting, which means that $R_k = \{\sigma(1), \ldots, \sigma(k)\}$ for some σ a (random) permutation that orders the p-values. Recall Remark 3.3.21: we assume that ζ_k is of the form $f\left(p_{(k)}\right) \wedge k$ for some measurable, nondecreasing function f that only takes integer values, which is a stronger assumption than in Proposition A.3.1.

The construction of the k-FWER family is the following. By convention, we let $p_{(0)} = -1$ and $f(-1) = 0$. We then let, for each $s \in [\![1, m]\!]$,

$$\nu(s) = \max\left\{k \in [\![0, m]\!] : f\left(p_{(k)}\right) \leq s - 1\right\}.$$

For any $s \in [\![1,m]\!]$, we then let $t_s = p_{(\nu(s))}$ and $R_s^\nu = \{i : p_i \leq t_s\}$. Note that $\nu(\cdot)$ is nondecreasing because $f\left(p_{(\nu(s))}\right) \leq s-1 \leq (s+1)-1$, and so the sequence $(t_s)_{s\in[\![1,m]\!]}$ is nondecreasing. Finally, we let $\mathfrak{R}^\nu = (R_s^\nu, s-1)_{1\leq s\leq m}$, which is indeed built according to the k-FWER setting.

Proposition A.3.5. *Let $\zeta_k = f\left(p_{(k)}\right) \wedge k$ for all $k \in [\![1,m]\!]$, where f is any measurable, nondecreasing function with integer values.*

Then, for all $S \subseteq [\![1,m]\!]$, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S) = \hat{V}_{\mathfrak{R}^\nu}^{\mathrm{JER}}(S)$.

Like Proposition A.3.1, this Proposition is purely deterministic. The proof uses arguments similar to the proof of Proposition A.3.1, that is, the use of Equations (3.12) and (3.13).

Proof. First note that for any $s \in [\![1,m]\!]$, if $\nu(s) \neq 0$, $R_s^\nu = R_{\nu(s)}$. Indeed, if $i \in R_s^\nu$, then $p_i \leq p_{(\nu(s))}$ and $i = \sigma(\ell)$ for some $\ell \in [\![1,m]\!]$. Hence $p_{(\ell)} \leq p_{(\nu(s))}$, $f\left(p_{(\ell)}\right) \leq f\left(p_{(\nu(s))}\right) \leq s-1$ because f is nondecreasing, and $\ell \leq \nu(s)$ by definition of $\nu(s)$, hence $i \in R_{\nu(s)}$. Conversely, if $i \in R_{\nu(s)}$, this means that $i = \sigma(\ell)$ for some $\ell \leq \nu(s)$, and so $p_i = p_{(\ell)} \leq p_{(\nu(s))}$ and $i \in R_s^\nu$.

Let $A \in \mathfrak{A}(\mathfrak{R})$, and $s \in [\![1,m]\!]$. If $\nu(s) = 0$, $|A \cap R_s^\nu| = |A \cap \varnothing| = 0 \leq s-1$. If $\nu(s) \neq 0$, by the previous remark,

$$\begin{aligned} |A \cap R_s^\nu| &= \left|A \cap R_{\nu(s)}\right| \\ &\leq \zeta_{\nu(s)} \text{ because } A \in \mathfrak{A}(\mathfrak{R}) \\ &\leq f\left(p_{(\nu(s))}\right) \\ &\leq s-1, \end{aligned}$$

so, in any case, $A \in \mathfrak{A}(\mathfrak{R}^\nu)$.

Let $A \in \mathfrak{A}(\mathfrak{R}^\nu)$, and let $k \in [\![1,m]\!]$. Note that $|A \cap R_k| \leq |R_k| = k$, so if $f\left(p_{(k)}\right) \geq k$, $\zeta_k = k$ and $|A \cap R_k| \leq \zeta_k$. Now assume that $f\left(p_{(k)}\right) \leq k-1$, which implies that $k \leq \nu(k)$. Let

$$s^* = \min\{s \in [\![1,m]\!] : k \leq \nu(s)\},$$

which is well-defined, because the set $\{s \in [\![1,m]\!] : k \leq \nu(s)\}$ is non-empty because it contains k, as we just emphasized. We distinguish cases, between $s^* = 1$ and $s^* > 1$. If $s^* = 1$, $k \leq \nu(1)$ and so $0 \leq f\left(p_{(k)}\right) \leq f\left(p_{(\nu(1))}\right) \leq 0$ and so $f\left(p_{(k)}\right) = 0$. If $s^* > 1$, we have $\nu(s^*-1) < k \leq \nu(s^*)$ and so $s^*-2 < f\left(p_{(k)}\right) \leq s^*-1$. Given that f only takes integer values, in both cases, $f\left(p_{(k)}\right) = s^*-1$. Also note that $k \leq \nu(s^*)$ implies that both $R_k \subseteq R_{\nu(s^*)}$ and $\nu(s^*) \neq 0$, so $R_{\nu(s^*)} = R_{s^*}^\nu$ by the previous remark. Finally,

$$\begin{aligned} |A \cap R_k| &\leq |A \cap R_{s^*}^\nu| \\ &\leq s^*-1 \text{ because } A \in \mathfrak{A}(\mathfrak{R}^\nu) \\ &= f\left(p_{(k)}\right), \end{aligned}$$

so $|A \cap R_k| \leq \zeta_k$ and $A \in \mathfrak{A}(\mathfrak{R})$.

In the end, $\mathfrak{A}(\mathfrak{R}) = \mathfrak{A}(\mathfrak{R}^\nu)$ and so $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(\cdot) = \hat{V}_{\mathfrak{R}^\nu}^{\mathrm{JER}}(\cdot)$. □

Remark A.3.6. *The previous construction just comes back to identifying where the sequence $\left(f\left(p_{(k)}\right)\right)_{k\in[\![1,m]\!]}$ makes its jumps.*

The two Simes envelopes are the same

In this Section, we prove that the two reference families built using the Simes family that we encountered in Section 3.3.3 yield the exact same confidence envelope, as a corollary of Section A.3.2.

Namely, let $\mathfrak{R}$ the reference family in the k-FWER setting described in Theorem 3.3.14: $\mathfrak{R} = \left(\left\{i : p_i \leq \frac{\alpha k}{m}\right\}, k-1\right)_{k\in[\![1,m]\!]}$. Let $\mathfrak{R}^\top = (R_\ell^\sigma, \zeta_\ell)_{\ell\in[\![1,m]\!]}$ the reference family in the top-k setting described in Theorem 3.3.19, item 3, which means that $R_\ell^\sigma = \{\sigma(1), \dots, \sigma(\ell)\}$ for σ a (random) permutation that orders the p-values, and $\zeta_\ell = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge \ell$.

Corollary A.3.7. *For all $S \subseteq [\![1,m]\!]$, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\top}(S)$.*

Proof. First note that, just like we proved (A.1), $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\top}(\cdot) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\ddagger}(\cdot)$, where

$$\mathfrak{R}^\ddagger = \left(R_\ell^\sigma, \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m\right)_{\ell\in[\![1,m]\!]},$$

that is, we just replaced the "$\wedge\,\ell$" by "$\wedge\, m$". The remainder of the proof consists in showing that $\mathfrak{R}^\ddagger$ is exactly the family $\mathfrak{R}^\sigma$ constructed in Section A.3.2 from $\mathfrak{R}$ (hence $K = m$ here), that is, that $\zeta_\ell^\sigma = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m$ for all $\ell \in [\![1,m]\!]$. The result will then ensue by applying Proposition A.3.2.

Let $\ell \in [\![1,m]\!]$. Let $k \in [\![1, m+1]\!]$ the unique integer such that $\ell \in (\!(\ell_{k-1}, \ell_k]\!]$. We distinguish the three cases $p_{(\ell)} = 0$, $k \leq m$ with $p_{(\ell)} > 0$, and $k = m+1$.

First, if $p_{(\ell)} = 0$, $\mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m = 0$. Given that $p_{(\ell)} = 0 \leq \frac{\alpha\times 1}{m}$, $\ell \leq \ell_1$, $k = 1$, and so $\zeta_\ell^\sigma = 1 - 1 = 0$ by definition. So, indeed, $\zeta_\ell^\sigma = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m$.

Second, assume that $k \leq m$ and $p_{(\ell)} > 0$. Then $\zeta_\ell^\sigma = k - 1$ by definition. Given that $\ell_{k-1} < \ell \leq \ell_k$, by definition of ℓ_k, we have $p_{(\ell)} \leq p_{(\ell_k)} \leq \frac{\alpha k}{m}$. If $k \geq 2$, we also have, by definition of ℓ_{k-1}, that $p_{(\ell)} > \frac{\alpha(k-1)}{m}$, and that is also true if $k = 1$ because we assumed $p_{(\ell)} > 0$. Then $k \geq \frac{mp_{(\ell)}}{\alpha}$ and $k - 1 < \frac{mp_{(\ell)}}{\alpha}$, which means that $k = \left\lceil \frac{mp_{(\ell)}}{\alpha}\right\rceil$ and, by Remark 3.3.22, $\zeta_\ell^\sigma = k - 1 = \left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor$. Furthermore, given that $k \leq m$, $\zeta_\ell^\sigma = k - 1 < m$ and so, indeed, $\zeta_\ell^\sigma = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m$.

Third, if $k = m+1$, then $\zeta_\ell^\sigma = m$ by definition. Furthermore, by definition of ℓ_m, $p_{(\ell)} > \frac{\alpha m}{m} = \alpha$. In particular, $\mathbb{1}\{p_{(\ell)} > 0\} = 1$. We also have that $\frac{mp_{(\ell)}}{\alpha} > m$ and so $\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \geq m$. So, indeed, $\zeta_\ell^\sigma = \mathbb{1}\{p_{(\ell)} > 0\}\left\lfloor \frac{mp_{(\ell)}^-}{\alpha}\right\rfloor \wedge m$. □

The following is a counterexample showing that the equality can fail if we forget to use the left-limit of the floor function instead of the floor function in the top-k Simes envelope.

Example A.3.8. *Let $m = 2$, $p_1 = \frac{\alpha}{2}$, $p_2 = \alpha$, which can happen with large probability under the alternative because we do not assume that the distributions of the p-values are continuous. Then $\zeta_1 = 0$ and $\zeta_2 = 1$ and, for example, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\top}(\{1\}) = 0$. But, without the left-limit, we would get $\zeta_1 = 1$, $\zeta_2 = 2$ and $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}^\top}(\{1\}) = 1$.*

A.4 Uniformization of super-uniform p-values

We present the following Lemma A.4.1, that explains how to transform any random variable into a uniform variable, which is, additionally in the case of super-uniformity,

smaller than the first one. This construction encompasses the classical "randomized p-value" construction, as stated by Remark A.4.2.

Note that this Lemma also offers an alternative way to extend the validity domain of various concentration inequalities from uniform variables to super-uniform variables (see Propositions 3.3.2 and 3.3.3), by providing a different coupling construction than in Lemma 3.3.1.

Lemma A.4.1 (Uniformization of any random variable)**.** *Let Y any real-valued random variable, let F its cdf and F^- the left-limit function of F. Assume that the probability space where Y is defined is rich enough so that there exists U another random variable independent from Y, with uniform distribution: $U \sim \mathcal{U}([0,1])$. Define $\widetilde{Y}$ as*

$$\widetilde{Y} = F^-(Y) + U(F(Y) - F^-(Y)).$$

Then $\widetilde{Y} \sim \mathcal{U}([0,1])$.

Furthermore, if Y is super-uniform: $Y \succeq \mathcal{U}([0,1])$, then $\widetilde{Y} \leq Y$ almost surely.

Proof. We prove that the cdf $\widetilde{F}$ of $\widetilde{Y}$ is $x \mapsto 0 \vee (x \wedge 1)$.

First note that $\widetilde{Y} \in [0,1]$ a.s., so $\widetilde{F}(x) = 1$ for all $x \in [1,\infty)$ and $\widetilde{F}(x) = 0$ for all $x \in (-\infty, 0)$. So let $x \in [0,1)$ and let us show that $\widetilde{F}(x) = x$. We distinguish two cases: if $x \in F(\mathbb{R})$ or not.

In the first case, there exists $\beta \in \mathbb{R}$ such that $x = F(\beta)$. We remark the following way to write $\widetilde{F}$:

$$\begin{aligned}\widetilde{F}(x) &= \mathbb{P}\big(F^-(Y) + U(F(Y) - F^-(Y)) \leq x\big) \\ &= \mathbb{E}\big[\mathbb{1}\{F(Y) \leq x\} + \mathbb{1}\big\{F^-(Y) \leq x < F(Y)\big\}\mathbb{1}\big\{U(F(Y) - F^-(Y)) \leq x - F^-(Y)\big\}\big]\end{aligned}$$

Now, if $F^-(Y) \leq x < F(Y)$ is true, then $F(\beta) < F(Y)$. Because F is nondecreasing, necessarily $\beta < Y$, which in turn implies that $F(\beta) \leq F^-(Y)$: in the end, $x = F(\beta) = F^-(Y)$, and $F(Y) > F^-(Y)$, so $\mathbb{1}\{U(F(Y) - F^-(Y)) \leq x - F^-(Y)\} = \mathbb{1}\{U = 0\} = 0$ a.s. So $\mathbb{1}\{F^-(Y) \leq x < F(Y)\}\mathbb{1}\{U(F(Y) - F^-(Y)) \leq x - F^-(Y)\} = 0$ a.s., and

$$\begin{aligned}\widetilde{F}(x) &= \mathbb{E}[\mathbb{1}\{F(Y) \leq x\}] \\ &= \mathbb{P}\big(F(Y) \leq x\big) \\ &= x,\end{aligned}$$

where the last equality comes from Proposition 2 of Chapter 1 of Shorack and Wellner (1986), thanks to x being in the range of F.

Now we turn to the case where $x \notin F(\mathbb{R})$. Let q_x the x-level quantile of F: $q_x = \inf\{t \in \mathbb{R} : F(t) \geq x\}$. By the usual properties of quantiles, $F^-(q_x) \leq x$ and $F(q_x) \geq x$, and we even have $F(q_x) > x$ because $x \notin F(\mathbb{R})$.

We now show that $\widetilde{Y} \leq x$ if and only if either $Y < q_x$, or $Y = q_x$ and $U(F(q_x) - F^-(q_x)) \leq x - F^-(q_x)$. Assume that $\widetilde{Y} \leq x$. Note that $\widetilde{Y} \in [F^-(Y), F(Y)]$ and so $F^-(Y) \leq x$. If we had $Y > q_x$, then we would have $F^-(Y) \geq F(q_x)$, and so $x \geq F^-(Y) \geq F(q_x) > x$ which would be a contradiction. So $Y \leq q_x$ and the implication follows. Conversely, assume the other proposition. Obviously, if $Y = q_x$ and $U(F(q_x) - F^-(q_x)) \leq x - F^-(q_x)$ then $\widetilde{Y} \leq x$. If $Y < q_x$, then $\widetilde{Y} \leq F(Y) \leq F^-(q_x) \leq x$.

Thanks to the equivalence, we have

$$\begin{aligned}\widetilde{F}(x) &= \mathbb{P}(Y < q_x) + \mathbb{P}\Big(\{Y = q_x\} \cap \big\{U(F(q_x) - F^-(q_x)) \leq x - F^-(q_x)\big\}\Big) \\ &= F^-(q_x) + \mathbb{P}(Y = q_x)\mathbb{P}\big(U(F(q_x) - F^-(q_x)) \leq x - F^-(q_x)\big),\end{aligned}$$

by independence of Y and U. Note that $F^-(q_x) \leq x < F(q_x)$, so $\frac{x-F^-(q_x)}{F(q_x)-F^-(q_x)}$ is well defined and in $[0,1]$. Then,

$$\begin{aligned}\widetilde{F}(x) &= F^-(q_x) + \big(F(q_x) - F^-(q_x)\big)\mathbb{P}\left(U \leq \frac{x - F^-(q_x)}{F(q_x) - F^-(q_x)}\right) \\ &= F^-(q_x) + \big(F(q_x) - F^-(q_x)\big)\frac{x - F^-(q_x)}{F(q_x) - F^-(q_x)} \\ &= x.\end{aligned}$$

This concludes that $\widetilde{Y} \sim \mathcal{U}([0,1])$.

Now assume that $Y \succeq \mathcal{U}([0,1])$, which means that $F(x) \leq 0 \vee (x \wedge 1)$ for all $x \in \mathbb{R}$. In particular, $F(x) \leq x$ for all $x \geq 0$, and $\mathbb{P}(Y \leq 0) = 0$, so $Y \geq 0$ a.s. Then

$$\begin{aligned}\widetilde{Y} &= F^-(Y) + U(F(Y) - F^-(Y)) \\ &\leq F(Y) \text{ a.s.} \\ &\leq Y \text{ a.s.},\end{aligned}$$

which concludes. □

Remark A.4.2. *The construction of Lemma A.4.1 covers the classical definition of "random p-values". Indeed, assume that we have at hand, for each $i \in [\![1,m]\!]$, a test statistic T_i such that for all $\mu \in H_i$, $X \sim \mu$, the distribution of $T_i(X)$ does not depend on μ and is known. The classical unilateral p-value derived from T_i is either $p_i^{(1)}(X) = \mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) \geq T_i(X)\big)$, either $p_i^{(2)}(X) = \mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) \leq T_i(X)\big)$, where the notation $\mathbb{P}_{X_0\sim\mu_i}$ denotes that the integration takes place with respect to a random variable X_0 drawn according to some $\mu_i \in H_i$. Then the corresponding classical random p-value is either $\tilde{p}_i^{(1)}(X) = \mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) > T_i(X)\big) + U\mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) = T_i(X)\big)$, either $\tilde{p}_i^{(2)}(X) = \mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) < T_i(X)\big) + U\mathbb{P}_{X_0\sim\mu_i}\big(T_i(X_0) = T_i(X)\big)$, where $U \sim \mathcal{U}([0,1])$ is independent of X. The key observation is that $\tilde{p}_i^{(1)}(X)$ and $\tilde{p}_i^{(2)}(X)$ follow exactly from the construction of Lemma A.4.1, applied respectively to the random variables $-T_i(X)$ and $T_i(X)$, and their cdf if $X \sim \mu_i$.*

A.5 Complements on Katsevich and Ramdas (2020) and Meah et al. (2024) confidence envelopes

We present in this section a slight refinement of the adaptive top-k method from Meah et al. (2024) obtained by application of the formalism introduced in Section 3.4. Let us consider the confidence bound for the ecdf introduced by Katsevich and Ramdas (2020); let $\delta \in (0, 0.31)$, then, for $U_1, \ldots, U_n \sim \mathcal{U}\big([0,1]\big)$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{U_i \leq t\} \geq C_\delta(1/n + t)\right) \leq \delta,$$

where $C_\delta = \frac{\log(1/\delta)}{\log(1+\log(1/\delta))}$.
Hence we derive for all $A \subseteq [\![1,m]\!]$ and $\alpha \in (0, 0.31)$ a local test of level α :

$$\varphi_A = \sup_{t\in[0,1]} \mathbb{1}\big\{i_A(t) \geq f_{|A|}(t)\big\},$$

where for all $t \in [0,1]$ and $n \in [\![1,m]\!]$, $f_n(t) = C_\delta(1 + tn)$.
For all $t \in [0,1]$ and $n \in [\![1,m]\!]$, $\varphi_{[\![1,m]\!],n,t} = \mathbb{1}\{f_n(t) \leq n + i(t) - m\}$. Therefore:

- If $t \geq \frac{1}{C_\delta}$, $\varphi_{[\![1,m]\!],n,t} = 0$,

- If $t < \frac{1}{C_\delta}$ and $\varphi_{[\![1,m]\!],n,t} = 1$, $f_{n+1}(t) = C_\delta\big(1+(n+1)t\big) = f_n(t)+C_\delta t \leq n+1+i(t)-m$. Hence $\varphi_{[\![1,m]\!],n+1,t} = 1$.

Thus, $\hat{V}_\varphi^{\mathrm{SC1}}\big([\![1,m]\!]\big) = \hat{V}_\varphi^{\mathrm{SC2}}\big([\![1,m]\!]\big)$ because φ verifies the assumptions from Proposition 3.4.4 for $S = [\![1,m]\!]$.
We now compute $b(t) = \max\big\{n \in [\![0,m]\!] : \varphi_{[\![1,m]\!],n,t} = 0\big\}$:

- If $t \geq \frac{1}{C_\delta}$, the first point above implies $b(t) = m$.

- If $t < \frac{1}{C_\delta}$, then:

$$\begin{aligned} b(t) &= \max\{n \in [\![0,m]\!] : C_\delta(1+tn) > n + i(t) - m\} \\ &= \max\left\{n \in [\![0,m]\!] : n > \frac{m - i(t) + C_\delta}{1 - tC_\delta}\right\} \\ &= m \wedge \left(\left\lceil \frac{m - i(t) + C_\delta}{1 - tC_\delta} \right\rceil - 1\right). \end{aligned}$$

Hence, $\hat{V}_\varphi^{\mathrm{SC1}}\big([\![1,m]\!]\big) = m \wedge \left(\min\limits_{t\in[0,1/C_\delta)} \left\lceil \frac{m-i(t)+C_\delta}{1-tC_\delta} \right\rceil - 1\right)$.
From Proposition 3.4.11, we conclude that

$$\hat{V}_\varphi^{\mathrm{IP}}(S) = \min_{0\leq k\leq |S|} \Big(\big\lceil f_{\hat{m}_0}\big(p_{(k:S)}\big)\big\rceil - 1 + \big|S \setminus R\big(p_{(k:S)}\big)\big|\Big),$$

where $\hat{m}_0 = m \wedge \left(\min\limits_{t\in[0,1/C_\delta)} \left\lceil \frac{m-i(t)+C_\delta}{1-tC_\delta} \right\rceil - 1\right)$.

Remark A.5.1. *The careful reader may remark that the local test from Proposition 3.4.11 are expressed with a " $>$ " instead of a " $\geq$ " and that we replaced here "$\lfloor\cdot\rfloor$" by "$\lceil\cdot\rceil - 1$" in the expression of $\hat{V}_\varphi^{\mathrm{IP}}(S)$. The proof can easily be adjusted, since the f_n are bijective in the specific case of KR inequality.*

Meah et al. (2024) already proposed an over-estimator on m_0, that we denote $\tilde{m}_0$, similar to the one we expose here. Precisely, the following holds:

$$\hat{m}_0 = \big\lceil \tilde{m}_0 \wedge (m - i(0) + C_\delta) - 1 \big\rceil.$$

It could be legitimate to question the usefulness of such refinements as the one described above or the one detailed by Corollary 3.4.8 on vanilla Simes confidence bounds. However, Figure 20 from Meah et al. (2024) shows that it reduces conservativeness, as the closed testing envelopes — that are equal to the inversion procedure envelopes — appear to be less conservative. We have in turn run these simulation while integrating our refinements; the adapted interpolated envelopes and the close-testing envelopes happened to be the same, as expected from our theoretical developments.

A.6 Proofs of Chapter 3

A.6.1 Proof of Lemma 3.3.1

Let $Q_1, \dots, Q_N$ the respective quantile functions of $U_1, \dots, U_N$. Let $\widetilde{U}_1, \dots, \widetilde{U}_N$ be N independent random variables that are $\mathcal{U}[0,1]$ (possibly defined on another probability space). By classical quantile properties, we know that $V_i = Q_i(\widetilde{U}_i)$ has the same distribution as U_i for any $i \in [\![1, N]\!]$, and so, by independence, $(V_i)_{i\in[\![1,N]\!]}$ has the same distribution as $(U_i)_{i\in[\![1,N]\!]}$. Furthermore, by super-uniformity, $V_i = Q\left(\widetilde{U}_i\right) \geq \widetilde{U}_i$ almost surely for all $i \in [\![1, N]\!]$.

A.6.2 Proofs of Proposition 3.3.2 and Proposition 3.3.3

We first prove Proposition 3.3.2. We know that the statement is true if the random variables are uniform, by Reeve (2024). Using the coupled variables $(\widetilde{U}_i, V_i)_{i\in[\![1,N]\!]}$ as constructed in Lemma 3.3.1, it holds

$$\begin{aligned}
\mathbb{P}\left(\sqrt{N}\sup_{t\in[0,1]}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\} - t\right) > \lambda\right) &= \mathbb{P}\left(\sqrt{N}\sup_{t\in[0,1]}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{V_i \leq t\} - t\right) > \lambda\right) \\
&\leq \mathbb{P}\left(\sqrt{N}\sup_{t\in[0,1]}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left\{\widetilde{U}_i \leq t\right\} - t\right) > \lambda\right) \\
&\leq \exp\left(-2\lambda^2\right).
\end{aligned}$$

To prove Proposition 3.3.3, we know that the statement is true if $U_1, \dots, U_N$ are uniform, by Meah et al. (2024). We then use the same trick as in the proof of Proposition 3.3.2.

A.6.3 Proof of Proposition 3.3.6

We consider that the event $\left\{\forall k \in [\![1, N]\!] : U_{(k)} > \frac{k\delta}{N}\right\}$ is realized, which happens with probability at least $1-\delta$ by Proposition 3.3.4. For any $t > \delta$, $\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\} \leq 1 < \frac{t}{\delta}$. Now let $j \in \{0, \dots, N-1\}$ and $t \in \left(\frac{j\delta}{N}, \frac{(j+1)\delta}{N}\right]$. Then $\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\}$ is equal to $\mathrm{card}\{i : U_i \leq t\}$, and given that $U_{(j+1)} > \frac{(j+1)\delta}{N} \geq t$, $\mathrm{card}\{i : U_i \leq t\} \leq j < \frac{tN}{\delta}$ and finally $\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\} < \frac{t}{\delta}$.

A.6.4 Proof of Theorem 3.3.14

We recall here the short proof of Theorem 3.3.14:
Let $\mu \in \mathfrak{F}$, $\alpha \in (0,1)$, and $X \sim \mu$.

$$\begin{aligned}
1 - \mathrm{JER}(\mathfrak{R}) &= \mathbb{P}\left(\exists k \leq m, |R_k \cap \mathcal{H}_0| \geq k\right) \\
&= \mathbb{P}\left(\exists k \leq m_0, p_{(k:\mathcal{H}_0)} \leq \frac{\alpha k}{m}\right) \\
&\leq \mathbb{P}\left(\exists k \leq m_0, p_{(k:\mathcal{H}_0)} \leq \frac{\alpha k}{m_0}\right) \\
&\leq \alpha,
\end{aligned}$$

where the last inequality is obtained by applying Proposition 3.3.4 to the random variables $(p_i)_{i\in\mathcal{H}_0}$.

A.6.5 Proof of Proposition 3.3.18

We fix a $\delta \in (0,1]$, and a $\mu \in \mathfrak{F}$. The proof is exactly the same as in Meah et al. (2024): we apply Propositions 3.3.2, 3.3.3 and 3.3.6 to $N = m_0(\mu)$ and $\{U_1,\dots,U_N\} = \{p_i, i \in \mathcal{H}_0(\mu)\}$. We then use monotonicity properties and the fact that $m_0(\mu) \le m$ to replace $m_0(\mu)$ by m in the obtained inequalities. Note that for Wellner's inequality, the desired monotonicity property is the last statement of Lemma 43 of Meah et al. (2024).

A.6.6 Proof of Proposition 3.3.23

First note that, for any $1 \le k \le m$, by nestedness of the top-k reference family:

$$\begin{aligned}\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_k) &= \min_{1\le i\le m} (\zeta_i + |R_k \setminus R_i|) \wedge k, \text{ by (3.14);}\\ &= \min_{1\le i\le m} (\zeta_i + 0 \vee (k-i)) \wedge k, \text{ using (3.15).}\end{aligned}$$

By assumption on f, the sequence $(\zeta_k)_{1\le k\le m}$ is nondecreasing, so for all $i \ge k$,

$$\zeta_i + 0 \vee (k-i) = \zeta_i \ge \zeta_k = \zeta_k + 0 \vee (k-k),$$

and so

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_k) = \min_{1\le i\le k} (\zeta_i + k - i) \wedge k. \tag{A.3}$$

Second, we show that, for all $0 \le k \le m$,

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_{k+1}) = \min\Big(\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_k) + 1, \zeta_{k+1}\Big), \tag{A.4}$$

where $R_0 = \varnothing, \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_0) = 0$. Indeed, if $k = 0$,

$$\min\Big(\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_0) + 1, \zeta_{0+1}\Big) = \min(0+1, \zeta_1),$$

which coincides with (A.3) for $k = 1$. And if $k \ge 1$, we have

$$\begin{aligned}\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_{k+1}) &= \min_{1\le i\le k+1} (\zeta_i + k + 1 - i) \wedge (k+1), \text{ by (A.3),}\\ &= \left(\min_{1\le i\le k} (\zeta_i + k + 1 - i)\right) \wedge (k+1) \wedge \zeta_{k+1}\\ &= \left(\left(\min_{1\le i\le k} (\zeta_i + k - i) \wedge k\right) + 1\right) \wedge \zeta_{k+1}\\ &= \min\Big(\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_k) + 1, \zeta_{k+1}\Big), \text{ again by (A.3),}\end{aligned}$$

which concludes the proof that (A.4) holds.

With that, we can now prove that $\tilde{\zeta}_k = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_k)$ for all $0 \le k \le m$ by an induction scheme going over the integers visited by the `while` loop of Algorithm 1. First define formally $\tilde{\zeta}_0 = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_0) = 0$. Now let k be an integer visited by the `while` loop and assume that $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(R_i) = \tilde{\zeta}_i$ for all $i \le k-1$. We distinguish the two cases $\zeta_k > \tilde{\zeta}_{k-1}$ and $\zeta_k \le \tilde{\zeta}_{k-1}$ as in the `if` condition of Algorithm 1.

First assume that $\zeta_k > \tilde{\zeta}_{k-1}$, let $j = \left(\zeta_k - \tilde{\zeta}_{k-1}\right) \wedge (m-k+1)$, so that $\zeta_k \geq \tilde{\zeta}_{k-1} + j$ and the next integer visited by the loop is $k+j$ (note that $j \geq 1$). In that case, then, we aim to prove that $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_i) = \tilde{\zeta}_i$ for all $i \leq k+j-1$. By definition, $\tilde{\zeta}_{k+i-1} = \tilde{\zeta}_{k-1} + i$ for all $i \in [\![1, j]\!]$. Note that, for each such i,

$$\zeta_{k+i-1} \geq \zeta_k \geq \tilde{\zeta}_{k-1} + j \geq \tilde{\zeta}_{k-1} + i = \tilde{\zeta}_{k+i-1}. \tag{A.5}$$

To conclude this first case, we show by recursion that $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k+i-1}) = \tilde{\zeta}_{k+i-1}$ for each $i \in [\![0, j]\!]$. For $i = 0$, this is already true by the induction assumption. Now, if $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k+i-1}) = \tilde{\zeta}_{k+i-1}$ for some $i \in [\![0, j-1]\!]$, then

$$\begin{aligned}
\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k+i}) &= \left(\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k+i-1}) + 1\right) \wedge \zeta_{k+i}, \text{ by (A.4)}, \\
&= (\tilde{\zeta}_{k+i-1} + 1) \wedge \zeta_{k+i} \\
&= (\tilde{\zeta}_{k-1} + i + 1) \wedge \zeta_{k+i} \\
&= \tilde{\zeta}_{k+i} \wedge \zeta_{k+i} = \tilde{\zeta}_{k+i}, \text{ by (A.5)}.
\end{aligned}$$

In the second case, assume that $\zeta_k \leq \tilde{\zeta}_{k-1}$, hence $\tilde{\zeta}_k = \tilde{\zeta}_{k-1}$ by construction, and the next integer visited by the loop is $k+1$. In that case, then, we aim to prove that $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_i) = \tilde{\zeta}_i$ for all $i \leq k$. First note that

$$\begin{aligned}
\tilde{\zeta}_{k-1} &= \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k-1}) \\
&= \max_{A \in \mathfrak{A}(\mathfrak{R})} |A \cap R_{k-1}| \\
&\leq \zeta_{k-1} \text{ because } A \in \mathfrak{A}(\mathfrak{R}) \text{ implies } |A \cap R_{k-1}| \leq \zeta_{k-1} \\
&\leq \zeta_k,
\end{aligned}$$

so actually $\zeta_k = \tilde{\zeta}_{k-1}$. To conclude this second case, we just need to show that $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k) = \tilde{\zeta}_k$. Indeed,

$$\begin{aligned}
\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_k) &= \left(\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k-1}) + 1\right) \wedge \zeta_k, \text{ by (A.4)}, \\
&= (\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(R_{k-1}) + 1) \wedge \tilde{\zeta}_{k-1} \\
&= (\tilde{\zeta}_{k-1} + 1) \wedge \tilde{\zeta}_{k-1}, \text{ by the induction assumption}, \\
&= \tilde{\zeta}_{k-1} \\
&= \tilde{\zeta}_k.
\end{aligned}$$

A.6.7 Proof of Lemma 3.4.1

Suppose that $\max_{i \in [\![1, |A|]\!]} \mathbb{1}\{p_{(i:A)} \leq \ell_{i:|A|}\} = 1$. Then there is some i such that $p_{(i:A)} \leq \ell_{i:|A|}$. Denote $j = i_A\left(p_{(i:A)}\right) \geq i$, we have $p_{(i:A)} \leq \ell_{i:|A|} \leq \ell_{j:|A|}$, thus there is some t such that $\mathbb{1}\{t \leq \ell_{i_A(t):|A|}\} = 1$.
Suppose now that $\max_{t \in [0,1]} \mathbb{1}\{t \leq \ell_{i_A(t):|A|}\} = 1$. Then there is $t \in [0,1]$ such that $t \leq \ell_{i_A(t):|A|}$, thus $p_{(i_A(t):A)} \leq t \leq \ell_{i_A(t):|A|}$.

A.6.8 Proof of Proposition 3.4.2

For $n \in [\![0, |S|]\!]$, denote $\mathcal{N}_S^n = \{A \in \mathcal{N}_\varphi : |A \cap S| = n\}$, where $\mathcal{N}_\varphi = \{A \subseteq [\![1, m]\!] : \varphi_A = 0\}$. Then, $\hat{V}_\varphi^{\mathrm{IP}}(S) = \max_{A \in \mathcal{N}} |A \cap S| = \max\{n \in [\![0, |S|]\!] : \mathcal{N}_S^n \neq \varnothing\}$.

Let $\varphi_{S,n} = \min_{\substack{A\subseteq[\![1,m]\!]\\|A\cap S|=n}} \varphi_A$ be the test statistic testing whether $\mathcal{N}_S^n$ is empty or not; indeed, $\varphi_{S,n} = 0$, if and only if there is some $A \subseteq [\![1,m]\!]$ such that $|A \cap S| = n$ and $\varphi_A = 0$. For all $t \in [0,1]$ and $A \subseteq [\![1,m]\!]$, $\varphi_{A,t} \leq \varphi_A$, thus, take consecutively the minimum over $A \subseteq [\![1,m]\!]$ such that $|A \cap S| = n$ and the maximum over $t \in [0,1]$ in this inequality to obtain:

$$\varphi_{S,n} \geq \max_{t\in[0,1]} \varphi_{S,n,t}.$$

Thus, $\varphi_{S,n} = 0$ implies $\max_{t\in[0,1]} \varphi_{S,n,t} = 0$, thus

$$\{n \in [\![0,|S|]\!] : \varphi_{S,n} = 0\} \subseteq \left\{n \in [\![0,|S|]\!] : \max_{t\in[0,1]} \varphi_{S,n,t} = 0\right\},$$

which implies the first inequality. For all $t \in [0,1]$, by definition,

$$\max_{t\in[0,1]} \varphi_{S,n,t} \geq \varphi_{S,n,t},$$

thus $\left\{n \in [\![0,|S|]\!] : \max_{t\in[0,1]} \varphi_{S,n,t} = 0\right\} \subseteq \{n \in [\![0,|S|]\!] : \varphi_{S,n,t} = 0\}$, implying

$$\max\left\{n \in [\![0,|S|]\!] : \max_{t\in[0,1]} \varphi_{S,n,t} = 0\right\} \leq \max\{n \in [\![0,|S|]\!] : \varphi_{S,n,t} = 0\}.$$

Take the minimum over t in this inequality to get the second inequality.

A.6.9 Proof of Proposition 3.4.3

- The first equality is straightforward ; if $\varphi_A = 1$ then there is some $t \in [0,1]$ such that $\varphi_{A,t} = 1$. It follows by assumption that $\varphi_{A,p_{(i_A(t):A)}} = 1$.
- To show the second equality, suppose that $\max_{t\in[0,1]} \varphi_{S,n,t} = 1$. Then there is some $t \in [0,1]$ such that $\varphi_{S,n,t} = 1$. Thus, for all $A \subseteq [\![1,m]\!]$ such that $|A \cap S| = n$, $\varphi_{A,t} = 1$, followed, by assumption, by $\varphi_{A,s} = 1$ for all $s \in \left[p_{(i_A(t):A)}, t\right]$. Hence, $\varphi_{A,p_{(i(t))}} = 1$ with $s = p_{(i(t))} \in \left[p_{(i_A(t):A)}, t\right]$ for all $A \subseteq [\![1,m]\!]$.
- Finally, to show the last equality, we get from the above item that if $n \in [\![0,|S|]\!]$ is such that $\min_{\substack{A\subseteq[\![1,m]\!]\\|A\cap S|=n}} \varphi_{A,p_{(i)}} = 0$, then for all $t \in \left[p_{(i)}, p_{(i+1)}\right)$, $\varphi_{S,n,t} = 0$, hence, for all $t \in \left[p_{(i)}, p_{(i+1)}\right)$,

$$\max\left\{n \in [\![0,|S|]\!] : \min_{\substack{A\subseteq[\![1,m]\!]\\|A\cap S|=n}} \varphi_{A,p_{(i)}} = 0\right\} \leq \max\{n \in [\![0,|S|]\!] : \varphi_{S,n,t} = 0\}.$$

A.6.10 Proof of Proposition 3.4.4

- $\hat{V}_\varphi^{\mathrm{SC1}}(S) \leq \hat{V}_\varphi^{\mathrm{SC2}}(S)$ by Proposition 3.4.2.
- Let $k_0 = \hat{V}_\varphi^{\mathrm{SC2}}(S)$. Then, for all $t \in [0,1]$, for all $n \leq k_0$, $\varphi_{S,n,t} = 0$ hence, for all $n \leq k_0$, $\max_{t\in[0,1]} \varphi_{S,n,t} = 0$. Finally, $\hat{V}_\varphi^{\mathrm{SC1}}(S) = \max\left\{n \in [\![0,|S|]\!] : \max_{t\in[0,1]} \varphi_{S,n,t} = 0\right\} \geq k_0$.

A.6.11 Proof of Lemma 3.4.5

Let $t \in [0,1]$, $n \in [\![0, |S|]\!]$ and $A \subseteq [\![1,m]\!]$ such that $|A \cap S| = n$. Let $k = i_{A \cap S^c}(t)$. Then $\mathbb{1}\{t \leq \ell_{i_A(t):|A|}\} = \mathbb{1}\{t \leq \ell_{(i_{A\cap S}(t)+k):(n+k+|A\cap S^c \cap R(t)^c|)}\}$ is minimal if $S^c \cap R(t)^c \subseteq A$ and $S \cap R(t)^c \subseteq A$ (if n is large enough), thus

$$\min_{\substack{A \subseteq [\![1,m]\!] \\ |A\cap S|=n, i_{A\cap S^c}(t)=k}} \varphi_{A,t} = \mathbb{1}\{t \leq \ell_{((n+i_S(t)-|S|)\vee 0+k):(n+k+|S^c\cap R(t)^c|)}\}.$$

A.6.12 Proof of Corollary 3.4.8

Let $n \in [\![1,m]\!]$ and $i \leq n$. Then

$$\begin{aligned}
\ell_{i+1:n+1} - \ell_{i:n} &= \left(\frac{1}{n+1} - \frac{1}{n}\right) i\alpha + \frac{1}{n+1}\alpha \\
&= \frac{-i}{n(n+1)}\alpha + \frac{1}{n+1}\alpha \\
&\geq 0.
\end{aligned}$$

We then compute $b(t) = \max\{n \in [\![1,m]\!] : \varphi_{[\![1,m]\!],n,t} = 0\}$ for all $t \in (0,1]$. By Example 3.4.6, for all $t \in (0,\alpha)$,

$$b(t) = \max\left\{n \in [\![1,m]\!] : t > \frac{n+i(t)-m}{n}\alpha\right\} = \max\left\{n \in [\![1,m]\!] : n < \frac{m-i(t)}{1-t/\alpha}\right\},$$

therefore $b(t) = \left(\left\lceil \frac{m-i(t)}{1-t/\alpha} \right\rceil - 1\right) \wedge m$. For $t \in [\alpha, 1]$ and $n \in [\![1,m]\!]$, $\varphi_{[\![1,m]\!],n,t} = 0$ thus $b(t) = m$, and for $t = 0$, $b(0) = m - i(0)$.

A.6.13 Proof of Proposition 3.4.9

We first show the control of the joint error rate:

$$\begin{aligned}
\mathrm{JER}(\mathfrak{R}) &= \mathbb{P}\big(\exists k \in [\![1, \hat{m}_0^{\mathrm{SC1}} + 1]\!] : V(R_k) \geq k\big) \\
&= \mathbb{P}\Big(\big(\exists k \in [\![1, m_0 \wedge \hat{m}_0^{\mathrm{SC1}}]\!] : p_{(k:\mathcal{H}_0)} \leq \ell_{k:\hat{m}_0^{\mathrm{SC1}}}\big) \cup \big(m_0 > \hat{m}_0^{\mathrm{SC1}}\big)\Big) \\
&\leq \mathbb{P}\Big(\big(\exists k \in [\![1, m_0]\!] : p_{(k:\mathcal{H}_0)} \leq \ell_{k:m_0}\big) \cup \big(m_0 > \hat{m}_0^{\mathrm{SC1}}\big)\Big) = \mathbb{P}(\varphi_{\mathcal{H}_0} = 1) \\
&\leq \alpha.
\end{aligned}$$

Then, recall that for all $S \subseteq [\![1,m]\!]$, $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S) = \max_{A \in \mathcal{A}(\mathfrak{R})} |A \cap S|$, where

$$\mathcal{A}(\mathfrak{R}) = \big\{A \subseteq [\![1,m]\!] : \forall k \in [\![1, \hat{m}_0^{\mathrm{SC1}} + 1]\!], |A \cap R_k| \leq k-1\big\}.$$

- Let $A \in \mathcal{N}_\varphi$, then $\varphi_A = 0$ thus for all $i \in [\![1, |A|]\!]$, $p_{(i:A)} > \ell_{i:|A|}$ *i.e.* $p_{[i:A]} > \ell_{(|A|-i+1):|A|}$ therefore, by definition of $\hat{m}_0^{\mathrm{SC1}}$, $|A| \leq \hat{m}_0^{\mathrm{SC1}}$ thus $|A \cap R_{\hat{m}_0^{\mathrm{SC1}}+1}| \leq \hat{m}_0^{\mathrm{SC1}}$.
  Let $i \in [\![1, \hat{m}_0^{\mathrm{SC1}}]\!]$, $|A \cap R_i| \leq i-1$ because $p_{(i:A)} > \ell_{i:|A|} \geq \ell_{i:\hat{m}_0^{\mathrm{SC1}}}$ thus $\mathcal{N}_\varphi \subseteq \mathcal{A}(\mathfrak{R})$, therefore
  $$\hat{V}_\varphi^{\mathrm{IP}} \leq \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}.$$

- Let $S \subseteq [\![1, m]\!]$. If $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S) = \min_{1 \leq u \leq \hat{m}_0^{\mathrm{SC1}}} \left\{ \left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| + u - 1 \right\} \wedge \hat{m}_0^{\mathrm{SC1}} = \hat{m}_0^{\mathrm{SC1}}$, for all $u \in [\![1, \hat{m}_0^{\mathrm{SC1}}]\!]$, $\left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| \geq \hat{m}_0^{\mathrm{SC1}} - u + 1$, therefore we can recursively construct $i_{\hat{m}_0^{\mathrm{SC1}}}, \ldots, i_1$ such that $p_{i_{\hat{m}_0^{\mathrm{SC1}}-u+1}}$ is maximal for $i_{\hat{m}_0^{\mathrm{SC1}}-u+1} \in \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \setminus \left\{ i_{\hat{m}_0^{\mathrm{SC1}}}, \ldots, i_{\hat{m}_0^{\mathrm{SC1}}-u+2} \right\}$. Thus, $A = \left\{ i_1, \ldots, i_{\hat{m}_0^{\mathrm{SC1}}} \right\}$ is such that $A \in \mathcal{N}_\varphi$ and $|A \cap S| = \hat{m}_0^{\mathrm{SC1}}$.

  Assume now that $\min_{1 \leq u \leq \hat{m}_0^{\mathrm{SC1}}} \left\{ \left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| + u - 1 \right\} \wedge \hat{m}_0^{\mathrm{SC1}} < \hat{m}_0^{\mathrm{SC1}}$, and let u_0 be minimal such that $v_0 = \left| \left\{ i \in S : p_i > \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| + u_0 - 1$ is equal to

$$\min_{1 \leq u \leq \hat{m}_0^{\mathrm{SC1}}} \left\{ \left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| + u - 1 \right\}.$$

  The idea from this part of the proof will be to construct a set $A \in \mathcal{N}_\varphi$ of cardinality $\hat{m}_0^{\mathrm{SC1}}$ such that $|A \cap S| = v_0$. From minimality property of u_0 and v_0, we get:

  - $\forall u \leq u_0 - 1, u_0 - u < \left| S \cap R\left(\ell_{u,\hat{m}_0^{\mathrm{SC1}}}\right)^c \cap R\left(\ell_{u_0,\hat{m}_0^{\mathrm{SC1}}}\right) \right|$,
  - $\forall u \geq u_0, u - u_0 \geq \left| S \cap R\left(\ell_{u,\hat{m}_0^{\mathrm{SC1}}}\right) \cap R\left(\ell_{u_0,\hat{m}_0^{\mathrm{SC1}}}\right)^c \right|$ $(\star)$.

  By the first item, there exists indices $i_1, i_2, \ldots, i_{u_0-1} \in S$ such that $p_{i_1} \leq p_{i_2} \leq \cdots \leq p_{i_{u_0-1}}$ and $\ell_{u_0,\hat{m}_0^{\mathrm{SC1}}} \geq p_{i_u} > \ell_{u,\hat{m}_0^{\mathrm{SC1}}}$ for all $u \leq u_0 - 1$. Let B be the set of indices matching with the $\hat{m}_0^{\mathrm{SC1}} - v_0$ greatest p-values from $\left\{ i \in S^c : p_i > \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} \right\}$ and define $A = \{i_1, \ldots, i_{u_0-1}\} \cup \left\{ i \in S : p_i > \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} \right\} \cup B$. Then, for $j \in [\![1, \hat{m}_0^{\mathrm{SC1}}]\!]$:

  - If $j \leq u_0 - 1$, $p_{(j:A)} = p_{i_j} > \ell_{j:\hat{m}_0^{\mathrm{SC1}}}$.
  - If $j \geq u_0$ is such that $R\big(p_{(j:A)}\big)^c \subseteq A$, then let $k = \left| R\big(p_{(j:A)}\big)^c \right| \leq \hat{m}_0^{\mathrm{SC1}} - j$, and by definition of $\hat{m}_0^{\mathrm{SC1}}$, $p_{(j:A)} = p_{[k+1]} > \ell_{\hat{m}_0^{\mathrm{SC1}}-k:\hat{m}_0^{\mathrm{SC1}}} \geq \ell_{j:\hat{m}_0^{\mathrm{SC1}}}$.
  - Else $\left\{ i \in A : \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} < p_i \leq p_{(j:A)} \right\} \subseteq S$, because if

$$i \in \left\{ i \in A : \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} < p_i \leq p_{(j:A)} \right\}$$

    verifies $i \in B$ although $R\big(p_{(j:A)}\big)^c \not\subseteq A$, there exists $i' \in A^c$ such that $p_i < p_{i'}$ which contradicts B's definition. Thus

$$j - u_0 \geq \left| \left\{ i \in A : \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} < p_i \leq \ell_{j:\hat{m}_0^{\mathrm{SC1}}} \right\} \right|$$

    by $(\star)$. Furthermore, $\left| \left\{ i \in A : \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} < p_i \leq p_{(j:A)} \right\} \right| \geq j - u_0 + 1$ because $\left| \left\{ i \in A : p_i \leq \ell_{u_0:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| = u_0 - 1$, therefore $p_{(j:A)} > \ell_{j:\hat{m}_0^{\mathrm{SC1}}}$.

  Thus, $A \in \mathcal{N}_\varphi^{\mathrm{IP}}$ and, since $B \cap S = \varnothing$,

$$|S \cap A| = v_0 = \min_{1 \leq u \leq \hat{m}_0^{\mathrm{SC1}}} \left\{ \left| \left\{ i \in S : p_i > \ell_{u:\hat{m}_0^{\mathrm{SC1}}} \right\} \right| + u - 1 \right\},$$

  hence

$$\hat{V}_\varphi^{\mathrm{IP}}(S) \geq \hat{V}_{\mathfrak{R}}^{\mathrm{JER}}(S).$$

Remark A.6.1. *One interesting by-product from this proof is that the inversion procedure bound is always attained for a set of cardinality $\hat{m}_0^{\mathrm{SC1}}$.*

A.6.14 Proof of Proposition 3.4.11

If f_n is nondecreasing, then the assumption from Proposition 3.4.3 is fulfilled, and the expression of φ_A follows.
Assume now that f_n is right continuous for all $n \in [\![1, m]\!]$ and (f_n) is nonincreasing. Then, define the generalized inverse of f_n by:

$$\forall t \in [0,1], f_n^{-1}(t) = \min\{x \in \mathbb{R} : f_n(x) \geq t\},$$

and note that $\varphi_A = \max_{j \in [\![1,|A|]\!]} \mathbb{1}\{f_n^{-1}(j) > p_{(j:A)}\}$, and that $\big(f_n^{-1}(j)\big)$ is nondecreasing with respect to j and nonincreasing with respect to n. Thus, by letting $\ell_{j:n} = f_n^{-1}(j)$, we can apply the version of Proposition 3.3.12 suited to local tests defined by strict inequalities, and we get

$$\begin{aligned}
\hat{V}_\varphi^{\mathrm{IP}}(S) &= \min_{1 \leq u \leq |S|} \Big\{ \big| \{ i \in S : p_i \geq f_{\hat{m}_0}^{-1}(u) \} \big| + u - 1 \Big\} \\
&= \min_{1 \leq u \leq |S|} \Big\{ \big| \{ i \in S : f_{\hat{m}_0}(p_i) \geq u \} \big| + u - 1 \Big\} \\
&= \min_{0 \leq k \leq |S|} \Big\{ \big| \{ i \in S : f_{\hat{m}_0}(p_i) \geq \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1 \} \big| + \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor \Big\} \\
&= \min_{0 \leq k \leq |S|} \psi(k),
\end{aligned}$$

with $\hat{m}_0$ as defined in Equation (3.28) and $\psi(k) = \big| \{ i \in S : f_{\hat{m}_0}(p_i) \geq \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1 \} \big| + \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor$ for all $0 \leq k \leq |S|$, where the third equality holds because

$$\big| \{ i \in S : f_{\hat{m}_0}(p_i) \geq u \} \big| + u - 1$$

is nondecreasing with respect to u on $[\![\big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1, \big\lfloor f_{\hat{m}_0}\big(p_{(k+1:S)}\big) \big\rfloor]\!]$ when

$$\big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor \neq \big\lfloor f_{\hat{m}_0}\big(p_{(k+1:S)}\big) \big\rfloor.$$

We now finally prove Equation (3.27). For all $k \in [\![0, |S|]\!]$, let

$$k^* = \max\big\{ \ell \in [\![0, |S|]\!] : \big\lfloor f_{\hat{m}_0}\big(p_{(\ell:S)}\big) \big\rfloor = \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor \big\}.$$

Notably, $k^* \geq k$ and $\psi(k^*) = \psi(k)$. By construction and by $f_{\hat{m}_0}$ being nondecreasing, for all $\ell \in [\![0, |S|]\!]$,

$$\begin{aligned}
\ell > k^* &\Leftrightarrow \big\lfloor f_{\hat{m}_0}\big(p_{(\ell:S)}\big) \big\rfloor \geq \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1 \\
&\Leftrightarrow f_{\hat{m}_0}\big(p_{(\ell:S)}\big) \geq \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1.
\end{aligned}$$

So,

$$\begin{aligned}
\{ i \in S : f_{\hat{m}_0}(p_i) \geq \big\lfloor f_{\hat{m}_0}\big(p_{(k:S)}\big) \big\rfloor + 1 \} &= \{ i \in S : p_i = p_{(\ell:S)} \text{ for some } \ell > k^* \} \\
&= \{ i \in S : p_i > p_{(k^*:S)} \} \\
&= S \setminus R\big(p_{(k^*:S)}\big),
\end{aligned}$$

where the second inequality holds because if $p_i > p_{(k^*:S)}$, let ℓ such that $p_i = p_{(\ell:S)}$, necessarily $\ell > k^*$, and reciprocally if $p_i = p_{(\ell:S)}$ with $\ell > k^*$, then $\big\lfloor f_{\hat{m}_0}\big(p_{(\ell:S)}\big) \big\rfloor >$

$\lfloor f_{\hat{m}_0}(p_{(k:S)}) \rfloor = \lfloor f_{\hat{m}_0}(p_{(k^*:S)}) \rfloor$ and so $p_i > p_{(k^*:S)}$ again by monotonicity. Thus,

$$\begin{aligned}\psi(k) &= \psi(k^*) \\ &= \lfloor f_{\hat{m}_0}(p_{(k^*:S)}) \rfloor + |S \setminus R(p_{(k^*:S)})| \\ &\leq \lfloor f_{\hat{m}_0}(p_{(k:S)}) \rfloor + |S \setminus R(p_{(k:S)})|,\end{aligned}$$

where the last inequality holds because $k^* \geq k$, hence $S \setminus R(p_{(k^*:S)}) \subseteq S \setminus R(p_{(k:S)})$.

Finally, let $k_0 \in \operatorname*{arg\,min}_{0 \leq k \leq |S|} \psi(k)$, for all $k \in [\![0, |S|]\!]$ we have:

$$\begin{aligned}\lfloor f_{\hat{m}_0}(p_{(k_0^*:S)}) \rfloor + |S \setminus R(p_{(k_0^*:S)})| = \psi(k_0^*) = \psi(k_0) &= \hat{V}_\varphi^{\mathrm{IP}}(S) \\ \leq \psi(k) \leq \lfloor f_{\hat{m}_0}(p_{(k:S)}) \rfloor + |S \setminus R(p_{(k:S)})|,&\end{aligned}$$

which entails Equation (3.27).

Remark A.6.2. *The proposition and its proof prove that local tests of the form* (3.26) *can be rewritten in the form* (3.7) *by letting* $\ell_{i:n} = f_n^{-1}(i)$*. Furthermore, the* $\hat{m}_0$ *given by Equation* (3.28) *is then exactly the* $\hat{m}_0$ *given by Equation* (3.9)*.*

A.6.15 Proofs of Lemma 3.4.12 and Proposition 3.4.14

For all $n \in [\![1, m]\!]$ f_n is nondecreasing and $(f_n)_{n \in [\![1,m]\!]}$ is nonincreasing. Applying Proposition 3.4.11 and Lemma 3.4.7, for all $S \subseteq [\![1, m]\!]$,

$$\hat{V}_\varphi^{\mathrm{IP}}(S) = \min_{0 \leq k \leq |S|} \Big(\Big\lfloor f_{\hat{m}_0^{\mathrm{SC1}}}(p_{(k:S)}) \Big\rfloor + |S \setminus R(p_{(k:S)})| \Big),$$

with

$$\hat{m}_0^{\mathrm{SC1}} = \max \Big\{ n \in [\![0, m]\!] : \forall t \in [0,1], f_n(t) \geq n + i(t) - m \Big\}.$$

For all $n \in [\![0, m]\!]$, all $t \in [0,1]$ such that $\varphi_{[\![1,m]\!],n,t} = \mathbb{1}\{f_n(t) < n + i(t) - m\} = 1$,

$$\begin{aligned}f_{n+1}(t) &= (n+1)t + \sqrt{n+1}\lambda_\alpha \\ &= f_n(t) + t + \Big(\sqrt{n+1} - \sqrt{n}\Big)\lambda_\alpha \\ &< n + i(t) - m + 1 + \frac{i(t)}{n} - \frac{m}{n} - \left(\sqrt{n} + \frac{1}{\sqrt{n}} - \sqrt{n+1}\right)\lambda_\alpha \\ &< n + 1 + i(t) - m,\end{aligned}$$

thus $\varphi_{[\![1,m]\!],n+1,t} = 1$. Applying Proposition 3.4.4,

$$\hat{m}_0^{\mathrm{SC1}} = \min_{t \in [0,1]} \max \Big\{ n \in [\![0, m]\!] : f_n(t) \geq n + i(t) - m \Big\},$$

and the expression (3.30) follows by studying the roots of a second-degree polynomial in $\sqrt{n}$.

A.6.16 Proof of Proposition 3.5.1

Let $\bar{J} : x \mapsto \frac{1}{N}\sum_{i=1}^{N} J_i(x)$. We first prove that, for all $\lambda \geq 0$,

$$\mathbb{P}\left(\sqrt{N}\sup_{t\in\mathbb{R}}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{X_i \leq t\} - \bar{J}(t)\right) > \lambda\right) \leq e \cdot \exp\left(-2\lambda^2\right). \tag{A.6}$$

From Shorack and Wellner (1986), we know that (A.6) holds, with $ce \cdot \exp(-2\lambda^2)$ as the right-hand side of the inequation, for any constant c such that for all $\nu \geq 0$,

$$\mathbb{P}\left(\sqrt{N}\sup_{t\in[0,1]}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\{U_i \leq t\} - t\right) > \nu\right) \leq c\exp\left(-2\nu^2\right), \tag{A.7}$$

with $U_1, \ldots, U_N$ independent, uniform, random variables. Reeve (2024) established that (A.7) holds with $c = 1$. This gives (A.6). Equation (3.31) follows immediately from (A.6) by a simple inclusion of events.

Note that, compared to (A.6), (3.31) is suited to the general case of Assumption 3.3, where the exact cdf of the p-values under the null is not necessarily known.

Remark A.6.3. *Massart (1990) showed that (A.7) holds with $c = 1$ but only for $\nu \geq \sqrt{\frac{\log 2}{2}}$, which does not allow the proof of Shorack and Wellner (1986) to use $c = 1$ and get (A.6), as that proof really requires (A.7) to be true for all $\nu \geq 0$. Prior to the work of Reeve (2024), we could have used $c = 2$, as (A.7) holds for $c = 2$ and any $\nu \geq 0$, by the two-sided version of the DKW inequality proven by Massart (1990). Hence, we divide the upper bound on the probability by a factor 2 thanks to very recent works.*

A.6.17 Proof of Corollary 3.5.2

Let $\mu \in H_A$. Equivalently, $A \subseteq \mathcal{H}_0$, so for each $i \in A$, the cdf of p_i is bounded by F_i, by Assumption 3.3. Applying (3.31) to $(p_i)_{i\in A}$ and $\tilde{\lambda}_\alpha$, it comes that:

$$\begin{aligned}
\alpha = e \cdot \exp\left(-2\tilde{\lambda}_\alpha^2\right) &\geq \mathbb{P}_\mu\left(\sqrt{|A|}\sup_{t\in\mathbb{R}}\left(\frac{1}{|A|}\sum_{i\in A}\mathbb{1}\{p_i \leq t\} - \frac{1}{|A|}\sum_{i\in A}F_i(t)\right) > \tilde{\lambda}_\alpha\right) \\
&= \mathbb{P}_\mu\left(\sup_{t\in\mathbb{R}}\left(\sum_{i\in A}\mathbb{1}\{p_i \leq t\} - \sum_{i\in A}F_i(t)\right) > \tilde{\lambda}_\alpha\sqrt{|A|}\right) \\
&= \mathbb{P}_\mu\left(\exists t \in \mathbb{R} : \sum_{i\in A}\mathbb{G}_i(t) > \tilde{\lambda}_\alpha\sqrt{|A|}\right) \\
&= \mathbb{P}_\mu(\varphi_A = 1).
\end{aligned}$$

This proves that φ_A is indeed a local test at level α.

A.6.18 Proof of Lemma 3.5.4

Let $t \in (0, 1]$. Then to minimize $\varphi_{A,t}$ with respect to $A \subseteq [\![1, m]\!]$ such that $|A \cap S| = n$, we let $|A| = n + j$ and then minimize $\sum_{k\in A}\mathbb{G}_k(t) = \sum_{k\in A\cap S}\mathbb{G}_k(t) + \sum_{k\in A\cap S^c}\mathbb{G}_k(t)$. The first sum is lower bounded by $\sum_{1\leq k\leq n}\big(\mathbb{G}(t)\big)_{(k:S)}$, while the second one is lower bounded by $\sum_{1\leq k\leq j}\big(\mathbb{G}(t)\big)_{(k:S^c)}$.

A.6.19 Proof of Proposition 3.5.6

By assumption,

$$\left\{\forall t \in [0,1], m_0^{-1/2} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \tilde{\lambda}_\alpha\right\} \subseteq \left\{m_0 \leq \hat{m}_0\right\},$$

thus,

$$\begin{aligned}
&\left\{\forall t \in [0,1], m_0^{-1/2} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \tilde{\lambda}_\alpha\right\} \\
=&\left\{\forall t \in [0,1], \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq t} \leq \sum_{i \in \mathcal{H}_0} F_i(t) + \sqrt{m_0}\tilde{\lambda}_\alpha\right\} \bigcap \left\{m_0 \leq \hat{m}_0\right\} \\
\subseteq&\left\{\forall t \in [0,1], \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq t} \leq \sum_{1 \leq i \leq \hat{m}_0} \big(\mathbf{F}(t)\big)_{[i]} + \sqrt{\hat{m}_0}\tilde{\lambda}_\alpha\right\}.
\end{aligned}$$

Then, as $\tilde{\lambda}_\alpha$ verifies the equality $e \cdot \exp\left(-2\tilde{\lambda}_\alpha^2\right) = \alpha$, it follows from Proposition 3.5.1 that

$$\begin{aligned}
&\mathbb{P}\left(\forall t \in [0,1], \sum_{i \in \mathcal{H}_0} \mathbb{1}_{p_i \leq t} \leq \sum_{1 \leq i \leq \hat{m}_0} \big(\mathbf{F}(t)\big)_{[i]} + \sqrt{\hat{m}_0}\tilde{\lambda}_\alpha\right) \\
&\geq \mathbb{P}\left(\forall t \in [0,1], m_0^{-1/2} \sum_{i \in \mathcal{H}_0} \mathbb{G}_i(t) \leq \tilde{\lambda}_\alpha\right) \\
&\geq 1 - \alpha.
\end{aligned}$$

A.6.20 Proof of Lemma 3.5.12

Let $U \subseteq V$. The threshold family τ satisfies (C1) with respect to V, thus, for all $k \in [\![1, |U|]\!]$ and $A \subseteq V$ such that $|A| = |V| - k + 1$, $\sum_{i \in A} \mathbf{H}\big(\tau_{|V|}, \tau_k\big) \leq k\alpha$, therefore

$$\begin{aligned}
\sum_{1 \leq i \leq |U|-k+1} \Big(\mathbf{H}\big(\tau_{|U|}, \tau_k\big)\Big)_{[i:U]} &\leq \sum_{1 \leq i \leq |U|-k+1} \Big(\mathbf{H}\big(\tau_{|V|}, \tau_k\big)\Big)_{[i:U]} \\
&\leq \sum_{1 \leq i \leq |V|-k+1} \Big(\mathbf{H}\big(\tau_{|V|}, \tau_k\big)\Big)_{[i:V]} \\
&\leq k\alpha,
\end{aligned}$$

where the first inequality comes because τ is a nondecreasing family of thresholds and $\mathbf{H}$ is coordinate-wise nondecreasing on its first parameter. The second inequality comes because all the terms of the smaller sum are terms of the greater sum. Thus $\tau_k \in \left\{t \in \mathcal{A}, t \leq \lambda : \big(\mathbf{H}(\lambda, t)\big)_{[1:U]} + \cdots + \big(\mathbf{H}(\lambda, t)\big)_{[|U|-k+1:U]} \leq k\alpha\right\}$ therefore $\tau_k \leq \tilde{\tau}_k$. Note that the previous sequence of inequalities with $k = |U|$ grants that $\tau_{|U|}$ verifies (3.39). Thus $\tilde{\tau}$ satisfies (C1) at level α with respect to U, and for all $k \in [\![1, m]\!]$, $\tau_k \leq \tilde{\tau}_k$, therefore τ satisfies (C1) at level α with respect to U.

A.6.21 Proof of Corollary 3.5.15

Let $\tilde{\tau}$ be defined by Example 3.5.11 with $U = A$. Then, $\tilde{\tau}$ satisfies (C1) at level α with respect to A, thus by Theorem 3.5.13, $\mathbb{P}\big(\exists i \in [\![1, |A|]\!], p_{(i:A)} \leq \tilde{\tau}_i\big) \leq \alpha$, and (3.42) arises

from the following equality:

$$\{p_{(i:A)} \leq \tilde{\tau}_i\} = \left\{ \sum_{\ell=1}^{k-i+1} \left(\mathbf{H}(\lambda, p_{(i:A)})\right)_{[\ell:A]} \leq i\alpha \right\} \bigcap \{p_{(i:A)} \leq \lambda\},$$

thus φ_A is indeed a local test of level α.
We now prove that expressions (3.42) and (3.43) are equal. Suppose that expression (3.42) is equal to 1. Then, there exists $i \in [\![1, k]\!]$ such that $\sum_{\ell=1}^{k-i+1} \left(\mathbf{H}(\lambda, p_{(i:A)})\right)_{[\ell:A]} \leq i\alpha$ and $p_{(i:A)} \leq \lambda$.
Let $j = i(p_{(i:A)})$. Then $p_{(j:A)} = p_{(i:A)} \leq \lambda$, $j \geq i$, thus $\sum_{\ell=1}^{k-j+1} \left(\mathbf{H}(\lambda, p_{(j:A)})\right)_{[\ell:A]} \leq j\alpha$.
Thus, expression (3.43) is equal to 1, with $t = p_{(j:A)}$.
Assume now that expression (3.43) is equal to 1, meaning there exists $t \leq \lambda$ such that $\sum_{\ell=1}^{k-i_A(t)+1} \left(\mathbf{H}(\lambda, t)\right)_{[\ell:A]} \leq i_A(t)\alpha$. Then, because $\mathbf{H}(\lambda, \cdot)$ is coordinate-wise nondecreasing and $p_{(i_A(t):A)} \leq t \leq \lambda$, $\sum_{\ell=1}^{k-i_A(t)+1} \left(\mathbf{H}(\lambda, p_{(i_A(t):A)})\right)_{[\ell:A]} \leq i_A(t)\alpha$, thus expression (3.42) is equal to 1.

A.6.22 Proof of Lemma 3.5.17

We show that if $A \subseteq [\![1, m]\!]$ is such that $|A \cap S| = n$ and $\varphi_{A,t} = 0$, then $B = A \cup (R(t)^c \cap S^c)$ is such that $\varphi_{B,t} = 0$. Assume that $A^c \cap R(t)^c \cap S^c \neq \varnothing$ and let $j \in A^c \cap R(t)^c \cap S^c$ and $\tilde{A} = A \cup \{j\}$. We have $\varphi_{A,t} = 0$, thus

$$\sum_{k=1}^{|A|-i_A(t)+1} \left(\mathbf{H}(\lambda, t)\right)_{[k:A]} > i_A(t)\alpha,$$

therefore

$$\sum_{k=1}^{|\tilde{A}|-i_{\tilde{A}}(t)+1} \left(\mathbf{H}(\lambda, t)\right)_{[k:\tilde{A}]} > i_{\tilde{A}}(t)\alpha$$

because $i_{\tilde{A}}(t) = i_A(t)$ and $|\tilde{A}| = |A| + 1$.
Thus $\varphi_{S,n,t}$ is achieve on a set A such that $R(t)^c \cap S^c \subseteq A$.
We now assume that $A \cap R(t) \cap S^c \neq \varnothing$ and let $j \in A \cap R(t) \cap S^c$ and $\tilde{A} = A \setminus \{j\}$. We have $\varphi_{A,t} = 0$, thus

$$\sum_{k=1}^{|A|-i_A(t)+1} \left(\mathbf{H}(\lambda, t)\right)_{[k:A]} > i_A(t)\alpha,$$

therefore,

$$\sum_{k=1}^{|\tilde{A}|-i_{\tilde{A}}(t)+1} \left(\mathbf{H}(\lambda, t)\right)_{[k:\tilde{A}]} > i_{\tilde{A}}(t)\alpha,$$

because $i_{\tilde{A}}(t) = i_A(t) - 1$ and all the two sums are on the same subset of A except possibly for j. In this case, the distance between the last term of the second sum and $\mathbf{H}_j(\lambda, t)$ is at most α by definition of λ. Thus $\varphi_{S,n,t}$ is achieve on a set A such that $A \cap R(t) \cap S^c = \varnothing$.
Note that by similar arguments considering $\tilde{A} = A \cup \{j_2\} \setminus \{j_1\}$ with $j_1 \in A \cap S \cap R(t)$

and $j_2 \in A^c \cap S \cap R(t)^c$, $\varphi_{S,n,t}$ is achieve on a set A such that $R(t)^c \cap S \subseteq A$.
To summarize, $\varphi_{S,n,t}$ is attained on a set A such that $R(t)^c \subseteq A$ and $A \cap R(t) \cap S^c = \varnothing$.
Thus, defining $k_S(t) = (n - |R(t)^c \cap S|) \vee 0$ and $S(t) = R(t) \cap S$,

$$\varphi_{S,n,t} = \mathbb{1}\left\{ \sum_{k=1}^{m-i(t)+1} \left(\mathbf{H}_{R(t)^c}(\lambda, t), \big(\mathbf{H}(\lambda,t)\big)_{[1:S(t)]}, \ldots, \big(\mathbf{H}(\lambda,t)\big)_{[k_S(t):S(t)]} \right)_{[k]} \leq k_S(t)\alpha \right\}.$$

A.6.23 Proof of Example 3.6.3

The conditions on α and ε are merely assumptions implying positive $a_1 < 1$ and $a_2 < 1$.
We first show that $\hat{m}_0 = \max\left\{ n \in [\![0,4]\!] : \forall i \in [\![1,4]\!], \sum_{1\leq j\leq n} \big(\mathbb{G}(p_i)\big)_{(j)} \leq \tilde{\lambda}_\alpha \sqrt{n} \right\} = 3.$

$$\begin{aligned} 3 - \sum_{1\leq j\leq 4} \mathbf{F}_j(p_3) = 3 - a_1 - 2a_2 &= 3 - \left(\frac{2 - \tilde{\lambda}_\alpha \sqrt{3}}{2} - \varepsilon \right) - \left(2 - \tilde{\lambda}_\alpha \sqrt{3} \right) \\ &= \tilde{\lambda}_\alpha \left(\frac{3\sqrt{3}}{2} - \sqrt{4} \right) + \sqrt{4}\tilde{\lambda}_\alpha + \varepsilon \\ &> \sqrt{4}\tilde{\lambda}_\alpha. \end{aligned} \tag{A.8}$$

The inequality $\sum_{1\leq j\leq 3} \big(\mathbb{G}(p_i)\big)_{(j)} \leq \tilde{\lambda}_\alpha \sqrt{3}$ is straightforward for $i \in \{1,4\}$, because the left-hand side of this inequality is nonpositive. For $i \in \{2,3\}$, $\sum_{1\leq j\leq 3} \big(\mathbb{G}(p_i)\big)_{(j)} = 2 - 2a_2 = \tilde{\lambda}_\alpha \sqrt{3}$. Thus, with $\zeta_1 = 1$, $\zeta_2 = 2 \wedge \left\lfloor a_1 + 2a_2 + \sqrt{3}\tilde{\lambda}_\alpha \right\rfloor = 2$, $\zeta_3 \geq 2$ and $\zeta_4 = 3$ defined by (3.47), we get $\hat{V}_{\mathfrak{R}}^{\mathrm{JER}}\big(\{1,2,4\}\big) = 3$.
We now show that $\hat{V}_{\varphi}^{\mathrm{SC1}}\big(\{1,2,4\}\big) \leq 2$. Recall that, from Proposition 3.5.5,

$$\hat{V}_{\varphi}^{\mathrm{SC1}}(\{1,2,4\}) = \max\left\{ n \in [\![0,3]\!] : \forall i \in [\![1,4]\!], \exists j \leq 1, \sum_{1\leq k\leq n} \big(\mathbb{G}(p_i)\big)_{(k:\{1,2,4\})} + \sum_{1\leq k\leq j} \mathbb{G}_3(p_i) \leq \tilde{\lambda}_\alpha \sqrt{n+j} \right\}.$$

Thus, if we test the condition for $n = 3$ in p_3,

$$\begin{aligned} \sum_{k\in\{1,2,4\}} \mathbb{G}_k(p_3) &= 2 - F_1(p_3) - F_2(p_3) - F_4(p_3) \\ &= 2 - a_1 - a_2 \\ &= \sqrt{3}\tilde{\lambda}_\alpha + \varepsilon > \sqrt{3}\tilde{\lambda}_\alpha, \end{aligned}$$

and,

$$\begin{aligned} \sum_{k\in\{1,2,4\}} \mathbb{G}_k(p_3) + \mathbb{G}_3(p_3) &= 3 - \sum_{1\leq j\leq 4} \mathbf{F}_j(p_3) \\ &> \sqrt{4}\tilde{\lambda}_\alpha, \end{aligned}$$

from (A.8). Therefore, $\hat{V}_{\varphi}^{\mathrm{SC1}}\big(\{1,2,4\}\big) \leq 2$.

A.6.24 Proof of Proposition 3.6.5

The expression of φ_A^{hom} as a local test of level α arises from upper bounding $\sum_{i\in A}\mathbf{F}_i(t)$ by $\sum_{1\le i\le|A|}\big(\mathbf{F}(t)\big)_{[i]}$.
Furthermore, φ_A^{hom} has the expression (3.26) and verifies that each individual f_n is non-decreasing for all $n\in[\![1,m]\!]$ and $(f_n)_{n\in[\![1,m]\!]}$ is nondecreasing as a sequence of functions, thus, by Proposition 3.4.11,

$$\hat{V}^{\mathrm{IP}}_{\varphi^{hom}}(S)=\min_{0\le k\le|S|}\Big(\Big\lfloor f_{\hat{m}_0^{hom}}\big(p_{(k:S)}\big)\Big\rfloor+\big|S\setminus R\big(p_{(k:S)}\big)\big|\Big)$$

where

$$\hat{m}_0^{hom}=\max\Big\{s\in[\![0,m]\!]:\forall i\in[\![1,s]\!],f_s\big(p_{[i]}\big)\ge s-i+1\Big\}.$$

However, if $\hat{m}_0^{\mathrm{SC1}}$ is defined by (3.36), we have $\hat{m}_0^{\mathrm{SC1}}\le\hat{m}_0^{hom}$ because $\varphi^{hom}_{[\![1,m]\!],n,t}\le\varphi_{[\![1,m]\!],n,t}$. Therefore, for all $S\subseteq[\![1,m]\!]$, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)\le\hat{V}^{\mathrm{IP}}_{\varphi^{hom}}(S)$ because

$$\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)=\min_{0\le k\le|S|}\Big(\Big\lfloor f_{\hat{m}_0^{\mathrm{SC1}}}\big(p_{(k:S)}\big)\Big\rfloor+\big|S\setminus R\big(p_{(k:S)}\big)\big|\Big)\wedge\hat{m}_0^{\mathrm{SC1}}.$$

A.6.25 Proof of Proposition 3.6.9 and Remark 3.6.10

Note that $\hat{m}_0^{hom}$ is more conservative than the over estimator $\hat{m}_0^{\mathrm{SC1}}$ defined by (3.45), because $\hat{m}_0^{hom}=\hat{V}^{\mathrm{SC1}}_{\varphi^{hom}}\big([\![1,m]\!]\big)$ and for all $A\subseteq[\![1,m]\!]$ and $t\in[0,1]$, $\varphi_{A,t}\ge\varphi^{hom}_{A,t}$ hence $\varphi_{[\![1,m]\!],n,t}\ge\varphi^{hom}_{[\![1,m]\!],n,t}$ for all $n\in[\![0,m]\!]$. Therefore, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}\le\hat{V}^{\mathrm{SC1}}_{\varphi^{hom}}$.

We then compute the expression of $\hat{m}_0^{hom}$. Note that $\ell_{i+1:n+1}=\max\big\{t\in\mathcal{A},t\le\lambda:\big(\mathbf{H}(\lambda,t)\big)_{[1]}+\cdots+\big(\mathbf{H}(\lambda,t)\big)_{[n-i+1]}\le(i+1)\alpha\big\}\ge\ell_{i:n}$, thus $\hat{m}_0^{hom}=\hat{V}^{\mathrm{SC2}}_{\varphi^{hom}}\big([\![1,m]\!]\big)$ by Lemma 3.4.7.
Let $t\in(0,1]$. Since $\varphi^{hom}_{[\![1,m]\!],n,t}=\mathbb{1}\big\{t\le\ell_{(n+i(t)-m):n}\big\}$ for all $n\in[\![1,m]\!]$ by Lemma 3.4.5, then $b(t)=\max\Big\{n\in[\![0,m]\!]:\varphi^{hom}_{[\![1,m]\!],n,t}=0\Big\}=\max\big\{n\in[\![0,m]\!]:t>\ell_{(n+i(t)-m):n}\big\}$. Thus, by definition of $(\ell_{i:n})$, $b(t)=\max\Big\{n\in[\![0,m]\!]:\sum_{k=1}^{m-i(t)+1}\big(\mathbf{H}(\lambda,t)\big)_{[k]}>(n+i(t)-m)\alpha\Big\}$, and the expression of $\hat{m}_0^{hom}$ follows.

Appendix B

Supplementary material for Chapter 6

Contents

B.1 Deferred proofs from Chapter 6 157
B.1.1 Proof of Proposition 6.3.3 157
B.1.2 Proof of Proposition 6.2.5 158
B.1.3 Proof of Proposition 6.3.4 159
B.2 An example of a coherent confidence envelope that cannot be written as a closed testing procedure 159

B.1 Deferred proofs from Chapter 6

B.1.1 Proof of Proposition 6.3.3

We first state a lemma that will be useful in the proof of Proposition 6.3.3.

Lemma B.1.1. *Let $\mathfrak{R} = (R_k, \zeta_k)_{k\in\mathcal{K}}$ be a reference family and $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ the confidence envelope associated to $\mathfrak{R}$. Then we have $\mathrm{op}_{\mathrm{JER}}\big(\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}\big) = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$.*

Proof. We show that $\mathcal{A}(\mathfrak{R}) = \mathcal{A}\Big(\mathfrak{R}_{\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}}\Big)$. The key property that we use here comes from Proposition 2.1 from Blanchard et al. (2020a): for any $A \subseteq [\![1,m]\!]$, $A \in \mathcal{A}(\mathfrak{R})$ if and only if for all $S \subseteq [\![1,m]\!]$, $|A \cap S| \leq \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)$. Hence, we have $\mathcal{A}(\mathfrak{R}) = \mathcal{A}\Big(\mathfrak{R}_{\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}}\Big)$ since $\mathcal{A}\Big(\mathfrak{R}_{\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}}\Big)$ is defined as the set of all $A \subseteq [\![1,m]\!]$ such that for all $S \subseteq [\![1,m]\!]$, $|A \cap S| \leq \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S)$. □

Let $\hat{V}$ be a confidence envelope and $S \subseteq [\![1,m]\!]$. Denote by $\mathfrak{R}_{\hat{V}} = \Big(S, \hat{V}(S)\Big)_{S\subseteq[\![1,m]\!]}$ the reference family associated to $\hat{V}$. We first show that $\mathrm{op}_{\mathrm{JER}}$ is an idempotent operator. By Lemma B.1.1, we have, since $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}}} = \mathrm{op}_{\mathrm{JER}}\big(\hat{V}\big)$ by definition of $\mathrm{op}_{\mathrm{JER}}$ in (6.8), that $\mathrm{op}_{\mathrm{JER}}\Big(\mathrm{op}_{\mathrm{JER}}\big(\hat{V}\big)\Big) = \mathrm{op}_{\mathrm{JER}}\big(\hat{V}\big)$.

We then show that $\mathrm{op}_{\mathrm{DBNR}}$ is an idempotent operator. We denote by $\mathfrak{R}_{\mathrm{op}_{\mathrm{DBNR}}\left(\hat{V}\right)} = \Big(\big(S, \mathrm{op}_{\mathrm{DBNR}}\big(\hat{V}\big)(S)\big)\Big)_{S\subseteq[\![1,m]\!]}$ the reference family associated to $\mathrm{op}_{\mathrm{DBNR}}\big(\hat{V}\big)$. Then, for $S \subseteq [\![1,m]\!]$, we have

- $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\mathrm{op}_{\mathrm{DBNR}}(\hat{V})}}(S) \leq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}}}(S)$ by definition of $\mathrm{op}_{\mathrm{DBNR}}$ in (6.9) with $\mathcal{B} = \{S\}$.

- Let $\mathcal{B} \subseteq \mathcal{P}(\llbracket 1, m \rrbracket)$ such that $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\mathrm{op}_{\mathrm{DBNR}}(\hat{V})}}(S) = \sum_{T \in B} \mathrm{op}_{\mathrm{DBNR}}(\hat{V})(T)$ and for all $T \in \mathcal{B}$, let $\mathcal{B}_T \subseteq \mathcal{P}(\llbracket 1, m \rrbracket)$ such that $\mathrm{op}_{\mathrm{DBNR}}(\hat{V})(T) = \sum_{U \in \mathcal{B}_T} \hat{V}(U)$. Recall that, by (6.10), such sets exist and we have that for all $T \in \mathcal{B}$, $T \subseteq \bigcup_{U \in \mathcal{B}_T} U$. Then

$$\begin{aligned}
\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\mathrm{op}_{\mathrm{DBNR}}(\hat{V})}}(S) &= \sum_{T \in \mathcal{B}} \mathrm{op}_{\mathrm{DBNR}}(\hat{V})(T) \\
&= \sum_{T \in \mathcal{B}} \sum_{U \in \mathcal{B}_T} \hat{V}(U) \\
&\geq \sum_{U \in \bigcup_{T \in \mathcal{B}} \mathcal{B}_T} \hat{V}(U) \\
&\geq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}}}(S),
\end{aligned}$$

where the last inequality comes from the fact that $S \subseteq \bigcup_{T \in \mathcal{B}} T \subseteq \bigcup_{U \in \bigcup_{T \in \mathcal{B}} \mathcal{B}_T} U$. Hence, $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\mathrm{op}_{\mathrm{DBNR}}(\hat{V})}}(S) = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}}}(S)$ and we have $\mathrm{op}_{\mathrm{DBNR}}(\hat{V}) = \mathrm{op}_{\mathrm{DBNR}}\Big(\mathrm{op}_{\mathrm{DBNR}}(\hat{V})\Big)$.

B.1.2 Proof of Proposition 6.2.5

Let $\mathfrak{R} = (R_k, \zeta_k)_{k \in \mathcal{K}}$ be a reference family. We show that $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ and $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ are coherent. This proof relies on Lemma 3 from Goeman et al., 2021: a confidence envelope $\hat{V}$ is coherent if and only if for all disjoint $S_1, S_2 \subseteq \llbracket 1, m \rrbracket$,

$$\hat{V}(S_1) \leq \hat{V}(S_1 \cup S_2) \leq \hat{V}(S_1) + \hat{V}(S_2). \tag{B.1}$$

Then, with $S_1, S_2 \subseteq \llbracket 1, m \rrbracket$ disjoint, we have:

- $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S_1) \leq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S_1 \cup S_2)$ since for all $Q \subseteq \mathcal{K}$,

$$\left| S_1 \cup S_2 \setminus \bigcup_{k \in Q} R_k \right| \geq \left| S_1 \setminus \bigcup_{k \in Q} R_k \right|.$$

- $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S_1 \cup S_2) \leq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S_1) + \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S_2)$ since for all $Q_1, Q_2 \subseteq \mathcal{K}$,

$$\left| S_1 \cup S_2 \setminus \bigcup_{k \in Q_1 \cup Q_2} R_k \right| \leq \left| S_1 \setminus \bigcup_{k \in Q_1} R_k \right| + \left| S_2 \setminus \bigcup_{k \in Q_2} R_k \right|,$$

  and $\sum_{k \in Q_1 \cup Q_2} \zeta_k \leq \sum_{k \in Q_1} \zeta_k + \sum_{k \in Q_2} \zeta_k$, in particular with Q_1 and Q_2 respectively achieving $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ for S_1 and S_2.

Hence, $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ is coherent. Remark that this proof does not use the fact that S_1 and S_2 are disjoint; the disjoint property of the sets is not required for Lemma 3 from Goeman et al., 2021, since if $\hat{V}$ verifies (B.1) for all disjoint $S_1, S_2 \subseteq \llbracket 1, m \rrbracket$, then it also verifies it for all $S_1, S_2 \subseteq \llbracket 1, m \rrbracket$.
To show that $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ is coherent, we use the same lemma as above with the following inequalities:

- $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1) \leq \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1 \cup S_2)$ since for all $A \in \mathcal{A}(\mathfrak{R})$, $|A \cap S_1| \leq |A \cap (S_1 \cup S_2)|$.

- $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1 \cup S_2) \leq \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1) + \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_2)$ since for all $A \in \mathcal{A}(\mathfrak{R})$ such that $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1 \cup S_2) = |A \cap (S_1 \cup S_2)|$,

$$|A \cap (S_1 \cup S_2)| \leq |A \cap S_1| + |A \cap S_2| \leq \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_1) + \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}(S_2),$$

  where the last inequality comes from the definition of $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$.

Hence, $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ is coherent.

B.1.3 Proof of Proposition 6.3.4

Let $S \subseteq [\![1, m]\!]$ and $\mathfrak{R} = (R_k, \zeta_k)_{k \in \mathcal{K}}$ a reference family. We show that $\tilde{\mathfrak{R}} = \big(\mathfrak{R}, (S, |S|)\big)$ is such that $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}} = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ and $\hat{V}^{\mathrm{JER}}_{\tilde{\mathfrak{R}}} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$.
We first show that $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}} = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$. We have $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}} \leq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ trivially since $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}}$ is the minimum of a larger set than $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$. Let $Q \subseteq \mathcal{K}$. Then, we have, for all $T \subseteq [\![1, m]\!]$,

$$\begin{aligned}\left|T \setminus \Big(\bigcup_{k \in Q} R_k \cup S\Big)\right| + \sum_{k \in Q} \zeta_k + |S| &= \left|T \setminus \bigcup_{k \in Q} R_k\right| + \sum_{k \in Q} \zeta_k + |S| - \left|T \setminus \Big(\bigcup_{k \in \tilde{Q}} R_k \cup S^c\Big)\right| \\ &\geq \left|T \setminus \bigcup_{k \in Q} R_k\right| + \sum_{k \in Q} \zeta_k,\end{aligned}$$

hence $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}} \geq \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ and we have $\hat{V}^{\mathrm{DBNR}}_{\tilde{\mathfrak{R}}} = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$.
The equality $\hat{V}^{\mathrm{JER}}_{\tilde{\mathfrak{R}}} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ is trivial since $\mathcal{A}(\tilde{\mathfrak{R}}) = \mathcal{A}(\mathfrak{R})$ by definition of $\mathcal{A}(\cdot)$. Hence, reasoning by induction, we have that $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}_{\hat{V}^{(0)}_{\mathfrak{R}}}} = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ and $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}_{\hat{V}^{(0)}_{\mathfrak{R}}}} = \hat{V}^{\mathrm{JER}}_{\mathfrak{R}}$ for all reference family $\mathfrak{R}$.

B.2 An example of a coherent confidence envelope that cannot be written as a closed testing procedure

Consider Example 6.2.9. As shown in Example 6.2.9, we have $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}([\![1, m]\!]) = 2$ and $\hat{V}^{\mathrm{JER}}_{\mathfrak{R}}([\![1, m]\!]) = 1$. We show that the closed testing procedure associated to $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ defined in Corollary 1 from Goeman et al. (2021), that is $\hat{V}^{\mathrm{CT}}_{\varphi}$ with $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$ and $\varphi_S = \mathbb{1}_{\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}(S) < |S|}$, is different from $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$. We can easily check that for all $S \subseteq [\![1, m]\!]$ such that $|S| \leq 1$, $\varphi_S = 0$ and that for all $S \subseteq [\![1, m]\!]$ such that $|S| \geq 2$, $\varphi_S = 1$. Hence, we have

$$\begin{aligned}\hat{V}^{\mathrm{CT}}_{\varphi}\big([\![1, m]\!]\big) &= \max_{A \subseteq [\![1,m]\!]} \left\{|A|, \inf_{U \supseteq A} \varphi_U = 0\right\} \\ &= \max_{A \subseteq [\![1,m]\!]} \{|A|, |A| \leq 1\} \\ &= 1.\end{aligned}$$

Hence, we have $\hat{V}^{\mathrm{CT}}_{\varphi}([\![1, m]\!]) = 1 \neq 2 = \hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}([\![1, m]\!])$, and we have an example of a coherent confidence envelope that cannot be written as a closed testing procedure by the methodology of Corollary 1 from Goeman et al. (2021), because by Proposition 6.2.5, for all reference family $\mathfrak{R}$, $\hat{V}^{\mathrm{DBNR}}_{\mathfrak{R}}$ is a coherent confidence envelope. This result obviously shows that Corollary 1 from Goeman et al. (2021) is erroneous.

Furthermore, let us show that $\hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ cannot be written as a closed testing procedure associated to any family of local tests.
Let $\varphi = (\varphi_S)_{S \subseteq [\![1,m]\!]}$ be a family of local tests and $\hat{V}_{\varphi}^{\mathrm{CT}}$ the associated closed testing procedure. Assume that $\hat{V}_{\varphi}^{\mathrm{CT}} \leq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$. We show that $\hat{V}_{\varphi}^{\mathrm{CT}}([\![1,m]\!]) < \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}([\![1,m]\!])$.
We have for all $i \in \{1,2,3\}$, $\hat{V}_{\varphi}^{\mathrm{CT}}(R_i) \leq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}(R_i) = \zeta_i = 1$. Hence, for all $i \in \{1,2,3\}$, for all $A \subseteq R_i$, $|A| \geq 2$ implies $\inf_{U \supseteq A} \varphi_U = 1$. It implies that for all $A \subseteq [\![1,m]\!]$ such that $|A| \geq 2$, $\inf_{U \supseteq A} \varphi_U = 1$. Therefore, we have

$$\hat{V}_{\varphi}^{\mathrm{CT}}([\![1,m]\!]) = \max_{A \subseteq [\![1,m]\!]} \left\{ |A|, \inf_{U \supseteq A} \varphi_U = 0 \right\} \leq 1 < 2 = \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}([\![1,m]\!]).$$

Hence, $\hat{V}_{\varphi}^{\mathrm{CT}} \neq \hat{V}_{\mathfrak{R}}^{\mathrm{DBNR}}$ for all families of local tests φ.

Appendix C

A note on $\mathrm{BH}+$ procedure from Chen (2020)

Contents

C.1 The $\mathrm{BH}+$ procedure 161
C.2 Generalization of the $\mathrm{BH}+$ procedure 162

This appendix discusses the procedure $\mathrm{BH}+$ from Chen (2020) and its relation with the vanilla BH procedure. It takes place in the heterogeneous multiple testing setting.

C.1 The $\mathrm{BH}+$ procedure

In Chen (2020), the author constructs a sequence of thresholds based on information on the null distributions of p-values in a heterogeneous setting, and then incorporates these thresholds in a step-up procedure. The thresholds $(\gamma_k)_{1\leq k\leq m}$ are defined by the following equation:

$$\forall k \leq m, \gamma_k = \max\left\{t \in S^* : F^*(t) \leq \frac{\alpha k}{m}\right\},$$

where $F^* = \max_{1\leq i\leq m} F_i$ is the maximum of the cumulative distribution functions (CDF) under the null hypotheses — hence F_i is the CDF of p_i when $i \in \mathcal{H}_0$ contrary to the assumptions Assumption 3.3 and Assumption 5.1 from Chapter 3 and Chapter 5 which consider an upper bound, and S^* is the support of F^*. Furthermore, in the simulations, all the CDF are supposed to have a finite support and there is a finite number of CDF, hence they verify that $F_i(t) = t$ when t is in the support of F_i.
We claim that this procedure reduces to applying the vanilla BH procedure on the set of p-values $(F^*(p_i))_{1\leq i\leq m}$.
Indeed, one can easily check the equality of the events $\{p_{(k)} \leq \gamma_k\}$ and $\{F^*(p_{(k)}) \leq \frac{\alpha k}{m}\}$. Furthermore, with the nondecreasing property of $u \mapsto \max\{t \in S^* : F^*(t) \leq u\}$, the wPRDS property on $(p_i)_{1\leq i\leq m}$ easily derives a wPRDS property on $(F^*(p_i))_{1\leq i\leq m}$ from Theorem C.2.1.
It follows that the super-uniform case of Chen (2020) rejects the same set than vanilla BH, because, in this simulation setting, $F^*(p_{(k)}) = p_{(k)}$, hence $\mathrm{BH}+$ rejects the same set than the one BH rejects when applied to $(F^*(p_i))_{1\leq i\leq m} = (p_i)_{1\leq i\leq m}$, .

From this analysis, Theorem 1 from Chen (2020), that states that BH+ is a conservative procedure, simply stems from the conservativeness from BH procedure. Furthermore, its Theorem 2, that states properties on the rejection set of BH+, also stems from the definition of vanilla BH rejection set and classical properties on it.

C.2 Generalization of the $\mathrm{BH}+$ procedure

To generalize this procedure, we prove first a more general result on wPRDS property. Let $X_1, \dots, X_n$ be $n \in \mathbb{N}$ random variables and $F_1, \dots, F_n$ their respective CDF. We assume that $X_1, \dots, X_m$ is wPRDS on $\mathcal{H}_0$ *i.e.*

$$\forall D \subseteq \mathbb{R}^n \text{ nondecreasing}, \forall i \in \mathcal{H}_0,\ u \mapsto \mathbb{P}((X_1, \dots, X_m) \in D \mid X_i \leq u) \tag{C.1}$$

is nondecreasing on $\{u : \mathbb{P}(X_i \leq u) > 0\}$, where a nondecreasing set is defined by the following property; for all $x, y \in \mathbb{R}^n$, $(x \leq y) \implies (x \in D \implies y \in D)$, where $x \leq y$ means that for all $i \in [\![1, n]\!], x_i \leq y_i$. Note that the nondecreasing property of $u \mapsto \mathbb{P}((X_1, \dots, X_m) \in D \mid X_i \leq u)$ is equivalent to the following inequality for all $u', v' \in \{w : \mathbb{P}(X_i \leq w) > 0\}$ such that $u' \leq v'$:

$$\frac{\mathbb{P}\big(X_i \leq u' \mid E(D)\big)\mathbb{P}\big(E(D)\big)}{\mathbb{P}(X_i \leq u')} \leq \frac{\mathbb{P}(X_i \leq v' \mid E(D))\mathbb{P}(E(D))}{\mathbb{P}(X_i \leq v')}. \tag{C.2}$$

where $E(D)$ is the event $\big\{(X_1, \dots, X_m) \in D\big\}$.

Theorem C.2.1. *Let $G_1, \dots, G_m$ be m nondecreasing functions. Then $G(X_1, \dots, X_n) = \big(G_1(X_1), \dots, G_m(X_m)\big)$ is wPRDS on $\mathcal{H}_0$.*

Proof. Let $D \subseteq \mathbb{R}^n$ nondecreasing and $i \in \mathcal{H}_0$. Let $u, v \in \{w : \mathbb{P}(G_i(X_i) \leq w) > 0\}$ and assume that $u \leq v$. Finally note $A_u = \sup G_i^{-1}\big((-\infty, u]\big)$. Remark that $A_u \leq A_v$, and that $G^{-1}(D)$ is a nondecreasing set.
We first show that if $G_i^{-1}\big((-\infty, u]\big) = (-\infty, A_u)$, then,

$$\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq u\big) \leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_u\big).$$

Note that $\{G_i(X_i) \leq u\} = \{X_i < A_u\} = \bigcup_{\varepsilon > 0} \{X_i \leq A_u - \varepsilon\}$, hence from (C.2) with $u' = A_u - \varepsilon$ and $v' = A_u$, and by measure continuity, we have $\mathbb{P}\big(E(G^{-1}(D)) \mid G_i(X_i) \leq u\big) \leq \mathbb{P}(E(G^{-1}(D)) \mid X_i \leq A_u)$. Therefore, since this inequality is trivially verified when $G_i^{-1}\big((-\infty, u]\big) = (-\infty, A_u]$, the inequality

$$\mathbb{P}(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq u) \leq \mathbb{P}(E\big(G^{-1}(D)\big) \mid X_i \leq A_u), \tag{C.3}$$

holds in all cases.

We then consider the following cases:

- Assume $A_u = A_v$. Then $G_i^{-1}\big((-\infty, u]\big) \subseteq G_i^{-1}\big((-\infty, v]\big)$. If
$$G_i^{-1}\big((-\infty, u]\big) \subset G_i^{-1}\big((-\infty, v]\big),$$
then $G_i^{-1}\big((-\infty, u]\big) = (-\infty, A_u)$ and $G_i^{-1}\big((-\infty, v]\big) = (-\infty, A_u]$. Hence
$$\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq u\big) \leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq v\big)$$
holds by (C.3). Otherwise, if $G_i^{-1}\big((-\infty, u]\big) = G_i^{-1}\big((-\infty, v]\big)$, then
$$\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq u\big) = \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq v\big).$$

- Assume $A_u < A_v$. Then there is some $\epsilon > 0$ such that $A_u < A_v - \epsilon$. We show that $\mathbb{P}(E(G^{-1}(D)) \mid X_i \leq A_v - \epsilon) \leq \mathbb{P}(E(G^{-1}(D)) \mid G_i(X_i) \leq v)$. Indeed, if $G_i^{-1}\big((-\infty, v]\big) = (-\infty, A_v]$, then it is true by wPRDS property. Otherwise, if $G_i^{-1}\big((-\infty, v]\big) = (-\infty, A_v)$, then for all $\epsilon > \delta > 0$,

$$\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_v - \epsilon\big) \leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_v - \delta\big).$$

  By measure continuity, when δ goes to 0 in this last inequality, we have

$$\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_v - \epsilon\big) \leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq v\big).$$

  Hence we have

$$\begin{aligned}\mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq u\big) &\leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_u\big) \\ &\leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid X_i \leq A_v - \epsilon\big) \\ &\leq \mathbb{P}\big(E\big(G^{-1}(D)\big) \mid G_i(X_i) \leq v\big),\end{aligned}$$

  where the second inequality follows from the wPRDS property of $(X_i)_{1 \leq i \leq n}$.

Hence $G(X_1, \ldots, X_n)$ is wPRDS on $\mathcal{H}_0$. □

The previous theorem allows to generalize the $\mathrm{BH}+$ procedure to any nondecreasing functions $G_1, \ldots, G_m$ instead of $G = (F^*, \ldots, F^*)$ by a simple application of the Benjamini-Hochberg procedure on the p-values $(G_1(p_1), \ldots, G_m(p_m))$, assuming that for all i, $G_i(p_i)$ is a valid p-value, that is, a super-uniform random variable under the null hypothesis H_i. Furthermore, it allows to apply a different function for all p-values and then apply the BH procedure.

Appendix D

How to fission a Bernoulli variable?

This appendix is dedicated to the fission of Bernoulli random variables, that is the construction of two independent Bernoulli random variables from a single Bernoulli random variable such that one can recover the original Bernoulli random variable from the two new ones, in the same spirit than Leiner et al. (2025).

Theorem D.0.1 (Symmetric version)**.** *Let $p_0 \in (0,1)$, $\rho_0 =:= \frac{1-p_0}{p_0}$ and $B \sim \mathcal{B}(p_0)$. Then, for $(i,j) \in \{0,1\}$, denote by $\alpha_{i,j}$ scalars verifying the following properties :*

- $\alpha_{0,0} \in [0,1]$,
- $\alpha_{1,0} = \alpha_{0,1} = \rho_0\Big(\sqrt{1 + \frac{1+\alpha_{0,0}}{\rho_0} + \frac{\alpha_{0,0}}{\rho_0^2}} - 1\Big) - \alpha_{0,0}$,
- $\alpha_{1,1} = 1 - (2\alpha_{1,0} + \alpha_{0,0})$.

It follows that, for $U \sim \mathcal{U}([0,1])$ independent of B,

$$B_1 = B\big(\mathbb{1}_{\{\alpha_{0,0} \le U \le \alpha_{0,0}+\alpha_{1,0}\}} + \mathbb{1}_{\{\alpha_{0,0}+2\alpha_{1,0} \le U \le 1\}}\big), \text{ and}$$

$$B_2 = B\mathbb{1}_{\{\alpha_{0,0}+\alpha_{1,0} \le U \le 1\}}$$

are independent Bernoulli variables with parameter $p_0\big(1 - (\alpha_{0,0} + \alpha_{1,0})\big)$.

Proof. Observe that, because B_1 and B_2 are Bernoulli variables, B_1 and B_2 are independent if and only if $\mathbb{P}(B_2 = 0 \mid B_1 = 1) = \mathbb{P}(B_2 = 0 \mid B_1 = 0)$.

$$\begin{aligned}
\mathbb{P}(B_2 = 0 \mid B_1 = 1) &= \frac{\mathbb{P}(B_1 = 1, B_2 = 0)}{\mathbb{P}(B_1 = 1)} \\
&= \frac{\mathbb{P}\big(U \in [\alpha_{0,0}, \alpha_{0,0} + \alpha_{0,1}], B = 1\big)}{\mathbb{P}\big(U \in [\alpha_{0,0}, \alpha_{0,0} + \alpha_{0,1}] \cup [\alpha_{0,0} + 2\alpha_{0,1}, 1], B = 1\big)} \\
&= \frac{\alpha_{0,1}}{1 - (\alpha_{0,0} + \alpha_{0,1})}, \text{ using independence between } B \text{ and } U. \\
\mathbb{P}(B_2 = 0 \mid B_1 = 0) &= \frac{\mathbb{P}(B_1 = 0, B_2 = 0)}{\mathbb{P}(B_1 = 0)} \\
&= \frac{\mathbb{P}\big(U < \alpha_{0,0}, B = 1\big) + \mathbb{P}\big(B = 0\big)}{\mathbb{P}\big(B = 0\big) + \mathbb{P}\big(U \in [0, \alpha_{0,0}) \cup (\alpha_{0,0} + \alpha_{1,0}, \alpha_{0,0} + 2\alpha_{1,0}), B = 1\big)} \\
&= \frac{p_0\alpha_{0,0} + (1 - p_0)}{(1 - p_0) + p_0(\alpha_{0,0} + \alpha_{0,1})}.
\end{aligned}$$

Denote $S_0 := \alpha_{0,0} + \alpha_{1,0}$.

Then B_1 and B_2 are independent *iif* $\frac{S_0-\alpha_{0,0}}{1-S_0} = \frac{p_0\alpha_{0,0}+(1-p_0)}{(1-p_0)+p_0S_0}$ *iif* $S_0^2 + 2\rho_0 S_0 - (\alpha_{0,0} + \alpha_{0,0}\rho_0 + \rho_0) = 0$.
This equation admits a unique nonnegative solution, $S_0 = -\rho_0 + \rho_0\sqrt{1 + \frac{1+\alpha_{0,0}}{\rho_0} + \frac{\alpha_{0,0}}{\rho_0^2}}$.

Finally, we show that $S_0 - \alpha_{0,0} \in [0,1]$ and that $1 - 2S_0 + \alpha_{0,0} \in [0,1]$.
When ρ_0 goes to $+\infty$, S_0 goes to $\frac{1}{2}(1+\alpha_{0,0})$ and when ρ_0 goes to 0, S_0 goes to $\sqrt{\alpha_{0,0}}$.
Studies on S_0 derivative with respect to ρ_0, $\frac{\partial S_0}{\partial \rho_0}(\rho_0) = \frac{1+\frac{\alpha_{0,0}+1}{2\rho_0}}{\sqrt{1+\frac{\alpha_{0,0}+1}{\rho_0}+\frac{\alpha_{0,0}}{\rho_0^2}}} - 1$, leads to a nondecreasing property of S_0 *w.r.t.* ρ_0 on $[0,+\infty)$ (because nonnegativeness of this derivative is equivalent of the nonnegativeness of $(\alpha_{0,0}-1)^2$). Hence,

$$S_0 - \alpha_{0,0} \in \left[\sqrt{\alpha_{0,0}}\big(1-\sqrt{\alpha_{0,0}}\big), \frac{1}{2}(1-\alpha_{0,0})\right] \subseteq [0,1]$$

and

$$1 - 2S_0 + \alpha_{0,0} \in \left[0, \big(1-\sqrt{\alpha_{0,0}}\big)^2\right] \subseteq [0,1].$$

thus B_1 and B_2 are independent Bernoulli variables by definition of $\alpha_{1,0}$.

□

Remark D.0.2. *When $\alpha_{0,0} = 0$ and one can recover B by the formula $B = \max(B_1, B_2)$. Further, $\alpha_{0,1} = \rho_0\sqrt{1+\frac{1}{\rho_0}} - \rho_0 < 1/2$ by increasing property of S_0 with respect to ρ_0, hence it is possible to have $B_1 = 1 = B_2$ without losing any information on B.*

Theorem D.0.3 (Asymmetric version)**.** *Let $p_0 \in (0,1)$, $\rho_0 =:= \frac{1-p_0}{p_0}$ and $B \sim \mathcal{B}(p_0)$. Then, for $(i,j) \in \{0,1\}$, denote by $\alpha_{i,j}$ nonnegative scalars verifying the following properties :*

- $\alpha_{0,0} \in [0,1]$,
- $\alpha_{1,0} \leq \frac{1}{2}\Big(\sqrt{(\alpha_{0,0}-\rho_0)^2 + 4(\alpha_{0,0}+\rho_0-\alpha_{0,0}^2)} - (\alpha_{0,0}+\rho_0)\Big)$,
- $\alpha_{0,1} = 1 - \alpha_{0,0} - \alpha_{1,0}\frac{\rho_0+\alpha_{0,0}+\alpha_{1,0}}{\rho_0+\alpha_{0,0}}$,
- $\alpha_{1,1} = 1 - (\alpha_{0,1}+\alpha_{1,0}+\alpha_{0,0})$.

It follows that, for $U \sim \mathcal{U}([0,1])$ independent of B, $B_1 = B\big(\mathbb{1}_{\{\alpha_{0,0}\leq U\leq \alpha_{0,0}+\alpha_{1,0}\}} + \mathbb{1}_{\{1-\alpha_{1,1}\leq U\leq 1\}}\big)$ and $B_2 = B\mathbb{1}_{\{\alpha_{0,0}+\alpha_{1,0}\leq U\leq 1\}}$ are independent Bernoulli variables with parameters $p_0\big(1-(\alpha_{0,0}+\alpha_{0,1})\big)$ and $p_0\big(1-(\alpha_{0,0}+\alpha_{1,0})\big)$ respectively.

The proof of this theorem is similar to the proof of the symmetric version, and is omitted for brevity.